Compton scattering at one loop: A detailed derivation

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Abstract

Roman N. Lee, Matthew D. Schwartz and Xiaoyuan Zhang published an analytic result for the Compton total cross section at next-to-leading order in 2021 [2]. This work independently recalculates and verifies that result. The calculation starts from the QED amplitudes and performs their integrations explicitly. The result includes the one-loop virtual correction to electron–photon scattering and the emission of a second final-state photon, integrated over their full final-state phase spaces. The electron mass is kept exactly. Electron–positron pair production is a separate process and is excluded from this total.

Starting from the Feynman rules, we carry out the loop and phase-space integrations analytically. A photon mass regulates the infrared singularities, whose cancellation is shown after matching the virtual and real contributions in a common convention. On-shell renormalization fixes the mass, charge, and external-field normalizations. The final cross section is a finite combination of logarithms and polylogarithms, including generalized polylogarithms defined by iterated integrals of specified kernels. No loop-parameter or phase-space integral remains to be evaluated numerically.

We give the intermediate primitives and endpoint terms, simplify the analytic result, and derive its low- and high-energy expansions with explicit remainder orders. Published total-cross-section formulas enter only in the subsequent analytic comparison. Independent numerical integration provides a separate cross-check of the analytic result.

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Chapter 1

Physical setup and principal results

The analytic Compton total cross section at next-to-leading order is already known from the work of Roman N. Lee, Matthew D. Schwartz and Xiaoyuan Zhang [2]. Our purpose is an independent recalculation and verification of their result. We derive the virtual and real amplitudes from the Feynman rules and integrate them with exact electron-mass dependence. The resulting expression in logarithms and polylogarithms contains no unevaluated loop-parameter or phase-space integral. This chapter states the observable, conventions, and complete result before the detailed derivation. The published coefficients are used only for the final comparison, never as boundary data or inputs to the integration.

1.1 Observable and conventions

We consider unpolarized scattering on a free electron,

$$e^-(p)+\gamma(k)\longrightarrow e^-(p')+\gamma(k'),\qquad e^-(p)+\gamma(k)\longrightarrow e^-(p')+\gamma(k_1)+\gamma(k_2). \tag{1.1}$$

The electron pole mass is m, its charge is $-e_{\mathrm{T}}$, and $\alpha_{\mathrm{T}}=e_{\mathrm{T}}^2/(4\pi)$ is defined in the Thomson limit. We average over the two incoming electron spins and two photon polarizations and sum over all final polarizations. All allowed photon energies and directions are integrated. The distinct pair-production channel $e^-\gamma\to e^-e^-e^+$ is excluded. This is the same observable as the photonic total of Reference [2]. The massive polarized calculation of Denner and Dittmaier also treats pair production separately [1].

We use natural units $\hbar=c=1$ and the metric $\eta_{\mu\nu}=\mathrm{diag}(1,-1,-1,-1)$. Define the physical invariants and the rest-frame photon energy ω by

$$s_{\mathrm{phys}}=(p+k)^2=m^2+2m\omega,\qquad t=(p-p')^2,\qquad u=(p-k')^2, \tag{1.2}$$

$$X=\frac{s_{\mathrm{phys}}}{m^2}-1=\frac{2\omega}{m}>0,\qquad Y=\sqrt{\frac{X}{X+4}},\qquad q=\frac{\sqrt{X+4}-\sqrt{X}}{\sqrt{X+4}+\sqrt{X}}. \tag{1.3}$$

The variable q is unrelated to electric charge. All displayed square roots are positive. For the two-body process, $s_{\mathrm{phys}}+t+u=2m^2$.

Ultraviolet integrals are regulated in $d=4-2\epsilon_{\mathrm{UV}}$ dimensions with scale μ. A photon mass $0<\lambda\ll m$ regulates virtual soft momenta and the matching real soft integral. In Feynman gauge the propagators and electron vertex are

$$S_F(P)=\frac{\mathrm{i}(\not{P}+m)}{P^2-m^2+\mathrm{i}0},\qquad D_{\rho\sigma}(\ell)=\frac{-\mathrm{i}\eta_{\rho\sigma}}{\ell^2-\lambda^2+\mathrm{i}0},\qquad +\mathrm{i}\mu^{\epsilon_{\mathrm{UV}}}e_{\mathrm{T}}\gamma^\rho. \tag{1.4}$$

Here $\not{P} = \gamma^\mu P_\mu$, with $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$. The photon symbol omits F but uses the same Feynman boundary conditions. External hard photons are massless. Terms vanishing as $\lambda/m \to 0$ are omitted only after the singular terms have been identified. The real and virtual contributions are combined before removing the infrared regulator.

Dimensional and integration variables. Chapters 1 and 2 use dimensional momenta and photon mass. In the integration chapters, $s = s_{\mathrm{phys}}/m^2$, $t = t_{\mathrm{phys}}/m^2$, $u = u_{\mathrm{phys}}/m^2$, and $\lambda = \lambda_{\mathrm{phys}}/m$. The external energy variables satisfy $H = X = s - 1$. In formulas for elastic angular integration, c denotes the center-of-mass photon scattering cosine. The laboratory angle is denoted by θ. Lowercase integration coordinates are defined locally and are not identified with X. The final evaluator uses $C_1(X)$, while the sector assembly uses $C_{\mathrm{total}}(s) = C_1(s-1)$, in both cases multiplying $\alpha_{\mathrm{T}}^3/m^2$. The labels UV, UW, and VW in the real-emission calculation name pairs of angular denominators. They do not denote ultraviolet counterterms.

1.2 On-shell renormalization constants

The renormalized electron propagator has its pole at m with unit residue. The transverse photon residue is normalized at zero momentum squared, and the charge is fixed by the zero-momentum-transfer electron vertex. These are the QED on-shell conditions of the general formalism reviewed in Section 3 of Denner [4]. For one electron species, use the bare-to-renormalized definitions

$$\psi_0 = Z_2^{1/2}\psi, \qquad A_{0\mu} = Z_3^{1/2}A_\mu, \qquad m_0 = Z_m m, \tag{1.5}$$

$$e_0 = \mu^{\epsilon_{\mathrm{UV}}} Z_e e_{\mathrm{T}}, \qquad Z_1 = Z_e Z_2 Z_3^{1/2}, \qquad Z_j = 1 + \delta_j + \mathcal{O}(\alpha_{\mathrm{T}}^2). \tag{1.6}$$

Here Z_1 multiplies the electron–photon interaction vertex. Thus Z_m relates the bare mass to the pole mass, while the mass coefficient in the Lagrangian is $Z_2 Z_m$. Define

$$U_m = \frac{1}{\epsilon_{\mathrm{UV}}} - \gamma_E + \ln 4\pi + \ln\frac{\mu^2}{m^2}, \qquad C = \frac{\alpha_{\mathrm{T}}}{4\pi}. \tag{1.7}$$

Here γ_E is Euler's constant. In the photon-mass convention the one-loop constants are

$$\delta_m = -C(3U_m + 4) + \alpha_{\mathrm{T}} o(1), \tag{1.8}$$

$$\delta_2 = -C\left[U_m + 4 + 2\ln\frac{\lambda^2}{m^2}\right] + \alpha_{\mathrm{T}} o(1), \tag{1.9}$$

$$\delta_1 = \delta_2, \qquad \delta_3 = -\frac{\alpha_{\mathrm{T}}}{3\pi}U_m, \qquad \delta_e = \frac{\alpha_{\mathrm{T}}}{6\pi}U_m. \tag{1.10}$$

The $o(1)$ terms vanish as $\lambda/m \to 0$ at fixed μ/m. The Ward identity gives $Z_1 = Z_2$ and hence $Z_e = Z_3^{-1/2}$. The field constants above refer to Feynman gauge. They are not separately observable quantities. These constants agree with the one-electron QED specialization of Denner's on-shell formulas, Equations (3.19), (3.20), and (3.32) of Reference [4]. His photon field counterterm δZ_{AA} is our δ_3, and in QED his charge relation reduces to $\delta_e = -\delta_3/2$.

The signs can be checked directly against the subtraction constants used later. With $\Sigma(P) = a(P^2)\not{P} + m b(P^2)$, let

$$c_m = m[a(m^2) + b(m^2)], \qquad c_Z = a(m^2) + 2m^2[a'(m^2) + b'(m^2)]. \tag{1.11}$$

Primes differentiate the squared argument. The counterterm insertion is $\mathrm{i}[\delta_2(\not{P}-m)-m\delta_m]$, so

$$m\delta_m = -c_m, \qquad \delta_2 = c_Z, \qquad \Sigma_R = \Sigma - c_m - (\not{P}-m)c_Z. \tag{1.12}$$

In this convention the subtraction constants are $c_m/m = C(3U_m+4)+\alpha_{\mathrm{T}}o(1)$ and $c_Z = -C[U_m+4+2\ln(\lambda^2/m^2)]+\alpha_{\mathrm{T}}o(1)$. The vertex counterterm is $c_Z\gamma^\rho$ after factoring out the charge. No further external-electron residue is applied to amplitudes already normalized this way. For the two external photons the charge and photon factors obey $Z_e^2 Z_3 = 1$. There is no internal Born photon on which to insert vacuum polarization in Compton scattering.

1.3 The complete total cross section

Let $\mathcal{M}_0$, $\mathcal{M}_1$, and $\mathcal{M}_{\gamma\gamma}$ denote the Born, renormalized one-loop, and double-Compton amplitudes. Their contributions have the normalization

$$\begin{aligned}\sigma_0+\sigma_1 = \frac{1}{4p\cdot k}\lim_{\lambda\to 0}\Bigg\{ &\int \mathrm{d}\Phi_2 \left[\overline{|\mathcal{M}_0|^2} + 2\,\mathrm{Re}\,\overline{\mathcal{M}_0^*\mathcal{M}_1}\right] \\ &+ \frac{1}{2!}\int \mathrm{d}\Phi_3 \overline{|\mathcal{M}_{\gamma\gamma}|^2}\Bigg\}.\end{aligned} \tag{1.13}$$

The bar denotes the spin and polarization average and sum specified above, $\mathrm{d}\Phi_n$ is Lorentz-invariant n-body phase space, and the real soft region uses the same photon-mass convention as the virtual term. The factor $1/2!$ accounts for the identical final photons. For the coefficient $A(s)$ of the virtual infrared logarithm, with $s = 1+X = s_{\mathrm{phys}}/m^2$,

$$\frac{m^2\sigma_{1,V}}{\alpha_{\mathrm{T}}^3} = A(1+X)\ln\frac{\lambda}{m} + V_{\mathrm{fin}}(X), \qquad \frac{m^2\sigma_{1,R}}{\alpha_{\mathrm{T}}^3} = -A(1+X)\ln\frac{\lambda}{m} + R_{\mathrm{fin}}(X). \tag{1.14}$$

Here V_{fin} and R_{fin} are finite coefficients in the same convention, and terms vanishing with λ are understood. The finite terms refer to a common photon-mass regulator. Their sum has the form

$$\boxed{\sigma_{\mathrm{inc}}^{\mathrm{NLO}}(X) = \frac{\pi\alpha_{\mathrm{T}}^2}{m^2}B(X) + \frac{\alpha_{\mathrm{T}}^3}{m^2}C_1(X) + \frac{1}{m^2}O(\alpha_{\mathrm{T}}^4).} \tag{1.15}$$

The subscript inc denotes the sum over the one-photon and two-photon final states specified above. The remainder is at fixed $X > 0$. Put $r_\omega = X/2$. The Born coefficient is

$$\begin{aligned}B(X) = 2\Bigg[&\frac{1+r_\omega}{r_\omega^2}\left(\frac{2(1+r_\omega)}{1+2r_\omega} - \frac{\ln(1+2r_\omega)}{r_\omega}\right) \\ &+ \frac{\ln(1+2r_\omega)}{2r_\omega} - \frac{1+3r_\omega}{(1+2r_\omega)^2}\Bigg].\end{aligned} \tag{1.16}$$

This is the integrated Klein–Nishina result, also written in Equation (5) of Reference [2].

To state the entire NLO coefficient, let w denote an ordered list of kernel labels, called a word, and let $\varnothing$ denote the empty list. Define the generalized polylogarithms (GPLs) recursively by

$$G(a,w;X) = \int_0^X \mathrm{d}t\, f_a(t)G(w;t), \qquad G(\varnothing;X) = 1, \tag{1.17}$$

$$f_a(t) = \frac{1}{t-a} \quad (a = 0,1,-1,-2,-4), \qquad f_r(t) = \frac{1}{\sqrt{t(t+4)}}, \tag{1.18}$$

$$G(\underbrace{0,\ldots,0}_{n};X) = \frac{\ln^n X}{n!}. \tag{1.19}$$

The last line fixes the logarithmically singular all-zero case. The length of a word is its weight. Products of these integrals obey shuffle identities: interleave the two lists in all ways preserving the order within each list, counting multiplicities. This follows by partitioning the product of their ordered integration domains. The letter r labels a square-root kernel, not a number to substitute for a. The integrals follow the positive real axis. In the words containing 1, the apparent singularity is removable because the inner factor is $\ln t/(t-1)$, with limit one at $t=1$. All other nonzero letters lie outside the integration path. Classical polylogarithms are defined by $\mathrm{Li}_n(z)=\sum_{k=1}^{\infty} z^k/k^n$ for $|z|<1$ and continuation elsewhere. Their real boundary values below agree with these real iterated integrals.

Products of elementary logarithms can make the collected result shorter without introducing additional polylogarithms. Define $L=\ln(1+X)$ and $M=\ln X$. The two GPLs with the same coefficient combine into

$$\mathcal{P}(X)=G(0,-1,0;X)+G(0,1,0;X)=\tfrac{1}{2}\left[\mathrm{Re}\,\mathrm{Li}_3(X^2)-M\,\mathrm{Re}\,\mathrm{Li}_2(X^2)\right]. \tag{1.20}$$

For the derivation, first take $0<X<1$. The shuffle relation $G(0)G(0,a)=2G(0,0,a)+G(0,a,0)$, with $G(0,a;X)=-\mathrm{Li}_2(X/a)$ and $G(0,0,a;X)=-\mathrm{Li}_3(X/a)$, gives

$$G(0,a,0;X)=2\,\mathrm{Li}_3(X/a)-M\,\mathrm{Li}_2(X/a),\qquad a=\pm1. \tag{1.21}$$

Summing the two signs and using $\mathrm{Li}_n(z)+\mathrm{Li}_n(-z)=2^{1-n}\,\mathrm{Li}_n(z^2)$ gives the stated result. Both shuffle multiplicities and this duplication factor are needed for its factor $1/2$. For $X>1$, continue all functions on the same edge of the principal cut. On either edge, the imaginary parts of $\mathrm{Li}_2(X)$ and $\mathrm{Li}_3(X)$ are respectively $\pm\pi\ln X$ and $\pm\pi\ln^2 X/2$, with the same sign. These follow by integrating their derivatives from $X=1$, and they cancel in $2\,\mathrm{Li}_3(X)-M\,\mathrm{Li}_2(X)$. The analogous cancellation holds for the paired expression at argument X^2. Thus both edges give the same real function and the continuation adds no branch constant. The following equivalent form uses only real polylogarithms with arguments in $(0,1]$:

$$\mathcal{P}(X)=\begin{cases}\frac{1}{2}[\mathrm{Li}_3(X^2)-M\,\mathrm{Li}_2(X^2)], & 0<X\le 1,\\ \frac{1}{2}[\mathrm{Li}_3(X^{-2})+M\,\mathrm{Li}_2(X^{-2})]+\frac{1}{3}M^3+\frac{1}{6}\pi^2M, & X\ge 1.\end{cases} \tag{1.22}$$

To derive the second line, put $t=\ln X>0$, $D(t)=\mathrm{Li}_2(e^{-2t})$ and $T(t)=\mathrm{Li}_3(e^{-2t})$. Their derivatives are $D'(t)=2\ln(1-e^{-2t})$ and $T'(t)=-2D(t)$. The identity $\ln(e^{2t}-1)=2t+\ln(1-e^{-2t})$ then gives

$$\mathrm{Re}\,\mathrm{Li}_2(e^{2t})=\frac{\pi^2}{3}-2t^2-D(t), \tag{1.23}$$

$$\mathrm{Re}\,\mathrm{Li}_3(e^{2t})=T(t)-\frac{4}{3}t^3+\frac{2\pi^2}{3}t. \tag{1.24}$$

Indeed, the derivatives of the two sides are respectively $-2\ln(e^{2t}-1)$ and $2\,\mathrm{Re}\,\mathrm{Li}_2(e^{2t})$. Write $\zeta(n)=\sum_{k=1}^{\infty}k^{-n}$ for the Riemann zeta function at integers $n>1$. Their values at $t=0$ agree because $\mathrm{Li}_2(1)=\pi^2/6$ and $\mathrm{Li}_3(1)=\zeta(3)$. These derivative and boundary conditions fix both identities and give $\mathcal{P}(1)=\zeta(3)/2$. The sole apparent rational pole at $X=1$ can also be removed before evaluation. Define the continuous elementary function

$$R_{\log}(X)=\begin{cases}\ln X/(X-1), & X\ne 1,\\ 1, & X=1,\end{cases}\qquad R_{\log}(1+h)=1-\frac{h}{2}+\frac{h^2}{3}+O(h^3). \tag{1.25}$$

The series follows by dividing the Taylor series of $\ln(1+h)$ by h, which converges for $|h|<1$. Extracting the common power X^{-3} gives the exact total coefficient as

$$C_1(X)=\frac{1}{X^3}\left[R_{\log}(X)+\mathcal{E}(X,L,M)+\sum_{j=1}^{13}b_j(X,Y,L,M)\mathcal{F}_j(X)\right]. \tag{1.26}$$

The elementary part is

$$\mathcal{E}=e_0L^3+e_1L^2M+e_2L^2+e_3LM+e_4L+e_5M+e_6. \tag{1.27}$$

$$e_0=-\frac{X^2-16X-23}{3}. \tag{1.28}$$

$$e_1=4X^2-2X-12. \tag{1.29}$$

$$e_2=-2X^3-\frac{11X^2}{2}+\frac{133X}{2}+36+\frac{96}{X+4}-\frac{6}{X+1}+\frac{1}{(X+1)^2}. \tag{1.30}$$

$$e_3=X^2+5X-10+\frac{12}{X+1}-\frac{4}{(X+1)^2}. \tag{1.31}$$

$$e_4=\frac{37X^2}{4}-32X-30-\frac{15}{X+1}-\frac{1}{(X+1)^2}. \tag{1.32}$$

$$e_5=-\frac{14X^5+31X^4+14X^3-6X^2-8X-2}{2\,(X+1)^3}. \tag{1.33}$$

$$e_6=-\frac{X\left(273X^3-982X^2-2960X-1744\right)}{24\,(X+1)^2}. \tag{1.34}$$

The thirteen remaining functions and their coefficients are

$$\mathcal{F}_1=G(0,-1;X). \tag{1.35}$$

$$b_1=X^3+X^2-43X+6-\frac{96}{X+4}-\frac{6}{X+1}+\frac{2}{(X+1)^2}-2LX\,(2X+5). \tag{1.36}$$

$$\mathcal{F}_2=G(-2,-1;X). \tag{1.37}$$

$$b_2=-4X^2-80X-68-\frac{128}{X+4}+\frac{8}{X+1}-\frac{4}{(X+1)^2}. \tag{1.38}$$

$$\mathcal{F}_3=G(1,0;X),\qquad b_3=-X^2+X-2. \tag{1.39}$$

$$\mathcal{F}_4=G(r,-1;X). \tag{1.40}$$

$$b_4=Y\left(3X^3+18X^2+44X-8-\frac{64}{X}\right)-3MY\,(X-2)\,(X+4). \tag{1.41}$$

$$\mathcal{F}_5=G(0,-1,-1;X),\qquad b_5=-5X^2-8X+32. \tag{1.42}$$

$$\mathcal{F}_6=G(-1,-2,-1;X),\qquad b_6=8\left(X^2-4X-6\right). \tag{1.43}$$

$$\mathcal{F}_7=G(-4,r,-1;X),\qquad b_7=Y\left(-6X^2+72+\frac{48}{X}\right). \tag{1.44}$$

$$\mathcal{F}_8=\mathcal{P}(X),\qquad b_8=2X^2+4X-12. \tag{1.45}$$

$$\mathcal{F}_9=G(r,0,-1;X),\qquad b_9=Y\left(-5X^2+14X+88+\frac{48}{X+4}+\frac{48}{X}\right). \tag{1.46}$$

$$\mathcal{F}_{10} = G(r, -1, 0; X), \qquad b_{10} = -3Y\,(X-2)\,(X+4)\,. \tag{1.47}$$

$$\mathcal{F}_{11} = G(r, -1, -1; X), \qquad b_{11} = Y\left(15X^2 - 18X - 216 - \frac{96}{X+4} - \frac{96}{X}\right). \tag{1.48}$$

$$\mathcal{F}_{12} = G(r, -2, -1; X), \qquad b_{12} = Y\left(-16X^2 + 32X + 256 + \frac{64}{X+4} + \frac{192}{X}\right). \tag{1.49}$$

$$\mathcal{F}_{13} = G(r, r, -1; X), \qquad b_{13} = 9X^2 - 24. \tag{1.50}$$

At $X = 1$ the elementary quotient $\ln X/(X-1)$ is assigned its continuous value one. These definitions specify the full physical function for every $X > 0$. Equation (1.26) follows from our integrated sectors by shuffle identities and rational coefficient collection. After conversion to a common basis, it agrees coefficient by coefficient with Equation (6) of Lee, Schwartz and Zhang [2]. This equality is a result of the recalculation, not an input to it.

1.4 Asymptotic limits

The following limits are taken in the complete infrared-finite result. For $X \to 0^{+}$,

$$\sigma_0 = \frac{\pi\alpha_{\mathrm{T}}^2}{m^2}\left[\frac{8}{3} - \frac{8}{3}X + \frac{52}{15}X^2 - \frac{133}{30}X^3 + \mathcal{O}(X^4)\right], \tag{1.51}$$

$$\begin{aligned}\sigma_1 = \frac{\alpha_{\mathrm{T}}^3}{m^2}\Bigg\{ & X^2\left[-\frac{16}{9}\ln X + \frac{7}{15}\right] + X^3\left[\frac{4}{3}\ln X - \frac{113}{90}\right] \\ & + X^4\left[-\frac{25}{9}\ln X + \frac{203}{125}\right] + \mathcal{O}\Big(X^5|\ln X|\Big)\Bigg\}.\end{aligned} \tag{1.52}$$

The correction vanishes, leaving the Thomson cross section $8\pi\alpha_{\mathrm{T}}^2/(3m^2)$. This is consistent with Thirring's low-energy theorem, whose all-orders formulation and generalizations are discussed by Dittmaier [5]. The $X^2\ln X$ and X^2 coefficients agree with Equation (10) of Lee, Schwartz and Zhang [2]. The two further orders shown here follow from the full expression stated above.

For $X \to \infty$, write $\ell = \ln X$. Then

$$\sigma_0 = \frac{\pi\alpha_{\mathrm{T}}^2}{m^2}\left[\frac{2\ell+1}{X} + \frac{18 - 8\ell}{X^2} + \mathcal{O}\Big(\frac{\ell}{X^3}\Big)\right], \tag{1.53}$$

$$\sigma_1 = \frac{\alpha_{\mathrm{T}}^3}{m^2}\left\{\frac{1}{X}\left[\frac{\ell^3}{3} - \frac{\ell^2}{2} + \frac{17\ell}{4} - \frac{75}{8} - \frac{\pi^2}{2} + 4\zeta(3)\right] + \mathcal{O}\left(\frac{\ell^3}{X^2}\right)\right\}. \tag{1.54}$$

Here $\zeta(3) = \sum_{n=1}^{\infty} n^{-3}$. All logarithmic coefficients and the finite constant agree with Equation (11) of Reference [2]. Replacing X by $s_{\mathrm{phys}}/m^2 = 1 + X$ in its leading high-energy term changes only the stated remainder. The finite constant is fixed by the endpoint values of our integrated GPLs. In particular,

$$\frac{\sigma_1}{\sigma_0} = \frac{\alpha_{\mathrm{T}}}{6\pi}\ell^2 + \mathcal{O}(\alpha_{\mathrm{T}}\ell). \tag{1.55}$$

Infrared cancellation therefore leaves an energy-dependent radiative correction. These are fixed-order results, with no resummation of the large energy logarithms.

Chapter 2

From the Feynman rules to the NLO integrands

The principal results in Chapter 1 summarize the calculation. They are not assumptions in the derivation that starts here. This chapter derives the Compton amplitudes and their parameter integrals from the QED vertices. The loop, energy, and angular integrations are performed on these amplitudes. The total in Equation (6) of Lee, Schwartz and Zhang [2] is reserved for a final comparison. No coefficient from that total is used to determine an integral or its integration constant here.

2.1 Conventions and diagram counting

Let p, k denote incoming electron and photon momenta, and p', k' the outgoing momenta. The electron mass is m, its charge is $-e_{\mathrm{T}}$, and $\alpha_{\mathrm{T}} = e_{\mathrm{T}}^2/(4\pi)$ is the Thomson coupling. Use the mostly-minus metric, the vertex $+\mathrm{i}e_{\mathrm{T}}\gamma^\mu$, and

$$p + k = p' + k', \qquad p^2 = p'^2 = m^2, \qquad k^2 = k'^2 = 0, \tag{2.1}$$

$$s = (p+k)^2, \qquad t = (p-p')^2, \qquad u = (p-k')^2. \tag{2.2}$$

We write $\epsilon = \epsilon_{\mathrm{UV}}$ and regulate ultraviolet integrals in $d = 4 - 2\epsilon$ dimensions, with coupling $\mu^\epsilon e_{\mathrm{T}}$, and soft photons by a mass λ. The propagators are

$$S_F(P) = \mathrm{i}S_0(P), \qquad S_0(P) = \frac{\not{P} + m}{P^2 - m^2 + \mathrm{i}0}, \qquad D_{\rho\sigma}(\ell) = \frac{-\mathrm{i}\eta_{\rho\sigma}}{\ell^2 - \lambda^2 + \mathrm{i}0}. \tag{2.3}$$

The photon symbol omits F but has Feynman boundary conditions. External photons remain massless. The Born amplitude is

$$\mathcal{M}_0 = -e_{\mathrm{T}}^2 \epsilon_\mu \epsilon'^*_\nu \bar{u}(p') R_0^{\nu\mu} u(p), \qquad R_0^{\nu\mu} = \gamma^\nu S_0(p+k)\gamma^\mu + \gamma^\mu S_0(p-k')\gamma^\nu. \tag{2.4}$$

At one loop the electron line has four photon vertices. Choosing the positions of the two external photons gives six choices and ordering those photons gives two choices. Of these twelve diagrams, four are external-electron self-energy diagrams. On-shell external normalization accounts for those four. The remaining eight are two internal self-energies, four vertices and two spanning photons. There is no exchanged Born photon on which to insert a vacuum polarization. The physical charge and real-photon normalization account for the photon two-point function at zero momentum squared.

2.2 Tensor loop integration

Define an integral whose factor 1/i removes the Minkowski loop factor. Here Γ denotes Euler's gamma function.

$$I_n(\Delta) = \frac{\mu^{2\epsilon}}{\mathrm{i}} \int \frac{\mathrm{d}^d L}{(2\pi)^d} \frac{1}{(L^2 - \Delta + \mathrm{i}0)^n} = \frac{(-1)^n \mu^{2\epsilon}}{(4\pi)^{d/2}} \frac{\Gamma(n - d/2)}{\Gamma(n)} (\Delta - \mathrm{i}0)^{d/2-n}. \tag{2.5}$$

To derive the last expression, first take $\Delta > 0$ and $\mathrm{Re}\, n > \mathrm{Re}\, d/2$. A Wick rotation supplies $\mathrm{i}(-1)^n$, and the Euclidean angular integral followed by $v = L_E^2$ gives

$$I_n(\Delta) = \frac{(-1)^n \mu^{2\epsilon}}{(4\pi)^{d/2}\Gamma(d/2)} \int_0^\infty \frac{v^{d/2-1} \mathrm{d}v}{(v + \Delta)^n}. \tag{2.6}$$

Scaling $v = \Delta z$ reduces the remaining integral to Euler's identity

$$\int_0^\infty \frac{z^{a-1} \mathrm{d}z}{(1+z)^{a+b}} = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \qquad \mathrm{Re}\, a, \mathrm{Re}\, b > 0, \tag{2.7}$$

with $a = d/2$ and $b = n - d/2$. Analytic continuation in d and in $\Delta - \mathrm{i}0$ then gives (2.5) with its causal boundary value. Odd powers of L vanish. Rotational invariance of the regulated integral gives

$$\int \mathrm{d}^d L\, L_\rho L_\sigma f(L^2) = \frac{\eta_{\rho\sigma}}{d} \int \mathrm{d}^d L\, L^2 f(L^2), \tag{2.8}$$

$$\frac{L^2}{(L^2 - \Delta)^n} = \frac{1}{(L^2 - \Delta)^{n-1}} + \frac{\Delta}{(L^2 - \Delta)^n}. \tag{2.9}$$

The scalar integral therefore also determines the quadratic numerators. Combine n denominators D_j with

$$\frac{1}{\prod_{j=1}^n D_j} = \Gamma(n) \int_{a_j \geq 0} \prod_j \mathrm{d}a_j \, \frac{\delta(1 - \sum_j a_j)}{(\sum_j a_j D_j)^n}. \tag{2.10}$$

The simplex measure is normalized to $1/(n-1)!$. This normalization is needed when checking pole coefficients.

2.3 Self-energy and residue subtraction

The insertion $-\mathrm{i}\Sigma(P)$ yields

$$\Sigma(P) = \frac{e_{\mathrm{T}}^2 \mu^{2\epsilon}}{\mathrm{i}} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{\gamma^\rho (\not{P} - \not{\ell} + m) \gamma_\rho}{[(P-\ell)^2 - m^2 + \mathrm{i}0](\ell^2 - \lambda^2 + \mathrm{i}0)}. \tag{2.11}$$

Use v for the electron denominator. Completing the square gives

$$v[(P-\ell)^2 - m^2] + (1-v)(\ell^2 - \lambda^2) = (\ell - vP)^2 - \Delta_\Sigma, \quad \Delta_\Sigma = vm^2 + (1-v)\lambda^2 - v(1-v)P^2. \tag{2.12}$$

After the shift $L = \ell - vP$, the gamma identities $\gamma^\rho \gamma^\alpha \gamma_\rho = (2-d)\gamma^\alpha$ and $\gamma^\rho \gamma_\rho = d$ leave

$$\Sigma(P) = \frac{e_{\mathrm{T}}^2 \mu^{2\epsilon}}{(4\pi)^{d/2}} \Gamma(\epsilon) \int_0^1 \mathrm{d}v \, \frac{(2-d)(1-v)\not{P} + dm}{(\Delta_\Sigma - \mathrm{i}0)^\epsilon} = a(P^2)\not{P} + mb(P^2). \tag{2.13}$$

For an explicit expansion define

$$C = \frac{\alpha_{\mathrm{T}}}{4\pi}, \qquad \mathcal{L}(\Delta) = \frac{1}{\epsilon} - \gamma_E + \ln 4\pi + \ln \frac{\mu^2}{\Delta - \mathrm{i}0}, \tag{2.14}$$

where γ_E is Euler's constant. Through the finite terms,

$$a(P^2) = C \int_0^1 \mathrm{d}v \, [-2(1-v)\mathcal{L}(\Delta_\Sigma) + 2(1-v)], \tag{2.15}$$

$$b(P^2) = C \int_0^1 \mathrm{d}v \, [4\mathcal{L}(\Delta_\Sigma) - 2]. \tag{2.16}$$

The last terms come from multiplying $d-4$ by the ultraviolet pole. They would be lost by doing the numerator algebra in four dimensions before subtraction.

To determine the pole and its residue, regard $\not{P}$ as a variable w near m. Then

$$\begin{aligned} a(w^2)w + mb(w^2) = {} & m[a(m^2) + b(m^2)] \\ & + (w-m)\{a(m^2) + 2m^2[a'(m^2) + b'(m^2)]\} \\ & + \mathcal{O}((w-m)^2). \end{aligned} \tag{2.17}$$

Here the primes differentiate the scalar squared argument. Subtract the first two coefficients,

$$c_m = m[a(m^2) + b(m^2)], \qquad c_Z = a(m^2) + 2m^2[a'(m^2) + b'(m^2)], \tag{2.18}$$

$$\Sigma_{\mathrm{R}}(P) = \Sigma(P) - c_m - (\not{P} - m)c_Z. \tag{2.19}$$

The renormalized propagator has a pole at m with unit residue. Writing $\Delta_0 = v^2 m^2 + (1-v)\lambda^2$ gives explicit subtraction integrals,

$$c_m = mC \int_0^1 \mathrm{d}v \, [2(1+v)\mathcal{L}(\Delta_0) - 2v], \tag{2.20}$$

$$c_Z = C \int_0^1 \mathrm{d}v \, \left[-2(1-v)\mathcal{L}(\Delta_0) + 2(1-v) + \frac{4m^2 v(1-v)(1+v)}{\Delta_0}\right]. \tag{2.21}$$

At fixed $\lambda > 0$ these are ordinary one-dimensional parameter integrals. The last term in c_Z displays the logarithmic soft sensitivity as $\lambda \to 0$.

2.3.1 Evaluation of the on-shell constants

The logarithmic integrals in the mass subtraction are integrable at $v = 0$, whereas the rational term in the residue subtraction is not integrable there after setting the photon mass to zero. To separate these two cases, put

$$r_\lambda = \frac{\lambda}{m}, \qquad d_0(v) = v^2 + (1-v)r_\lambda^2, \qquad U_m = \frac{1}{\epsilon} - \gamma_E + \ln 4\pi + \ln \frac{\mu^2}{m^2}. \tag{2.22}$$

Then $\mathcal{L}(\Delta_0) = U_m - \ln d_0(v)$. For the logarithmic terms the limit can be taken under the integral, giving

$$\frac{c_m}{mC} = 3U_m - 4\int_0^1 (1+v)\ln v \, \mathrm{d}v - 1 + o(1) \tag{2.23}$$

$$= 3U_m + 4 + o(1). \tag{2.24}$$

For the regular part of c_Z/C, define and evaluate

$$I_{\mathrm{reg}} = \int_0^1 [-2(1-v)\mathcal{L}(\Delta_0) + 2(1-v)]\,\mathrm{d}v \tag{2.25}$$

$$= -U_m + 4\int_0^1 (1-v)\ln v\,\mathrm{d}v + 1 + o(1) \tag{2.26}$$

$$= -U_m - 2 + o(1). \tag{2.27}$$

Here $o(1)$ means a term vanishing as $r_\lambda \to 0$ at fixed μ/m. The elementary moments used are $\int_0^1 \ln v\,\mathrm{d}v = -1$ and $\int_0^1 v\ln v\,\mathrm{d}v = -1/4$.

The remaining term must be integrated before this limit. Replacing $d_0(v)$ by $v^2 + r_\lambda^2$ changes its integral by $O(r_\lambda)$. For example, split the interval at $v = r_\lambda$ and $v = 1/2$. The denominator difference is vr_λ^2, so the absolute integrand difference is bounded by a constant times $r_\lambda^2 v^2/(v^2+r_\lambda^2)^2$ on the first two intervals and by a constant times r_λ^2 on the last. Its integral therefore vanishes. Consequently,

$$4\int_0^1 \frac{v(1-v^2)}{d_0(v)}\,\mathrm{d}v = 4\int_0^1 \frac{v-v^3}{v^2+r_\lambda^2}\,\mathrm{d}v + o(1) \tag{2.28}$$

$$= 2(1+r_\lambda^2)\ln\frac{1+r_\lambda^2}{r_\lambda^2} - 2 + o(1) \tag{2.29}$$

$$= -4\ln r_\lambda - 2 + o(1). \tag{2.30}$$

Combining the regular and singular parts fixes the finite constant as well as the infrared logarithm,

$$c_Z = -C[U_m + 4 + 4\ln r_\lambda] + C\,o(1). \tag{2.31}$$

To relate these subtractions to the Lagrangian, write $Z_j = 1 + \delta_j + O(\alpha_{\mathrm{T}}^2)$, $\psi_0 = Z_2^{1/2}\psi$, and $m_0 = Z_m m$. The electron quadratic counterterm is

$$\delta\mathcal{L}_e = \overline{\psi}[\delta_2(\mathrm{i}\not{\partial} - m) - m\delta_m]\psi. \tag{2.32}$$

Its insertion is $\mathrm{i}[\delta_2(\not{P} - m) - m\delta_m]$. Combining it with $-\mathrm{i}\Sigma$ gives $\Sigma_R = \Sigma - \delta_2(\not{P} - m) + m\delta_m$. Comparison with the pole and residue subtractions thus determines

$$\delta_m = -\frac{c_m}{m}, \qquad \delta_2 = c_Z. \tag{2.33}$$

This also explains why the field counterterm multiplies both the kinetic term and the renormalized mass term.

These mass and field counterterms agree with the diagonal QED limit of Denner's on-shell expressions, Equation (3.20) of Reference [4]. In his self-energy convention the left- and right-chiral scalar coefficients both equal $-a$, and the mass coefficient equals $-b$. His additive mass counterterm therefore equals $-m(a+b) = m\delta_m$, and either diagonal field counterterm equals $a + 2m^2(a'+b') = \delta_2$. This comparison is made after evaluating our parameter integrals.

2.4 The off-shell vertex

At a photon insertion let the electron momenta be P and P', with $q = P' - P$. The correction to the dimensionless vertex is

$$V^\rho(P', P) = \frac{e_{\mathrm{T}}^2\mu^{2\epsilon}}{\mathrm{i}}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\frac{\gamma^\alpha S_0(P'-\ell)\gamma^\rho S_0(P-\ell)\gamma_\alpha}{\ell^2 - \lambda^2 + \mathrm{i}0}. \tag{2.34}$$

Introduce $x, y, z \geq 0$, $x + y + z = 1$, for the two electron lines and the photon. With $\int_{\mathcal{S}_2} = \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y$,

$$K = xP' + yP, \qquad \Delta = (x+y)m^2 + z\lambda^2 - xzP'^2 - yzP^2 - xyq^2. \tag{2.35}$$

Let $A = \not{P}' - \not{K} + m$ and $B = \not{P} - \not{K} + m$. After shifting $\ell = L + K$, the numerator is

$$N^\rho(L+K) = \gamma^\alpha(A - \not{L})\gamma^\rho(B - \not{L})\gamma_\alpha \longrightarrow N^\rho(K) + \frac{(d-2)^2}{d} L^2 \gamma^\rho. \tag{2.36}$$

The arrow denotes symmetric loop integration. Equations (2.5) and (2.10) then give

$$V^\rho = \frac{e_{\mathrm{T}}^2 \mu^{2\epsilon}}{(4\pi)^{d/2}} \int_{\mathcal{S}_2} \left[-\frac{\Gamma(1+\epsilon) N^\rho(K)}{(\Delta - \mathrm{i}0)^{1+\epsilon}} + \frac{(d-2)^2 \Gamma(\epsilon) \gamma^\rho}{2(\Delta - \mathrm{i}0)^\epsilon} \right]. \tag{2.37}$$

Equivalently, its finite expansion is

$$V^\rho = C \int_{\mathcal{S}_2} \left[-\frac{N_4^\rho(K)}{\Delta - \mathrm{i}0} + \gamma^\rho \{ 2\mathcal{L}(\Delta) - 4 \} \right]. \tag{2.38}$$

The subscript 4 permits four-dimensional gamma algebra in the nondivergent numerator term. Add the counterterm $V_{\mathrm{R}}^\rho = V^\rho + c_Z \gamma^\rho$. The pole of V is $C\gamma^\rho/\epsilon$, since the simplex has area $1/2$. The pole of c_Z is $-C/\epsilon$. Their cancellation also removes the common ultraviolet scale dependence.

The relative sign follows independently from

$$S_0(A)(\not{A} - \not{B})S_0(B) = S_0(B) - S_0(A), \qquad q_\rho V^\rho(P', P) = \Sigma(P) - \Sigma(P'). \tag{2.39}$$

On adding the counterterms,

$$q_\rho V_{\mathrm{R}}^\rho(P', P) = \Sigma_{\mathrm{R}}(P) - \Sigma_{\mathrm{R}}(P'). \tag{2.40}$$

A decomposition into only the on-shell Dirac and Pauli form factors cannot satisfy this off-shell identity for general P, P'.

2.4.1 Photon normalization and the charge counterterm

The off-shell Ward identity fixes the vertex counterterm to the same c_Z as the electron field counterterm. In the notation $Z_j = 1 + \delta_j + O(\alpha_{\mathrm{T}}^2)$ this gives

$$\delta_1 = \delta_2 = c_Z. \tag{2.41}$$

To determine the charge counterterm, define $A_{0\mu} = Z_3^{1/2} A_\mu$ and $e_0 = \mu^\epsilon Z_e e_{\mathrm{T}}$. Substituting these and the bare electron field into the interaction gives

$$Z_1 = Z_e Z_2 Z_3^{1/2}, \qquad \delta_1 = \delta_e + \delta_2 + \tfrac{1}{2}\delta_3, \qquad \delta_e = -\tfrac{1}{2}\delta_3. \tag{2.42}$$

Thus only the photon residue remains to be calculated.

For one electron species, the closed electron loop contributes

$$\mathrm{i}\Pi^{\mu\nu}(q) = -e_{\mathrm{T}}^2 \mu^{2\epsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{\mathrm{tr}[\gamma^\mu(\not{\ell} + m)\gamma^\nu(\not{\ell} + \not{q} + m)]}{(\ell^2 - m^2 + \mathrm{i}0)[(\ell+q)^2 - m^2 + \mathrm{i}0]}. \tag{2.43}$$

The sign includes the minus sign of a closed fermion loop. The trace is

$$4\{\ell^\mu(\ell+q)^\nu + \ell^\nu(\ell+q)^\mu - \eta^{\mu\nu}[\ell \cdot (\ell+q) - m^2]\}. \tag{2.44}$$

Combine the denominators with a parameter v and shift $L = \ell + vq$. Their quadratic form is $L^2 - \Delta_\gamma + \mathrm{i}0$, where $\Delta_\gamma = m^2 - v(1-v)q^2$. Odd terms vanish, and symmetric integration replaces $L^\mu L^\nu$ by $\eta^{\mu\nu}L^2/d$. The remaining terms proportional to the metric are reduced using

$$0 = \int \mathrm{d}^d L \, \frac{\partial}{\partial L^\rho} \frac{L^\rho}{L^2 - \Delta_\gamma + \mathrm{i}0} = \int \mathrm{d}^d L \left[\frac{d}{L^2 - \Delta_\gamma + \mathrm{i}0} - \frac{2L^2}{(L^2 - \Delta_\gamma + \mathrm{i}0)^2} \right]. \tag{2.45}$$

This identity follows in a convergent domain of dimensional regularization and extends by analytic continuation. Together with the scalar loop integral it gives the transverse form

$$\mathrm{i}\Pi^{\mu\nu}(q) = \mathrm{i}(q^\mu q^\nu - q^2\eta^{\mu\nu})\Pi_\gamma(q^2), \tag{2.46}$$

$$\Pi_\gamma(q^2) = 8C \int_0^1 v(1-v)\mathcal{L}(\Delta_\gamma)\,\mathrm{d}v + O(\epsilon), \tag{2.47}$$

$$\Pi_\gamma(0) = \frac{4C}{3} U_m. \tag{2.48}$$

There is no soft photon in this loop, so its zero-momentum value has no photon-mass logarithm. With this sign convention the transverse quadratic action has coefficient $1 + \delta_3 + \Pi_\gamma(q^2)$. Unit residue at $q^2 = 0$ therefore requires

$$\delta_3 = -\Pi_\gamma(0) = -\frac{\alpha_\mathrm{T}}{3\pi} U_m, \qquad \delta_e = \frac{\alpha_\mathrm{T}}{6\pi} U_m. \tag{2.49}$$

These relations agree with the QED limit of Denner's photon and charge counterterms, Equations (3.19) and (3.32) of Reference [4]. For a Compton amplitude the two charge factors and the two external photon field factors combine to $Z_e^2 Z_3 = 1$ at this order. This cancellation accounts for photon normalization even though there is no internal photon propagator in the Born amplitude.

2.5 The spanning photon integral

Put $Q = p + k$. The remaining s-ordered matrix is

$$W_s^{\nu\mu} = \frac{e_\mathrm{T}^2}{\mathrm{i}} \int \frac{\mathrm{d}^4\ell}{(2\pi)^4} \frac{N^{\nu\mu}(\ell)}{D_0 D_1 D_2 D_3}, \tag{2.50}$$

where

$$(D_0, D_1, D_2, D_3) = (\ell^2 - \lambda^2, (p-\ell)^2 - m^2, (Q-\ell)^2 - m^2, (p'-\ell)^2 - m^2) + \mathrm{i}0, \tag{2.51}$$

$$N^{\nu\mu}(\ell) = \gamma^\alpha(\not{p}' - \not{\ell} + m)\gamma^\nu(\not{Q} - \not{\ell} + m)\gamma^\mu(\not{p} - \not{\ell} + m)\gamma_\alpha. \tag{2.52}$$

Each component of the denominator tuple carries $+\mathrm{i}0$. The loop is ultraviolet convergent by power counting. For simplex parameters $a_0, \ldots, a_3$ define

$$K = a_1 p + a_2 Q + a_3 p', \qquad \Delta = K^2 + a_0\lambda^2 + a_2(m^2 - s), \qquad \sum_j a_j D_j = (\ell - K)^2 - \Delta + \mathrm{i}0. \tag{2.53}$$

Expand the cubic numerator about K. Only its constant and second derivative contribute,

$$N(L + K) \longrightarrow N(K) + \frac{L^2}{8} \Box_\ell N(K), \qquad \Box_\ell = \frac{\partial}{\partial \ell^\rho} \frac{\partial}{\partial \ell_\rho}. \tag{2.54}$$

The factor 1/8 is the Taylor factor 1/2 times the tensor average 1/4. More explicitly, with $A_1 = \not{p} - \not{K} + m$, $A_2 = \not{Q} - \not{K} + m$, $A_3 = \not{p}' - \not{K} + m$,

$$\begin{aligned}\Box_\ell N(K) = 2\gamma^\alpha \big[&\gamma^\rho\gamma^\nu\gamma_\rho\gamma^\mu A_1 + \gamma^\rho\gamma^\nu A_2\gamma^\mu\gamma_\rho \\ &+ A_3\gamma^\nu\gamma^\rho\gamma^\mu\gamma_\rho\big]\gamma_\alpha.\end{aligned} \tag{2.55}$$

There are three ways to differentiate two different electron numerators. All have a positive sign because each derivative brings a minus sign. The needed scalar integrals at $d = 4$ are

$$3!I_4 = \frac{1}{16\pi^2\Delta^2}, \qquad 3!(I_3 + \Delta I_4) = \frac{-2}{16\pi^2\Delta}. \tag{2.56}$$

Thus the complete parameter representation is

$$W_s^{\nu\mu} = C\int_{\mathcal{S}_3} \mathrm{d}^4a \left[\frac{N^{\nu\mu}(K)}{(\Delta - \mathrm{i}0)^2} - \frac{\Box_\ell N^{\nu\mu}(K)}{4(\Delta - \mathrm{i}0)}\right], \qquad \int_{\mathcal{S}_3}\mathrm{d}^4a = \int_0^1 \prod_{j=0}^{3}\mathrm{d}a_j\,\delta(1 - \sum_j a_j). \tag{2.57}$$

Physical s-channel cuts require the displayed complex boundary value. Replacing it by a positive real denominator would remove physical absorptive terms.

2.6 Assembly and external Ward identities

The s ordering is

$$\begin{aligned}R_{1,s}^{\nu\mu} = V_{\mathrm{R}}^\nu(p', Q)S_0(Q)\gamma^\mu + \gamma^\nu S_0(Q)V_{\mathrm{R}}^\mu(Q, p) \\ + \gamma^\nu S_0(Q)\Sigma_{\mathrm{R}}(Q)S_0(Q)\gamma^\mu + W_s^{\nu\mu}.\end{aligned} \tag{2.58}$$

Interchange (μ, k) and $(\nu, -k')$ to obtain $R_{1,u}$, and set $R_1 = R_{1,s} + R_{1,u}$. Its amplitude is $\mathcal{M}_1 = -e_{\mathrm{T}}^2\epsilon_\mu\epsilon'^*_\nu\bar{u}(p')R_1^{\nu\mu}u(p)$. For the incoming Ward identity, write $\doteq$ for equality between external on-shell spinors. The four s-ordered terms contract to

$$k_\mu V_{\mathrm{R}}^\nu S_0(Q)\gamma^\mu \doteq V_{\mathrm{R}}^\nu(p', Q), \tag{2.59}$$

$$k_\mu\gamma^\nu S_0(Q)V_{\mathrm{R}}^\mu(Q, p) \doteq -\gamma^\nu S_0(Q)\Sigma_{\mathrm{R}}(Q), \tag{2.60}$$

$$k_\mu\gamma^\nu S_0(Q)\Sigma_{\mathrm{R}}(Q)S_0(Q)\gamma^\mu \doteq \gamma^\nu S_0(Q)\Sigma_{\mathrm{R}}(Q), \tag{2.61}$$

$$k_\mu W_s^{\nu\mu} = V^\nu(p', p) - V^\nu(p', Q). \tag{2.62}$$

Their sum is $V_{\mathrm{R}}^\nu(p', p)$. The u ordering yields its negative. Hence $k_\mu R_1^{\nu\mu} \doteq 0$. The outgoing identity $k'_\nu R_1^{\nu\mu} \doteq 0$ follows in the same way. In particular, omitting the spanning graph spoils the cancellation.

Define $\overline{A} = \gamma^0 A^\dagger \gamma^0$ and the bilinear

$$\mathcal{T}(A, B) = \frac{1}{4}\eta_{\mu\mu'}\eta_{\nu\nu'}\,\mathrm{tr}[(\not{p}' + m)A^{\nu\mu}(\not{p} + m)\overline{B}^{\nu'\mu'}]. \tag{2.63}$$

The factor 1/4 averages over the two incoming electron spins and photon polarizations. For amplitudes satisfying both external Ward identities, the two physical polarization sums can be replaced by $-\eta_{\mu\mu'}$ and $-\eta_{\nu\nu'}$. Their product gives the positive metric contraction displayed here. The virtual interference is $V = 2e_{\mathrm{T}}^4\,\mathrm{Re}\,\mathcal{T}(R_1, R_0)$, where the factor two comes from the two cross terms in the squared amplitude. Explicitly,

$$V = \frac{e_{\mathrm{T}}^4}{2}\,\mathrm{Re}\,\eta_{\mu\mu'}\eta_{\nu\nu'}\,\mathrm{tr}[(\not{p}' + m)R_1^{\nu\mu}(\not{p} + m)\overline{R}_0^{\nu'\mu'}], \tag{2.64}$$

$$B_{\mathrm{amp}} = 2e_{\mathrm{T}}^4\left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2\theta\right), \qquad \omega' = \frac{\omega}{1 + (\omega/m)(1 - \cos\theta)}, \tag{2.65}$$

$$\frac{\mathrm{d}\sigma_{\mathrm{Born+virt}}}{\mathrm{d}\Omega} = \frac{1}{64\pi^2m^2}\left(\frac{\omega'}{\omega}\right)^2(B_{\mathrm{amp}} + V). \tag{2.66}$$

The rest-frame incident energy is $\omega = k^0$, and θ is the angle between the incoming and outgoing photons in that frame. Here $B_{\text{amp}} = \overline{|\mathcal{M}_0|^2}$ is the Born squared amplitude, including its coupling factors. These expressions average over both initial spin and initial photon polarization. No square of R_1 belongs at this order.

2.6.1 Integration of the Born cross section

The Born total follows directly from the preceding differential cross section. Define $X = 2\omega/m$ and $y_B = 2\omega'/m$ in the electron rest frame. Energy conservation and the angular Jacobian give

$$\frac{X}{1+X} \le y_B \le X, \qquad 1 - \cos\theta = 2\left(\frac{1}{y_B} - \frac{1}{X}\right), \qquad \frac{\mathrm{d}\cos\theta}{\mathrm{d}y_B} = \frac{2}{y_B^2}. \tag{2.67}$$

Thus the dimensionless squared amplitude and the cross section are

$$\mathcal{B}(X, y_B) = \frac{B_{\text{amp}}}{e_{\text{T}}^4} = 2\left[\frac{X}{y_B} + \frac{y_B}{X} - 4\left(\frac{1}{y_B} - \frac{1}{X}\right) + 4\left(\frac{1}{y_B} - \frac{1}{X}\right)^2\right], \tag{2.68}$$

$$\frac{\mathrm{d}\sigma_0}{\mathrm{d}y_B} = \frac{\pi\alpha_{\text{T}}^2}{m^2 X^2}\mathcal{B}(X, y_B). \tag{2.69}$$

A primitive of $\mathcal{B}/2$ is

$$P_X(y) = \frac{y^2}{2X} + \left(\frac{4}{X} + \frac{4}{X^2}\right)y + \left(X - 4 - \frac{8}{X}\right)\ln y - \frac{4}{y}. \tag{2.70}$$

Substitution of the two physical endpoints gives

$$\sigma_0 = \frac{\pi\alpha_{\text{T}}^2}{m^2}B(X), \qquad B(X) = \frac{2}{X^2}\left[P_X(X) - P_X\left(\frac{X}{1+X}\right)\right]. \tag{2.71}$$

Expanding this endpoint difference yields the Born function stated in Chapter 1. Its normalization has therefore been obtained from the amplitude and phase space.

2.7 Real emission and phase space

Let k_1, k_2 be outgoing photon momenta. Put $\kappa = (k, -k_1, -k_2)$ and $E = (\not{\epsilon}, \not{\epsilon}_1^*, \not{\epsilon}_2^*)$. The six electron-line orders give

$$\mathcal{M}_{\gamma\gamma} = e_{\text{T}}^3 \bar{u}(p')\mathcal{K}u(p), \qquad \mathcal{K} = \sum_{(a,b,c)\in\mathfrak{S}_3} E_c S_0(p + \kappa_a + \kappa_b) E_b S_0(p + \kappa_a) E_a. \tag{2.72}$$

Here $\mathfrak{S}_3$ permutes $(0, 1, 2)$. To evaluate its unpolarized square, sum over the two transverse polarizations of each photon,

$$\overline{|\mathcal{M}_{\gamma\gamma}|^2} = \frac{e_{\text{T}}^6}{4}\sum_{\epsilon,\epsilon_1,\epsilon_2} \operatorname{tr}[(\not{p}' + m)\mathcal{K}(\not{p} + m)\overline{\mathcal{K}}]. \tag{2.73}$$

The factor $1/4$ averages the initial electron and photon states. This trace includes all 36 interferences, rather than six separate squares. Replacing any one polarization by its momentum telescopes the six permutations to zero. Relabelling $1 \leftrightarrow 2$ simply permutes the sum.

For $P = p + k$, introduce $Q = k_1 + k_2$ and the standard invariant phase space

$$\mathrm{d}\Phi_n(P) = (2\pi)^4\delta^{(4)}(P - \textstyle\sum_j p_j)\prod_j \frac{\mathrm{d}^3\boldsymbol{p}_j}{(2\pi)^3 2p_j^0}, \tag{2.74}$$

$$\mathrm{d}\Phi_3(P; p', k_1, k_2) = \frac{\mathrm{d}Q^2}{2\pi}\mathrm{d}\Phi_2(P; p', Q)\mathrm{d}\Phi_2(Q; k_1, k_2), \tag{2.75}$$

$$\mathrm{d}\Phi_2(P; a, b) = \frac{\sqrt{\Lambda(P^2, a^2, b^2)}}{32\pi^2 P^2}\mathrm{d}\Omega, \qquad \Lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz. \tag{2.76}$$

The solid angle is in the rest frame of its decaying momentum. For two massless photons this gives

$$\mathrm{d}\Phi_3 = \frac{\sqrt{\Lambda(s, m^2, Q^2)}}{2048\pi^5 s}\mathrm{d}Q^2\mathrm{d}\Omega_Q\mathrm{d}\Omega_*, \qquad 0 \le Q^2 \le (\sqrt{s} - m)^2. \tag{2.77}$$

Integrating both labelled photons requires the factor 1/2! in the cross section. Restricting to one photon ordering instead would remove that divisor.

2.8 The soft subtraction in the electron rest frame

When one outgoing photon is soft with momentum ℓ, only the external-line denominators generate an inverse power of its energy. For example, before mass regularization,

$$\overline{u}(p')\gamma^\rho\frac{\not{p}' + \not{\ell} + m}{(p' + \ell)^2 - m^2} = \frac{p'^\rho}{p' \cdot \ell}\overline{u}(p') + O((\ell^0)^0). \tag{2.78}$$

The on-shell Dirac equation and $\gamma^\rho\not{p}' = 2p'^\rho - \not{p}'\gamma^\rho$ give the numerator $2p'^\rho$ at leading order. Emission from the incoming leg similarly gives $-p^\rho/(p \cdot \ell)$ because its denominator is $-2p \cdot \ell$. The other insertions are finite in this limit. Their sum therefore gives

$$\mathcal{M}_{\gamma\gamma} \longrightarrow e_{\mathrm{T}}\epsilon^*(\ell) \cdot J(\ell)\mathcal{M}_0, \qquad J = \frac{p'}{p' \cdot \ell} - \frac{p}{p \cdot \ell}, \qquad \ell \cdot J = 0. \tag{2.79}$$

A common sign does not affect the rate. Define the relative soft correction by

$$\delta_{\mathrm{s}}(\varepsilon, \lambda) = e_{\mathrm{T}}^2\int_{E_\ell < \varepsilon}\frac{\mathrm{d}^3\boldsymbol{\ell}}{(2\pi)^3\, 2E_\ell}[-J(\ell)^2], \qquad E_\ell = \sqrt{|\boldsymbol{\ell}|^2 + \lambda^2}. \tag{2.80}$$

Here ε is an auxiliary photon-energy boundary satisfying $\lambda \ll \varepsilon$ and small compared with the hard energies. Current conservation $\ell \cdot J = 0$ removes longitudinal terms from the photon polarization sum. The expression is exact in λ within the leading soft current. Terms beyond that current vanish when the soft-energy boundary is removed. In the rest frame of p, choose the polar axis along $\boldsymbol{p}'$. Set $E' = p'^0$, $v = |\boldsymbol{p}'|/E'$, $q = |\boldsymbol{\ell}|$, $E_\ell = \sqrt{q^2 + \lambda^2}$, $\beta = q/E_\ell$, and $c = \widehat{\boldsymbol{p}'} \cdot \widehat{\boldsymbol{\ell}}$. Then

$$-J^2 = \frac{1}{E_\ell^2}\left[\frac{2}{1 - v\beta c} - 1 - \frac{1 - v^2}{(1 - v\beta c)^2}\right]. \tag{2.81}$$

The azimuthal integral is 2π. For $a = v\beta$ the two polar integrals are

$$\int_{-1}^{1}\frac{\mathrm{d}c}{1 - ac} = \frac{2\,\mathrm{artanh}\,a}{a}, \qquad \int_{-1}^{1}\frac{\mathrm{d}c}{(1 - ac)^2} = \frac{2}{1 - a^2}. \tag{2.82}$$

Their values at $a = 0$ are understood by continuity. Substituting them and $e_{\mathrm{T}}^2 = 4\pi\alpha_{\mathrm{T}}$ gives

$$\delta_{\mathrm{s}}(\varepsilon, \lambda) = \frac{\alpha_{\mathrm{T}}}{2\pi} \int_0^{\sqrt{\varepsilon^2-\lambda^2}} \frac{q^2 \mathrm{d}q}{E_\ell^3} \left[\frac{4}{v\beta} \operatorname{artanh}(v\beta) - 2 - \frac{2(1-v^2)}{1-v^2\beta^2} \right]. \tag{2.83}$$

This one-dimensional integral supplies the finite terms as well as the logarithm. For $q \gg \lambda$ its bracket approaches $4(b-1)$, where $b = \operatorname{artanh}(v)/v$ and $v = \sqrt{1 - m^4/(p \cdot p')^2}$. Therefore

$$\delta_{\mathrm{s}} = \frac{2\alpha_{\mathrm{T}}}{\pi}(b-1) \ln \frac{\varepsilon}{\lambda} + \text{finite terms}. \tag{2.84}$$

The soft region of the virtual graph has the same two eikonal lines. Its interference has the opposite coefficient of $\ln \lambda$,

$$V = B_{\mathrm{amp}} \frac{2\alpha_{\mathrm{T}}}{\pi}(b-1) \ln \frac{\lambda}{m} + V_{\mathrm{finite}}. \tag{2.85}$$

Consequently $V + B_{\mathrm{amp}}\delta_{\mathrm{s}}$ is independent of λ in its zero-mass limit. The finite remainder can be evaluated with the parameter integrals above and (2.83).

For dimensionless measurement functions F_2, F_3, evaluated on the two-body and three-body events respectively, require symmetry under exchange of the two final photons and $F_3 \to F_2$ as either photon becomes soft. The observable is

$$\begin{aligned} \sigma^{\mathrm{NLO}}[F] = \frac{1}{4p \cdot k} \lim_{\varepsilon \to 0} \Bigg\{ & \int \mathrm{d}\Phi_2 F_2 [B_{\mathrm{amp}} + \lim_{\lambda \to 0} (V + B_{\mathrm{amp}}\delta_{\mathrm{s}})] \\ & + \frac{1}{2} \int_{\min(k_1^0, k_2^0) > \varepsilon} \mathrm{d}\Phi_3 F_3 \overline{|\mathcal{M}_{\gamma\gamma}|^2} \Bigg\} + \frac{1}{m^2} \mathcal{O}(\alpha_{\mathrm{T}}^4). \end{aligned} \tag{2.86}$$

The total considered in this document has $F_2 = F_3 = 1$. All separator energies in this expression are in the same electron rest frame. The two soft corners cancel the divisor 2! in the soft part. The hard integral cancels the separator logarithm. At a fixed nonzero separator, there are power corrections from replacing the soft event by its two-body limit. Both limits are needed before interpreting the result as an inclusive observable.

Chapter 3

Integration of the amplitudes

3.1 Loop integration

The calculation proceeds from loop parameters to elastic angles, then to soft matching and real-emission energies. All boundary constants are fixed by the original integrals and their causal boundary values.

3.1.1 Analytic integration of the radial loop parameter

We now integrate the preceding amplitudes independently of the total cross section of Lee, Schwartz and Zhang [2]. In this section all momenta and photon masses are divided by the electron mass, so $m = 1$. The final correction is restored as $\sigma_1 = \alpha_{\mathrm{T}}^3 C_{\mathrm{total}}(s)/m^2$, with the argument s here denoting the dimensionless ratio s_{phys}/m^2. Likewise t, u, and λ in this chapter stand for t_{phys}/m^2, u_{phys}/m^2, and $\lambda_{\mathrm{phys}}/m$. Thus $s + t + u = 2$ in elastic kinematics. The incident-energy variables X and H used below both equal $s - 1$. Integration coordinates with a lowercase letter are defined locally and must not be identified with these external invariants. Remove the common factor $\alpha_{\mathrm{T}}/(4\pi)$ from the loop kernel and denote the resulting matrix by $\widehat{R}_1$. The notation $\mathcal{C}$ with a sector subscript denotes that sector's contribution to C_{total}. The complete coefficient satisfies $C_{\mathrm{total}}(s) = C_1(s-1)$. Throughout this chapter the interference trace is $\mathcal{T} = \mathcal{T}(\widehat{R}_1, R_0)$, using (2.63) with $m = 1$.

For the spanning graph write the three electron parameters as

$$a_1 = \rho(1-b)z, \qquad a_2 = \rho b, \qquad a_3 = \rho(1-b)(1-z), \qquad a_0 = 1-\rho, \quad 0 \le \rho, b, z \le 1. \tag{3.1}$$

The simplex measure becomes $\rho^2(1-b)\mathrm{d}\rho\mathrm{d}b\mathrm{d}z$. Let Q be the internal electron momentum and define

$$D = Q^2 - 1, \qquad \overline{K} = bQ + (1-b)[zp + (1-z)p'], \tag{3.2}$$

$$M_z = 1 - tz(1-z), \qquad C = 1 - t(1-b)^2 z(1-z), \tag{3.3}$$

$$A = \overline{K}^2 = C + bD, \qquad B = -bD, \qquad \Delta = \rho(A\rho + B) + (1-\rho)\lambda^2. \tag{3.4}$$

Here $t = (p - p')^2 \le 0$, and $C = A + B > 0$. This last inequality makes the analytic continuation particularly simple. The internal s ordering has $D > 0$, while the u ordering has $D < 0$.

Expand the numerator and its second derivative in ρ,

$$N(\rho\overline{K}) = \sum_{n=0}^{3} \rho^n N_n, \qquad (\Box_\ell N)(\rho\overline{K}) = L_0 + \rho L_1. \tag{3.5}$$

These coefficients are fixed without any integration by

$$N_n = \frac{(\overline{K}\cdot\partial_\ell)^n}{n!}N(\ell)\big|_{\ell=0}, \qquad L_n = \frac{(\overline{K}\cdot\partial_\ell)^n}{n!}\Box_\ell N(\ell)\big|_{\ell=0} \quad (n=0,1). \tag{3.6}$$

Only the N_0 term is infrared divergent. All the other terms may be integrated with $\lambda = 0$. Their radial integrals are

$$F_n(A,B) = \int_0^1 \frac{\rho^n \mathrm{d}\rho}{A\rho + B - \mathrm{i}0}, \qquad R_n(A,B) = \int_0^1 \frac{\rho^n \mathrm{d}\rho}{(A\rho + B - \mathrm{i}0)^2}, \tag{3.7}$$

$$F_0 = \frac{\ln C - \ln(B - \mathrm{i}0)}{A}, \qquad F_n = \frac{1}{nA} - \frac{B}{A}F_{n-1} \quad (n \geq 1), \tag{3.8}$$

$$R_0 = \frac{1}{BC}, \qquad R_n = \frac{F_{n-1} - BR_{n-1}}{A} \quad (n \geq 1). \tag{3.9}$$

Polynomial division gives the recurrence for F_n, and the identity $\rho = [(A\rho + B) - B]/A$ gives that for R_n. For $B < 0$, use $\ln(B - \mathrm{i}0) = \ln|B| - \mathrm{i}\pi$. Keeping this term is essential for the physical cut. For $|A/B| < 1$, expanding the denominator as a geometric series gives the following convergent representations. They also avoid cancellation when the recurrence divides a small difference by A,

$$F_n = \frac{1}{B}\sum_{j=0}^{\infty}\frac{(-A/B)^j}{n+j+1}, \qquad R_n = \frac{1}{B^2}\sum_{j=0}^{\infty}\frac{(j+1)(-A/B)^j}{n+j+1}. \tag{3.10}$$

3.1.2 Evaluation of the infrared scalar box

The scalar coefficient of N_0 is

$$D_0 = \int_0^1 \mathrm{d}z \int_0^1 \mathrm{d}b \int_0^1 \mathrm{d}\rho \, \frac{\rho^2(1-b)}{[\rho(A\rho + B) + (1-\rho)\lambda^2 - \mathrm{i}0]^2}. \tag{3.11}$$

Its infrared endpoint is $b = \rho = 0$. The Feynman boundary value means that $-\mathrm{i}0$ is first replaced by $-\mathrm{i}\epsilon$, with $\epsilon \downarrow 0$ at each fixed $\lambda > 0$. We next take $\lambda \to 0$ at fixed nonzero D and fixed t. A direct estimate on the physical side is useful because an asymptotic expansion in the region $D < 0$ need not remain uniform on a cut.

For $D > 0$, deform the b contour to Γ, parametrized by

$$b(v) = v + \mathrm{i}\eta v(1-v), \qquad 0 \leq v \leq 1, \qquad 0 < \eta \ll 1. \tag{3.12}$$

Here η is fixed independently of λ. Along this contour, with $h = M_z - 1 \geq 0$, the exact denominator obeys

$$\mathrm{Im}\,\Delta = -\eta v(1-v)\rho[D(1-\rho) + 2h\rho(1-v)] \leq 0. \tag{3.13}$$

Thus the deformation crosses no displaced pole. At $\rho = 1$ and $h = 0$ the denominator is 1, so the vanishing imaginary part there causes no singularity. The undeformed endpoint $b = 1$ is understood by the same Feynman limit. For $D < 0$ the real contour suffices, and we set $\eta = 0$. Choose a temporary boundary $v = \delta \ll 1$ and write $b_* = b(\delta)$. The two contour segments define $D_0^<$ and $D_0^>$.

On the segment $v \geq \delta$, the radial integral has the uniform small-mass limit $R_0 = 1/(BC)$. To see this directly for $D > 0$, factor its quadratic denominator. Its roots are

$$\rho_- = \frac{\lambda^2}{Db} + O_\delta(\lambda^4), \qquad \rho_+ = \frac{Db}{A} + O_\delta(\lambda^2). \tag{3.14}$$

The roots remain separated, and b, A, and $C = A - Db$ stay away from zero for sufficiently small fixed η. In particular, ρ_+ stays away from both radial endpoints, uniformly in z and $v \geq \delta$. In the partial fractions, the double-pole and simple-pole coefficients at ρ_- are respectively $O_\delta(\lambda^4)$ and $O_\delta(\lambda^2)$. Their integrals are therefore $O_\delta(\lambda^2)$ and $O_\delta(\lambda^2|\ln\lambda|)$. The coefficients at ρ_+ change by $O_\delta(\lambda^2)$, and their endpoint primitives remain bounded on the chosen branch. These statements also hold as $v \to 1$, where a displaced pole approaches the real radial interval but never either of its endpoints. They are uniform at $h = 0$, including $z = 0, 1$ and $t = 0$. Thus the radial error is $O_\delta(\lambda^2|\ln\lambda|)$. At zero mass the primitive is $-1/[A(A\rho + B - \mathrm{i}0)]$, which gives $1/(BC)$ also when $B < 0$. For $D < 0$ no radial pole occurs, and the same limit follows directly. Consequently,

$$D_0^{>} = -\frac{1}{D}\int_0^1 \mathrm{d}z \int_{\Gamma^>} \mathrm{d}b\, \frac{1-b}{b[1+(M_z-1)(1-b)^2]} + o_\lambda(1). \tag{3.15}$$

On $0 \leq v \leq \delta$, put $G = \rho^2 + v\rho + \lambda^2$ and $F = M_z\rho^2 - Db\rho + \lambda^2$. The fixed contour angle gives $|F| \geq cG$, with $c > 0$ uniform in z, v, ρ, and small λ. For $D < 0$ positivity gives the same bound. Moreover,

$$\Delta - F = O(v\rho^2 + \rho\lambda^2), \qquad |\Delta| \geq c'G. \tag{3.16}$$

The second inequality follows for small δ and λ, since $|\Delta - F|/G = O(v + \lambda)$. Replacing $1 - b$ by 1 and Δ by F consequently changes this segment by at most a constant times

$$\int_0^\delta \mathrm{d}v \int_0^1 \mathrm{d}\rho \left[\frac{v\rho^2}{G^2} + \frac{\lambda^2\rho^3}{G^3}\right] = O(\delta) + O(\lambda). \tag{3.17}$$

For the first term use $G \geq \rho(\rho + v)$ and integrate ρ. For the second integrate v first, leaving a bound proportional to $\lambda^2 \int_0^1 \rho^2(\rho^2+\lambda^2)^{-2}\mathrm{d}\rho = O(\lambda)$. This controls the endpoint approximation on the physical branch, including its subsequent z integral.

The model denominator is linear in b, so its contour integral gives

$$D_0^{<} = \frac{1}{D}\int_0^1 \mathrm{d}z \int_0^1 \rho\,\mathrm{d}\rho \left[\frac{1}{M_z\rho^2 - Db_*\rho + \lambda^2} - \frac{1}{M_z\rho^2 + \lambda^2}\right] + O(\delta) + o_\lambda(1). \tag{3.18}$$

At fixed b_* the first radial integral tends to

$$\frac{1}{M_z}\big[\ln(M_z - Db_*) - \ln(-Db_*)\big], \tag{3.19}$$

whereas the second is exactly $(2M_z)^{-1}\ln[(M_z + \lambda^2)/\lambda^2]$. For $D > 0$, $-Db_*$ lies below the negative real axis and $\ln(-Db_*) = \ln(-D - \mathrm{i}0) + \ln b_*$. This fixes the physical branch directly, without continuing a remainder from $D < 0$. Expanding only after the fixed-δ mass limit gives

$$D_0^{<} = \frac{1}{D}\int_0^1 \frac{\mathrm{d}z}{M_z}\left[\ln\lambda + \frac{1}{2}\ln M_z - \ln b_* - \ln(-D - \mathrm{i}0)\right] + O(\delta) + o_\lambda(1). \tag{3.20}$$

The regular part of the outer contour integral can be deformed back to the real interval. Its singular part is $-\ln b_*/M_z$, so the complex cutoff cancels just as a real cutoff would. The finite part of the remaining b integral is elementary. With $h = M_z - 1 \geq 0$ and $r = 1 - b$,

$$\int_0^1 \mathrm{d}b \left[\frac{1-b}{b[1+h(1-b)^2]} - \frac{1}{bM_z}\right] = \frac{1}{M_z}\int_0^1 \frac{hr-1}{1+hr^2}\mathrm{d}r \tag{3.21}$$

$$= \frac{1}{M_z}\left[\frac{1}{2}\ln M_z - \frac{\arctan\sqrt{h}}{\sqrt{h}}\right]. \tag{3.22}$$

Adding the two regions cancels b_*. After the mass limit the remaining error is bounded by $O(\delta)$, and therefore vanishes as $\delta \to 0$. We obtain

$$\boxed{D_0 = \frac{1}{D}\left[g_0(t)\ln\frac{\lambda}{-D-\mathrm{i}0} + g_a(t)\right] + o(1),} \tag{3.23}$$

where the remainder vanishes as $\lambda \to 0$ at fixed nonzero D and fixed t, and

$$g_0(t) = \int_0^1 \frac{\mathrm{d}z}{M_z} = \frac{4\,\mathrm{arsinh}(\sqrt{-t}/2)}{\sqrt{(-t)(4-t)}}, \tag{3.24}$$

$$g_a(t) = \int_0^1 \frac{\mathrm{d}z}{M_z}\frac{\arctan\sqrt{M_z-1}}{\sqrt{M_z-1}}. \tag{3.25}$$

Both functions have value 1 at $t = 0$. The remaining integral is smooth and real, and can also be performed analytically. Define

$$a = \frac{\sqrt{-t}}{2}, \qquad h_a = \sqrt{1+a^2}, \qquad \kappa = \frac{a}{h_a}, \qquad 0 \le \kappa < 1. \tag{3.26}$$

Introduce an auxiliary angle ϑ, set $z = (1+\cos\vartheta)/2$, and use reflection symmetry about $\vartheta = \pi/2$. Next set $\tan\vartheta = \tan\varphi/h_a$. These substitutions give

$$g_a(t) = \frac{1}{a}\int_0^{\pi/2} \frac{\arctan(a\sin\vartheta)}{1+a^2\sin^2\vartheta}\mathrm{d}\vartheta = \frac{J(\kappa)}{ah_a}, \tag{3.27}$$

$$J(\kappa) = \int_0^{\pi/2} \arcsin(\kappa\sin\varphi)\mathrm{d}\varphi. \tag{3.28}$$

Differentiating the last expression gives an elementary integral,

$$J'(\kappa) = \int_0^1 \frac{\mathrm{d}y}{\sqrt{1-\kappa^2+\kappa^2y^2}} = \frac{\mathrm{artanh}\,\kappa}{\kappa}. \tag{3.29}$$

The boundary condition $J(0) = 0$ then fixes the answer in terms of the dilogarithm Li_2,

$$\boxed{g_a(t) = \frac{\mathrm{Li}_2(\kappa) - \mathrm{Li}_2(-\kappa)}{2ah_a}.} \tag{3.30}$$

The continuous limit at $a = 0$ is 1. All arguments in this formula are real and lie between -1 and 1, so no branch continuation is needed for the physical spacelike momentum transfer. The imaginary part of D_0 for $D > 0$ is $\pi g_0(t)/D$. This derivation fixes both the infrared logarithm and the finite part without importing a Compton correction factor.

The complete spanning matrix, with its common coupling removed, is therefore

$$\widehat{W} = N_0D_0 + \int_0^1 \mathrm{d}b \int_0^1 \mathrm{d}z\,(1-b)\left[\sum_{n=1}^{3} N_nR_n - \frac{1}{4}(L_0F_1 + L_1F_2)\right]. \tag{3.31}$$

The logarithmic $b = 0$ endpoint of the finite terms is integrable. The matrices N_n, L_n follow simply by multiplying the three linear electron numerators given above.

3.1.3 Integration of the self-energy and vertex

For the self-energy let $S = P^2$, and in the moments (3.9) set $A = S$, $B = 1 - S$. Integrating the two logarithms in $\Delta_\Sigma = v[1 - (1 - v)S]$ gives, after removing the common ultraviolet constant,

$$\widehat{a}(S) = -\frac{1}{2} - 2SF_1 + SF_2, \qquad \widehat{b}(S) = 2 + 4SF_1, \tag{3.32}$$

$$\widehat{c}_m = 4, \qquad \widehat{c}_Z = -4 - 4\ln\lambda, \tag{3.33}$$

$$\widehat{\Sigma}_{\mathrm{R}}(P) = \widehat{a}(S)\not{P} + [\widehat{b}(S) - 4]\mathbf{1} - (\not{P} - 1)\widehat{c}_Z. \tag{3.34}$$

A hat means that $\alpha_{\mathrm{T}}/(4\pi)$ has been factored out. The expression is the photon-mass asymptotic value, including all terms that survive the zero-mass limit.

At the vertex write $x = \rho v$, $y = \rho(1 - v)$ and $z = 1 - \rho$. Because the external photon has $q^2 = 0$,

$$\overline{K} = vP' + (1 - v)P, \qquad A = \overline{K}^2 = vP'^2 + (1 - v)P^2, \qquad B = 1 - A. \tag{3.35}$$

Expand $N(\rho\overline{K}) = N_0 + \rho N_1 + \rho^2 N_2$. Integration by parts gives $\int_0^1 \rho \ln(A\rho + B - \mathrm{i}0)\mathrm{d}\rho = -AF_2/2$, since $A + B = 1$. The renormalized vertex becomes the single integral

$$\widehat{V}^\mu_{\mathrm{R}} = \int_0^1 \mathrm{d}v \left[-\sum_{n=0}^{2} N^\mu_n F_n + \gamma^\mu \left(-\frac{3}{2} + AF_2 \right) \right] + \widehat{c}_Z \gamma^\mu. \tag{3.36}$$

The constant $-3/2$ includes the finite terms from the d-dimensional numerator. These integrated matrices are inserted into the same eight-diagram sum derived earlier.

3.1.4 Closed evaluation of the vertex parameter

Each real-photon vertex has one on-shell electron momentum P_a and one off-shell momentum P_b. Put $P_a^2 = 1$, $P_b^2 = S$, $D = S - 1$, and choose v to multiply the off-shell momentum. The change $v \mapsto 1 - v$ exchanges the original parameter ordering when necessary. Then

$$\overline{K} = P_a + v(P_b - P_a), \qquad A = 1 + Dv, \qquad B = -Dv. \tag{3.37}$$

Because the numerator is at most quadratic in v, only six scalar moments are needed,

$$J_{nk}(S) = \int_0^1 v^k F_n(1 + Dv, -Dv)\mathrm{d}v, \qquad (n,k) = (0,0),(1,0),(1,1),(2,0),(2,1),(2,2). \tag{3.38}$$

Changing variable from v to A yields an analytic primitive,

$$J_{nk}(S) = \frac{P_{nk}(S) - P_{nk}(1)}{(S-1)^{k+1}}, \qquad P'_{nk}(A) = (A-1)^k F_n(A, 1 - A). \tag{3.39}$$

The radial recurrences reduce this derivative to rational functions and rational functions times $\ln(1 - A - \mathrm{i}0)$. For example, $F_0 = -\ln(1 - A - \mathrm{i}0)/A$ integrates to $\mathrm{Li}_2(A + \mathrm{i}0)$. Integration by parts handles the higher inverse powers of A.

Write $\ell = \ln(1 - A - \mathrm{i}0)$ and $d_2 = \mathrm{Li}_2(A + \mathrm{i}0)$. The six primitives are

$$P_{00}(A) = d_2, \tag{3.40}$$

$$P_{10}(A) = d_2 + (1 - A^{-1})\ell, \tag{3.41}$$

$$P_{11}(A) = 2A - 2d_2 + (-A + A^{-1})\ell - 1, \tag{3.42}$$

$$P_{20}(A) = d_2 + \left(\frac{3}{2} - \frac{2}{A} + \frac{1}{2A^2}\right)\ell + \frac{1}{2A}, \tag{3.43}$$

$$P_{21}(A) = \frac{5}{2}A - 3d_2 + \left(-A - \frac{3}{2} + \frac{3}{A} - \frac{1}{2A^2}\right)\ell - 1 - \frac{1}{2A}, \tag{3.44}$$

$$P_{22}(A) = A^2 - \frac{15}{2}A + 6d_2 + \left(-\frac{A^2}{2} + 4A - \frac{4}{A} + \frac{1}{2A^2}\right)\ell + 4 + \frac{1}{2A}. \tag{3.45}$$

Their continuous boundary values are

$$(P_{00}, P_{10}, P_{11})(1) = \left(\frac{\pi^2}{6}, \frac{\pi^2}{6}, 1 - \frac{\pi^2}{3}\right), \tag{3.46}$$

$$(P_{20}, P_{21}, P_{22})(1) = \left(\frac{1}{2} + \frac{\pi^2}{6}, 1 - \frac{\pi^2}{2}, -2 + \pi^2\right). \tag{3.47}$$

Although individual terms have inverse powers of A, each primitive is regular at $A = 0$. Its values there are respectively $0, 1, -2, 7/4, -15/4, 31/4$. These limits allow the crossed channel to pass through $S = 0$ without a spurious singularity. For $S > 1$, the logarithm is $\ln(S - 1) - \mathrm{i}\pi$ and the dilogarithm is approached from the upper half-plane. For $S < 1$, both functions are real. The point $S = 1$ is an infrared endpoint and requires the regulator before taking the limit.

The branch choices follow directly from the loop denominator. After the momentum integration its quadratic form is $\Delta - \mathrm{i}0$. At fixed $0 < \rho, v < 1$, taking $S \to S + \mathrm{i}\eta$, with $\eta > 0$, gives

$$\Delta = \rho[\rho - (S - 1)v(1 - \rho)], \qquad \operatorname{Im}\Delta = -\eta\rho v(1 - \rho) < 0. \tag{3.48}$$

Thus the continuation approaches the physical cut from the same side as the original Feynman denominators. In the radial result this is $A \to A + \mathrm{i}0$ and $1 - A \to 1 - A - \mathrm{i}0$. For $A > 1$, the latter quantity approaches the negative real axis from below, fixing its argument to $-\pi$. Finally,

$$\frac{\mathrm{d}}{\mathrm{d}A}\,\mathrm{Li}_2(A + \mathrm{i}0) = -\frac{\ln(1 - A - \mathrm{i}0)}{A}, \qquad \operatorname{Im}\mathrm{Li}_2(A + \mathrm{i}0) = \pi \ln A \quad (A > 1). \tag{3.49}$$

The imaginary part follows by integration from $A = 1$, where it vanishes. It is therefore fixed by the causal boundary condition, not by selecting a sign to match a reference value.

To assemble the vertex, expand the numerator coefficients as

$$N_1^\mu(v) = U_0^\mu + vU_1^\mu, \qquad N_2^\mu(v) = V_0^\mu + vV_1^\mu + v^2V_2^\mu. \tag{3.50}$$

These matrices are obtained by inserting $\overline{\not{K}} = \not{P}_a + v(\not{P}_b - \not{P}_a)$ in the numerator of (3.36). The fully integrated vertex is

$$\begin{aligned}\widehat{V}_{\mathrm{R}}^\mu = {} & -N_0^\mu J_{00} - U_0^\mu J_{10} - U_1^\mu J_{11} - V_0^\mu J_{20} - V_1^\mu J_{21} - V_2^\mu J_{22} \\ & + \gamma^\mu\left[-\frac{3}{2} + J_{20} + DJ_{21} + \widehat{c}_Z\right].\end{aligned} \tag{3.51}$$

Thus no vertex parameter integral remains. The counterterm $\widehat{c}_Z = -4 - 4\ln\lambda$ supplies the on-shell subtraction in the same convention as the self-energy above.

3.1.5 Closed evaluation of the spanning matrix

The finite two-parameter terms in (3.31) can be integrated without constructing a separate primitive for each numerator coefficient. The numerator is a polynomial of degree three in the loop momentum. Differences of propagator denominators therefore reduce its tensor integrals to a small set of scalar integrals. We derive this reduction before listing the scalar answers. All momenta remain in units of the electron mass.

Write the four denominators as

$$d_i = (\ell - q_i)^2 - m_i^2 + \mathrm{i}0, \qquad (q_0, q_1, q_2, q_3) = (0, p, Q, p'), \qquad (m_0^2, m_1^2, m_2^2, m_3^2) = (0, 1, 1, 1). \tag{3.52}$$

The photon mass is kept in the infrared-divergent scalar integrals. Setting it to zero in the algebraic coefficients omits only terms that vanish as $\lambda \to 0$. For an ordered subset I of denominator indices define

$$T_I^{\mu_1 \cdots \mu_r} = \frac{e^{\gamma_E \epsilon} \mu^{2\epsilon}}{\mathrm{i}\pi^{d/2}} \int \mathrm{d}^d \ell \, \frac{\ell^{\mu_1} \cdots \ell^{\mu_r}}{\prod_{i \in I} d_i}, \qquad d = 4 - 2\epsilon. \tag{3.53}$$

The factor $e^{\gamma_E \epsilon}$ fixes the normalization so that the common pole is $U = 1/\epsilon$, without an accompanying $-\gamma_E$. Since the full box is ultraviolet finite, this common factor does not change its four-dimensional value. We choose $\mu = m$ in the dimensionless finite parts below.

Contractions from denominator differences. Choose an index $a \in I$ as the origin and put $L = \ell - q_a$. Let $v_i = q_i - q_a$ for $i \in I \setminus \{a\}$. Define the Gram matrix, its dual vectors and the transverse projector by

$$G_{ij} = v_i \cdot v_j, \qquad w_i^\mu = \sum_j (G^{-1})_{ij} v_j^\mu, \qquad P^{\mu\nu} = \eta^{\mu\nu} - \sum_i w_i^\mu v_i^\nu. \tag{3.54}$$

These formulas first apply when the vectors v_i are linearly independent and $\det G \neq 0$. Singular limits are taken in the combined tensor, not in isolated inverse-Gram coefficients. Thus $w_i \cdot v_j = \delta_{ij}$ and $P^\mu{}_\mu = d - (|I| - 1)$. The two required identities are

$$2L \cdot v_i = d_a - d_i + f_i, \qquad L^2 = d_a + m_a^2, \qquad f_i = v_i^2 + m_a^2 - m_i^2. \tag{3.55}$$

For the following formulas every tensor uses the numerator L. A pinched tensor uses this same L, even when its denominator set no longer contains a. This convention avoids an implicit shift of the numerator in the reduction.

Suppressing the set I, the first identity gives

$$V_i^{\mu_2 \cdots \mu_r} \equiv v_{i\mu_1} T^{\mu_1 \cdots \mu_r} = \frac{1}{2} \left[T_{I\setminus\{a\}}^{\mu_2 \cdots \mu_r} - T_{I\setminus\{i\}}^{\mu_2 \cdots \mu_r} + f_i T_I^{\mu_2 \cdots \mu_r} \right]. \tag{3.56}$$

The loop rank falls by one on the right-hand side. Define

$$H_i = V_i, \qquad H_{ij} = v_{j\nu} V_i^\nu, \qquad H_{ijk} = v_{j\nu} v_{k\rho} V_i^{\nu\rho}, \qquad n = |I| - 1. \tag{3.57}$$

Rotations in the space transverse to all v_i then determine ranks one through three com-

pletely,

$$T^{\mu} = \sum_i w_i^{\mu} H_i, \tag{3.58}$$

$$T^{\mu\nu} = \sum_{ij} w_i^{\mu} w_j^{\nu} H_{ij} + P^{\mu\nu} Z, \tag{3.59}$$

$$Z = \frac{T_{I\setminus\{a\}} + m_a^2 T_I - \sum_{ij} (G^{-1})_{ij} H_{ij}}{d-n}, \tag{3.60}$$

$$T^{\mu\nu\rho} = \sum_{ijk} w_i^{\mu} w_j^{\nu} w_k^{\rho} H_{ijk} + \sum_i \left(P^{\mu\nu} w_i^{\rho} + P^{\mu\rho} w_i^{\nu} + P^{\nu\rho} w_i^{\mu} \right) Z_i, \tag{3.61}$$

$$Z_i = \frac{v_{i\mu} (T^{\mu}_{I\setminus\{a\}} + m_a^2 T_I^{\mu}) - \sum_{jk} (G^{-1})_{jk} H_{ijk}}{d-n}. \tag{3.62}$$

The equations for Z and Z_i follow by taking a Lorentz trace and using the second identity in (3.55). To change the origin in a pinched integral, expand each factor $L = \widetilde{L} + (q_b - q_a)$ in its numerator. This is a finite binomial expansion of tensors of lower or equal rank. Consequently a rank-three box requires only rank-two triangles, rank-one bubbles and scalar tadpoles.

Although the original box is ultraviolet finite, some pinched integrals are not. Their poles must be carried through the recurrence before taking $d = 4$. In particular,

$$\frac{x_{-1}/\epsilon + x_0}{c - 2\epsilon} = \frac{x_{-1}}{c\epsilon} + \frac{x_0}{c} + \frac{2x_{-1}}{c^2} + \mathcal{O}(\epsilon). \tag{3.63}$$

The last term is needed for the finite box. This also specifies how the calculation handles the rational terms without assuming that a four-dimensional tensor reduction is sufficient.

Scalar integrals. Let $S = Q^2$, $D = S - 1$, $T = -t \geq 0$, and define

$$a = \frac{\sqrt{T}}{2}, \qquad h = \sqrt{1 + a^2}, \qquad r = \operatorname{arsinh} a, \qquad \kappa = \frac{a}{h}. \tag{3.64}$$

For clarity, denote scalar integrals by $A_i = T_{\{i\}}$, $B_{ij} = T_{\{i,j\}}$, $C_{ijk} = T_{\{i,j,k\}}$ and $D_{0123} = T_{\{0,1,2,3\}}$. The tadpoles and bubbles are

$$A_0 = 0, \qquad A_1 = A_2 = A_3 = U + 1, \tag{3.65}$$

$$B_{01} = B_{03} = U + 2, \qquad B_{12} = B_{23} = U, \tag{3.66}$$

$$B_{02} = U + 2 + \frac{1 - S}{S} \ln(1 - S - \mathrm{i}0), \tag{3.67}$$

$$B_{13} = U + 2 - \frac{2h}{a} r. \tag{3.68}$$

The continuous values at $S = 0$ and $T = 0$ are used. These expressions follow by integrating

$$B_{ij} = U - \int_0^1 \mathrm{d}x \, \ln\left[x m_i^2 + (1-x) m_j^2 - x(1-x)(q_i - q_j)^2 - \mathrm{i}0 \right]. \tag{3.69}$$

For equal masses the rank-one bubble is exactly $(q_i + q_j)^{\mu} B_{ij}/2$. This formula is also valid when $(q_i - q_j)^2 = 0$, where a one-vector Gram inverse would be singular. For unequal masses use

$$T^{\mu}_{\{i,j\}} = q_i^{\mu} B_{ij} + (q_j - q_i)^{\mu} \frac{A_j - A_i + [(q_j - q_i)^2 + m_i^2 - m_j^2] B_{ij}}{2(q_j - q_i)^2}. \tag{3.70}$$

For $i=0$, $m_j=1$, the coefficient in the second term approaches $U/2+1/4$ as $(q_j-q_0)^2\to 0$.

The four triangles reduce to three distinct functions,

$$C_{012}=C_{023}=-\frac{\mathrm{Li}_2(S+\mathrm{i}0)-\pi^2/6}{S-1}, \tag{3.71}$$

$$C_{123}=-\frac{2r^2}{T}, \tag{3.72}$$

$$C_{013}=\frac{r}{ah}\ln\lambda-\frac{1}{2}\mathcal{H}(T), \tag{3.73}$$

$$\mathcal{H}(T)=\frac{2r\ln(2h)-r^2-\mathrm{Li}_2(-e^{-2r})+\mathrm{Li}_2(-1)}{ah}. \tag{3.74}$$

The overall minus signs in the finite triangles follow from $\int \mathrm{d}^d\ell/(\mathrm{i}\pi^{d/2})$ with three Feynman denominators. The first line is the previously integrated vertex moment $-J_{00}$. For the pure-electron triangle, integrate its simplex denominator $1+Txz$ to obtain $2r^2/T$ before applying this sign. The infrared triangle follows by writing the sum of its two electron parameters as ρ,

$$C_{013}=-\int_0^1\mathrm{d}z\int_0^1\frac{\rho\mathrm{d}\rho}{\rho^2[1+Tz(1-z)]+(1-\rho)\lambda^2-\mathrm{i}0} \tag{3.75}$$

$$=g_0(t)\ln\lambda-\frac{1}{2}\int_0^1\frac{\ln[1+Tz(1-z)]}{1+Tz(1-z)}\mathrm{d}z+o(1). \tag{3.76}$$

To evaluate the last integral, set $2z-1=y$ and then $\kappa y=\tanh u$. Its upper boundary is $u=r$, giving

$$\mathcal{H}(T)=\frac{1}{ah}\left[2r\ln h-2\int_0^r\ln\cosh u\,\mathrm{d}u\right], \tag{3.77}$$

$$\int_0^r\ln\cosh u\,\mathrm{d}u=\frac{r^2}{2}-r\ln 2+\frac{\mathrm{Li}_2(-e^{-2r})-\mathrm{Li}_2(-1)}{2}. \tag{3.78}$$

This proves the last line of (3.74). Its boundary values are $C_{123}\to -1/2$ and $\mathcal{H}(T)\to 0$ at $T=0$. Finally the scalar box is the result already derived in (3.30),

$$D_{0123}=\frac{1}{D}\left[\frac{r}{ah}\ln\frac{\lambda}{-D-\mathrm{i}0}+\frac{\mathrm{Li}_2(\kappa)-\mathrm{Li}_2(-\kappa)}{2ah}\right]+o(1). \tag{3.79}$$

No parameter integral remains in this list.

Physical boundary values. The parameter denominators have the boundary value $\Delta-\mathrm{i}0$ inherited from the original propagators. For $S>1$ this gives $\ln(1-S-\mathrm{i}0)=\ln(S-1)-\mathrm{i}\pi$. Since $\mathrm{d}\,\mathrm{Li}_2(S)/\mathrm{d}S=-\ln(1-S)/S$, the corresponding dilogarithm is approached from the upper half-plane. This fixes its imaginary part by continuation from $S<1$, without choosing a sign from a numerical comparison. The functions of $T\geq 0$ in the other lines are real. In particular $-e^{-2r}\in[-1,0)$ and $\kappa\in[0,1)$, so they do not cross a dilogarithm cut.

Assembly of the matrix. Expand the original numerator as

$$N(\ell)=\sum_{r=0}^{3}\mathcal{N}_{\mu_1\cdots\mu_r}\ell^{\mu_1}\cdots\ell^{\mu_r}=\gamma^\alpha(\not{p}'-\not{\ell}+1)B(\not{Q}-\not{\ell}+1)A(\not{p}-\not{\ell}+1)\gamma_\alpha, \tag{3.80}$$

where A and B are the incident and outgoing photon vertex matrices in the chosen electron ordering. The complete closed spanning matrix is then

$$\boxed{\widehat{W} = \sum_{r=0}^{3} \mathcal{N}_{\mu_1\cdots\mu_r}\, T^{\mu_1\cdots\mu_r}_{\{0,1,2,3\}}\Big|_{\epsilon^0}\,.} \tag{3.81}$$

Equations (3.56)– (3.74) give every coefficient through finite algebra and logarithms or dilogarithms. The ultraviolet poles cancel in each box tensor. At exactly forward or backward kinematics the three-vector Gram matrix is singular. The physical answer is the continuous limit of the closed result at interior scattering angles. This completes the loop-parameter integrations of the spanning matrix. The subsequent spin interference and elastic-angle integration are separate steps toward the total cross section.

3.2 Virtual interference and angular integration

3.2.1 Closed interference coefficients for the box

For the angular integration, introduce positive variables $x = s-1$ and $y = 1-u$, so $t = y-x$. Contract each external photon pair with $-\eta_{\mu\nu}$ and average over the two initial electron spins and two initial photon polarizations. Individual diagram groups do not satisfy the photon Ward identity. The same covariant polarization contraction must therefore be used for the box, vertex and self-energy groups before they are added.

Let $\mathcal{T}_{\mathrm{box},s}$ denote the bilinear between the s-ordered spanning matrix and the full Born matrix, with $\alpha_{\mathrm{T}}/(4\pi)$ removed from the former. Define finite scalar parts $\overline{A}_1 = A_1 - U$ and $\overline{B}_{ij} = B_{ij} - U$. Direct Clifford traces and the preceding reduction give

$$\mathcal{T}_{\mathrm{box},s} = f_A\overline{A}_1 + f_a\overline{B}_{01} + f_b\overline{B}_{02} + f_t\overline{B}_{13} + f_cC_{012} + f_dC_{013} + f_eC_{123} + f_DD_{0123} + f_0. \tag{3.82}$$

The coefficients are real rational functions,

$$f_A = \frac{4(xy - x + y)}{xy(x+1)}, \tag{3.83}$$

$$f_a = -\frac{8P_a}{x^2y(x-y+4)}, \tag{3.84}$$

$$f_b = \frac{4P_b}{x^2y(x+1)}, \tag{3.85}$$

$$f_t = -\frac{4(x-y)(xy-x+y)}{xy(x-y+4)}, \tag{3.86}$$

$$f_c = \frac{4(2x^2 - 2xy - 2x + y^2 - 8)}{y}, \tag{3.87}$$

$$f_d = -\frac{2(2x^3 - 4x^2y + 2x^2 + xy^2 - 6xy - 12x + y^3 - 4y)}{xy}, \tag{3.88}$$

$$f_e = -\frac{2(x-y)(2x^2 - 2xy - 2x + y^2 - 2y - 4)}{xy}, \tag{3.89}$$

$$f_D = \frac{2P_D}{xy}, \qquad f_0 = \frac{4(xy - x + y)}{xy}. \tag{3.90}$$

Here the polynomials in the longer coefficients are

$$P_a = x^3y - 2x^2y^2 + 6x^2y + xy^3 - 5xy^2 - 2xy + 8x + y^3 - 2y^2 - 8y, \tag{3.91}$$

$$P_b = 3x^3y - x^3 - 2x^2y^2 + 3x^2y + 4x^2 - 4xy^2 - 4xy + 4x - 2y^2 - 4y, \tag{3.92}$$

$$P_D = 2x^4 - 4x^3y + 3x^2y^2 + 2x^2y - 24x^2 - xy^3 - 2xy^2 + 24xy - 16x - 8y^2 + 16y. \tag{3.93}$$

The u-ordered coefficient of each corresponding scalar integral is obtained by the simultaneous replacement $(x, y) \mapsto (-y, -x)$. In that ordering the off-shell invariant in $\overline{B}_{02}$, C_{012} and D_{0123} is $S = 1 - y$. This relation follows from exchanging the two photon insertions in the original numerator. It also follows by independent trace evaluation of that ordering.

The calculation of these coefficients uses only the Clifford identity $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ and the tensor recurrences above. For example, the scalar trace numerator is at most cubic in ℓ. Each factor of ℓ^2 cancels d_0, and each contraction of ℓ with an external momentum is reduced using (3.55). There are seventeen scalar monomials before this reduction in each photon ordering. The coefficients multiplying U cancel exactly. Thus (3.82) is an explicit finite expression, including its infrared logarithm.

3.2.2 A rational variable for the elastic angle

For fixed $x > 0$ the physical interval is $x/(1 + x) \le y \le x$. The substitution

$$v = \sqrt{\frac{x - y}{x - y + 4}}, \qquad y = \frac{x - (x + 4)v^2}{1 - v^2}, \qquad 0 \le v \le \frac{x}{x + 2}, \qquad -\mathrm{d}y = \frac{8v}{(1 - v^2)^2}\mathrm{d}v \tag{3.94}$$

removes the square root from every spacelike scalar integral. In particular $r = \operatorname{artanh} v$ and $ah = v/(1-v^2)$. This substitution follows directly from the quadratic $T(T+4)$ in the scalar box, and its upper boundary follows from the backward-scattering value $T = x^2/(1 + x)$.

The generalized polylogarithms (GPLs) used here are defined recursively by

$$G(\varnothing; v) = 1, \tag{3.95}$$

$$G(a_1, \ldots, a_n; v) = \int_0^v \frac{\mathrm{d}t}{t - a_1} G(a_2, \ldots, a_n; t), \tag{3.96}$$

$$G(\underbrace{0, \ldots, 0}_{n}; v) = \frac{\ln^n v}{n!}. \tag{3.97}$$

The last line specifies the logarithmically singular lower boundary rather than an ordinary convergent integral. The ordered list $(a_1, \ldots, a_n)$ is called a word, its entries are letters, and its weight is its length n. The alphabet is the finite set of letters needed by a given expression. These terms describe the iterated integrals, not additional physical assumptions. For a real letter outside the integration interval, $G(a; v) = \ln(1 - v/a)$. Letters on the path inherit their displacement from the original Feynman denominators, as specified for the crossed triangle below.

At a logarithmic endpoint, a regularized value means the constant term after explicitly separating powers of the endpoint logarithm. This choice is also called a tangential base point. At the origin it is fixed by the all-zero formula above and by the vanishing limit of words containing a nonzero letter. A full convergent combination is formed before taking an endpoint limit. Define

$$R(v) = \frac{G(-1; v) - G(1; v)}{2}, \qquad Z = \sqrt{\frac{x}{x + 4}}. \tag{3.98}$$

Then the needed logarithms and spacelike functions have the explicit word representations

$$\ln y = \ln x + G(Z;v) + G(-Z;v) - G(1;v) - G(-1;v), \tag{3.99}$$

$$g_0(t) = \frac{1-v^2}{v} R(v), \tag{3.100}$$

$$g_a(t) = \frac{1-v^2}{2v} \left[G(0,-1;v) - G(0,1;v) \right], \tag{3.101}$$

$$\mathcal{H}(T) = \frac{1-v^2}{2v} \left[G(1,1;v) - G(1,-1;v) + G(-1,1;v) - G(-1,-1;v) \right], \tag{3.102}$$

$$C_{123} = -\frac{1-v^2}{2v^2} R(v)^2, \qquad \overline{B}_{13} = 2 - \frac{2R(v)}{v}. \tag{3.103}$$

Products of words are reduced by interleaving their letters while preserving the order within each word. This identity follows by partitioning the product of their ordered integration domains. For example $G(a)G(b) = G(a,b) + G(b,a)$. These relations turn the closed box interference into rational functions of v multiplying words of weight at most two. They provide the starting point for its remaining elastic-angle integral.

3.2.3 Integrating the crossed rational subgroup first

Before integrating the crossed box in the rationalized angle, separate the scalar terms whose coefficients and function arguments are rational in the original invariant y. Transforming those terms to the angle variable would introduce an auxiliary root $\rho_u = \sqrt{(x-1)/(x+3)}$, whose apparent threshold at $x = 1$ is absent from the original scalar integral. This root is introduced by applying the spacelike variable v to a subgroup that is already rational in the original invariant y. This subgroup can be integrated before that change of variable. Here $x = s - 1 > 0$ and $x/(1+x) \le y \le x$, in the mass units used for the box calculation.

For this subgroup write $A = \overline{A}_1$, $B_s = \overline{B}_{02}$, and $C_s = C_{012}$, evaluated at the crossed invariant $S = 1 - y$. The subscript s in these local scalar names does not label the photon ordering. Include also the explicit constant in the crossed-box trace. The scalar functions are

$$A = 1, \qquad B_s = 2 + \frac{y \ln y}{1-y}, \qquad C_s = \frac{\mathrm{Li}_2(1-y) - \pi^2/6}{y}. \tag{3.104}$$

Define the two GPLs

$$F(y) = G(1,0;y) = \frac{\pi^2}{6} - \mathrm{Li}_2(1-y), \qquad W(y) = G(0,1,0;y). \tag{3.105}$$

Their differential equations are $F'(y) = \ln y/(y-1)$ and $W'(y) = F(y)/y$. In particular, F is real and analytic for every $y > 0$. The apparent pole of its derivative at $y = 1$ is removable. These statements fix the continuation of the GPL combination without assigning separate imaginary parts at an artificial pole.

The selected crossed-box integrand is

$$I_{\mathrm{rat}}(x,y) = R(x,y) + Q(x,y) \ln y - D(x,y) F(y), \tag{3.106}$$

where exact partial fractions of the trace coefficients give

$$R = \frac{16(x-2)}{y^2} + \frac{28x+12}{x} + \frac{4}{x(y-1)} - \frac{8(2x^2+x-4)}{xy}, \tag{3.107}$$

$$Q = \frac{8(x-2)}{y} + \frac{4(x-6)}{x(y-1)} - \frac{12x+4}{x} - \frac{4}{x(y-1)^2}, \tag{3.108}$$

$$D = -\frac{8y}{x} + \frac{8x-8}{x} - \frac{4(x^2-8)}{xy}. \tag{3.109}$$

The only apparent double pole is integrated with its accompanying simple pole. The exact identity

$$\frac{1}{y-1}-\frac{\ln y}{(y-1)^2}=\frac{\mathrm{d}}{\mathrm{d}y}\frac{\ln y}{y-1}+\frac{1}{y} \tag{3.110}$$

shows explicitly why no isolated $\ln(1-y)$ is needed. Integration by parts of the two polynomial coefficients of F then gives the finite primitive

$$\begin{aligned}P(x,y)=&-\frac{16(x-2)}{y}+\frac{(32x+28)y+y^2}{x} &(3.111)\\ &+\frac{-16x^2-8x+36-(4x+16)y-2y^2}{x}\ln y+4(x-2)\ln^2 y &(3.112)\\ &+\frac{4}{x}\frac{\ln y}{y-1}+\frac{4}{x}[y^2+(2-2x)y+3x-9]F(y) &(3.113)\\ &+\frac{4(x^2-8)}{x}W(y). &(3.114)\end{aligned}$$

Direct differentiation, using the two displayed GPL differential equations, gives $\partial_y P = I_{\mathrm{rat}}$ as a rational identity in the independent symbols $\ln y, F, W$. The contribution to the unnormalized crossed angular integral is therefore exactly

$$J_{u,\mathrm{rat}}(x)=P(x,x)-P\left(x,\frac{x}{1+x}\right). \tag{3.115}$$

Since $\mathrm{d}c = 2s\,\mathrm{d}y/x^2$ and the interference contributes $(4s)^{-1}\int \mathrm{Re}\,\mathcal{T}\,\mathrm{d}c$, this subgroup enters the cross-section coefficient as $J_{u,\mathrm{rat}}/(2x^2)$. At $x=1$ the upper endpoint quotient is assigned its analytic value $\ln y/(y-1)=1$, so the same expression covers the apparent root collision there.

The lower endpoint introduces no new alphabet. Write $f(x)=x/(1+x)$. The first identity below follows directly from the logarithm of this quotient. For the other two, differentiation and the boundary values $F(0)=W(0)=0$ prove

$$\begin{aligned}\ln f(x)&=G(0;x)-G(-1;x), &(3.116)\\ F(f(x))&=-G(-1,0;x)+G(-1,-1;x), &(3.117)\\ W(f(x))&=-G(0,-1,0;x)+G(0,-1,-1;x) &(3.118)\\ &\quad+G(-1,-1,0;x)-G(-1,-1,-1;x). &(3.119)\end{aligned}$$

Thus this entire subgroup has constant letters $\{-1,0,1\}$. The remaining crossed masters are $B_a=\overline{B}_{01}$, $B_t=\overline{B}_{13}$, $C_t=C_{013}$, $C_m=C_{123}$, and $D_b=D_{0123}$. Their rational trace coefficients have denominators built only from x, y, and $x-y+4$. In particular none has a pole at $y=1$. After the spacelike substitution their words involve $0,\pm1,\pm\sqrt{x/(x+4)}$ and do not involve ρ_u. Consequently ρ_u disappears within the crossed box itself, before adding soft or real-radiation contributions. This removes an auxiliary root within the virtual calculation.

There is also no remaining positive interior letter in the angular remainder. If $V=x/(x+2)$ and $z=\sqrt{x/(x+4)}$, then

$$z^2-V^2=\frac{4x}{(x+4)(x+2)^2}>0. \tag{3.120}$$

Thus $0<V<z<1$. The remainder can be integrated along the real v interval without detouring around an artificial crossed root.

To fix the lower boundary in the subsequent function-basis conversion, the series at positive $x \to 0$ follows from

$$F(y) = \sum_{n=1}^{\infty} y^n \left(\frac{1}{n^2} - \frac{\ln y}{n} \right), \tag{3.121}$$

$$W(y) = \sum_{n=1}^{\infty} y^n \left(\frac{2}{n^3} - \frac{\ln y}{n^2} \right). \tag{3.122}$$

They give

$$J_{u,\mathrm{rat}}(x) = x(32 \ln x - 32) + x^2 \left(\frac{92}{3} - 24 \ln x \right) \tag{3.123}$$

$$+ x^3 \left(\frac{88}{3} \ln x - \frac{236}{9} \right) + O(x^4 \ln x). \tag{3.124}$$

In particular this unnormalized subgroup vanishes at the lower boundary. The series fixes the boundary constant from the derived integrand itself, without numerical fitting or a literature value.

3.2.4 Angular primitives and endpoint assembly

The preferred construction applies the finite primitive rules below to the direct box and to the seventeen-word crossed remainder, adding the rational subgroup boundary obtained above. For an independent algebraic representation, one may instead transform the entire crossed integrand. This introduces twenty-nine words. We give its continuation explicitly so that both representations have a fully specified physical branch.

The crossed real-photon triangle is also a weight-two expression. Put $\eta^2 = (x-1)/(x+3)$ and define coefficients

$$(c_Z, c_{-Z}, c_1, c_{-1}) = (1, 1, -1, -1), \qquad (d_\eta, d_{-\eta}, d_1, d_{-1}) = (-1, -1, 1, 1). \tag{3.125}$$

Differentiating its dilogarithm gives

$$\frac{\mathrm{d}}{\mathrm{d}v} \operatorname{Li}_2(1 - y(v) + \mathrm{i}0) = \frac{y'(v)}{1 - y(v) + \mathrm{i}0} \ln(y(v) - \mathrm{i}0), \qquad \frac{y'}{1-y} = \sum_{a \in \{\eta, -\eta, 1, -1\}} \frac{d_a}{v - a}. \tag{3.126}$$

Integrating each simple kernel therefore gives the explicit result

$$\begin{aligned} \operatorname{Li}_2(1 - y(v) + \mathrm{i}0) - \frac{\pi^2}{6} = {} & \operatorname{Li}_2(1 - x) - \frac{\pi^2}{6} \\ & + \sum_a d_a \left[\ln x \, G(a; v) + \sum_b c_b G(a, b; v) \right]. \end{aligned} \tag{3.127}$$

The sums have exactly the four entries listed above. For $x > 1$ the root $\eta > 0$ lies inside the angular interval. Indeed, with $v_{\max} = x/(x+2)$,

$$v_{\max}^2 - \eta^2 = \frac{4}{(x+2)^2(x+3)} > 0, \qquad Z^2 - v_{\max}^2 = \frac{4x}{(x+4)(x+2)^2} > 0. \tag{3.128}$$

Thus the only positive auxiliary root crossed before the endpoint is η, while the zero Z of y is outside the path. It is an intermediate zero of $1 - y$, not a singularity of the triangle:

$\ln y$ vanishes there. Its displacement follows from $1 - y + \mathrm{i}0 = 0$. Since $y'(\eta) < 0$, the positive root is below the real axis and the negative root is above it. More explicitly,

$$y'(v) = -\frac{8v}{(1-v^2)^2}, \qquad v_{\mathrm{pole}} = \eta + \frac{\mathrm{i}\varepsilon}{y'(\eta)} + O(\varepsilon^2), \qquad \varepsilon > 0. \tag{3.129}$$

Here ε is the infinitesimal Feynman displacement, not a soft-energy separator. The GPL terms in (3.127) use those boundary values together. Their apparent singularity cancels. For $x < 1$ the two roots are off the real integration interval. The value at $x = 1$ is obtained by continuity of the combined expression. The reality of the combined result can also be proved without evaluating any GPL. Throughout the path $y > 0$, hence $1 - y < 1$ is on the real branch of the dilogarithm. At $y = 1$, $\ln y/(1-y) \to -1$, so its displayed derivative is regular. That derivative and the real initial value at $v = 0$ uniquely fix a real function on the full interval. Consequently the imaginary parts introduced by the separated GPL terms must cancel.

Finite integration of the words. The remaining rational coefficients need no new special functions. For a rational function $F(v)$, separate its polynomial and pole parts,

$$F(v) = \sum_{k\geq 0} p_k v^k + \sum_a \sum_{j=1}^{m_a} \frac{r_{aj}}{(v-a)^j} = P'(v) + \sum_a \frac{r_{a1}}{v-a}, \tag{3.130}$$

where all sums are finite and

$$P(v) = \sum_{k\geq 0} \frac{p_k v^{k+1}}{k+1} - \sum_a \sum_{j=2}^{m_a} \frac{r_{aj}}{(j-1)(v-a)^{j-1}}. \tag{3.131}$$

The coefficients r_{aj} are fixed algebraically by derivatives of $(v-a)^{m_a} F(v)$ at $v = a$. This form is the rational Hermite reduction, written here as partial fractions. It gives the following finite primitive for the empty word and for words of lengths one and two,

$$\Phi_\varnothing[F](v) = P(v) + \sum_\alpha r_{\alpha 1} G(\alpha; v), \tag{3.132}$$

$$\Phi_a[F](v) = P(v)G(a; v) + \sum_\alpha r_{\alpha 1} G(\alpha, a; v) - \Phi_\varnothing[P/(v-a)](v), \tag{3.133}$$

$$\Phi_{a,b}[F](v) = P(v)G(a, b; v) + \sum_\alpha r_{\alpha 1} G(\alpha, a, b; v) - \Phi_b[P/(v-a)](v). \tag{3.134}$$

Each recursive use shortens the word. The displayed formulas therefore terminate after at most three algebraic steps. Differentiating them gives $\Phi'_w[F](v) = F(v)G(w; v)$, since the derivative of a nonempty word removes its first letter and divides by its kernel denominator. Thus these formulas describe closed GPL expressions of weight at most three, not a residual parameter quadrature.

Endpoint expression for the box contribution. For each photon ordering $j \in \{s, u\}$, the preceding explicit scalar formulas uniquely specify the finite set of rational coefficients $F_{jw}(v; x)$ through

$$\frac{8v}{(1-v^2)^2} \operatorname{Re} \mathcal{T}_{\mathrm{box},j}(x, y(v)) = \sum_{|w|\leq 2} F_{jw}(v; x) G(w; v). \tag{3.135}$$

There are nine input words in the s ordering and twenty-nine in the u ordering when products have been interleaved as above. Apply (3.134) to these rational coefficients and define the resulting finite GPL sums by

$$\mathcal{F}_j(v;x) = \sum_w \Phi_w[F_{jw}](v). \tag{3.136}$$

All partial-fraction coefficients used here have the explicit algebraic definition

$$r_{aj} = \frac{1}{(m_a-j)!}\left(\frac{\mathrm{d}}{\mathrm{d}v}\right)^{m_a-j}\left[(v-a)^{m_a}F(v)\right]\Big|_{v=a}. \tag{3.137}$$

Thus a coefficient need not be expanded into a long quotient of polynomials and square roots in order to define or evaluate the closed answer.

The lower boundary is also determined by finite algebra. Write the rational coefficient of a word in $\mathcal{F}_j$ as $f_w(v)$, with pole order p_w at zero. For a word that does not end in zero, let $g_{w,n}$ be its Taylor coefficient of v^n. The coefficients required at this boundary follow from

$$g_{\varnothing,n} = \delta_{n0}, \qquad g_{aw,0} = 0, \tag{3.138}$$

$$g_{aw,n} = -\frac{1}{n}\sum_{k=0}^{n-1}\frac{g_{w,k}}{a^{n-k}} \quad (a \neq 0), \tag{3.139}$$

$$g_{0w,n} = \frac{g_{w,n}}{n} \quad (n \geq 1). \tag{3.140}$$

Only coefficients up to $n = p_w$ are needed. The single logarithm $G(0;v) = \ln v$ has zero constant term. Its divergent coefficient cancels in the combined primitive, as required by the regular original integrand at forward scattering. Define

$$\mathcal{L}_j(x) = \sum_w \sum_{n=0}^{p_w} \frac{g_{w,n}}{(p_w-n)!}\left(\frac{\mathrm{d}}{\mathrm{d}v}\right)^{p_w-n}\left[v^{p_w}f_w(v)\right]\Big|_{v=0}. \tag{3.141}$$

This is an explicit finite sum of derivatives of rational functions. It does not require an integration or a numerical endpoint limit. The integrated box contribution to $\sigma_1 = \alpha_{\mathrm{T}}^3\mathcal{C}/m^2$ is

$$\boxed{\mathcal{C}_{\mathrm{box}}(x;\lambda) = \frac{1}{2x^2}\sum_{j=s,u}\left[\mathcal{F}_j\left(\frac{x}{x+2};x\right) - \mathcal{L}_j(x)\right].} \tag{3.142}$$

The factor $1/(2x^2)$ follows from $\mathrm{d}c = 2(x+1)\mathrm{d}y/x^2$ and the interference normalization $\mathcal{C}_{\mathrm{virt}} = (4s)^{-1}\int \mathrm{d}c\,\mathrm{Re}\,\mathcal{T}$. For the preferred shorter construction let $\mathcal{F}_{u,\mathrm{rem}}$ be the same finite primitive operation applied only to the remainder specified in Section 3.2.3, and let $\mathcal{L}_{u,\mathrm{rem}}$ be its lower boundary computed by the same finite series rule. The full crossed boundary difference obeys

$$\begin{aligned}\mathcal{F}_u(v_{\max};x) - \mathcal{L}_u(x) = {} & P(x,x) - P\left(x,\frac{x}{1+x}\right)\\ & + \mathcal{F}_{u,\mathrm{rem}}(v_{\max};x) - \mathcal{L}_{u,\mathrm{rem}}(x).\end{aligned} \tag{3.143}$$

Here $P(x,y)$ is the rational-subgroup primitive derived above. The equality follows by differentiating the two constructions and using the same original angular endpoints. It reduces the crossed angular input from twenty-nine to seventeen words and avoids an internal ρ_u pole in this construction.

The expression still contains the infrared logarithm. It is combined with the other virtual diagrams and the real contribution before removing the photon mass.

Continuation through the apparent crossed-channel pole. For $x > 1$, the positive roots obey $\eta < v_{\max} < Z$ because

$$v_{\max}^2 - \eta^2 = \frac{4}{(x+2)^2(x+3)} > 0, \qquad Z^2 - v_{\max}^2 = \frac{4x}{(x+4)(x+2)^2} > 0. \tag{3.144}$$

The original triangle requires the root $\eta - \mathrm{i}0$. An equivalent GPL evaluation starts at zero and deforms the path into the upper half-plane before returning to $v_{\max}$. In particular,

$$G(\eta - \mathrm{i}0; v_{\max}) = \ln\left(\frac{v_{\max}}{\eta} - 1\right) - \mathrm{i}\pi. \tag{3.145}$$

This follows by writing the weight-one integral as $\log(v_{\max} - \eta + \mathrm{i}0) - \log(-\eta + \mathrm{i}0)$: the first argument has phase zero and the second has phase $+\pi$. Discarding this imaginary part term by term would change the analytic continuation. Instead all words use the same deformed path. Their imaginary parts cancel in the complete real crossed-channel interference.

3.2.5 Angular integration of the vertex and self-energy terms

The closed vertex matrices can now be interfered with the Born amplitude and integrated over the elastic scattering angle. Define $x = s - 1$ and $y = 1 - u = x + t$, with c the center-of-mass photon scattering cosine. Since $t = -x^2(1-c)/(2s)$, the physical limits and Jacobian are

$$y_- = \frac{x}{1+x}, \qquad y_+ = x, \qquad \mathrm{d}c = \frac{2s}{x^2}\mathrm{d}y. \tag{3.146}$$

Let T_s and T_u be the spin-averaged interferences from the two vertex insertions and the self-energy insertion in each ordering, with the common loop factor removed as above. We contract each external photon pair with $-\eta_{\mu\nu}$. These diagram subsets are not separately gauge invariant. Physical transverse polarization sums can replace these contractions only after including the spanning graphs, when the full Ward identities apply.

The traces use only the Clifford anticommutation relation and on-shell momentum identities. They are recorded by `compton_virtual_trace_analytic.py`. Both vertex insertions in a given ordering give the same interference. For the s ordering,

$$T_s(x,y) = \sum_f f(1+x)\left[\frac{a_f(x)}{y} + b_f(x) + c_f(x)y\right], \tag{3.147}$$

where the symbols f in the first column of Table 3.1 mean the moments and self-energy coefficients already evaluated above. The entries $\widehat{c}_Z$ and 1 are constants, not functions of $1+x$. There is no J_{22} contribution after the trace. The integral is therefore elementary once the moments have been evaluated,

$$I_s = \sum_f f(1+x)\left[a_f \ln(1+x) + b_f\frac{x^2}{1+x} + \frac{c_f x^2}{2}\left(1 - \frac{1}{(1+x)^2}\right)\right]. \tag{3.148}$$

For the crossed ordering the moment arguments depend on the integration variable, $S = 1 - y$. Substituting the explicit moments and self-energy coefficients reduces T_u to rational functions multiplying 1, $\ln y$ and $\mathrm{Li}_2(1-y)$. Partial fractions and integration by parts give the following closed primitive. Define the real functions

$$H_{10}(y) = \frac{\pi^2}{6} - \mathrm{Li}_2(1-y), \tag{3.149}$$

$$H_{010}(y) = 2\,\mathrm{Li}_3(y + \mathrm{i}0) - \ln y\,\mathrm{Li}_2(y + \mathrm{i}0). \tag{3.150}$$

f	a_f	b_f	c_f
J_{00}	$16(x^2+2x+4)/x$	$-8(x^3+2x^2+8x+8)/x^2$	0
J_{10}	$-8(x^2+2x+8)/x$	$8(x^3+x^2+6x+8)/x^2$	$16/x$
J_{11}	$-8x$	$8(x+2)$	$-8/x$
J_{20}	$-4(2x^2+7x+12)/x$	$4(2x^2+13x+12)/x^2$	$-4/x$
J_{21}	$4(x-4)$	$-4(2x^2+x-4)/x$	$4(x+2)/x$
$\widehat{a}$	$-4(x+2)^2/x^2$	$2(3x^2+12x+8)/x^3$	$2(x+1)/x^2$
$\widehat{b}$	$-2(x^2+4x+8)/x^2$	$16(x+1)/x^3$	$2/x^2$
$\widehat{c}_Z$	$-2(x+4)/x$	$2(3x+4)/x^2$	$2/x$
1	$2(7x^2+28x+32)/x^2$	$-2(x+4)(9x+8)/x^3$	$-2(3x+4)/x^2$

Table 3.1: Exact trace coefficients for the two s-channel vertex insertions and the self-energy insertion together.

Their derivatives are $H'_{10} = \ln y/(y-1)$ and $H'_{010} = H_{10}/y$. For $y > 1$, continuation from the upper half-plane gives imaginary parts $\pi \ln y$ and $\pi \ln^2 y/2$ for the dilogarithm and trilogarithm respectively. They cancel in H_{010}. Thus these expressions are real for all $y > 0$, including when the physical interval crosses $y = 1$.

The primitive is

$$F_u(x,y) = -\frac{2(x^2+8)}{x}\ln^2 y + \frac{8(x+1)}{x}H_{010}(y) + P(x,y)\ln y + Q(x,y)H_{10}(y) + R(x,y), \tag{3.151}$$

$$P(x,y) = -\frac{2}{y-1} - \frac{4(x+8)}{y} + \frac{8x+12}{x}y + \frac{2\widehat{c}_Z(x^2-3x-4)+2x^2+10x}{x}, \tag{3.152}$$

$$Q(x,y) = -\frac{4x}{y^2} + \frac{16}{y} + \frac{2x^2-18x+20}{x}, \tag{3.153}$$

$$R(x,y) = \frac{4(-2\widehat{c}_Z+x-8)}{y} + \frac{2\widehat{c}_Z-12x-16}{x}y. \tag{3.154}$$

The apparent pole multiplying $\ln y$ at $y = 1$ is removable, since $\ln y/(y-1) \to 1$. Direct differentiation recovers the complete crossed-ordering interference, including its finite on-shell subtraction. Consequently the angular integral of this virtual subset is

$$\boxed{\mathcal{C}_{\text{vertex}+\Sigma} = \frac{1}{2x^2}\,\mathrm{Re}[I_s + F_u(x,x) - F_u(x,x/(1+x))].} \tag{3.155}$$

This is an analytic contribution to the total cross section, with no loop-parameter or angular integral left. It still contains $\widehat{c}_Z = -4 - 4\ln\lambda$. Its regulator dependence is canceled only in the complete sum with the spanning and real contributions.

3.2.6 Algebraic cancellation of the infrared logarithm

The explicit trace coefficients allow the infrared cancellation to be checked as an identity before any angular integration. Write $x = s-1$, $y = 1-u$, $t = y-x$ and

$$\mathcal{B} = 2\left[\frac{x}{y} + \frac{y}{x} - 4\left(\frac{1}{y} - \frac{1}{x}\right) + 4\left(\frac{1}{y} - \frac{1}{x}\right)^2\right], \qquad z_e = 1 + \frac{x-y}{2}. \tag{3.156}$$

This is the Born spin average divided by e_{T}^4. For comparison with the soft current, define

$$b_e = \frac{z_e \operatorname{arcosh} z_e}{\sqrt{z_e^2-1}}, \qquad b_e\big|_{z_e=1} = 1. \tag{3.157}$$

The six vertex and self-energy insertions have total coefficient $\mathcal{B}$ multiplying $\widehat{c}_Z$ in their interference. Since $\widehat{c}_Z = -4 - 4\ln\lambda$, their infrared part is

$$\mathcal{T}_{\text{vertex}+\Sigma}|_{\ln\lambda} = -4\mathcal{B}\ln\lambda. \tag{3.158}$$

In each spanning graph only the scalar triangle $C_t = C_{013}$ and scalar box $D_b = D_{0123}$ contain $\ln\lambda$. Their coefficients are respectively $g_0(t)$ and $g_0(t)/(Q^2-1)$. Let $c_{t,s}$ and $c_{b,s}$ denote the rational trace coefficients multiplying these two scalar integrals in the s ordering, and use subscript u for the crossed ordering. The explicit coefficients derived in the spanning calculation satisfy the rational identity

$$c_{t,s} + \frac{c_{b,s}}{x} + c_{t,u} - \frac{c_{b,u}}{y} = 4\mathcal{B}z_e. \tag{3.159}$$

The script `check_compton_infrared_analytic.py` checks this equation symbolically, giving an identically zero numerator. Thus the full interference has infrared part

$$\operatorname{Re}\mathcal{T}|_{\ln\lambda} = 4\mathcal{B}[z_e g_0(t) - 1]\ln\lambda = 4\mathcal{B}(b_e - 1)\ln\lambda. \tag{3.160}$$

The last equality follows from $z_e = \cosh[2\,\operatorname{arsinh}(\sqrt{-t}/2)]$ and the explicit formula for $g_0(t)$. It is not an inference from numerical agreement.

After angular integration, the coefficient of $\ln\lambda$ in $\mathcal{C}_{\text{virt}}$ is therefore exactly

$$A(s) = \frac{1}{s}\int_{-1}^{1} \mathcal{B}(b_e - 1)\mathrm{d}c. \tag{3.161}$$

The soft calculation below evaluates this angular integral in closed form. Its real-emission matching contribution has coefficient $-A(s)\ln\lambda$, canceling the virtual term. This fixes the regulator-dependent part of the total before the remaining finite real-energy integrations are performed.

3.3 Soft subtraction and its normalization

3.3.1 Local subtraction in the real phase space

Work in the center-of-mass frame, and define

$$E = \frac{s+1}{2\sqrt{s}}, \qquad q = \frac{s-1}{2\sqrt{s}}, \qquad p = (E, 0, 0, -q), \qquad k = (q, 0, 0, q). \tag{3.162}$$

Order the outgoing photons by energy. Let $k_1 = \omega n_1$ be the lower-energy photon, $k_2 = \omega_2 n_2$, and $n_i = (1, \widehat{\boldsymbol{n}}_i)$. Restricting to $\omega < \omega_2$ accounts for identical photons without a factor $1/2!$. Put $c_{12} = \widehat{\boldsymbol{n}}_1 \cdot \widehat{\boldsymbol{n}}_2$ and

$$D_\gamma = \sqrt{s} - \omega(1 - c_{12}), \qquad \omega_2 = \frac{s - 1 - 2\sqrt{s}\,\omega}{2D_\gamma}, \tag{3.163}$$

$$\omega_* = \frac{s-1}{2\sqrt{s} + \sqrt{4s - 2(1-c_{12})(s-1)}}, \qquad 0 < \omega < \omega_*. \tag{3.164}$$

The on-shell condition for $p' = p + k - k_1 - k_2$ determines ω_2. The upper limit follows by setting $\omega = \omega_2$. Integrating the electron momentum delta function gives

$$\mathrm{d}\Phi_3 = \frac{\omega\omega_2}{8(2\pi)^5 D_\gamma}\mathrm{d}\omega\mathrm{d}\Omega_1\mathrm{d}\Omega_2. \tag{3.165}$$

Rotational symmetry about the incoming photon integrates the azimuth of n_2 to 2π. Denote its remaining polar cosine by c_2 and the relative azimuth of n_1 by ϕ.

Let $\mathcal{R} = \overline{|\mathcal{M}_{\gamma\gamma}|^2}/e_{\mathrm{T}}^6$ be computed from the six diagrams, and $\mathcal{B} = \overline{|\mathcal{M}_0|^2}/e_{\mathrm{T}}^4$. The elastic electron paired with direction n_2 is $p_B' = (E, -q\widehat{\boldsymbol{n}}_2)$. Define

$$j_n = \frac{p_B'}{p_B' \cdot n_1} - \frac{p}{p \cdot n_1}, \qquad \mathcal{A}_{\mathrm{loc}} = \mathcal{B}(-j_n^2)\frac{q}{\sqrt{s}}. \tag{3.166}$$

Then $(\omega\omega_2/D_\gamma)\mathcal{R} \to \mathcal{A}_{\mathrm{loc}}/\omega$. Subtract this term at every phase-space point, rather than integrating two separately large soft contributions. With $u = \omega/\omega_*$, the finite real part is

$$\mathcal{C}_{\mathrm{R}} = \frac{1}{4\pi(s-1)}\int_{-1}^{1}\mathrm{d}c_2\int\mathrm{d}\Omega_1\Big\{\int_0^1\mathrm{d}u\left[\frac{\omega_*^2 u\,\omega_2}{D_\gamma}\mathcal{R} - \frac{\mathcal{A}_{\mathrm{loc}}}{u}\right] + \mathcal{A}_{\mathrm{loc}}\ln\omega_*\Big\}. \tag{3.167}$$

The logarithms have dimensionless arguments because energies are measured in units of m. The subtracted integral is finite at $u = 0$. Both hard recoil and all spin-dependent emission terms remain.

3.3.2 A finite integral for the total correction

For the elastic momenta let $z_e = p \cdot p_B' = 1 - t/2$ and $b_e = z_e\,\mathrm{arcosh}(z_e)/\sqrt{z_e^2 - 1}$. The soft angular integrals in the center-of-mass frame reduce to one real parameter integral. Define

$$M_x = 1 - tx(1-x), \quad v_x = \sqrt{1 - M_x/E^2}, \quad \beta = q/E, \tag{3.168}$$

$$K = z_e\int_0^1\frac{\mathrm{d}x}{M_x}\frac{\operatorname{artanh} v_x}{v_x} - \frac{\operatorname{artanh}\beta}{\beta}. \tag{3.169}$$

Let ε be an auxiliary soft-photon energy boundary in this frame, with $\lambda \ll \varepsilon$ and ε small relative to the hard energies. The massive-photon radial integral is

$$\int_0^{\sqrt{\varepsilon^2-\lambda^2}}\frac{r^2\mathrm{d}r}{\sqrt{r^2+\lambda^2}(M_x r^2 + E^2\lambda^2)} = \frac{1}{M_x}\left[\operatorname{arcosh}\frac{\varepsilon}{\lambda} - \frac{\operatorname{artanh}(v_x\sqrt{1-\lambda^2/\varepsilon^2})}{v_x}\right]. \tag{3.170}$$

Thus $\delta_{\mathrm{s}}/\alpha_{\mathrm{T}} = 2[(b_e - 1)\ln(2\varepsilon/\lambda) - K]/\pi$ in the zero-mass limit. Let $\mathcal{T} = \mathcal{T}(\widehat{R}_1, R_0)$ be the spin-averaged bilinear in (2.63), with $m = 1$. Combining the soft integral with the analytic subtraction in (3.167) cancels ε before the remaining integrations. The result is

$$\mathcal{C}_{\mathrm{VS}} = \frac{1}{s}\int_{-1}^{1}\mathrm{d}c_2\left[\frac{1}{4}\operatorname{Re}\mathcal{T} + \mathcal{B}\left\{(b_e-1)\ln\frac{2}{\lambda} - K\right\}\right], \tag{3.171}$$

$$\sigma_1 = \boxed{\frac{\alpha_{\mathrm{T}}^3}{m^2}(\mathcal{C}_{\mathrm{VS}} + \mathcal{C}_{\mathrm{R}}).} \tag{3.172}$$

In the normalization used here, the soft part of the virtual interference obeys

$$\operatorname{Re}\mathcal{T}|_{\ln\lambda} = 4\mathcal{B}(b_e - 1)\ln\lambda. \tag{3.173}$$

It cancels the coefficient of $\ln\lambda$ in $\mathcal{C}_{\mathrm{VS}}$ point by point in the elastic angle. There is no photon-energy acceptance cut or numerical soft separator in this final expression. Its ingredients are the independently derived amplitudes and the displayed integrations. The analytic total in Equation (6) of Lee, Schwartz and Zhang [2] supplies a separate final comparison.

3.3.3 The finite soft function in dilogarithms

The soft-photon calculation leaves one finite parameter integral at fixed elastic scattering angle. It can be integrated before performing the elastic angular integral. In this section momenta and energies are in units of the electron mass, so $s > 1$. Define

$$E = \frac{s+1}{2\sqrt{s}}, \qquad q = \frac{s-1}{2\sqrt{s}}, \qquad \beta = \frac{q}{E}, \qquad t = -2q^2(1-c), \tag{3.174}$$

$$z_e = 1 - \frac{t}{2}, \qquad M_x = 1 - tx(1-x), \qquad v_x = \sqrt{1 - M_x/E^2}. \tag{3.175}$$

Here c is the cosine of the elastic photon scattering angle in the center-of-mass frame. The function to be integrated is

$$K = z_e I - \frac{\operatorname{artanh}\beta}{\beta}, \qquad I = \int_0^1 \frac{\operatorname{artanh} v_x}{v_x M_x}\,\mathrm{d}x. \tag{3.176}$$

The ratio $\operatorname{artanh} v_x / v_x$ equals one at $v_x = 0$ by continuity. Every quantity in this integral is real in the physical angular interval.

Rationalizing the square root. For an interior angle introduce

$$a = \frac{\sqrt{-t}}{2}, \qquad \kappa = \frac{\sqrt{q^2 - a^2}}{E}, \qquad 0 < a < q, \qquad 0 < \kappa < \beta < 1. \tag{3.177}$$

Symmetry about $x = 1/2$ allows the substitution $x = (1+y)/2$ with $0 \le y \le 1$. Then

$$M_x = 1 + a^2 - a^2 y^2, \qquad v_x^2 = \kappa^2 + \frac{a^2 y^2}{E^2}, \qquad I = \frac{1}{aE}\int_\kappa^\beta \frac{\operatorname{artanh} v}{(1-v^2)\sqrt{v^2-\kappa^2}}\,\mathrm{d}v. \tag{3.178}$$

Set $v = \kappa\cosh u = \kappa(r + r^{-1})/2$ with $r = e^u$. The result contains only rational functions and logarithms,

$$I = \frac{1}{aE}\int_1^R \frac{2r}{P_+(r)P_-(r)} \ln\frac{P_+(r)}{P_-(r)}\,\mathrm{d}r, \tag{3.179}$$

$$P_+(r) = \kappa r^2 + 2r + \kappa, \qquad P_-(r) = -\kappa r^2 + 2r - \kappa, \tag{3.180}$$

$$R = \frac{q+a}{\sqrt{q^2-a^2}}, \qquad Y = R - 1. \tag{3.181}$$

This transformation selects the positive square root because $y \ge 0$ and $u \ge 0$. Thus it does not require analytic continuation from another kinematic region.

A real dilogarithmic primitive. Let $d = \sqrt{1-\kappa^2}$ and list the four roots with their signs,

$$(z_1, z_2, z_3, z_4) = \left(-\frac{1+d}{\kappa}, -\frac{\kappa}{1+d}, \frac{\kappa}{1+d}, \frac{1+d}{\kappa}\right), \tag{3.182}$$

$$(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4) = (1, 1, -1, -1), \qquad h_j = z_j - 1, \tag{3.183}$$

$$A_j = \frac{1}{4 - 2\kappa^2(z_j^2 + 1)}, \qquad L_0 = \ln\frac{1+\kappa}{1-\kappa}. \tag{3.184}$$

Partial fractions and factorization give

$$\frac{2r}{P_+(r)P_-(r)} = \sum_{j=1}^{4} \frac{A_j}{r - z_j}, \tag{3.185}$$

$$\ln \frac{P_+(1+y)}{P_-(1+y)} = L_0 + \sum_{j=1}^{4} \epsilon_j \ln(1 - y/h_j). \tag{3.186}$$

The two negative roots and the smaller positive root have $h_j < 0$. The remaining root satisfies $z_4 > R$, because $v(R) = \beta < 1$ whereas $v(z_4) = 1$. Consequently every factor $1 - y/h_j$ is positive for $0 \le y \le Y$. This proves that all logarithms in the transformed integral are real and that no pole lies on the integration path.

Define the real dilogarithm on the whole real axis by

$$\mathcal{L}_2(x) = \operatorname{Re} \operatorname{Li}_2(x + \mathrm{i}0), \qquad \frac{\mathrm{d}\mathcal{L}_2(x)}{\mathrm{d}x} = -\frac{\ln|1-x|}{x}. \tag{3.187}$$

For $x > 1$ it can be evaluated without a complex branch choice as

$$\mathcal{L}_2(x) = \frac{\pi^2}{6} - \ln x \ln(x-1) - \operatorname{Li}_2(1-x), \qquad x > 1. \tag{3.188}$$

This identity follows by differentiating both sides and matching their continuous limits at $x = 1$. For $x \le 1$ the ordinary real dilogarithm applies. Thus an argument larger than one does not introduce an imaginary contribution to K. For g and h drawn from the h_j, put

$$\mathcal{J}(Y; g, h) = \ln|1 - g/h| \ln|1 - Y/g| - \mathcal{L}_2\!\left(\frac{Y-g}{h-g}\right) + \mathcal{L}_2\!\left(-\frac{g}{h-g}\right), \qquad g \neq h, \tag{3.189}$$

$$\mathcal{J}(Y; g, g) = \frac{1}{2} \ln^2|1 - Y/g|. \tag{3.190}$$

Differentiation establishes the required primitive directly,

$$\frac{\partial \mathcal{J}(Y; g, h)}{\partial Y} = \frac{\ln(1 - Y/h)}{Y - g}, \qquad \mathcal{J}(0; g, h) = 0. \tag{3.191}$$

Indeed, for $g \neq h$ the two logarithms produced by differentiation combine to $\ln|(h - Y)/h|$. Positivity proved above removes the absolute value from this last expression. The closed result is therefore

$$W_j = L_0 \ln|1 - Y/h_j| + \sum_{k=1}^{4} \epsilon_k \mathcal{J}(Y; h_j, h_k), \tag{3.192}$$

$$K = \boxed{\frac{z_e}{aE} \sum_{j=1}^{4} A_j W_j - \frac{\operatorname{artanh} \beta}{\beta}}. \tag{3.193}$$

This is a finite sum of logarithms and dilogarithms with explicit arguments. No parameter quadrature remains in the finite soft function.

Angular endpoints. At $t = 0$, $M_x = 1$ and $v_x = \beta$ throughout the interval, which gives

$$K(s, 0) = 0. \tag{3.194}$$

At $t = -4q^2$ the rationalizing variable used above degenerates because $\kappa = 0$. The endpoint can instead be integrated directly. Set $w = \operatorname{artanh}\beta$. Then

$$I\big|_{t=-4q^2} = \frac{1}{\beta E^2}\int_0^\beta \frac{\operatorname{artanh} v}{v(1-v^2)}\,\mathrm{d}v = \frac{J_\beta}{\beta E^2}, \tag{3.195}$$

$$J_\beta = \frac{\mathrm{Li}_2(\beta) - \mathrm{Li}_2(-\beta)}{2} + w\ln(2\cosh w) - \frac{w^2}{2} - \frac{1}{2}\mathrm{Li}_2(-e^{-2w}) - \frac{\pi^2}{24}, \tag{3.196}$$

$$K(s,-4q^2) = \frac{1+2q^2}{\beta E^2}J_\beta - \frac{w}{\beta}. \tag{3.197}$$

To obtain the second line, split $1/[v(1-v^2)]$ into $1/v + v/(1-v^2)$. The first term integrates to the difference of two dilogarithms. In the second term $v = \tanh u$ gives $\int_0^w u\tanh u\,\mathrm{d}u$, which is integrated by parts using $\ln\cosh u = u + \ln(1+e^{-2u}) - \ln 2$. All dilogarithm arguments in this endpoint expression are less than one.

3.3.4 The integrated infrared coefficient

The coefficient multiplying the soft logarithm can also be integrated analytically over the elastic angle. Define the spin-averaged Born squared amplitude with its factor e_{T}^4 removed by $\mathcal{B}$. In terms of $x = s-1$ and $y = 1-u = x+t$, its invariant expression is

$$\mathcal{B} = 2\left[\frac{x}{y} + \frac{y}{x} - 4\left(\frac{1}{y} - \frac{1}{x}\right) + 4\left(\frac{1}{y} - \frac{1}{x}\right)^2\right]. \tag{3.198}$$

The forward limit is $\mathcal{B} = 4$. At threshold this expression tends to $2(1+c^2)$, which also checks the normalization. Write the required angular coefficient as

$$\mathcal{C}_{\mathrm{IR}}(s) = \int_{-1}^{1}\mathcal{B}(s,t)\,[b_e(t)-1]\,\mathrm{d}c, \qquad b_e(t) = \frac{z_e\operatorname{arcosh} z_e}{\sqrt{z_e^2-1}}. \tag{3.199}$$

The rapidity variable $r = \exp(\operatorname{arcosh} z_e)$ removes the square root in b_e. Reversing the integration limits gives

$$t = 2 - r - r^{-1}, \qquad y = s + 1 - r - r^{-1}, \qquad 1 \le r \le s, \tag{3.200}$$

$$\mathcal{C}_{\mathrm{IR}}(s) = \frac{2s}{x^2}\int_1^s [R_+(r)\ln r - R_-(r)]\,\mathrm{d}r, \tag{3.201}$$

$$R_\eta(r) = \mathcal{B}(s, 2-r-r^{-1})(1+\eta r^{-2}), \qquad \eta = \pm 1. \tag{3.202}$$

The last integral is a rational function times a single logarithm. For completeness all its partial-fraction coefficients can be stated explicitly. Put

$$A = x - 4 - \frac{8}{x}, \qquad B_0 = \frac{2}{x}\left(s + 5 + \frac{4}{x}\right), \tag{3.203}$$

$$z_\pm = \frac{s+1 \pm \sqrt{(s-1)(s+3)}}{2}, \tag{3.204}$$

$$d_z = -1 + z^{-2}, \qquad e_z = -2z^{-3}, \qquad f_{\eta z} = 1 + \eta z^{-2}. \tag{3.205}$$

Then

$$R_\eta(r) = \sum_{n=-3}^{1} p_{\eta n} r^n + \sum_{z=z_\pm} \left[\frac{c_{\eta z}}{r-z} + \frac{d_{\eta z}}{(r-z)^2} \right], \tag{3.206}$$

$$p_{\eta,1} = -\frac{2}{x}, \quad p_{\eta,0} = B_0, \quad p_{\eta,-3} = -\frac{2\eta}{x}, \quad p_{\eta,-2} = \eta B_0, \tag{3.207}$$

$$p_{\eta,-1} = -\frac{2}{x} + \eta \left(-\frac{2}{x} - 2A \right), \tag{3.208}$$

$$d_{\eta z} = \frac{8 f_{\eta z}}{d_z^2}, \qquad c_{\eta z} = f_{\eta z} \left(\frac{2A}{d_z} - \frac{8 e_z}{d_z^3} \right) - \frac{16\eta}{z^3 d_z^2}. \tag{3.209}$$

The coefficients follow by expanding y to second order at each of its zeros. Both zeros are outside the interval: $0 < z_- < 1 < s < z_+$. No pole displacement or continuation is required for this angular integral.

Define elementary power primitives and the two pole primitives by

$$U_n(r) = \frac{r^{n+1}}{n+1}, \qquad V_n(r) = r^{n+1} \left[\frac{\ln r}{n+1} - \frac{1}{(n+1)^2} \right], \qquad n \neq -1, \tag{3.210}$$

$$U_{-1}(r) = \ln r, \qquad V_{-1}(r) = \frac{1}{2} \ln^2 r, \tag{3.211}$$

$$H_1(r,z) = \ln|r-z|, \qquad H_2(r,z) = -\frac{1}{r-z}, \tag{3.212}$$

$$J_1(r,z) = \ln r \ln|1 - r/z| + \mathcal{L}_2(r/z), \tag{3.213}$$

$$J_2(r,z) = -\frac{\ln r}{r-z} + \frac{\ln|r-z| - \ln r}{z}. \tag{3.214}$$

Their derivatives are, respectively, r^n, $r^n \ln r$, $(r-z)^{-j}$, and $(r-z)^{-j} \ln r$. The integrated result is

$$\mathcal{F}_+(r) = \sum_{n=-3}^{1} p_{+,n} V_n(r) + \sum_{z=z_\pm} [c_{+z} J_1(r,z) + d_{+z} J_2(r,z)], \tag{3.215}$$

$$\mathcal{F}_-(r) = \sum_{n=-3}^{1} p_{-,n} U_n(r) + \sum_{z=z_\pm} [c_{-z} H_1(r,z) + d_{-z} H_2(r,z)], \tag{3.216}$$

$$\Delta\mathcal{F}_\eta = \mathcal{F}_\eta(s) - \mathcal{F}_\eta(1), \tag{3.217}$$

$$\mathcal{C}_{\mathrm{IR}}(s) = \boxed{\frac{2s}{(s-1)^2} [\Delta\mathcal{F}_+ - \Delta\mathcal{F}_-].} \tag{3.218}$$

This expression contains no angular integral. The cancellation of the total soft logarithm can therefore be checked with its coefficient already integrated analytically.

3.3.5 Exchanging the finite soft angular integrals

The remaining finite soft angular coefficient is

$$\mathcal{C}_K(s) = \int_{-1}^{1} \mathcal{B}(s,t) K(s,t) \, \mathrm{d}c. \tag{3.219}$$

Rather than integrate the individual dilogarithms in (3.193), it is useful to exchange the original parameter and angular integrations. Put $T = -t$, $h = 1 + T/4$, and

$$f(M) = \frac{\operatorname{artanh} \sqrt{1 - M/E^2}}{M \sqrt{1 - M/E^2}}, \qquad I(T) = \frac{1}{2\sqrt{h-1}} \int_1^h \frac{f(M)}{\sqrt{h-M}} \, \mathrm{d}M. \tag{3.220}$$

This is obtained from $M = h - (h-1)y^2$ in the original parameter integral. Since $1 \le h \le E^2$, interchanging the two integrations yields

$$\int_0^{4q^2} \mathcal{B}(T)(1+T/2)I(T)\,\mathrm{d}T = 2\int_1^{E^2} f(M)\,\mathrm{d}M \int_M^{E^2} \frac{\mathcal{B}(h)(2h-1)}{\sqrt{(h-1)(h-M)}}\,\mathrm{d}h. \tag{3.221}$$

All endpoint singularities here are integrable. The substitution $u = \sqrt{(h-M)/(h-1)}$ makes the inner integral rational. Its upper limit is $U = \sqrt{E^2 - M}/q$. Define

$$k^2 = \frac{1+xU^2}{s}, \qquad x = s-1, \qquad h = \frac{M-u^2}{1-u^2}. \tag{3.222}$$

The rational function that occurs is

$$\frac{\mathcal{B}(h)(2h-1)}{1-u^2} = \frac{a_1}{1-u^2} + \frac{a_2}{(1-u^2)^2} + \frac{a_3}{(1-u^2)^3} + \frac{b_1}{k^2-u^2} + \frac{b_2}{(k^2-u^2)^2}, \tag{3.223}$$

with explicit coefficients

$$a_1 = -\frac{x^3 - 6x^2 - 16x - 8}{x^2}, \qquad a_2 = \frac{(1-k^2)(x^2+2x+4)}{x}, \tag{3.224}$$

$$a_3 = -x(1-k^2)^2, \qquad b_1 = \frac{x^3 - 2x^2 - 16x - 8}{x^2}, \tag{3.225}$$

$$b_2 = \frac{4(1-k^2)(x+2)}{x^2}. \tag{3.226}$$

The elementary primitives needed at the upper limit are

$$F_1(U;k) = \frac{\operatorname{artanh}(U/k)}{k}, \tag{3.227}$$

$$F_2(U;k) = \frac{U}{2k^2(k^2-U^2)} + \frac{F_1(U;k)}{2k^2}, \tag{3.228}$$

$$F_3(U;k) = \frac{U}{4k^2(k^2-U^2)^2} + \frac{3F_2(U;k)}{4k^2}. \tag{3.229}$$

They vanish at $U=0$ and differentiate to $(k^2-U^2)^{-n}$ when k is held fixed. In the inner integration it is indeed fixed. Consequently

$$H(U) = a_1F_1(U;1) + a_2F_2(U;1) + a_3F_3(U;1) + b_1F_1(U;k) + b_2F_2(U;k), \tag{3.230}$$

$$\mathcal{C}_K(s) = \frac{4}{\beta}\int_0^1 \frac{\operatorname{artanh}(\beta U)}{E^2(1-\beta^2U^2)} H(U)\,\mathrm{d}U - \frac{\operatorname{artanh}\beta}{\beta}\mathcal{C}_0(s), \tag{3.231}$$

$$\mathcal{C}_0(s) = \int_{-1}^1 \mathcal{B}\,\mathrm{d}c = \frac{4s}{x^2}[P(x) - P(x/s)], \tag{3.232}$$

$$P(y) = \frac{y^2}{2x} + \left(\frac{4}{x} + \frac{4}{x^2}\right)y + \left(x - 4 - \frac{8}{x}\right)\ln y - \frac{4}{y}. \tag{3.233}$$

The square root k is positive, and $k^2 - U^2 = (1-U^2)/s > 0$ for $0 \le U < 1$. Both inverse hyperbolic tangents in H are therefore real. This reduction integrates the original elastic angle analytically before performing the remaining soft parameter integral.

The one remaining square root can itself be rationalized by

$$w = \sqrt{x}U + \sqrt{1+xU^2}, \qquad 1 \le w \le \sqrt{x} + \sqrt{s}, \tag{3.234}$$

$$U = \frac{w - w^{-1}}{2\sqrt{x}}, \qquad k = \frac{w + w^{-1}}{2\sqrt{s}}. \tag{3.235}$$

Thus (3.231) becomes an integral of rational functions of w multiplying one or two logarithms of rational functions. It requires at most weight-three multiple polylogarithms. The endpoints and all logarithm arguments have been obtained directly from the physical interval, without selecting branches by comparison with a numerical answer.

Reduction to a finite polylogarithmic expression. It is useful to state an algebraic reduction which avoids printing a large list of weight-three terms. It terminates after at most two reductions for the expression below and contains no quadrature. Set $z = w - 1$, $Z = \sqrt{s} + \sqrt{x} - 1$, and regard U and k as the rational functions of $w = 1 + z$ just obtained. Write

$$H = h_0 + h_1 \operatorname{artanh} U + h_k \operatorname{artanh}(U/k), \tag{3.236}$$

$$h_0 = \frac{a_2 U}{2(1-U^2)} + a_3 \left[\frac{U}{4(1-U^2)^2} + \frac{3U}{8(1-U^2)} \right] + \frac{b_2 U}{2k^2(k^2-U^2)}, \tag{3.237}$$

$$h_1 = a_1 + \frac{a_2}{2} + \frac{3a_3}{8}, \qquad h_k = \frac{b_1}{k} + \frac{b_2}{2k^3}, \tag{3.238}$$

$$R_j(z) = \frac{4U'(z) h_j}{\beta E^2 (1 - \beta^2 U^2)}, \qquad j = 0, 1, k. \tag{3.239}$$

Define the positive root

$$r = \sqrt{1 + (\sqrt{x}/\beta)^2} - \sqrt{x}/\beta, \qquad 0 < r < 1. \tag{3.240}$$

Each R_j is explicitly rational. Its poles in the w plane belong to

$$w = 0, \quad \pm\mathrm{i}, \quad \pm r, \quad \pm r^{-1}, \tag{3.241}$$

The orders at $w = 0$ are three for R_0 and four for R_1, and R_k has no pole there. At $w = \pm\mathrm{i}$, R_0 has simple poles and R_k has double poles. The four real poles are simple. The polynomial parts of R_0, R_1, and R_k have degrees one, two, and no polynomial part, respectively. These statements follow by substituting the displayed rational expressions for U and k. Apparent endpoint poles in h_0 cancel because $1 - k^2 = x(1 - U^2)/s$ and $k^2 - U^2 = (1 - U^2)/s$.

Reducing the kernel before integration. Put $X = x = s - 1$ in this paragraph. An integration by parts removes the poles at $w = \pm\mathrm{i}$ before any polylogarithmic primitive is constructed. It combines the elementary and root-dependent parts of the original soft kernel, so no integrated cross section is needed.

Keep $X = s - 1$, $\beta = X/(X + 2)$ and $E^2 = (X + 2)^2/(4s)$, and define

$$k^2 = \frac{1 + XU^2}{s}, \qquad d = 1 - \beta^2 U^2, \qquad a = \operatorname{artanh}(\beta U), \qquad f = \operatorname{artanh}(U/k). \tag{3.242}$$

In this section d denotes this rational function, rather than the spacetime dimension. On $0 < U < 1$, all square roots are positive and both functions are real. Separate the elementary part of H as

$$h_0 = \overline{h}_0 + \frac{2(X+2)U}{Xk^2}, \tag{3.243}$$

$$\overline{h}_0 = U \left[\frac{X^2 + 2X + 4}{2s} - \frac{X^3}{4s^2} - \frac{3X^3(1-U^2)}{8s^2} \right], \tag{3.244}$$

$$h_k = \frac{b_1}{k} + \frac{2(X+2)(1-U^2)}{Xsk^3}, \qquad b_1 = \frac{X^3 - 2X^2 - 16X - 8}{X^2}. \tag{3.245}$$

The identities

$$\frac{\mathrm{d}f}{\mathrm{d}U} = \frac{1}{k(1-U^2)}, \qquad p(U) = \frac{2(X+2)U(1-U^2)}{X} \tag{3.246}$$

give the exact integration-by-parts relation

$$\begin{aligned}\frac{a}{d}\left[\frac{2(X+2)U}{Xk^2} + h_k f\right] &= \frac{\mathrm{d}}{\mathrm{d}U}\left(\frac{apf}{dk}\right) \\ &\quad + \frac{f}{k}\left[a\left(\frac{b_1}{d} + \frac{16sU^2}{X(X+2)d^2}\right) - \frac{2U(1-U^2)}{d^2}\right].\end{aligned} \tag{3.247}$$

Its boundary term vanishes at both ends. At $U = 0$, a and f vanish linearly. At $U \to 1$, $p = O(1-U)$ while f grows only as $\ln(1-U)$, and $d \to 1-\beta^2 > 0$. There is therefore no boundary constant to supply from a different calculation.

Use the previous rational variable $U = (w - w^{-1})/(2\sqrt{X})$. The remaining root factor cancels from its differential,

$$\frac{\mathrm{d}U}{k} = \sqrt{\frac{s}{X}}\frac{\mathrm{d}w}{w}. \tag{3.248}$$

Writing $z = w - 1$, define four rational kernels

$$\widetilde{R}_0 = \frac{4}{\beta E^2}\frac{U'\overline{h}_0}{d}, \qquad \widetilde{R}_1 = \frac{4}{\beta E^2}\frac{U' h_1}{d}, \tag{3.249}$$

$$\widetilde{R}_k = \frac{4}{\beta E^2}\sqrt{\frac{s}{X}}\frac{1}{w}\left[\frac{b_1}{d} + \frac{16sU^2}{X(X+2)d^2}\right], \qquad \widetilde{R}_f = -\frac{8}{\beta E^2}\sqrt{\frac{s}{X}}\frac{U(1-U^2)}{wd^2}. \tag{3.250}$$

Here $U' = \mathrm{d}U/\mathrm{d}z$ and

$$h_1 = -\frac{X^3 - 6X^2 - 16X - 8}{X^2} + \frac{(1-k^2)(X^2+2X+4)}{2X} - \frac{3X}{8}(1-k^2)^2. \tag{3.251}$$

The logarithms are $L_\beta = \operatorname{artanh}(\beta U)$, $L_1 = \operatorname{artanh} U$, and $L_k = \operatorname{artanh}(U/k)$. The transformed integrand is

$$\widetilde{R}_0 L_\beta + \widetilde{R}_1 L_\beta L_1 + \widetilde{R}_k L_\beta L_k + \widetilde{R}_f L_k. \tag{3.252}$$

Every denominator now factors into powers of w and

$$Q_w = -Xw^4 + (4X^2 + 18X + 16)w^2 - X. \tag{3.253}$$

The roots of Q_w are the four real poles $\pm r, \pm r^{-1}$, where $r = \sqrt{1 + X/\beta^2} - \sqrt{X}/\beta$. The factor $1 + w^2$ is absent. Thus the entire GPL expression, including the letters introduced by partial fractions, has only real letters.

For a finite list of pairs (a, ℓ_a) write $L(z) = \sum_a \ell_a G(a; z)$, where G is the usual multiple polylogarithm with kernels $1/(z-a)$. The three lists needed here are

$$\mathcal{A}_\beta = \tfrac{1}{2}\{(r-1,+1), (-r^{-1}-1,+1), (r^{-1}-1,-1), (-r-1,-1)\}, \tag{3.254}$$

$$\mathcal{A}_1 = \tfrac{1}{2}\{(W^{-1}-1,+1), (-W-1,+1), (W-1,-1), (-W^{-1}-1,-1)\}, \tag{3.255}$$

$$\mathcal{A}_k = \tfrac{1}{2}\{(W^{-1}-1,+1), (-W^{-1}-1,+1), (W-1,-1), (-W-1,-1)\}, \tag{3.256}$$

$$W = \sqrt{s} + \sqrt{x}. \tag{3.257}$$

Here the factor $1/2$ multiplies the weights, not the letters. Factoring $1 \pm \beta U$, $1 \pm U$, and $k \pm U$ proves

$$L_\beta(z) = \operatorname{artanh}(\beta U), \qquad L_1(z) = \operatorname{artanh} U, \qquad L_k(z) = \operatorname{artanh}(U/k). \tag{3.258}$$

All three vanish at $z = 0$, which fixes the constants in the factorized logarithms. Their real letters lie outside $0 < z < Z$. The two functions with endpoint logarithms are defined by their limits from $z < Z$.

Here is an explicit algebraic rule for removing the rational factors. Let $\mathfrak{T}[R; \boldsymbol{a}](z)$ be linear in the rational function R. Expand R into its polynomial part and partial fractions. For a pole a of order m_a, the coefficient of $(z-a)^{-n}$ is explicitly

$$C_{a,n} = \frac{1}{(m_a - n)!} \frac{\mathrm{d}^{m_a - n}}{\mathrm{d}z^{m_a - n}} [(z-a)^{m_a} R(z)] \bigg|_{z=a}, \qquad 1 \le n \le m_a. \tag{3.259}$$

The polynomial part is obtained by polynomial division. The known pole list above makes all these operations finite. For a nonempty word $\boldsymbol{a} = (b, \boldsymbol{c})$ use

$$\mathfrak{T}[(z-a)^{-1}; \boldsymbol{a}](z) = G(a, \boldsymbol{a}; z), \tag{3.260}$$

$$\mathfrak{T}[(z-a)^{-n}; b, \boldsymbol{c}](z) = -\frac{G(b, \boldsymbol{c}; z)}{(n-1)(z-a)^{n-1}} + \frac{1}{n-1} \mathfrak{T}\left[\frac{1}{(z-a)^{n-1}(z-b)}; \boldsymbol{c}\right](z), \quad n \ge 2, \tag{3.261}$$

$$\mathfrak{T}[z^n; b, \boldsymbol{c}](z) = \frac{z^{n+1}}{n+1} G(b, \boldsymbol{c}; z) - \frac{1}{n+1} \mathfrak{T}\left[\frac{z^{n+1}}{z-b}; \boldsymbol{c}\right](z), \quad n \ge 0. \tag{3.262}$$

For the empty word, use the elementary rules

$$\mathfrak{T}[z^n; \varnothing](z) = \frac{z^{n+1}}{n+1}, \tag{3.263}$$

$$\mathfrak{T}[(z-a)^{-1}; \varnothing](z) = G(a; z), \tag{3.264}$$

$$\mathfrak{T}[(z-a)^{-n}; \varnothing](z) = \frac{(z-a)^{1-n} - (-a)^{1-n}}{1-n}, \qquad n \ge 2. \tag{3.265}$$

Every pole appearing here has $a \neq 0$. Differentiation proves that $\mathfrak{T}[R; \boldsymbol{a}]' = R\, G(\boldsymbol{a}; z)$ and its value at zero is zero. For repeated poles and polynomial terms the word length decreases. Simple poles terminate immediately. Hence these rules specify a finite combination of rational functions and ordinary multiple polylogarithms, rather than a new unevaluated integral.

The endpoint expression for the finite soft angular coefficient is

$$\mathcal{S}_0(z) = \sum_{a \in \mathcal{A}_\beta} \ell_a \mathfrak{T}[R_0; a](z), \tag{3.266}$$

$$\mathcal{S}_j(z) = \sum_{a \in \mathcal{A}_\beta} \sum_{b \in \mathcal{A}_j} \ell_a \ell_b \left\{ \mathfrak{T}[R_j; a, b](z) + \mathfrak{T}[R_j; b, a](z) \right\}, \qquad j = 1, k, \tag{3.267}$$

$$\mathcal{C}_K(s) = \lim_{z \to Z^-} [\mathcal{S}_0(z) + \mathcal{S}_1(z) + \mathcal{S}_k(z)] - \frac{\operatorname{artanh} \beta}{\beta} \mathcal{C}_0(s). \tag{3.268}$$

The two words in $\mathcal{S}_j$ are the shuffle identity for the product of two weight-one logarithms. Formula (3.268) is an algebraically specified weight-three polylogarithmic expression. An expanded list of its coefficients can be generated using only partial fractions and the finite rules above. The limit is finite: the original U integrand has at most an integrable logarithm at $U = 1$.

The reduced soft endpoint. For an explicit closed endpoint expression, apply the finite primitive operator $\mathfrak{T}$ just defined to the four reduced kernels and the logarithm lists

$\mathcal{A}_\beta, \mathcal{A}_1, \mathcal{A}_k$. The notation $\mathfrak{T}[R; L_a L_b]$ means the finite sum over their weights after applying $G(a)G(b) = G(a,b) + G(b,a)$. Put

$$\begin{aligned}\widetilde{\Psi}(z) = \mathfrak{T}[\widetilde{R}_0; L_\beta](z) + \mathfrak{T}[\widetilde{R}_1; L_\beta L_1](z) \\ + \mathfrak{T}[\widetilde{R}_k; L_\beta L_k](z) + \mathfrak{T}[\widetilde{R}_f; L_k](z).\end{aligned} \tag{3.269}$$

Each primitive is normalized to vanish at $z = 0$. The simplified answer is

$$\boxed{\mathcal{C}_K(s) = \lim_{z \to Z^-} \widetilde{\Psi}(z) - \frac{\operatorname{artanh}\beta}{\beta}\mathcal{C}_0(s), \qquad Z = \sqrt{s} + \sqrt{X} - 1.} \tag{3.270}$$

The combined endpoint is finite. All terms are GPLs of weight at most three with algebraic endpoint coefficients. There are two further exact residue cancellations. If P_1 denotes the primitive of the higher-order pole part of $\widetilde{R}_1$ at $w = 0$, then

$$\operatorname*{Res}_{w=0} \widetilde{R}_1 = 0, \qquad \operatorname*{Res}_{w=0}\left[\widetilde{R}_0 - P_1 \frac{\mathrm{d}L_1}{\mathrm{d}w}\right] = 0. \tag{3.271}$$

The first identity removes 32 weight-three words introduced by a vanishing simple-pole coefficient. The second removes four weight-two words. They follow by substituting the displayed rational kernels, so no small numerical coefficient is discarded. After these cancellations, the finite rule produces 190 terms instead of 306, and removes both imaginary letters. The same physical interval fixes all logarithms.

The finite endpoint limit. The convergence of the endpoint expression follows from its local expansion. Let $d = 1 - U$ and let $\mathcal{I}(U)$ denote the integrand including the factor $4/\beta$ in (3.231). Define

$$h_{0*} = \frac{x^2 + 2x + 4}{2s} - \frac{x^3}{4s^2} + \frac{2(x+2)}{x}, \tag{3.272}$$

$$B_* = h_{0*} + 2\ln 2 + \frac{b_1}{2}\ln s. \tag{3.273}$$

The rational coefficients have finite limits at $U = 1$, and $h_1 + h_k = 4$ there. Also $1 - U/k = d/s + O(d^2)$. Hence

$$\mathcal{I}(1-d) = \frac{4\operatorname{artanh}\beta}{\beta}[-2\ln d + B_*] + O(d\ln d), \tag{3.274}$$

$$\int_{1-d}^{1} \mathcal{I}(U)\,\mathrm{d}U = \frac{4\operatorname{artanh}\beta}{\beta} d[-2\ln d + 2 + B_*] + O(d^2 \ln d). \tag{3.275}$$

Since $d\ln d \to 0$, the omitted tail vanishes. This establishes the existence of the combined polylogarithmic endpoint limit.

3.3.6 Changing the soft energy frame

A useful independent check is to place the soft-photon energy bound in the final electron rest frame. Write $z_e = \cosh\eta$ and $v = \tanh\eta$. In this frame the final electron has energy one, the initial electron has energy z_e, and their interpolated momentum has

$$E_x = 1 + x(z_e - 1), \qquad M_x = 1 + 2(z_e - 1)x(1-x), \qquad v_x = \frac{x\sqrt{z_e^2 - 1}}{1 + x(z_e - 1)}. \tag{3.276}$$

In this paragraph x is a Feynman parameter. The finite cross term integrates directly after changing from x to v_x,

$$\int_0^1 \frac{\operatorname{artanh} v_x}{v_x M_x}\,\mathrm{d}x = \frac{J(v)}{\sqrt{z_e^2-1}}, \qquad J(v) = \int_0^v \frac{\operatorname{artanh} u}{u(1-u^2)}\,\mathrm{d}u. \tag{3.277}$$

The integral J is the dilogarithmic expression J_β above with β replaced by v and w by η. Including the two self terms gives the closed finite function

$$K_{\mathrm{rest}}(t) = \frac{J(v)}{v} - \frac{b_e+1}{2}, \qquad b_e = \frac{\eta}{v}. \tag{3.278}$$

Thus the difference between a rest-frame energy bound $\sqrt{s}\varepsilon$ and a center-of-mass energy bound ε is

$$\frac{\delta_{\mathrm{rest}} - \delta_{\mathrm{CM}}}{\alpha_{\mathrm{T}}} = \frac{2}{\pi}\left[(b_e-1)\ln\sqrt{s} + K - K_{\mathrm{rest}}\right]. \tag{3.279}$$

This finite change also follows directly from the angular integral of the eikonal factor multiplied by $\ln[\sqrt{s}/(p'\cdot n)]$, where $n=(1,\widehat{\boldsymbol{k}})$ in the center-of-mass frame.

For example, consider the asymmetric invariant cuts $2P\cdot k_1 > \delta$ and $2p'\cdot k_2 > \delta$, with $P = p+k$ and $\delta = 2\sqrt{s}\varepsilon$. Since $2P\cdot k_1 + 2p'\cdot k_2 = s-1$, their omitted soft regions do not overlap for sufficiently small δ. The two soft terms have weights $1/2$ from the identical-photon phase-space factor. Their average differs from the original center-of-mass soft term by half of (3.279). In the total coefficient multiplying $\alpha_{\mathrm{T}}^3/m^2$, the shift is

$$\Delta C_{\mathrm{VS}} = \frac{1}{2s}\left[\mathcal{C}_K - \mathcal{C}_{K,\mathrm{rest}} + \frac{\ln s}{2}\mathcal{C}_{\mathrm{IR}}\right]. \tag{3.280}$$

It cannot be omitted if this asymmetric cut is used.

The rest-frame angular coefficient has a short weight-three expression. Set $r = e^\eta$ and $z = r-1$. Since

$$\frac{\mathrm{d}J}{\mathrm{d}r} = \ln r\left[-\frac{1}{r} + \frac{1}{r-1} + \frac{1}{r+1}\right], \qquad J(0\text{ in }\eta) = 0, \tag{3.281}$$

its exact polylogarithmic form is

$$J = -G(-1,-1;z) + G(0,-1;z) + G(-2,-1;z). \tag{3.282}$$

Let $\widetilde{R}_+(z) = R_+(1+z)$, with R_+ the rational kernel used in the integrated infrared coefficient. The algebraic map $\mathfrak{T}$ then gives

$$\mathcal{Q}(z) = -\mathfrak{T}[\widetilde{R}_+;-1,-1](z) + \mathfrak{T}[\widetilde{R}_+;0,-1](z) + \mathfrak{T}[\widetilde{R}_+;-2,-1](z), \tag{3.283}$$

$$\mathcal{C}_{K,\mathrm{rest}}(s) = \frac{2s}{(s-1)^2}\mathcal{Q}(s-1) - \frac{1}{2}\mathcal{C}_{\mathrm{IR}}(s) - \mathcal{C}_0(s). \tag{3.284}$$

This alternative frame is not needed for the symmetric analytic regulator described next.

3.4 Real-emission integration

3.4.1 Real radiation in invariant energy variables

The energy ordering used above is convenient for numerical checks. For analytic integration it is useful to integrate over both photon labels and include the identical-particle factor

1/2!. All invariants in this section are again measured in units of m^2. In the center-of-mass frame define

$$H = s - 1, \qquad x = 2\sqrt{s}\,\omega_1, \qquad y = 2\sqrt{s}\,\omega_2, \qquad z = x + y - H. \tag{3.285}$$

Momentum conservation and the final electron mass shell give

$$2k_1 \cdot k_2 = z, \qquad 2p' \cdot k_1 = H - y, \qquad 2p' \cdot k_2 = H - x. \tag{3.286}$$

Since $0 \le 1 - \widehat{\boldsymbol{n}}_1 \cdot \widehat{\boldsymbol{n}}_2 \le 2$, $0 \le z \le xy/s$. Thus the exact integration region is

$$0 < x < H, \qquad H - x < y < \frac{s(H-x)}{s-x}. \tag{3.287}$$

This two-dimensional region in final-state energy invariants is called the Dalitz region. Its boundaries are rational functions. The limits $x = 0$ and $y = 0$ are the two soft endpoints, while $z = 0$ means that the final photons are parallel.

To obtain the measure, first integrate the three-momentum delta function over the final electron momentum. Let $c_{12} = \widehat{\boldsymbol{n}}_1 \cdot \widehat{\boldsymbol{n}}_2$ and let ϕ be the azimuth of the second photon around the first. The remaining energy delta function gives

$$E' = \sqrt{1 + \omega_1^2 + \omega_2^2 + 2\omega_1\omega_2 c_{12}}, \tag{3.288}$$

$$\mathrm{d}\Phi_3 = \frac{\omega_1\omega_2}{8(2\pi)^5 E'}\delta(\sqrt{s} - \omega_1 - \omega_2 - E')\mathrm{d}\omega_1\mathrm{d}\omega_2\mathrm{d}\Omega_1\mathrm{d}\phi\mathrm{d}c_{12}. \tag{3.289}$$

The absolute derivative of the delta-function argument with respect to c_{12} is $\omega_1\omega_2/E'$. Its integral therefore cancels precisely the energy-dependent prefactor in this measure. A root exists with $-1 \le c_{12} \le 1$ exactly in the Dalitz region already derived.

For fixed x, y, the relative angle is now fixed. Choose a frame attached to the two outgoing photon directions, and denote the incoming photon direction in that frame by $\widehat{\boldsymbol{n}}$. The orientation measure $\mathrm{d}\Omega_1\mathrm{d}\phi$ has total volume $8\pi^2$. Rotations about the incoming direction leave the unpolarized amplitude unchanged and integrate to 2π, leaving $\mathrm{d}\Omega_{\boldsymbol{n}}$. Finally $\mathrm{d}\omega_1\mathrm{d}\omega_2 = \mathrm{d}x\mathrm{d}y/(4s)$, so

$$\mathrm{d}\Phi_3 = \frac{\mathrm{d}x\,\mathrm{d}y}{128\pi^3 s}\frac{\mathrm{d}\Omega_{\boldsymbol{n}}}{4\pi}. \tag{3.290}$$

For example, its massless limit integrates over a triangle of area $s^2/2$ and gives $\Phi_3 = s/(256\pi^3)$. With $\mathcal{R} = \overline{|\mathcal{M}_{\gamma\gamma}|^2}/e_{\mathrm{T}}^6$, the real cross section is consequently

$$\sigma_{\mathrm{R}} = \frac{\alpha_{\mathrm{T}}^3}{8sHm^2}\int_0^H \mathrm{d}x \int_{H-x}^{s(H-x)/(s-x)} \mathrm{d}y\,\langle\mathcal{R}\rangle_{\boldsymbol{n}}, \qquad \langle f\rangle_{\boldsymbol{n}} = \int \frac{\mathrm{d}\Omega_{\boldsymbol{n}}}{4\pi} f. \tag{3.291}$$

This formula requires the same infrared subtraction as the earlier real contribution. Its unsubtracted integral diverges at the soft endpoints.

3.4.2 Analytic direction integrals

Introduce three positive electron–photon invariants,

$$U = 2p \cdot k_1 = \frac{x}{2s}[s + 1 + H\widehat{\boldsymbol{n}} \cdot \widehat{\boldsymbol{n}}_1], \tag{3.292}$$

$$V = 2p \cdot k_2 = \frac{y}{2s}[s + 1 + H\widehat{\boldsymbol{n}} \cdot \widehat{\boldsymbol{n}}_2], \tag{3.293}$$

$$W = 2p' \cdot k = U + V - z. \tag{3.294}$$

The labels UV, UW, and VW below identify pairs of these directional denominators. In particular, UV here does not mean ultraviolet. The six electron-line orderings have denominator products proportional to

$$H(H-y), \quad H(H-x), \quad U(H-x), \quad V(H-y), \quad UW, \quad VW. \tag{3.295}$$

The signs of the individual propagators follow directly from $(p-k_i)^2-1=-2p\cdot k_i$ and $(p'-k)^2-1=-2p'\cdot k$. Thus all directional denominators of the spin-summed squared amplitude are powers of U,V,W. Its numerator is polynomial in these invariants, since all remaining scalar products follow from momentum conservation.

Their affine relation gives a useful partial-fraction identity,

$$\frac{1}{UVW}=\frac{1}{z}\left(\frac{1}{UW}+\frac{1}{VW}-\frac{1}{UV}\right). \tag{3.296}$$

Repeated application after multiplication by further inverse powers reduces every term with three denominator types to terms with at most two. This operation is valid for $z>0$. Any apparent pole at $z=0$ introduced by this reduction must be combined before taking that limit.

The basic two-denominator direction integral has an elementary closed form. Write $A=a_0-\boldsymbol{a}\cdot\widehat{\boldsymbol{n}}$ and $B=b_0-\boldsymbol{b}\cdot\widehat{\boldsymbol{n}}$, where a and b are future timelike four-vectors. Define

$$d=a\cdot b, \qquad \Delta=d^2-a^2b^2. \tag{3.297}$$

The bounds $A\geq a_0-|\boldsymbol{a}|>0$ and $B\geq b_0-|\boldsymbol{b}|>0$ hold uniformly on the sphere. Their convex combinations are bounded below by $\min(a_0-|\boldsymbol{a}|,b_0-|\boldsymbol{b}|)>0$, so the Feynman parameter integral can be interchanged with the direction average. Direct polar integration of $[vA+(1-v)B]^{-2}$ gives $[va+(1-v)b]^{-2}$, where the latter square is a Minkowski square. Consequently,

$$J(a,b)=\left\langle\frac{1}{AB}\right\rangle_{\boldsymbol{n}}=\int_0^1\frac{\mathrm{d}v}{v^2a^2+2v(1-v)d+(1-v)^2b^2} \tag{3.298}$$

$$=\frac{1}{2\sqrt{\Delta}}\ln\frac{d+\sqrt{\Delta}}{d-\sqrt{\Delta}}. \tag{3.299}$$

For $\Delta>0$, the substitution $u=v/(1-v)$ evaluates the parameter integral explicitly:

$$J(a,b)=\int_0^\infty\frac{\mathrm{d}u}{a^2u^2+2du+b^2}=\left[\frac{1}{2\sqrt{\Delta}}\ln\frac{a^2u+d-\sqrt{\Delta}}{a^2u+d+\sqrt{\Delta}}\right]_{u=0}^{u=\infty}. \tag{3.300}$$

Future timelike vectors obey $d>0$, $\Delta\geq0$, and $d>\sqrt{\Delta}$. Thus both logarithm arguments are positive, and the primitive tends to zero at infinity. These endpoint values fix the real logarithm and its constant. At $\Delta=0$ the quadratic denominator is $(\sqrt{a^2}u+\sqrt{b^2})^2$, whose integral is $1/d$. The positive bounds on A and B persist in a neighborhood of any fixed future timelike pair. The integrand and its derivatives of any finite order are then uniformly bounded on the sphere, which permits differentiation under the integral. Holding $\boldsymbol{a},\boldsymbol{b}$ fixed, all higher denominator powers follow by differentiation,

$$\left\langle A^{-r}B^{-t}\right\rangle_{\boldsymbol{n}}=\frac{(-\partial_{a_0})^{r-1}(-\partial_{b_0})^{t-1}}{(r-1)!(t-1)!}J(a,b), \qquad r,t\geq1. \tag{3.301}$$

These derivatives are taken before substituting the physical invariants. They are finite algebraic differentiations of the logarithm in (3.299). For a single denominator the corresponding formula is

$$\langle A^{-1}\rangle_{\boldsymbol{n}} = \frac{1}{2|\boldsymbol{a}|}\ln\frac{a_0+|\boldsymbol{a}|}{a_0-|\boldsymbol{a}|}, \qquad \langle A^{-2}\rangle_{\boldsymbol{n}} = \frac{1}{a^2}. \tag{3.302}$$

The physical vectors corresponding to U, V, W satisfy

$$a_U^2 = x^2/s, \qquad a_V^2 = y^2/s, \qquad a_W^2 = H^2/s, \tag{3.303}$$

$$a_U \cdot a_V = xy/s + H^2 z/(2s), \tag{3.304}$$

$$a_U \cdot a_W = xH/s + zH(H-x)/(2s), \tag{3.305}$$

$$a_V \cdot a_W = yH/s + zH(H-y)/(2s). \tag{3.306}$$

In particular they remain timelike throughout the interior of the Dalitz region. These identities make each scalar angular master a closed logarithm of s, x, y.

The next section carries out this reduction for the full six-diagram trace and lists the resulting coefficients. The remaining x, y integrations still require the infrared subtraction.

3.4.3 Explicit completion of the direction integral

The preceding reduction can be carried out for the full six-diagram square. We describe the algebra and give its coefficients so that the result can be reconstructed without evaluating Dirac matrices. The ordered photon triples are

$$(a,b,c) = (0,1,2),(0,2,1),(1,0,2),(1,2,0),(2,0,1),(2,1,0). \tag{3.307}$$

Their signed denominator products are, in this order,

$$d_i = \big(H(H-x), H(H-y), -U(H-x), UW, -V(H-y), VW\big)_i. \tag{3.308}$$

Let T_{ij} be the numerator trace for the interference of orders i, j, including the initial average $1/4$ and three polarization sums $-\eta_{\mu\nu}$. Then

$$\mathcal{R} = \sum_{i\le j} \frac{(2-\delta_{ij})T_{ij}}{d_i d_j}. \tag{3.309}$$

There are 21 independent terms, including all 36 interferences. The trace reduction follows directly by moving the first contracted gamma matrix through the intervening string. For $\Gamma_n = \gamma^{\alpha_1}\cdots\gamma^{\alpha_n}$ in four dimensions,

$$\gamma_\mu \Gamma_n \gamma^\mu = 4(-1)^n \Gamma_n + 2\sum_{j=1}^{n}(-1)^{j-1}\gamma^{\alpha_1}\cdots\widehat{\gamma^{\alpha_j}}\cdots\gamma^{\alpha_n}\gamma^{\alpha_j}. \tag{3.310}$$

The hat denotes omission. Apply this relation to the three photon index pairs, using cyclicity of the trace to bring each pair to the two ends. The remaining traces contain only slashed momenta and obey

$$\operatorname{tr}(\not{v}_1\cdots\not{v}_{2n}) = \sum_{j=2}^{2n}(-1)^j (v_1\cdot v_j)\operatorname{tr}(\not{v}_2\cdots\widehat{\not{v}_j}\cdots\not{v}_{2n}), \qquad \operatorname{tr}\mathbf{1} = 4. \tag{3.311}$$

All scalar products are fixed by the Gram matrix of (p, k, k_1, k_2),

$$G = \frac{1}{2}\begin{pmatrix} 2 & H & U & V \\ H & 0 & x-U & y-V \\ U & x-U & 0 & z \\ V & y-V & z & 0 \end{pmatrix}. \tag{3.312}$$

Thus this is a finite polynomial calculation with no integration or input from a published correction factor.

After applying (3.296) and cancelling numerator factors, the result has 25 monomials,

$$\mathcal{R} = \sum_{\boldsymbol{e}} C_{\boldsymbol{e}} U^{e_1} V^{e_2} W^{e_3}, \qquad \boxed{\langle \mathcal{R} \rangle_{\boldsymbol{n}} = \sum_{\boldsymbol{e}} C_{\boldsymbol{e}} \mathfrak{M}_{\boldsymbol{e}}.} \tag{3.313}$$

The coefficients are listed below. The moments $\mathfrak{M}$ in this formula are the following closed expressions, not unevaluated angular integrals.

For two denominator types write $A = a_0 - \boldsymbol{a}\cdot\widehat{\boldsymbol{n}}$, $B = b_0 - \boldsymbol{b}\cdot\widehat{\boldsymbol{n}}$ and abbreviate $\alpha = a^2$, $\beta = b^2$, $d = a\cdot b$, $\Delta = d^2 - \alpha\beta$. Here α, β are auxiliary kinematic symbols and are not the electromagnetic coupling. Let J be the logarithm in (3.299). The required inverse-power moments are

$$\langle A^{-1}B^{-1}\rangle = J, \tag{3.314}$$

$$\langle A^{-2}B^{-1}\rangle = \frac{(b_0 d - a_0\beta)J + a_0 d/\alpha - b_0}{\Delta}, \tag{3.315}$$

$$\langle A^{-1}B^{-2}\rangle = \frac{(a_0 d - b_0\alpha)J + b_0 d/\beta - a_0}{\Delta}, \tag{3.316}$$

$$\langle A^{-2}B^{-2}\rangle = \frac{Q_J J + Q_0}{\Delta^2}, \tag{3.317}$$

$$Q_J = 2a_0 b_0(\alpha\beta + 2d^2) - d(\Delta + 3\alpha b_0^2 + 3\beta a_0^2), \tag{3.318}$$

$$Q_0 = \Delta + (2\alpha\beta + d^2)\left(\frac{a_0^2}{\alpha} + \frac{b_0^2}{\beta}\right) - 6a_0 b_0 d. \tag{3.319}$$

For positive powers of A and one denominator type B, define

$$r^2 = b_0^2 - \beta, \qquad h = \frac{a_0 b_0 - d}{r^2}, \qquad g = a_0 - h b_0, \tag{3.320}$$

$$v_\perp^2 = a_0^2 - \alpha - \frac{(a_0 b_0 - d)^2}{r^2}, \qquad L_B = \frac{1}{2r}\ln\frac{b_0 + r}{b_0 - r}, \tag{3.321}$$

$$n_0 = g^2 - \frac{\beta v_\perp^2}{2r^2}, \qquad n_1 = 2gh + \frac{b_0 v_\perp^2}{r^2}, \qquad n_2 = h^2 - \frac{v_\perp^2}{2r^2}. \tag{3.322}$$

Integration over the azimuth about $\boldsymbol{b}$ first gives the polynomials $g + hB$ and $n_0 + n_1 B + n_2 B^2$. Hence

$$\langle A/B\rangle = gL_B + h, \qquad \langle A/B^2\rangle = g/\beta + hL_B, \tag{3.323}$$

$$\langle A^2/B\rangle = n_0 L_B + n_1 + n_2 b_0, \tag{3.324}$$

$$\langle B^{-1}\rangle = L_B, \qquad \langle B^{-2}\rangle = 1/\beta, \qquad \langle A\rangle = a_0, \qquad \langle 1\rangle = 1. \tag{3.325}$$

These formulas cover every monomial in (3.313). For the physical substitutions,

$$(a_{U0}, a_{V0}, a_{W0}) = \left(\frac{x(s+1)}{2s}, \frac{y(s+1)}{2s}, \frac{(s-1)(2s-x-y)}{2s}\right). \tag{3.326}$$

Their squares and scalar products are given in (3.306). When Δ or r vanishes while the denominator vectors remain future timelike, the expressions mean their continuous limits. The same positive lower bounds justify taking these limits under the direction integral. Apparent inverse powers at these points are removable. Soft limits in which a denominator vector vanishes instead require the infrared subtraction.

The direction integration is now complete analytically. The remaining real contribution is a two-dimensional Dalitz integral of explicit rational functions and logarithms, with its two soft endpoints still to be combined with the virtual correction.

Coefficients of the integrated direction moments

The following coefficients use $y = H - x + z$ to shorten the polynomials. Every exponent triple not displayed has zero coefficient.

$$C_{-2,0,-2} = -32. \tag{3.327}$$

$$C_{-2,0,-1} = \frac{8\left(H^2 - 2Hx - 4H + x^2 + 4x - 8\right)}{-H + x}. \tag{3.328}$$

$$C_{-2,0,0} = -\frac{8\left(-Hz + 4H + xz - 4x + 4\right)}{\left(-H + x\right)^2}. \tag{3.329}$$

$$C_{-2,1,0} = \frac{8}{-H + x}. \tag{3.330}$$

$$C_{-1,-1,0} = \frac{P_{-1,-1,0}}{z^2\left(-H + x\right)\left(x - z\right)}. \tag{3.331}$$

$$\begin{aligned} P_{-1,-1,0} = {} & -64x^2 + 32z^3 - 24xz^3 - 4Hz^4 - 4H^3z^2 + 16Hz^3 + 16H^2z^2 \\ & + 24x^2z^2 + 32Hz^2 + 32H^2z + 64Hx + 64xz - 32Hxz^2 - 8H^2xz + 8Hx^2z. \end{aligned} \tag{3.332}$$

$$C_{-1,0,-2} = \frac{8\left(z^2 + 4z - 8\right)}{z}. \tag{3.333}$$

$$C_{-1,0,-1} = \frac{P_{-1,0,-1}}{Hz^2\left(-H + x\right)\left(x - z\right)}. \tag{3.334}$$

$$\begin{aligned} P_{-1,0,-1} = {} & -64z^3 + 32z^4 - 128Hz^3 - 64H^2x - 56H^2z^3 - 32H^3z - 8xz^4 - 4xz^5 \\ & - 4H^2z^4 - 4H^4z^2 - 4x^3z^3 + 4Hz^5 + 4H^3z^3 + 8x^2z^3 + 32xz^3 \\ & + 32x^2z^2 + 64Hx^2 + 128Hz^2 + 128H^2z^2 - 64Hxz - 64Hxz^2 \\ & - 56Hx^2z^2 - 12H^2xz^3 - 12H^2x^2z^2 - 8H^2x^2z + 4Hxz^4 \\ & + 4Hx^3z^2 + 8H^3xz + 12Hx^2z^3 + 12H^3xz^2 + 48H^2xz^2 + 64Hxz^3. \end{aligned} \tag{3.335}$$

$$C_{-1,0,0} = \frac{P_{-1,0,0}}{H\left(-H + x\right)^2\left(x - z\right)}. \tag{3.336}$$

$$\begin{aligned} P_{-1,0,0} = {} & -64z - 16x^3 - 8H^4 - 4x^4 + 32x^2 + 64H + 64H^3 + 160H^2 - 160Hx - 96H^2x \\ & - 64Hz - 24H^2z - 24H^2x^2 - 8H^2z^2 - 4H^3z - 4x^2z^2 + 16Hx^3 \\ & + 16x^2z + 20H^3x + 32xz + 40Hx^2 + 4Hx^2z + 12Hxz^2 + 16Hxz. \end{aligned} \tag{3.337}$$

$$C_{-1,1,0} = -\frac{4\left(2H^2 - Hx + Hz - 4H + xz + 4x - 2z - 8\right)}{H\left(-H + x\right)\left(x - z\right)}. \tag{3.338}$$

$$C_{-1,2,0} = -\frac{4\left(-2H+x\right)}{H\left(-H+x\right)\left(x-z\right)}. \tag{3.339}$$

$$C_{0,-2,-2} = -32. \tag{3.340}$$

$$C_{0,-2,-1} = -\frac{8\left(x^2-2xz-4x+z^2+4z-8\right)}{x-z}. \tag{3.341}$$

$$C_{0,-2,0} = \frac{8\left(xz-4x-z^2+4z-4\right)}{\left(x-z\right)^2}. \tag{3.342}$$

$$C_{0,-1,-2} = \frac{8\left(z^2+4z-8\right)}{z}. \tag{3.343}$$

$$C_{0,-1,-1} = \frac{P_{0,-1,-1}}{Hz^2\left(-H+x\right)\left(x-z\right)}. \tag{3.344}$$

$$\begin{aligned} P_{0,-1,-1} = & -64z^3-8z^6+96z^4-96Hz^3-96xz^3-64H^2x-56H^2z^3-32H^3z \\ & -16H^3z^2-12x^2z^4-8xz^4+4x^3z^3+8Hz^5+8x^2z^3+16Hz^4 \\ & +16xz^5+32x^2z^2+64Hx^2+96H^2z^2+128Hz^2-64Hxz-56Hx^2z^2 \\ & -16Hxz^4-8H^2x^2z-4Hx^3z^2+8H^3xz+12Hx^2z^3+32Hxz^3 \\ & +80H^2xz^2. \end{aligned} \tag{3.345}$$

$$C_{0,-1,0} = \frac{P_{0,-1,0}}{H\left(-H+x\right)\left(x-z\right)^2}. \tag{3.346}$$

$$\begin{aligned} P_{0,-1,0} = & -64H-64z^2-32H^2-32x^2-16x^3+4x^4+8H^3+8z^4+64z-96Hx \\ & -64H^2x-40Hz^2-24xz^3-16xz^2-16x^3z-8Hz^3-8H^2z^2-4H^3z \\ & +4H^3x+8Hx^2+28x^2z^2+32x^2z+56H^2z+96xz+128Hz-4Hx^2z \\ & +8H^2xz+12Hxz^2+32Hxz. \end{aligned} \tag{3.347}$$

$$C_{0,0,-2} = -16. \tag{3.348}$$

$$C_{0,0,-1} = \frac{P_{0,0,-1}}{H\left(-H+x\right)\left(x-z\right)}. \tag{3.349}$$

$$\begin{aligned} P_{0,0,-1} = & -32z-8z^2+4z^3+16H^2+32H+32x^2-32Hx-32xz-12Hz^2+4xz^2 \\ & +8H^2z+24Hz-4Hxz. \end{aligned} \tag{3.350}$$

$$C_{0,0,0} = \frac{P_{0,0,0}}{H^2\left(-H+x\right)^2\left(x-z\right)^2}. \tag{3.351}$$

$$\begin{aligned} P_{0,0,0} = & -32H^2-32H^3-32z^2-16x^4-72H^2z^2-64Hz^2-48H^3x-32H^2x \\ & -32x^2z-24x^2z^2-20H^3z^2-8Hz^3-4H^4z-4H^2x^3+4H^4x \\ & +8xz^3+32Hx^2+32Hx^3+32xz^2+32x^3z+32H^2x^2+48H^3z+64Hz \\ & +96H^2z-120Hx^2z-20H^2x^2z-4Hx^2z^2+4Hx^3z+20H^3xz \\ & +24H^2xz^2+40H^2xz+96Hxz^2. \end{aligned} \tag{3.352}$$

$$C_{0,1,0} = \frac{4\left(3Hx-3Hz+2H-2z\right)}{H\left(-H+x\right)\left(x-z\right)^2}. \tag{3.353}$$

$$C_{1,-2,0} = -\frac{8}{x-z}. \tag{3.354}$$

$$C_{1,-1,0} = \frac{4\left(-H^2 - Hx - Hz + xz + 4x - z^2 - 2z + 8\right)}{H\left(-H + x\right)\left(x - z\right)}. \tag{3.355}$$

$$C_{1,0,-1} = -\frac{4\left(-H + z\right)\left(-H + 2x - z\right)}{H\left(-H + x\right)\left(x - z\right)}. \tag{3.356}$$

$$C_{1,0,0} = \frac{4\left(-3H^2 + 3Hx - 2H + 2z\right)}{H\left(-H + x\right)^2\left(x - z\right)}. \tag{3.357}$$

$$C_{2,-1,0} = \frac{4\left(H + x - z\right)}{H\left(-H + x\right)\left(x - z\right)}. \tag{3.358}$$

3.4.4 Rationalization of the remaining logarithms

After the completed angle average is collected, its functional basis is particularly small,

$$\langle \mathcal{R} \rangle = R_0 + R_U L_U + R_V L_V + R_W L_W + R_{UV} J_{UV} + R_{UW} J_{UW} + R_{VW} J_{VW}. \tag{3.359}$$

Each R is rational in H, x, z. The elementary single-denominator functions obey $L_U = s \ln s/(xH)$ and $L_V = s \ln s/(yH)$. The coefficients follow by inserting the moment formulas above in (3.313). The square roots in the other four logarithms can each be rationalized before the remaining energy integrals. This operation is performed separately in each logarithmic term, with a common original integration domain.

For the UV term, the angular Gram determinant is

$$\Delta_{UV} = \frac{H^2 z(4xy + H^2 z)}{4s^2}. \tag{3.360}$$

Its logarithm is a hyperbolic angle. Defining its positive exponential v fixes the substitution uniquely by $v \geq 1$,

$$\frac{H^2 z}{xy} = \frac{(v-1)^2}{v}, \qquad D_v = H^2 v - x(v-1)^2, \tag{3.361}$$

$$z = \frac{x(H-x)(v-1)^2}{D_v}, \qquad y = \frac{H^2 v(H-x)}{D_v}, \tag{3.362}$$

$$\frac{\partial z}{\partial v} = \frac{x(H-x)H^2(v^2-1)}{D_v^2}, \qquad J_{UV} = \frac{2sv \ln v}{xy(v^2-1)}. \tag{3.363}$$

The exact upper Dalitz boundary gives $v = s$. Therefore

$$0 < x < H, \quad 1 < v < s, \qquad J_{UV} \mathrm{d}z = \frac{2s \ln v}{D_v} \mathrm{d}v. \tag{3.364}$$

No square root remains. At fixed v, the coefficient multiplying $\ln v$ is a rational function of x.

For the UW term the corresponding determinant and substitution are

$$\Delta_{UW} = \frac{H^2 z(H-x)[4x + z(H-x)]}{4s^2}, \tag{3.365}$$

$$z = \frac{x(v-1)^2}{(H-x)v}, \qquad \frac{\partial z}{\partial v} = \frac{x(v^2-1)}{(H-x)v^2}, \tag{3.366}$$

$$J_{UW} = \frac{2sv \ln v}{Hx(v^2-1)}, \qquad J_{UW} \mathrm{d}z = \frac{2s \ln v}{H(H-x)v} \mathrm{d}v. \tag{3.367}$$

Its transformed domain is $1 < v < s - x$. Exchanging the integration order gives

$$1 < v < s, \qquad 0 < x < s - v. \tag{3.368}$$

Again the inner integration is rational. The VW sector follows by exchanging x and y before making the same transformation.

Finally the logarithm L_W contains $\sqrt{(s+1-z)^2 - 4s}$. Use the positive solution

$$s + 1 - z = u + \frac{s}{u}, \qquad \sqrt{s} \le u \le s. \tag{3.369}$$

The original Dalitz bounds at fixed z become

$$s - u \le x \le s - \frac{s}{u}, \qquad L_W |\mathrm{d}z| = \frac{s}{Hu} \ln \frac{u^2}{s} \mathrm{d}u. \tag{3.370}$$

Thus every term has a rational inner integral with algebraic boundaries, followed by one logarithmic integration. The square roots and their signs have been fixed by positive physical energies throughout.

These separated logarithmic terms must not be assigned unrelated endpoint cutoffs. For example, a common center-of-mass soft boundary $\omega_i > \varepsilon$ means $x, y > \delta$, where $\delta = 2\sqrt{s}\,\varepsilon$. In (3.364) it becomes

$$\delta < x < \frac{H^2 v (H - \delta)}{H^2 v - \delta (v-1)^2}. \tag{3.371}$$

In (3.368) the same boundary is $x > \delta$ together with $H - x + x(v-1)^2/[(H-x)v] > \delta$. The finite terms generated by these transformed boundaries belong to the cross section. Replacing them by arbitrary small limits would change the finite answer. Apparent poles at $z = 0$ introduced by the partial fractions must likewise cancel after the rational and logarithmic terms are combined. The transformations must therefore be combined with the local Gram subtraction and the common regulator matching developed below.

3.4.5 Removal of artificial Gram singularities

The original angular average is regular when $z = 0$ at fixed $0 < x < H$. Individual logarithmic coefficients contain z^{-2} because of the partial-fraction reduction. To remove these artificial singularities locally, write $a = H - x$ and define

$$T_{UV}(z) = \frac{s}{xa} - \frac{s(6x + H^2)}{6x^2 a^2}\, z, \tag{3.372}$$

$$T_{UW}(z) = \frac{s}{xH} - \frac{sa}{6Hx^2}\, z, \tag{3.373}$$

$$T_{VW}(z) = \frac{s}{aH} - \frac{s(x+6)}{6Ha^2}\, z. \tag{3.374}$$

These are the constant and linear Taylor terms of the corresponding J functions, including $y = a + z$. They follow from

$$\frac{\operatorname{arcosh}(1+q)}{\sqrt{(1+q)^2 - 1}} = 1 - \frac{q}{3} + \mathcal{O}(q^2). \tag{3.375}$$

Replace each $R_{ij} J_{ij}$ by $R_{ij}(J_{ij} - T_{ij})$, and add the subtracted pieces to

$$R_0^{\mathrm{reg}} = R_0 + R_{UV} T_{UV} + R_{UW} T_{UW} + R_{VW} T_{VW}. \tag{3.376}$$

This is an exact algebraic identity. Each difference $J - T$ is $\mathcal{O}(z^2)$. On substituting the explicit coefficients, the reduced rational denominator of R_0^{reg} has no factor z. Thus all these contributions can be integrated across $z = 0$ without assigning separate unphysical endpoint cutoffs. This operation does not remove either physical soft endpoint.

3.4.6 Analytic regulation of the soft endpoints

A Bose-symmetric analytic regulator makes the remaining endpoint operations independent of frame-dependent energy boundaries. Multiply the real phase-space integrand by $(xy)^\eta$, initially with positive real η large enough for convergence, and then continue meromorphically to $\eta = 0$. Since $x, y > 0$ in the physical region, the real logarithm defines this factor without a branch ambiguity. This is an intermediate integration device. The virtual amplitude and its infrared matching still use the photon mass λ.

Define the dimensionless regulated real coefficient by

$$\mathcal{C}_{\mathrm{R}}(\eta) = \frac{1}{8sH}\int_0^H \mathrm{d}x \int_{H-x}^{s(H-x)/(s-x)} \mathrm{d}y\,(xy)^\eta \langle \mathcal{R}\rangle = \frac{A(s)}{\eta} + F(s) + \mathcal{O}(\eta). \tag{3.377}$$

The regulator is applied to the complete trace, including the identical-photon factor already present in its normalization. At one soft endpoint $x = 2\sqrt{s}\,\omega$ while $y \to H$. Consequently $(xy)^\eta \to (2\sqrt{s}H\omega)^\eta$. The other endpoint gives the same contribution. Each carries one half of the usual eikonal result because of 1/2!, so together

$$A(s) = \frac{1}{s}\int_{-1}^{1} \mathrm{d}c\,\mathcal{B}(s,c)[b_e(s,c) - 1]. \tag{3.378}$$

The functions $\mathcal{B}$ and b_e are the Born square and the relative-velocity factor defined earlier.

This relation also fixes the finite matching constant, rather than only the pole. Put $C = 2\sqrt{s}H$. Integration of the leading soft term below a center-of-mass energy ε gives

$$A(s)C^\eta \int_0^\varepsilon \omega^{\eta-1}\mathrm{d}\omega = \frac{A(s)}{\eta} + A(s)\ln(C\varepsilon) + \mathcal{O}(\eta). \tag{3.379}$$

Thus the real coefficient above that energy boundary is $F(s) - A(s)\ln(C\varepsilon) + o(1)$. The photon-mass soft term derived previously is

$$\mathcal{C}_{\mathrm{soft}}(\varepsilon,\lambda) = A(s)\ln\frac{2\varepsilon}{\lambda} - \frac{1}{s}\int_{-1}^{1} \mathcal{B}K\,\mathrm{d}c. \tag{3.380}$$

Adding it to the virtual contribution $\mathcal{C}_{\mathrm{virt}} = (4s)^{-1}\int \mathrm{Re}\,\mathcal{T}\,\mathrm{d}c$ therefore gives

$$\mathcal{C}_{\mathrm{total}} = \mathcal{C}_{\mathrm{virt}} + F(s) + A(s)\ln\frac{1}{\sqrt{s}H\lambda} - \frac{1}{s}\int_{-1}^{1} \mathcal{B}K\,\mathrm{d}c. \tag{3.381}$$

The coefficient of $\ln\lambda$ cancels the virtual coefficient $A(s)\ln\lambda$. Both the analytic pole and the temporary energy boundary have disappeared. Equation (3.381) fixes how the finite part $F(s)$ must be combined with the earlier photon-mass calculation. The following analytic primitives and endpoint formulas evaluate this finite part without phase-space quadrature.

3.4.7 A complete logarithmic-sector primitive

The part proportional to L_W can be integrated further without any phase-space quadrature. Its rational coefficient has the simple partial-fraction form

$$R_W = C_0(z) + C_1(z)\left(\frac{1}{x-z} - \frac{1}{x-H}\right), \tag{3.382}$$

where, with $D_z = (H-z)^2 - 4z$,

$$C_0(z) = -\frac{8}{HD_z}[H^2z - 4H^2 - 2Hz^2 + 12Hz + z^3 - 8z^2 + 16z], \tag{3.383}$$

$$C_1(z) = \frac{4}{HD_z}[-H^3z - 4H^3 + 3H^2z^2 + 8H^2z - 8H^2 \tag{3.384}$$

$$- 3Hz^3 + 2Hz^2 + 32Hz + z^4 - 6z^3 + 32z]. \tag{3.385}$$

At fixed z, the exact limits are $x_\pm = (H + z \pm \sqrt{D_z})/2$. Their substitution into the rational primitive gives

$$\int_{x_-}^{x_+} R_W \mathrm{d}x = C_0(z)\sqrt{D_z} + 2C_1(z) \ln \frac{H - z + \sqrt{D_z}}{H - z - \sqrt{D_z}}. \tag{3.386}$$

Use $z = s + 1 - u - s/u$ as above. Define two rational functions

$$A(u) = \frac{s}{Hu} C_0(s + 1 - u - s/u)(u - s/u), \qquad B(u) = \frac{2s}{Hu} C_1(s + 1 - u - s/u). \tag{3.387}$$

The remaining integrand of this sector is exactly

$$A(u) \ln \frac{u^2}{s} + B(u) \ln \frac{u^2}{s} \ln \frac{u(u-1)}{s-u}, \qquad \sqrt{s} < u < s. \tag{3.388}$$

Only simple real logarithms occur. The real branches are fixed by $u > 1$ and $s - u > 0$.

For a compact analytic primitive put $r = \sqrt{s} > 1$. Denote by $G_r(a_1, \ldots, a_n; u)$ the generalized polylogarithm based at r, with derivative

$$\frac{\mathrm{d}}{\mathrm{d}u} G_r(a_1, \ldots, a_n; u) = \frac{G_r(a_2, \ldots, a_n; u)}{u - a_1}, \qquad G_r(; u) = 1. \tag{3.389}$$

The integration path is the real interval from r to u. At a basepoint letter r use the continuous or combined tangential limit. All other letters below are $0, 1, -r, s$, which lie outside the open interval. Thus the functions that occur are real there. In particular, suppressing r, u in the notation,

$$\ln\Big(u^2/s\Big) = 2G(0), \qquad \ln[u(u-1)/(s-u)] = G(0) + G(1) - G(s). \tag{3.390}$$

Shuffle multiplication therefore reduces (3.388) to a finite weight-two integrand. Its primitive has weight at most three and is

$$\mathcal{F}_W = -\frac{16(r+1)^2}{r-1}[G(-r,0) + G(-r,1) - G(-r,s)] \tag{3.391}$$

$$+ \frac{16(r-1)^2}{r+1}[G(r,0) + G(r,1) - G(r,s)] \tag{3.392}$$

$$- \frac{32r^2(2r^2+1)}{(r^2-1)^2}[2G(0,0,0) + G(0,0,1) + G(0,1,0) - G(0,0,s) - G(0,s,0)] \tag{3.393}$$

$$+ A_{00}G(0,0) + A_{01}[G(0,1) - G(0,s)] + A_{10}G(1,0) + A_{s0}G(s,0) \tag{3.394}$$

$$+ A_0 G(0) + A_1[G(1) - G(s)] + A_{\mathrm{c}}. \tag{3.395}$$

The rational coefficients below are obtained by partial fractions and integration by parts. They contain no fitted constants. Differentiating this finite expression yields (3.388). The

upper soft endpoint must still be combined with the other sectors in the common analytic regulator. This primitive alone is not the finite inclusive real correction.

$$A_{00} = \frac{N_{00}}{u^2 (r-1)^2 (r+1)^2 (r-u)(r+u)}. \tag{3.396}$$

$$\begin{aligned} N_{00} = & -16r^8 + 64u^4 - 192r^2u^3 - 112r^4u^4 - 48r^6u^2 - 16r^2u^6 \\ & - 8r^6u^4 + 8r^8u^2 + 32r^6u + 32r^8u + 32r^2u^5 + 32r^4u^5 \\ & + 56r^4u^2 + 64r^4u^3 + 72r^2u^4. \end{aligned} \tag{3.397}$$

$$A_{01} = \frac{N_{01}}{u^2 (r-1)^2 (r+1)^2 (r-u)(r+u)}. \tag{3.398}$$

$$\begin{aligned} N_{01} = & -8r^8 + 32u^4 - 96r^2u^3 - 88r^4u^4 - 8r^2u^6 + 8r^6u^2 + 16r^6u \\ & + 16r^8u + 16r^2u^5 + 16r^4u^5 + 32r^4u^3 + 64r^2u^4. \end{aligned} \tag{3.399}$$

$$A_{10} = \frac{N_{10}}{u^2 (r-1)^2 (r+1)^2 (r-u)(r+u)}. \tag{3.400}$$

$$\begin{aligned} N_{10} = & -8r^8 - 96r^2u^3 - 56r^4u^4 - 24r^6u^2 - 8r^2u^6 - 8r^8u^2 \\ & + 8r^4u^2 + 8r^6u^4 + 16r^6u + 16r^8u + 16r^2u^5 + 16r^4u^5 \\ & + 32r^2u^2 + 32r^4u^3 + 56r^2u^4. \end{aligned} \tag{3.401}$$

$$A_{s0} = \frac{N_{s0}}{u^2 (r-1)^2 (r+1)^2 (r-u)(r+u)}. \tag{3.402}$$

$$\begin{aligned} N_{s0} = & -32u^4 + 8r^8 - 56r^4u^2 - 32r^4u^3 - 16r^6u - 16r^8u - 16r^2u^5 \\ & - 16r^4u^5 - 8r^2u^4 - 8r^8u^2 + 8r^2u^6 + 8r^6u^4 + 24r^4u^4 \\ & + 56r^6u^2 + 96r^2u^3. \end{aligned} \tag{3.403}$$

$$A_0 = \frac{12r^2 \left(r^4u + 2r^4 + 2r^2u^2 + 6r^2 + 6u^2 - u\right)}{u (r-1)^2 (r+1)^2}. \tag{3.404}$$

$$A_1 = \frac{N_1}{u^2 (r-1)^2 (r+1)^2}. \tag{3.405}$$

$$\begin{aligned} N_1 = & -4r^6 - 32r^4u^2 - 12r^2u^2 - 12r^6u^2 - 4r^2u^4 + 16r^4u + 16r^6u \\ & + 16r^2u^3 + 16r^4u^3. \end{aligned} \tag{3.406}$$

$$A_{\mathrm{c}} = \frac{4r^2 (r-u)(r+u)\left(7r^2 + 17\right)}{u (r-1)^2 (r+1)^2}. \tag{3.407}$$

3.4.8 The sector proportional to the energy logarithm

The single denominators U, V give another sector that can be integrated completely in the interior. Substituting $L_U = s \ln s/(xH)$ and $L_V = s \ln s/(yH)$ first combines their coefficients into a rational function. At fixed z its poles in x are only $0, z, H, H+z$, with orders at most three. The exact limits $x_\pm$ therefore produce rational functions and the two logarithms

$$\ln \frac{x_+}{x_-} = \ln \frac{s(u-1)}{u(s-u)}, \qquad \ln \frac{x_+ - z}{x_- - z} = \ln \frac{u(u-1)}{s-u}. \tag{3.408}$$

After $z = s + 1 - u - s/u$, no square root remains. The final primitive can be written with the same basepoint $r = \sqrt{s}$ as above. Define the conjugate letters

$$a_\pm = 1 \pm \mathrm{i}\sqrt{H}, \qquad H = s - 1 > 0. \tag{3.409}$$

They are the roots of $u^2 - 2u + s$. Both lie off the real path, and complex conjugation exchanges their GPLs. Thus their sum below is manifestly real, with no choice based on numerical values. The primitive of this entire logarithmic sector is $\ln s$ times

$$\mathcal{F}_{\ln s} = \frac{96s^2}{H^2} G(0,0) + \frac{32s(H^2-4H-6)}{H^2}[G(1,0) + G(s,0)] \tag{3.410}$$

$$+ \frac{32s(H^2-H-3)}{H^2}\Big[G(0,1) - G(0,s) \tag{3.411}$$

$$- \sum_{a=a_\pm} \{G(a,0) + G(a,1) - G(a,s)\}\Big] \tag{3.412}$$

$$+ P_0 G(0) + P_1[G(1) - G(s)] + P_{\mathrm{c}}. \tag{3.413}$$

There are fourteen GPL words, all of weight at most two. The rational coefficients are displayed below. Exact symbolic differentiation of every GPL coefficient recovers the rational and logarithmic integrand after the x integration. This is an independent check of the primitive, rather than a fit to a total cross section. As in the previous sector, its soft endpoint is to be combined in the common regulator before the finite part is taken.

$$P_0 = \frac{Q_0}{H^2 u^2 (H+1)}. \tag{3.414}$$

$$\begin{aligned} Q_0 = {} & 12 - 48u - 48u^3 - 32u^2 + 12H^4 + 12u^4 + 48H + 48H^3 + 72H^2 \\ & - 136Hu - 120H^2u - 88Hu^3 - 56Hu^2 - 32H^2u^3 - 24H^3u \\ & + 8H^4u + 8H^3u^3 + 12H^2u^4 + 24Hu^4 + 28H^2u^2 \\ & + 36H^4u^2 + 80H^3u^2. \end{aligned} \tag{3.415}$$

$$P_1 = \frac{Q_1}{H^4 u^2 (H+1)}. \tag{3.416}$$

$$\begin{aligned} Q_1 = {} & -112 - 1696u^2 - 796H^2 - 624H^3 - 480H - 216H^4 - 112u^4 - 16H^5 \\ & + 4H^6 + 960u + 960u^3 - 7448H^2u^2 - 5856Hu^2 - 4152H^3u^2 \\ & - 932H^4u^2 - 256Hu^4 - 172H^2u^4 - 112H^5u^2 - 36H^6u^2 \\ & - 24H^3u^4 + 4H^4u^4 + 24H^6u + 24H^5u^3 + 224H^4u^3 \\ & + 248H^5u + 1272H^3u^3 + 1496H^4u + 2816Hu^3 + 2928H^2u^3 \\ & + 3776Hu + 4200H^3u + 5744H^2u. \end{aligned} \tag{3.417}$$

$$P_{\mathrm{c}} = \frac{Q_{\mathrm{c}}}{H^3 u^2 (H+1)}. \tag{3.418}$$

$$\begin{aligned} Q_{\mathrm{c}} = {} & -56 - 368u^3 - 272H^2 - 206H - 144H^3 - 16H^4 + 6H^5 + 56u^4 + 368u \\ & - 880Hu^3 - 660H^2u^3 - 136H^3u^3 - 6H^3u^4 - 4H^5u \\ & + 4H^4u^3 + 28H^2u^4 + 94Hu^4 + 132H^4u + 796H^3u + 1248Hu \\ & + 1540H^2u. \end{aligned} \tag{3.419}$$

3.4.9 Factorization of both soft endpoints in the pair sector

The UW sector admits a rectangular domain in which the analytic regulator factorizes at both soft endpoints. Starting from (3.368), put

$$a = H - x, \qquad v = 1 + at, \qquad 0 < a < H, \quad 0 < t < 1, \tag{3.420}$$

$$z = \frac{a(H-a)t^2}{1+at}, \qquad y = \frac{a[1 + Ht^2 + at(1-t)]}{1+at}, \qquad \left|\frac{\partial(x,v)}{\partial(a,t)}\right| = a. \tag{3.421}$$

The product that regulates the two soft endpoints is therefore

$$xy = a(H-a)f(a,t), \qquad f(a,t) = \frac{1 + Ht^2 + at(1-t)}{1+at} > 0. \tag{3.422}$$

Both endpoints are now ordinary powers of a single variable. The regulator introduces no new physical branch.

Define two rational functions by the previously specified coefficient R_{UW} and Taylor subtraction T_{UW},

$$L(a,t) = a\, R_{UW}(x,z) \frac{2s}{H(H-x)v}\bigg|_{x=H-a,\, v=1+at}, \tag{3.423}$$

$$T(a,t) = a\, R_{UW}(x,z) T_{UW}(z) \frac{\partial z}{\partial v}\bigg|_{x=H-a,\, v=1+at}, \tag{3.424}$$

$$F(a,t) = L(a,t)\ln(1+at) - T(a,t). \tag{3.425}$$

The subtraction sign in F is fixed by the local removal of the artificial Gram singularity. The complementary rational term is already included in R_0^{reg}. The only poles of L, T in the complex a plane are

$$a = 0, \quad H, \quad -1/t, \quad -2/t, \quad -1/[t(1-t)]. \tag{3.426}$$

For $0 < t < 1$, the last three are negative and lie outside the integration interval. Their orders are finite, at most five. There is no polynomial denominator of degree higher than one after partial fractions in a.

For clarity, the endpoint conversion can be written without a hypergeometric expansion. Suppose

$$F(a,t) = \frac{f_1(t)}{a} + \mathcal{O}(1), \qquad F(H-b,t) = \frac{g_2(t)}{b^2} + \frac{g_1(t)}{b} + \mathcal{O}(1). \tag{3.427}$$

For the explicit coefficients above, $f_1(t) = 0$ identically. The Taylor subtraction has therefore also made the $a = 0$ soft corner finite in this particular sector. The other endpoint has the stated maximum pole orders. Let $\mathrm{FP}_a \int_0^H F\mathrm{d}a$ denote the constant term of the integral from δ to $H - \delta$ after all powers and logarithms of δ are subtracted. Expansion of (3.422) at each endpoint gives

$$\int_0^H [xy]^\eta F(a,t)\mathrm{d}a = \frac{f_1(t) + g_1(t)}{\eta} + \mathrm{FP}_a \int_0^H F(a,t)\mathrm{d}a \tag{3.428}$$

$$+ f_1(t)\ln[H(1+Ht^2)] + g_1(t)\ln H \tag{3.429}$$

$$+ g_2(t)\left[-\frac{1}{H} + \frac{t^2}{1+Ht}\right] + \mathcal{O}(\eta). \tag{3.430}$$

For example, the last term follows because the term linear in b in the logarithm of the regulator's smooth factor is $-1/H + t^2/(1+Ht)$. Its multiplication by η and a double endpoint pole produces a finite term. Dropping that term would change the answer even though the infrared pole remained correct.

Every primitive required on the right-hand side is elementary or a real dilogarithm. To see this explicitly, define

$$E_n(a;c) = \begin{cases} \ln|a-c|, & n=1, \\ -(a-c)^{1-n}/(n-1), & n \geq 2. \end{cases} \tag{3.431}$$

For a simple logarithmic pole, let $w = 1+at$ and $d = 1+ct$. When $c = -2/t$ or $c = -1/[t(1-t)]$, one has $d < 0$, and a primitive is

$$I_1(a;c) = \ln w \ln(1-w/d) + \mathrm{Li}_2(w/d). \tag{3.432}$$

Both logarithms and the dilogarithm are real. For $c = H$ the same formula applies with $0 < w/d < 1$ in the interior. The exceptional values are

$$I_1(a;0) = -\mathrm{Li}_2(-at), \qquad I_1(a;-1/t) = \tfrac{1}{2}\ln^2(1+at). \tag{3.433}$$

In each case $\partial_a I_1 = \ln(1+at)/(a-c)$. Higher poles follow by one integration by parts. With $b = -1/t$ and $n \geq 2$,

$$I_n(a;c) = -\frac{\ln(1+at)}{(n-1)(a-c)^{n-1}} + \frac{Q_{n-1}(a;c,b)}{n-1}, \tag{3.434}$$

$$Q_m(a;c,b) = \frac{E_1(a;b)}{(b-c)^m} - \sum_{j=1}^{m} \frac{E_j(a;c)}{(b-c)^{m-j+1}} \quad (c \neq b), \tag{3.435}$$

$$Q_m(a;b,b) = E_{m+1}(a;b). \tag{3.436}$$

Thus partial fractions of the explicitly defined rational functions L, T give the finite part in (3.430) as a finite sum of boundary values of I_n and E_n. No numerical integration enters this inner energy integration.

The nontrivial negative dilogarithm argument at $a = H$ is $-(1-t)(1+Ht)/t$. Its complementary argument contains

$$1 + Ht - Ht^2, \qquad t_\pm = \frac{1}{2}\left(1 \pm \sqrt{1+4/H}\right). \tag{3.437}$$

Here $t_- < 0$ and $t_+ > 1$. The square root $\sqrt{H(H+4)}$ therefore follows directly from the remaining integration alphabet. It has not been inferred from a published expression. The inner integration leaves one angular variable t. We first write its finite integrand explicitly, including its regulated endpoint terms, so that the remaining primitive can be constructed from its actual poles and logarithms.

Closed inner finite part. The following boundary formulas make the inner result a finite expression of logarithms and dilogarithms, with no integration operation left. Put $\ell = \ln(1+Ht)$ and write the partial fractions

$$L = l_0 + l_1 a + \sum_{c,n} \frac{L_{cn}}{(a-c)^n}, \qquad T = \sum_{j=0}^{3} t_j a^j + \sum_{c,n} \frac{T_{cn}}{(a-c)^n}. \tag{3.438}$$

The pole set is the five-element set displayed above. For a pole of order m_c, its coefficient is fixed algebraically,

$$L_{cn} = \frac{1}{(m_c-n)!}\left(\frac{\partial}{\partial a}\right)^{m_c-n}\left[(a-c)^{m_c}L(a,t)\right]\Big|_{a=c}, \tag{3.439}$$

with the same formula for T_{cn}. The polynomial parts follow by ordinary polynomial division. Define $\mathcal{I}_n(c)$ and $\mathcal{E}_n(c)$ to be the finite boundary differences of the explicit primitives I_n, E_n above. At a negative pole they are ordinary differences between $a = H$ and $a = 0$. At the two physical endpoints the required values are

$$\mathcal{E}_1(0) = \ln H, \qquad \mathcal{E}_1(H) = -\ln H, \qquad \mathcal{E}_2(0) = \mathcal{E}_2(H) = -H^{-1}, \tag{3.440}$$

$$\mathcal{I}_1(0) = -\operatorname{Li}_2(-Ht), \tag{3.441}$$

$$\mathcal{I}_2(0) = -\ell/H + t\ln H - t\ell + t, \tag{3.442}$$

$$\mathcal{I}_1(H) = -\ell\ln H - \operatorname{Li}_2(-Ht) - \ell^2/2, \tag{3.443}$$

$$\mathcal{I}_2(H) = -\frac{t}{1+Ht}(\ln H + \ell + 1). \tag{3.444}$$

The two polynomial logarithmic moments are

$$P_0 = \frac{(1+Ht)\ell - Ht}{t}, \tag{3.445}$$

$$P_1 = \frac{[(1+Ht)^2/2 - (1+Ht)]\ell - (1+Ht)^2/4 + (1+Ht) - 3/4}{t^2}. \tag{3.446}$$

The exact inner coefficient of η^0 is consequently

$$F_{UW}(t) = l_0P_0 + l_1P_1 - \sum_{j=0}^{3}\frac{t_jH^{j+1}}{j+1} + \sum_{c,n}[L_{cn}\mathcal{I}_n(c) - T_{cn}\mathcal{E}_n(c)] \tag{3.447}$$

$$+ g_1(t)\ln H + g_2(t)\left[-H^{-1} + \frac{t^2}{1+Ht}\right]. \tag{3.448}$$

This formula uses only finite algebraic differentiations and the real functions already displayed. The corresponding η^{-1} coefficient is $g_1(t)$.

For the last integration, every dilogarithm in this result has a fixed-letter GPL representation. Put $b = -1/H$, $d = -2/H$, and let $t_\pm$ be the two real roots above. With GPLs now based at zero,

$$\operatorname{Li}_2(-Ht) = -G(0,b;t), \tag{3.449}$$

$$\operatorname{Li}_2(-1-Ht) + \frac{\pi^2}{12} = -\ln 2\, G(b;t) - G(b,d;t). \tag{3.450}$$

$$\begin{aligned}&\operatorname{Li}_2\left(-\frac{(1-t)(1+Ht)}{t}\right) - \operatorname{Li}_2\left(-\frac{1-t}{t}\right)\\ &\quad = G(b,0;t) - \sum_{c=t_\pm}[G(1,c;t) + G(b,c;t) - G(0,c;t)].\end{aligned} \tag{3.451}$$

The last identity follows by differentiating both sides and using their common zero limit as $t \to 0^+$. It fixes its integration constant as well as its branches. This gives a weight-two GPL integrand with rational coefficients for the final energy integral.

3.4.10 Closed construction for the regular rational sector

The rational term R_0^{reg} also has an analytic primitive after both Dalitz integrations. The compact construction below specifies its coefficients by finite algebraic operations, without introducing a new integral. At fixed z, its partial fractions are

$$R_0^{\mathrm{reg}} = p_0(z) + \sum_{\rho\in\mathcal{P}} \sum_{n=1}^{m_\rho} \frac{c_{\rho n}(z)}{(x-\rho)^n}, \tag{3.452}$$

where there are sixteen nonzero pole terms and

$$\mathcal{P} = \left\{0, z, H, H+z, \frac{Hz}{z-4},\ H+z-\frac{Hz}{z-4}\right\}. \tag{3.453}$$

Every order m_ρ is at most three. A uniform formula, including vanishing coefficients when the actual order is smaller, is

$$c_{\rho n}(z) = \frac{1}{(3-n)!}\frac{\partial^{3-n}}{\partial x^{3-n}}[(x-\rho)^3 R_0^{\mathrm{reg}}(x,z)]\bigg|_{x=\rho}, \quad n=1,2,3, \tag{3.454}$$

$$p_0(z) = \lim_{x\to\infty} R_0^{\mathrm{reg}}(x,z). \tag{3.455}$$

The limit exists because its polynomial part is constant. This supplies an explicit algebraic definition from the twenty-five trace coefficients and the displayed Taylor subtractions.

Let $z = s+1-u-s/u$, $x_- = s-u$, $x_+ = s-s/u$, and $j = 1-s/u^2$. Define four rational functions by

$$R(u) = j\left[p_0(x_+ - x_-) + \sum_{\rho}\sum_{n=2}^{m_\rho} \frac{c_{\rho n}}{1-n}\{(x_+-\rho)^{1-n} - (x_- - \rho)^{1-n}\}\right], \tag{3.456}$$

$$K_1(u) = j[-c_{0,1} + c_{H+z,1}], \tag{3.457}$$

$$K_2(u) = j[-c_{H,1} + c_{0,1} + c_{z,1} - c_{H+z,1}], \tag{3.458}$$

$$K_3(u) = j[c_{Hz/(z-4),1} - c_{H+z-Hz/(z-4),1}]. \tag{3.459}$$

All quantities on the right are evaluated at $z = s+1-u-s/u$. The exact x boundary differences then give the final integrand

$$R(u) + K_1(u)L_1(u) + K_2(u)L_2(u) + K_3(u)L_3(u), \tag{3.460}$$

where

$$L_1 = \ln\Big(u^2/s\Big), \qquad L_2 = \ln[u(u-1)/(s-u)], \tag{3.461}$$

$$L_3 = \ln\frac{(u-1)(u+s)^2}{u(u+1)^2(s-u)}. \tag{3.462}$$

Each logarithm vanishes at $u=\sqrt{s}$. In terms of GPLs based there, the additional logarithm is

$$L_3 = G(1) + 2G(-s) - G(0) - 2G(-1) - G(s). \tag{3.463}$$

For precision, define the finite algebraic operation $\mathcal{P}[R;w]$ on a rational function R and an empty or one-letter word w. Decompose R into polynomial and pole terms. On a simple pole its value is

$$\mathcal{P}[c/(u-a);w] = cG(a,w;u). \tag{3.464}$$

For a pole of order $n \geq 2$ and an empty word it is $-c/[(n-1)(u-a)^{n-1}]$. For a one-letter word (b) it is

$$\mathcal{P}[c/(u-a)^n;(b)] = -\frac{cG(b;u)}{(n-1)(u-a)^{n-1}} + \frac{cQ_{n-1}(u;a,b)}{n-1}, \tag{3.465}$$

where Q_m is the finite elementary combination already given above, with logarithms interpreted as the same basepoint GPLs. Polynomial terms are fixed explicitly, for $k \geq 0$, by

$$\mathcal{P}[u^k;()] = \frac{u^{k+1}}{k+1}, \tag{3.466}$$

$$\mathcal{P}[u^k;(b)] = \frac{u^{k+1}-b^{k+1}}{k+1}G(b;u) - \frac{1}{k+1}\sum_{j=0}^{k}\frac{b^j u^{k+1-j}}{k+1-j}. \tag{3.467}$$

Linearity completes the definition for any rational coefficient. These operations terminate after one step because w has length at most one. Thus $\mathcal{P}$ denotes a finite sum of specified GPLs and rational functions.

The complete interior primitive of the regular rational sector is

$$\mathcal{F}_{\rm reg} = \mathcal{P}[R;()] + \mathcal{P}[2K_1+K_2-K_3;(0)] + \mathcal{P}[K_2+K_3;(1)] \tag{3.468}$$

$$- \mathcal{P}[K_2+K_3;(s)] + 2\mathcal{P}[K_3;(-s)] - 2\mathcal{P}[K_3;(-1)]. \tag{3.469}$$

Its expansion contains forty-eight GPL words of weight at most two. This form keeps the relatively long rational coefficient table specified by exact derivatives instead of reprinting it.

A possible internal letter in this representation is removable. The quadratic $u^2-(s-3)u+s$ corresponds to $z=4$. The original angular denominator $4x+z(H-x)$ is strictly positive in the physical interior, so $z=4$ is not a singularity of the amplitude. The pole $x=Hz/(z-4)$ merely moves through infinity at that point. For $z<4$, the pole $Hz/(z-4)$ is negative and its partner $H+z-Hz/(z-4)$ exceeds $H+z$. For $z>4$, the first exceeds H and the partner is smaller than z. Both therefore stay outside $z < x_- \leq x \leq x_+ < H$ on either side of $z=4$. The original finite inner integral is analytic at this point. At either root of the quadratic, $L_3=0$. Consequently the combination K_3L_3 has a removable limit there, even when one root lies inside the real u interval. The GPL combination must be formed before that limit is taken. Passing above or below the individual artificial letter gives the same complete expression, because its would-be discontinuity is proportional to this zero. This argument fixes the continuation from the original real integrand and does not add a physical threshold at $s=9$.

Exchange symmetry after the local subtraction. The full angular expression is invariant under the exchange of the identical final photons. This exchange sends x to $y=H-x+z$ and leaves z unchanged. The Taylor polynomials used to remove the Gram zero were instead expanded at fixed x. These operations do not commute beyond the order of the subtraction. In particular, direct substitution gives

$$\begin{aligned} T_{UW}(y,z) - T_{VW}(x,z) = & -\frac{sz^2}{6H(x-H)^2(x-H-z)^2} \\ & \times [-H^2-6H+x^2-xz+6x-6z]. \end{aligned} \tag{3.470}$$

Using the coefficient R_{VW} of J_{VW}, define the rational exchange correction

$$D(x,z) = R_{VW}(x,z)[T_{UW}(H-x+z,z) - T_{VW}(x,z)]. \tag{3.471}$$

The factor z^2 cancels the apparent Gram pole of R_{VW}. The symmetric regulator $(xy)^\eta$, the Dalitz domain, and its measure are all invariant under photon exchange. Changing variables in the subtracted VW integral therefore proves

$$I^{\text{sub}}_{VW}(\eta) = I^{\text{sub}}_{UW}(\eta) + I_D(\eta). \tag{3.472}$$

This identity holds first where the regulated integrals converge and then for their meromorphic continuations. It applies separately to the pole and finite Laurent coefficients. Consequently the real integral is assembled from the regular rational term, the two single-denominator logarithms, the W logarithm, the UV term, twice the UW term, and D. Omitting D would change both the infrared pole and the finite answer.

Endpoint values of the primitives. The preceding formulas specify finite combinations of GPLs rather than unevaluated phase-space integrals. Their endpoint subtraction can also be stated algebraically. Write $b = s - u$, let $f(u)$ denote one complete sector integrand, and let $\mathcal{F}'(u) = f(u)$. After combining all terms of that sector its expansion is

$$f(s-b) = \frac{A}{b} + \sum_{n=0}^{\infty} (c_n + d_n \ln b) b^n. \tag{3.473}$$

There are no higher negative powers or $\ln b/b$ terms in these three sectors. This follows by expanding the displayed rational coefficients and the logarithms before taking the endpoint. The coefficient A vanishes for the W sector. All series coefficients are obtained by polynomial division and the Taylor series of $\ln(1 - b/a)$.

Let $\mathcal{F}_0$ be the finite limit of the complete primitive at $u = \sqrt{s}$. It includes constant contributions from products of a vanishing GPL and a rational coefficient with a pole. They are obtained by multiplying their local series before setting $u = \sqrt{s}$.

The lower endpoint constants simplify considerably. Direct local expansion of the finite GPL combinations gives

$$\mathcal{F}_{0,W} = \mathcal{F}_{0,\ln s} = 0, \tag{3.474}$$

$$\mathcal{F}_{0,\text{reg}} = \frac{2s[-s^5 - 5s^4 + 18s^3 + 122s^2 + 151s - 45]}{3(s^2 + 2s - 3)}. \tag{3.475}$$

For the W primitive, the coefficients that have a simple pole at the basepoint multiply weight-two words. Those words vanish quadratically, so they do not contribute a constant. For the regular rational primitive the nonempty-word coefficients are regular there. Its lower endpoint constant is therefore the rational empty-word coefficient evaluated at $u = \sqrt{s}$.

Since

$$z = \frac{H}{s} b + O(b^2), \tag{3.476}$$

the finite part defined by a cut on z is exactly

$$F_z = \mathcal{F}(s-\delta) - \mathcal{F}_0 + A \ln \delta + A \ln(H/s) \tag{3.477}$$

$$+ \sum_{n=0}^{\infty} \delta^{n+1} \left[\frac{c_n}{n+1} + d_n \left(\frac{\ln \delta}{n+1} - \frac{1}{(n+1)^2} \right) \right]. \tag{3.478}$$

Here $\delta > 0$ lies inside the convergence disk of the endpoint series. The series is simply the analytic tail of the primitive. It is not a numerical phase-space integration. Equivalently, $F_z = \lim_{\epsilon \to 0} [\int_{z>\epsilon} f \, du + A \ln \epsilon]$. The subsequent conversion to $(xy)^\eta$ must still be included.

The artificial interior GPL letters can be bypassed during series continuation without choosing a new physical branch. For the regular rational sector the original finite x integral is analytic at $z = 4$: its denominators have no zero there for interior x. Differentiation of the specified primitive gives this same analytic function, including at the removable limit. Its integral around a small closed contour surrounding an artificial letter is therefore zero. Continuations above and below it agree when the entire GPL combination is formed. The starting value is fixed by $\mathcal{F}_0$ on the real interval. This proves path independence of the complete expression without dropping an imaginary part of an individual GPL.

For $1 < s < 9$ the additional roots are complex and do not meet the integration interval. For $s > 9$ the only root that can meet it is

$$q_+ = \frac{s - 3 + \sqrt{(s-1)(s-9)}}{2}. \tag{3.479}$$

The root at $s = 9$ coincides with the lower endpoint. The original x integral remains regular there, and the coincident GPL words are combined before their basepoint series is evaluated. Thus the same analytic continuation covers both sides of $s = 9$.

Exact symbolic differentiation verifies the displayed GPL primitives for all three sectors.

3.4.11 Analytic soft-endpoint terms in the inner energy integral

The UV logarithmic sector illustrates why the regulator must be kept until both endpoints have been integrated. Put

$$k = \frac{(v-1)^2}{Hv}, \qquad u = \frac{x}{H}, \qquad y = \frac{H(1-u)}{1-ku}, \qquad (xy)^\eta = H^{2\eta} u^\eta (1-u)^\eta (1-ku)^{-\eta}. \tag{3.480}$$

For $1 < v < s = 1 + H$, $0 < k < H/s < 1$. Thus the third pole $x = H/k$ is outside the entire inner interval $0 < x < H$. The rational coefficient of $\ln v$ has precisely the form

$$R(x, v) = C_0 + \frac{C_{01}}{x} + \frac{C_{02}}{x^2} + \frac{C_{H1}}{x - H} + \frac{C_{H2}}{(x-H)^2} + \frac{C_{a1}}{x - H/k} + \frac{C_{a2}}{(x - H/k)^2}. \tag{3.481}$$

The coefficients are obtained by ordinary partial fractions of the explicit R_{UV} above after the change of variables. The script `compton_real_inner_analytic.py` performs this algebra and verifies exact reconstruction of the original rational function. No kinematic fitting is involved.

Each integral is an Euler beta function times a Gauss hypergeometric function. For example, for sufficiently large $\mathrm{Re}\,\eta$,

$$\int_0^H \frac{(xy)^\eta}{x^n} \mathrm{d}x = H^{1-n+2\eta} B(1 + \eta - n, 1 + \eta)\, {}_2F_1(\eta, 1 + \eta - n; 2 + 2\eta - n; k), \tag{3.482}$$

$$\int_0^H \frac{(xy)^\eta}{(x-H)^n} \mathrm{d}x = (-1)^n H^{1-n+2\eta} B(1 + \eta, 1 + \eta - n)\, {}_2F_1(\eta, 1 + \eta; 2 + 2\eta - n; k), \tag{3.483}$$

$$\int_0^H \frac{(xy)^\eta}{(x - H/k)^n} \mathrm{d}x = (-H/k)^{-n} H^{1+2\eta} B(1 + \eta, 1 + \eta)\, {}_2F_1(n + \eta, 1 + \eta; 2 + 2\eta; k). \tag{3.484}$$

Here $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$. The Gauss function is

$${}_2F_1(a, b; c; k) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n n!} k^n, \qquad |k| < 1, \qquad (a)_n = \frac{\Gamma(a + n)}{\Gamma(a)}, \quad (a)_0 = 1. \tag{3.485}$$

Expanding $(1-ku)^{-a}$ and integrating each power against $u^{b-1}(1-u)^{c-b-1}$ gives the displayed beta integrals in their convergence domain. These expressions then uniquely define the continuation to $\eta = 0$. Their Laurent expansion gives, at fixed v in the interior,

$$\int_0^H (xy)^\eta R(x,v)\mathrm{d}x = \frac{C_{01}-C_{H1}}{\eta} + F_{UV}(v) + \mathcal{O}(\eta), \tag{3.486}$$

$$F_{UV}(v) = HC_0 + \frac{k-2}{H}C_{02} - \frac{2-k}{H(1-k)}C_{H2} + 2(C_{01}-C_{H1})\ln H \tag{3.487}$$

$$+ (C_{H1}+C_{a1})\ln(1-k) + \frac{k^2}{H(1-k)}C_{a2}. \tag{3.488}$$

The double-pole terms have finite limits but no $1/\eta$ pole. Their finite constants cannot be obtained by simply setting $\eta = 0$ in the integrand. For example,

$$(1-u)^\eta(1-ku)^{-\eta} = 1 + \eta(k-1)u + \mathcal{O}(u^2), \qquad \int_0^1 u^{\eta-2}\mathrm{d}u = \frac{1}{\eta-1}. \tag{3.489}$$

The linear term contributes $\eta(k-1)/\eta = k-1$, which combines with -1 from the first term to give $k-2$. This explicitly accounts for the finite endpoint contribution.

Equation (3.488) still has to be combined with the local $z = 0$ subtraction and integrated over v. It must not be integrated as an isolated term across a spurious Gram pole before those cancellations have been made.

3.5 Matching the real-energy endpoints

3.5.1 Soft endpoints of the real-energy integrals

The remaining rational and single-logarithm sectors have elementary soft residues. Put $H = s-1$, and use $e = x$ at the first soft endpoint and $e = y$ at the second. Here e is an energy invariant, not the charge. With $z = er$, the exact phase-space boundary is

$$0 < e < H, \qquad 0 < r < \rho(e) = \frac{H-e}{H+1-e}, \qquad xy = e[H+e(r-1)], \qquad \mathrm{d}x\,\mathrm{d}z = e\,\mathrm{d}e\,\mathrm{d}r. \tag{3.490}$$

To obtain this domain from $0 < z < x(H-x)/(s-x)$, set $x = e$ and divide the inequality by $e > 0$. At the other soft endpoint set $y = e$ and use $x = H - e + z$ in the same inequality. Both give $r < \rho(e)$. The two changes of variables have Jacobian magnitude e. A disjoint partition near the two soft corners allows their residues to be added without double counting. Away from both corners the regulators converge to one on compact subsets, so their difference has zero limit there.

Let $q = \rho(0) = H/(H+1)$. For a sector with a simple soft singularity, define its two residues by

$$f_X(r) = \lim_{e\to 0} e^2 R(H,e,er), \qquad f_Y(r) = \lim_{e\to 0} e^2 R(H,H+e(r-1),er). \tag{3.491}$$

The rational sector and the two terms multiplying $\ln s$ have this behavior. The coefficient of L_W is only of order e^{-1}, and $L_W(0)$ is finite, so this sector needs no soft matching term. All coefficients in this section precede the common normalization $1/(8sH)$ of the real cross section.

Write $f = f_X + f_Y$. The regulator $(xy)^\eta$ gives a pole A/η and finite part F_η. If the energy primitives are instead evaluated with $z > \zeta$, define their finite part by subtracting

$-A\ln\zeta$. The symbol FP denotes the constant term after that subtraction. Then

$$A = \int_0^q f(r)\,\mathrm{d}r, \tag{3.492}$$

$$F_\eta = \mathrm{FP}_{\zeta\to 0}\int_{z>\zeta} R\,\mathrm{d}x\,\mathrm{d}z + A\ln H - \int_0^q f(r)\ln r\,\mathrm{d}r. \tag{3.493}$$

Indeed, the soft measure is $f(r)\mathrm{d}e/e$, while $z>\zeta$ sets its lower endpoint to $e=\zeta/r$. The analytic regulator instead multiplies this measure by $(He)^\eta$ at leading order. The difference produces the two logarithmic terms above. The $O(e)$ change in $\rho(e)$ is integrable for a simple soft pole.

These residue integrals do not require numerical quadrature. Partial fractions give a polynomial and poles at $r=1$ and $r=-4/H$, both outside $0\le r\le q$. The necessary moments are

$$\int_0^q r^n\,\mathrm{d}r = \frac{q^{n+1}}{n+1}, \tag{3.494}$$

$$\int_0^q r^n\ln r\,\mathrm{d}r = q^{n+1}\left[\frac{\ln q}{n+1} - \frac{1}{(n+1)^2}\right], \tag{3.495}$$

$$\int_0^q \frac{\mathrm{d}r}{r-a} = \ln(1-q/a), \tag{3.496}$$

$$\int_0^q \frac{\ln r}{r-a}\,\mathrm{d}r = \ln q\ln(1-q/a) + \mathrm{Li}_2(q/a). \tag{3.497}$$

Here $n\ge 0$ and a is a pole outside the interval. Repeated poles follow by applying $\partial_a^{j-1}/(j-1)!$. Every logarithm above has a positive argument, and $q/a<1$. These formulas therefore specify real branches directly from the physical integration interval. The independently generated residue coefficients and their exact moments are recorded in `compton_real_soft_matching_integrated.json`.

Bose symmetry of the subtractions. The two mixed angular kernels J_{UW} and J_{VW} are exchanged by $x\leftrightarrow y$. Their local Taylor subtractions, however, were expanded at fixed x. Thus $T_{VW}(H,x,z)$ is not exactly $T_{UW}(H,H-x+z,z)$. To replace the subtracted VW sector by a second copy of UW, the rational sector must receive

$$D(H,x,z) = R_{VW}(H,x,z)\big[T_{UW}(H,H-x+z,z) - T_{VW}(H,x,z)\big]. \tag{3.498}$$

The bracket starts at z^2, so it introduces no artificial Gram pole. It can nevertheless change the soft residue because its expansion at fixed x is not uniform as $y\to 0$.

At this endpoint write $D(H,H+e(r-1),er) = a(r)e^{-3} + b(r)e^{-2} + O(e^{-1})$. The coefficients needed in this endpoint expansion are explicit:

$$a(r) = \frac{16(H+1)(Hr+2)(Hr-2H-6)}{3H(r-1)^3}, \tag{3.499}$$

$$b(r) = \frac{2(H+1)}{3H^2(r-1)^3(Hr+4)^2}\sum_{j=0}^{5} B_j(H)r^j. \tag{3.500}$$

The six polynomials are

$$B_0(H) = 256H, \tag{3.501}$$
$$B_1(H) = 64\left(H^3 + 6H^2 - 11H - 12\right), \tag{3.502}$$
$$B_2(H) = 16\left(H^4 + 2H^3 + 2H^2 + 128H + 108\right), \tag{3.503}$$
$$B_3(H) = -8\left(H^5 + 8H^4 - 19H^3 - 160H^2 - 192H - 72\right), \tag{3.504}$$
$$B_4(H) = -2H\left(H^5 + 3H^4 - 36H^3 - 108H^2 - 152H - 96\right), \tag{3.505}$$
$$B_5(H) = H^2(H+6)\left(H^3 - 4H^2 - 12H - 8\right). \tag{3.506}$$

These follow by substituting the exact soft coordinates into (3.498) and expanding the resulting rational numerator and denominator through the first two powers.

The moving angular boundary now contributes to the logarithmic soft coefficient. Since $\rho'(0) = -1/(H+1)^2$, it is

$$A_D = \int_0^q b(r)\,\mathrm{d}r - \frac{a(q)}{(H+1)^2}. \tag{3.507}$$

More explicitly, the radial density after the angular integration is

$$\int_0^{\rho(e)} eD\,\mathrm{d}r = \frac{1}{e^2}\int_0^q a(r)\,\mathrm{d}r + \frac{1}{e}\left[\int_0^q b(r)\,\mathrm{d}r + \rho'(0)a(q)\right] + O(1). \tag{3.508}$$

The first term is a power singularity, whose analytic continuation has a pole at $\eta = 1$. The coefficient of $1/e$ gives the pole at $\eta = 0$. This explains why expanding at a fixed angular upper limit would miss the second term in (3.507). The terms generated by the $O(\eta e)$ part of $(xy)^\eta$ affect its finite part and must not be discarded.

Its finite contribution must also be included when combining the real sectors. Omitting D would violate the integrated soft coefficient even though the original amplitude is Bose symmetric.

Integrated infrared check. Denote the poles of the rational, $\ln s$, UV, and UW sectors by $A_{\mathrm{rat}}, A_{\log}, A_{UV}, A_{UW}$, respectively. The required sum, before the common factor $1/(8sH)$, is

$$A_{\mathrm{rat}} + A_{\log} + A_{UV} + 2A_{UW} + A_D = 8H\int_{-1}^{1} \mathcal{B}(c)\,[b_e(c) - 1]\,\mathrm{d}c. \tag{3.509}$$

The right-hand side follows independently from the soft current and the Born trace. It equals the coefficient required to cancel the photon-mass logarithm of the virtual amplitude after the regulator matching.

3.5.2 Matching a moving soft boundary

The separate rational subtraction terms can have stronger soft poles than their sum. Their finite parts therefore require the exact phase-space boundary and the first subleading term of the regulator. This section derives that matching directly from the integration region.

Let $H > 0$, $s = H + 1$, and $y = H - x + z$. The dimensionless Dalitz region is

$$0 < x < H, \qquad 0 < z < \frac{x(H-x)}{s-x}. \tag{3.510}$$

At either soft corner introduce a variable e, which here denotes a kinematic variable rather than the electric charge, and $r = z/e$. The two choices are $e = x$ and $e = y$. In both cases the Jacobian is e and

$$xy = e[H + e(r-1)], \qquad 0 < e < H, \qquad 0 < r < R(e) = \frac{H-e}{s-e}, \tag{3.511}$$

$$R(e) = q + q_1 e + O(e^2), \qquad q = \frac{H}{s}, \qquad q_1 = -\frac{1}{s^2}. \tag{3.512}$$

Consider a kernel whose expansion at this corner is

$$K(e, er) = \frac{a(r)}{e^3} + \frac{b(r)}{e^2} + O(e^{-1}), \tag{3.513}$$

where a and b are smooth on $[0, q]$. Terms with $a = 0$ include the usual logarithmic soft singularity. Define the four moments

$$P = \int_0^q r a(r)\,\mathrm{d}r, \tag{3.514}$$

$$A = \int_0^q b(r)\,\mathrm{d}r + q_1 a(q), \tag{3.515}$$

$$B = \int_0^q b(r)\ln r\,\mathrm{d}r + q_1 a(q)\ln q, \tag{3.516}$$

$$C = \frac{1}{H}\int_0^q (1-r)a(r)\,\mathrm{d}r. \tag{3.517}$$

The moment A includes the displacement of the upper boundary. Indeed, after multiplication by the Jacobian, the singular part can be written, in the sense of integration against smooth functions of r, as

$$eK\,\theta(R(e) - r) = \frac{a(r)\theta(q-r)}{e^2} + \frac{b(r)\theta(q-r) + q_1 a(q)\delta(r-q)}{e} + O(1). \tag{3.518}$$

Here θ is the Heaviside step function and δ is the Dirac delta distribution. The last term includes distributions at $r = q$ with integrable coefficients in e. It contributes the same finite limit in the two regularizations.

Let $I_z(\zeta)$ be the integral with $z > \zeta > 0$. Let I_η be the integral multiplied by $(xy)^\eta$, initially at sufficiently large $\operatorname{Re}\eta$ and then continued to zero. Define their finite parts by

$$I_z(\zeta) = \frac{P}{\zeta} - A\ln\zeta + F_z + o(1), \tag{3.519}$$

$$I_\eta = \frac{A}{\eta} + F_\eta + O(\eta). \tag{3.520}$$

The relation between them is

$$\boxed{F_\eta = F_z + A\ln H - B - C.} \tag{3.521}$$

Contributions from distinct corners are added. All quantities here refer to the unnormalized integral, before the common cross-section prefactor.

To prove the relation, first replace the regulator by $(He)^\eta$. Integration of the e^{-1} term in (3.518) gives $A/\eta + A\ln H$. Changing the lower boundary from $e > \zeta/r$ to an r-independent lower boundary gives the logarithmic moment B. The e^{-2} term gives

P/ζ, with no additional finite shift for a fixed upper boundary. The remaining regulator dependence follows from the exact identity for xy:

$$(xy)^\eta = (He)^\eta \left[1 - \frac{\eta e(1-r)}{H} + O(\eta e^2) + O(\eta^2 e^2)\right]. \tag{3.522}$$

Its linear term multiplies a/e^2. The resulting power-regulated integral contains $\eta \int e^{\eta-1}\mathrm{d}e$, whose limit is one. This gives $-C$. Terms quadratic in e are integrable at $\eta = 0$ and give no finite regulator shift. Omitting the linear term would thus give an incorrect constant even if the residue A were correct.

The corner of the moving boundary does not add another constant. The lower intersection with $z = \zeta$ satisfies

$$e_{\min} = \frac{\zeta}{q} - \frac{q_1\zeta^2}{q^3} + O(\zeta^3). \tag{3.523}$$

The interval displaced from ζ/q has width $O(\zeta^2)$. After the leading angular interval has closed, its integrated density is at most $O(\zeta^{-1})$, so its contribution is $O(\zeta)$. Smoothness at $r = 0$ likewise makes the excluded strip contribute a vanishing remainder after the displayed singular terms have been removed. This establishes that the boundary term already present in A and B is sufficient.

A logarithmic test kernel. For $K = x^{-2}$, $a = 0$, $b = 1$, and $A = q$. The exact analytically regulated integral is

$$I_\eta^{(2)} = H^{2\eta}\frac{\Gamma(\eta-1)\Gamma(\eta+1)}{\Gamma(2\eta+1)}\left[{}_2F_1(\eta+1,\eta-1;2\eta+1;q) - 1\right]. \tag{3.524}$$

This follows by first integrating y^η between $H - x$ and $s(H-x)/(s-x)$ and then using the Euler beta integral. Let

$$x_\pm = \frac{H + \zeta \pm \sqrt{(H-\zeta)^2 - 4\zeta}}{2}. \tag{3.525}$$

The independent cutoff integral and the two finite parts are

$$I_z^{(2)} = q\ln\frac{x_+}{x_-} + \frac{1}{s}\ln\frac{s-x_+}{s-x_-} + \zeta\left(\frac{1}{x_+} - \frac{1}{x_-}\right), \tag{3.526}$$

$$F_z^{(2)} = 2q\ln H - \ln s - q, \tag{3.527}$$

$$F_\eta^{(2)} = 2q\ln H - \frac{\ln s}{s}. \tag{3.528}$$

Since $B = q(\ln q - 1)$, these expressions obey (3.521). In particular, the finite $-q$ in the cutoff integral cannot be discarded.

A power-divergent test kernel. For $K = x^{-3}$, $a = 1$, $b = 0$, and the residue comes entirely from the moving boundary: $A = -1/s^2$. Direct integration gives

$$I_\eta^{(3)} = H^{2\eta-1}\frac{\Gamma(\eta-2)\Gamma(\eta+1)}{\Gamma(2\eta)}\left[{}_2F_1(\eta+1,\eta-2;2\eta;q) - 1\right], \tag{3.529}$$

$$I_z^{(3)} = J(x_+) - J(x_-), \tag{3.530}$$

$$J(x) = -\frac{q}{x} - \frac{\ln x}{s^2} + \frac{\ln(s-x)}{s^2} + \frac{\zeta}{2x^2}. \tag{3.531}$$

Their expansions yield

$$P = \frac{q^2}{2}, \qquad B = -\frac{\ln q}{s^2}, \qquad C = \frac{q - q^2/2}{H}, \tag{3.532}$$

$$F_z^{(3)} = -\frac{1}{s} - \frac{2\ln H}{s^2}, \tag{3.533}$$

$$F_\eta^{(3)} = -\frac{1}{s} - \frac{2\ln H + \ln s}{s^2} - \frac{q - q^2/2}{H}. \tag{3.534}$$

Both the boundary term and C are necessary for agreement.

The Bose subtraction correction. For the correction $D = R_{VW}[T_{UW}(y,z) - T_{VW}(x,z)]$ constructed from the rational coefficients and their local subtraction terms, the x corner is integrable. At the y corner its leading coefficient is

$$a(r) = \frac{16(H+1)(Hr+2)(Hr-2H-6)}{3H(r-1)^3}. \tag{3.535}$$

The coefficient $b(r)$ is obtained by differentiating $e^3 D$ at $e = 0$, after substituting $y = e$, $z = er$, and $x = H + e(r-1)$. This is a finite algebraic operation on the exact rational subtraction. The only poles of these coefficients in r are at 1 and $-4/H$, outside $[0, q]$. Their moments are therefore closed logarithms and dilogarithms. For example, every repeated pole follows by differentiation with respect to its location u:

$$\int_0^q \frac{\mathrm{d}r}{(r-u)^n} = \frac{\partial_u^{n-1}}{(n-1)!} \ln(1 - q/u), \tag{3.536}$$

$$\int_0^q \frac{\ln r \,\mathrm{d}r}{(r-u)^n} = \frac{\partial_u^{n-1}}{(n-1)!} \left[\ln q \ln(1-q/u) + \mathrm{Li}_2(q/u)\right]. \tag{3.537}$$

The arguments make both expressions real for these pole locations.

Integrating the rational correction. Write the exact partial fraction decomposition of D in x as

$$D(x,z) = \sum_{a,n} \frac{c_{a,n}(z)}{(x - a(z))^n}. \tag{3.538}$$

There are nine terms, with (a,n) equal to

$$(H+z,1),\ (H+z,2),\ (H,1),\ (H,2),\ (H,3),\ (z,1),\ (z,2),\ (a_*,1),\ (a_*,2), \tag{3.539}$$

$$a_* = \frac{-4H + z^2 - 4z}{z-4}. \tag{3.540}$$

The coefficients are fixed algebraically by the residues of D, including derivatives for repeated poles. Thus no fit is involved. Rationalize the physical boundary by

$$z = s + 1 - u - \frac{s}{u}, \qquad x_- = s - u, \qquad x_+ = s - \frac{s}{u}, \qquad \sqrt{s} < u < s. \tag{3.541}$$

The inner integral is explicitly

$$\int_{x_-}^{x_+} D \,\mathrm{d}x = \sum_a c_{a,1}(z) \ln \frac{x_+ - a}{x_- - a} \tag{3.542}$$

$$+ \sum_{a,n>1} \frac{c_{a,n}(z)}{1-n} \left[(x_+ - a)^{1-n} - (x_- - a)^{1-n}\right]. \tag{3.543}$$

The ratios are positive in the physical interval, as follows directly from the positions of the poles relative to $[x_-, x_+]$. After multiplication by $1 - s/u^2$, the outer integrand consists of rational functions multiplying the following three real logarithms:

$$L_1 = \ln \frac{u^2}{s}, \tag{3.544}$$

$$L_2 = \ln \frac{u(u-1)}{s-u}, \tag{3.545}$$

$$L_3 = \ln \frac{(u-1)(u+s)^2}{u(u+1)^2(s-u)}. \tag{3.546}$$

Each logarithm vanishes at $u = \sqrt{s}$. Partial fractions followed by integration by parts therefore gives a finite sum of generalized polylogarithms of weight at most two, based at $\sqrt{s}$, with rational coefficients. The letters are the zeros of the rational denominators and of the logarithm arguments. Differentiating that finite sum checks the original outer integrand term by term.

The power divergence also affects the conversion of the outer endpoint. For $b = s - u$, the exact relation gives

$$\zeta = \frac{H}{s} b - \frac{b^2}{s^2} + O(b^3), \qquad \frac{P}{\zeta} = \frac{Ps}{Hb} + \frac{P}{H^2} + O(b). \tag{3.547}$$

Consequently, if F_b is the finite part after subtracting the b^{-1} and $\ln b$ terms, then

$$F_z = F_b + A \ln(H/s) - \frac{P}{H^2}. \tag{3.548}$$

The last constant is required before applying (3.521). Convergent Taylor series for the endpoint remainder and for the generalized polylogarithms evaluate this closed result without quadrature.

3.5.3 Cancelling a quadratic pole family before integration

The separate primitive constructions above also provide a way to check individual sectors. For the total, a shorter route combines their inner integrals before constructing the outer primitives. The four contributions use the same coordinate u, the same interval $\sqrt{s} < u < s$, and the same relation $z = s + 1 - u - s/u$ to the original endpoint. They can therefore be added at any fixed positive cutoff in z. This observation permits cancellations before either a generalized polylogarithm or an endpoint constant is evaluated.

For $A = R_0, D$, denote the rational coefficients of the corresponding inner integral, including the Jacobian $1 - s/u^2$, by $K_{A,j}$, with $j = 0, 1, 2, 3$. Thus its outer integrand is $K_{A,0} + \sum_{j=1}^{3} K_{A,j} L_j$, with the real logarithms L_j already defined above. Write the energy-logarithm sector as $\ell(K_{\log,0} + K_{\log,1} L_1 + K_{\log,2} L_2)$, where $\ell = \ln s$, and the single-W sector as $A_W L_1 + B_W L_1 L_2$. Here A_W and B_W are its two rational coefficients after the same change of variable. The combined integrand is exactly

$$f_4(u) = \overline{K}_0 + \overline{K}_1 L_1 + \overline{K}_2 L_2 + \overline{K}_3 L_3 + B_W L_1 L_2, \tag{3.549}$$

$$\overline{K}_0 = K_{R_0,0} + K_{D,0} + \ell K_{\log,0}, \tag{3.550}$$

$$\overline{K}_1 = K_{R_0,1} + K_{D,1} + \ell K_{\log,1} + A_W, \tag{3.551}$$

$$\overline{K}_2 = K_{R_0,2} + K_{D,2} + \ell K_{\log,2}, \tag{3.552}$$

$$\overline{K}_3 = K_{R_0,3} + K_{D,3}. \tag{3.553}$$

All these coefficients follow from the preceding rational partial fractions and inner boundary differences.

Consider the polynomial

$$Q_b(u) = u^2 + 2u + s = (u+1)^2 + s - 1. \tag{3.554}$$

For $s > 1$ it is irreducible over the real numbers. Its two roots $b_\pm = -1 \pm \mathrm{i}\sqrt{s-1}$ would introduce an additional pair of GPL letters if the sectors were integrated separately. Polynomial division shows that their entire pole family cancels. To specify this statement precisely, let $\nu_b(K)$ be the multiplicity of Q_b in the reduced denominator of a rational coefficient K. The inner integrals give

j	$\nu_b(K_{R_0,j})$	$\nu_b(K_{D,j})$	$\nu_b(K_{R_0,j} + K_{D,j})$
0	2	2	0
1	0	0	0
2	3	3	0
3	3	3	0

. (3.555)

The cancellation can be checked without factoring Q_b into its roots. For each row, put the two coefficients over a common denominator $Q_b^n d$, where d has no factor Q_b, and write their numerator sum as N. Euclidean division gives

$$N = Q_b^n M + R, \qquad \deg_u R < 2n, \qquad R = 0. \tag{3.556}$$

Cancelling Q_b^n then gives M/d. This is an exact identity over the rational functions of $H = s - 1$. The energy-logarithm and single-W coefficients have no Q_b factor, so none is reintroduced in (3.553).

The preferred outer integration applies partial fractions to these combined coefficients. Products of logarithms are converted to GPL words by the shuffle identity, and repeated poles are reduced by the integration-by-parts operations already given. The letters $b_\pm$ need never be introduced, and their rational pole terms and six weight-two functions need never be evaluated separately. This reduction also leaves no boundary constant to determine for the removed group: its integrand vanishes identically at every common finite cutoff. Linearity then preserves the identity when the endpoint finite parts and their previously derived cutoff conversions are taken. The exact coefficient construction and the polynomial divisions are implemented in `compton_real_combined_kernel.py` using only the inner rational kernels.

3.6 Completion of the pair sectors

3.6.1 The final coordinate of the UV difference

Throughout this calculation the electron mass is the unit of energy, $s = P^2 > 1$, $P = p + k_{\mathrm{in}}$ is the incoming total momentum, with k_{in} the incident photon momentum, and $H = s - 1$ is fixed and positive. The original Dalitz variables are

$$x = 2P \cdot k_1, \qquad y = 2P \cdot k_2, \qquad z = x + y - H = 2k_1 \cdot k_2. \tag{3.557}$$

The final electron is on shell. Positive final energies and the three-body kinematic boundary give

$$0 < x < H, \qquad 0 < z < \frac{x(H-x)}{s-x}, \qquad y = H - x + z. \tag{3.558}$$

The hyperbolic angle in the UV angular function is rationalized by its positive exponential v. Explicitly,

$$D_v = H^2 v - x(v-1)^2, \qquad z = \frac{x(H-x)(v-1)^2}{D_v}, \tag{3.559}$$

$$y = \frac{H^2 v(H-x)}{D_v}, \qquad \left|\frac{\partial(x,z)}{\partial(x,v)}\right| = \frac{x(H-x)H^2(v^2-1)}{D_v^2}. \tag{3.560}$$

At fixed x, the lower and upper Dalitz boundaries map to $v=1$ and $v=s$. The Jacobian is positive in the interior. Thus the new domain is the rectangle $0<x<H$, $1<v<s$. This is why the inner energy integration can be completed before the final v integration. In particular,

$$xy = \frac{x(H-x)}{1-kx/H}, \qquad k = \frac{(v-1)^2}{Hv}, \qquad 0 \le k \le \frac{H}{s} < 1. \tag{3.561}$$

The analytic regulator therefore has the positive real value $[x(H-x)]^\eta(1-kx/H)^{-\eta}$. The denominator is at least $1/s$ on the closed rectangle. No logarithmic branch is inferred from a numerical result.

After the inner energy integration, the UV angular difference sector is a finite sum of terms

$$R_{ab}(H,v)\ln^a v \ln^b(1-k), \qquad k = \frac{(v-1)^2}{Hv}, \qquad 1 \le v \le H+1. \tag{3.562}$$

Here $H = s-1 > 0$, and the R_{ab} are the rational functions obtained from the real-emission amplitude. Their coefficients may include $\ln H$, which is constant in this integration. The pole and finite parts of the analytic regulator are treated separately.

The endpoint Laurent coefficients. The separated rational energy kernels have poles of at most second order at $x = 0, H$. Other poles are at $x = H/k > H$ and hence outside the integration path. Start with $\operatorname{Re}\eta > 1$, where these endpoint integrals converge absolutely. Subtract the constant and linear Taylor terms of the smooth endpoint factor before analytically continuing to a neighborhood of $\eta = 0$. This operation gives an ordinary convergent remainder and explicit Mellin integrals for the removed terms. For example, near $x = 0$ the smooth factor obeys

$$(H-x)^\eta(1-kx/H)^{-\eta} = H^\eta\left[1+\eta\frac{k-1}{H}x + O(x^2)\right]. \tag{3.563}$$

The kernel x^{-2} consequently has the finite coefficient $-1/H + (k-1)/H = (k-2)/H$. The first term is the finite part of the unregulated double pole. The second follows from ηx multiplying $x^{-2+\eta}$. Although that term vanishes pointwise as $\eta \to 0$, its endpoint integral has a nonzero limit. At $x = H$, write $b = H - x$. The corresponding expansion is

$$x^\eta(1-kx/H)^{-\eta} = \left(\frac{H}{1-k}\right)^\eta\left[1 - \frac{\eta b}{H(1-k)} + O(b^2)\right]. \tag{3.564}$$

Including the signs in $(x-H)^{-1}=-b^{-1}$ gives the complete rules

$$\int_0^H \frac{[x(H-x)]^\eta}{x(1-kx/H)^\eta}\,\mathrm{d}x=\frac{1}{\eta}+2\ln H+O(\eta), \tag{3.565}$$

$$\int_0^H \frac{[x(H-x)]^\eta}{x^2(1-kx/H)^\eta}\,\mathrm{d}x=\frac{k-2}{H}+O(\eta), \tag{3.566}$$

$$\int_0^H \frac{[x(H-x)]^\eta}{(x-H)(1-kx/H)^\eta}\,\mathrm{d}x=-\frac{1}{\eta}+\ln(1-k)-2\ln H+O(\eta), \tag{3.567}$$

$$\int_0^H \frac{[x(H-x)]^\eta}{(x-H)^2(1-kx/H)^\eta}\,\mathrm{d}x=-\frac{2-k}{H(1-k)}+O(\eta). \tag{3.568}$$

The displayed integrals near $\eta=0$ mean their analytic continuations from the convergence half-plane. The other energy kernels are bounded on $0\le x\le H$ and can be integrated at $\eta=0$ for their finite coefficient. These rules explain every regulator-generated constant used in the inner integration. They do not discard power-divergent endpoint terms by hand.

Put $\zeta=v-1$ for the final polylogarithm coordinate. This ζ is distinct from the original Dalitz variable z. The two roots of $1-k$ are

$$r_\pm=\frac{H+2\pm\sqrt{H(H+4)}}{2},\qquad r_+r_-=1,\qquad 0<r_-<1<H+1<r_+. \tag{3.569}$$

Factorization, together with the value at $\zeta=0$, gives

$$\ln v=G(-1;\zeta), \tag{3.570}$$

$$\ln(1-k)=G(r_+-1;\zeta)+G(r_--1;\zeta)-G(-1;\zeta). \tag{3.571}$$

The normalization of the multiple polylogarithms is

$$G(\varnothing;\zeta)=1,\qquad G(a,\boldsymbol{b};\zeta)=\int_0^\zeta\frac{\mathrm{d}u}{u-a}G(\boldsymbol{b};u), \tag{3.572}$$

$$G(\underbrace{0,\dots,0}_{n};\zeta)=\frac{\ln^n\zeta}{n!}. \tag{3.573}$$

For words with a convergent lower endpoint, the path is the real segment. The last line fixes the normalization for logarithmic lower endpoints. All logarithms are real on $0\le\zeta\le H$. Their letters lie outside this interval. Products are expanded by the shuffle identity, and the rational factors are integrated by partial fractions and the finite polylogarithmic reduction already described for the soft angular term.

A removable origin. Individual rational coefficients can have poles at $\zeta=0$, even though their sum multiplied by the logarithms is regular. An origin pole must therefore be handled before setting the lower endpoint to zero. For an analytic word write

$$G(\boldsymbol{a};\zeta)=\sum_{j\ge1}g_j(\boldsymbol{a})\zeta^j. \tag{3.574}$$

Normalize each primitive to have zero finite part at $\zeta=0$, meaning zero constant term after separating its negative powers and logarithms. For $\boldsymbol{a}=(b,\boldsymbol{c})$ and $n\ge2$, integration by parts then gives

$$\mathfrak{T}[\zeta^{-n};b,\boldsymbol{c}](\zeta)=-\frac{G(b,\boldsymbol{c};\zeta)}{(n-1)\zeta^{n-1}}+\frac{g_{n-1}(b,\boldsymbol{c})}{n-1} \tag{3.575}$$

$$+\frac{1}{n-1}\mathfrak{T}\left[\frac{1}{\zeta^{n-1}(\zeta-b)};\boldsymbol{c}\right](\zeta). \tag{3.576}$$

The displayed constant is essential. It subtracts the finite part of the boundary term and is absent in the simpler case of a nonzero pole. For the empty word the origin rules are

$$\mathfrak{T}[\zeta^{-1};\varnothing](\zeta)=\ln\zeta,\qquad \mathfrak{T}[\zeta^{-n};\varnothing](\zeta)=\frac{\zeta^{1-n}}{1-n}\quad (n\geq 2). \tag{3.577}$$

Linearity then makes the normalized primitive of the full regular integrand vanish at zero. Negative powers and $\ln\zeta$ must cancel in the complete expression. This is a check on the independently derived rational coefficients, not a choice of a finite subtraction constant.

An elementary example displays the role of the constant without any numerical evaluation,

$$\int_0^Z \frac{\ln(1+\zeta)-\zeta+\zeta^2/2}{\zeta^3}\,\mathrm{d}\zeta=\left(\frac{1}{2}-\frac{1}{2Z^2}\right)\ln(1+Z)+\frac{1}{2Z}-\frac{1}{4}. \tag{3.578}$$

The right-hand side differentiates to the integrand and tends to zero as $Z\to 0^+$. The $-1/4$ is the finite-part normalization constant.

The script `compton_soft_uv_check.py` implements this reduction. It reads the two independently generated inner-energy coefficient files, forms their difference, and evaluates the resulting ordinary multiple polylogarithms by convergent power series and analytic continuation. The original coordinate integral is used only as a separate check. For general $H>0$, direct expansion of the rational coefficients gives

$$\mathcal{P}_{\mathrm{UV}}(H,1+\zeta)=-\frac{256(H+1)(H+5)}{15H^2}\zeta+O(\zeta^2), \tag{3.579}$$

$$\mathcal{F}_{\mathrm{UV}}(H,1+\zeta)=-\frac{256(H+1)}{15H^3}[2H(H+5)\ln H-H^2-5H-15]\zeta+O(\zeta^2). \tag{3.580}$$

These follow by expanding $\ln(1+\zeta)$ and $\ln[1-\zeta^2/(H(1+\zeta))]$ before multiplying the rational Laurent series. The coefficients of $\zeta^{-4},\dots,\zeta^{-1},\zeta^0$ vanish identically in both sums. Only finitely many Taylor coefficients are required to prove these cancellations, since the highest rational origin pole is fourth order. Thus the apparent origin poles cancel for every $H>0$.

Constant letters for the UV contribution

The moving letters of the integrated UV contribution can be removed without using the total cross section. Introduce

$$q=\frac{\sqrt{H+4}-\sqrt{H}}{\sqrt{H+4}+\sqrt{H}},\qquad H=\frac{(1-q)^2}{q},\qquad 0<q<1. \tag{3.581}$$

Write $z=v-1$. Its upper endpoint is H, and the two logarithms of the inner Laurent coefficient become

$$\ln v=G(-1;z), \tag{3.582}$$

$$\ln(1-k)=G(q-1;z)+G(q^{-1}-1;z)-G(-1;z),\qquad k=\frac{z^2}{H(1+z)}. \tag{3.583}$$

This follows by factoring $1-k$ and matching its value one at $z=0$. All factors are positive on $0\leq z\leq H$: the negative letter $q-1$ is outside the interval, and $q^{-1}-1>H$. Rational partial fractions and integration by parts therefore produce GPL words with letters $0,-1,q-1,q^{-1}-1$ and rational coefficients in q,z.

To replace their moving endpoint and letters by constants, differentiate the whole word with respect to q. For a word $w = (a_1, \ldots, a_n)$, let $w \setminus j$ mean deletion of its jth entry. The total differential is

$$\mathrm{d}G(w;H) = G(w \setminus 1;H)\,\mathrm{d}\ln(H - a_1) - G(w \setminus n;H)\,\mathrm{d}\ln(-a_n) \tag{3.584}$$

$$+ \sum_{j=1}^{n-1} [G(w \setminus (j+1);H) - G(w \setminus j;H)]\,\mathrm{d}\ln(a_j - a_{j+1}). \tag{3.585}$$

The last endpoint term is absent when $a_n = 0$. Adjacent equal letters are treated by continuity, with their paired difference equal to zero. Factoring each logarithmic differential gives only constant simple poles. In this case the complete alphabet is

$$\left\{0, 1, -1, \frac{1+\mathrm{i}\sqrt{3}}{2}, \frac{1-\mathrm{i}\sqrt{3}}{2}\right\}. \tag{3.586}$$

For example, $H + 1 = (q^2 - q + 1)/q$ explains the complex-conjugate pair. Its appearance does not change the real physical boundary value.

The recursive integration uses basepoint $q = 1$ approached from below. For every nonzero moving letter, $H/a_j \to 0$ there. Thus every original word containing a nonzero letter vanishes, including powers of logarithms multiplying that vanishing limit. Pure-zero words instead have

$$G(\underbrace{0, \ldots, 0}_{n}; H) = \frac{\ln^n H}{n!}, \qquad \ln H = 2\ln(1-q) - \ln q. \tag{3.587}$$

These relations fix all integration constants. The recursion checks that pure powers of $\ln(1-q)$ cancel for every nonzero-letter original word.

The lower endpoint of the angular primitive supplies an additional finite constant, even though the combined integral is convergent. With

$$P(q) = 3q^8 + 18q^7 + 12q^6 - 1874q^5 + 1666q^4 - 1874q^3 + 12q^2 + 18q + 3, \tag{3.588}$$

$$Q(q) = -25q^8 + 238q^7 - 436q^6 + 2274q^5 - 4294q^4 + 2274q^3 - 436q^2 + 238q - 25, \tag{3.589}$$

the constants to subtract from the upper primitive are

$$b_{UV,-1} = -\frac{(q^2 - q + 1)P(q)}{9q^3(q-1)^4}, \tag{3.590}$$

$$b_{UV,0} = -\frac{q^2 - q + 1}{18q^3(q-1)^4}[4P(q)\ln H + Q(q)]. \tag{3.591}$$

The subscripts label the pole and finite Laurent coefficients, respectively. These constants follow from the Taylor coefficients of each GPL multiplied by its rational prefactor at $z = 0$. Setting every GPL to zero before multiplying by the prefactor would lose them.

The resulting pole and finite coefficients contain respectively seven and 41 constant-letter words of weight at most three, before multiplication by $1/(8sH)$. The script `compton_soft_uv_constant_basis.py` generates their explicit coefficient lists, `soft_uv_constant_basis_pole.json` and `soft_uv_constant_basis_finite.json`, in `build/intermediate/`. The reduction script uses only the independent inner Laurent coefficients. The primitive follows the exact rational integration recurrence above. The primitive is

checked by differentiating each independent kernel term before summing. If its coefficient dictionary is $Q_w(y)$, the derivative follows from

$$\frac{\mathrm{d}}{\mathrm{d}y}[Q_w(y)G(w;y)] = Q'_w(y)G(w;y) + \frac{Q_w(y)}{y-a}G(\widehat{w};y), \qquad w = (a,\widehat{w}). \tag{3.592}$$

For the empty word only the first term occurs. Subtracting the input kernel leaves a rational coefficient for each GPL word. Every coefficient vanishes exactly, for both pole and finite parts. The calculation holds the external parameter q and $\ln H$ fixed while differentiating in $y = v - 1$. Canceling rational factors before differentiation makes this test finite in size. Linearity then proves the derivative identity for their sum. The reproducible checks are in `compton_uv_primitive_certificate.py`. Evaluation against the earlier moving-letter expression provides a separate numerical check of the conversion.

3.6.2 The UW endpoint matching

Use the original Dalitz coordinates defined above, and set $a = H - x$. The positive angular variable satisfies $1 < v < 1 + a$, so $t = (v-1)/a$ places its two endpoints at zero and one. The complete map is

$$x = H - a, \qquad v = 1 + at, \qquad z = \frac{a(H-a)t^2}{1+at}, \tag{3.593}$$

$$y = \frac{a[1 + Ht^2 + at(1-t)]}{1+at}, \tag{3.594}$$

$$\left|\frac{\partial(x,z)}{\partial(a,t)}\right| = \frac{a(H-a)t(2+at)}{(1+at)^2} > 0. \tag{3.595}$$

The new domain is exactly $0 < a < H$, $0 < t < 1$. Both the sign of the Jacobian and the range of t follow from the positive solution for v. In these rectangular coordinates the analytic regulator is $[a(H-a)f(a,t)]^\eta$, with

$$f(a,t) = \frac{1 + Ht^2 + at(1-t)}{1+at}, \qquad 0 < a < H, \quad 0 < t < 1. \tag{3.596}$$

The useful alternative form $f = 1 + (H-a)t^2/(1+at)$ shows that $1 \le f \le 1 + H$ on the rectangle. Its logarithm and endpoint derivatives are real and bounded for fixed $H > 0$. For a fixed interior t, start again at $\mathrm{Re}\,\eta > 1$ and continue analytically only after subtracting the endpoint Taylor terms. Write the unregulated difference as $F(a,t) = L(a,t)\ln(1+at) - T(a,t)$, where L and T are the rational coefficient and Taylor subtraction obtained after including the Jacobian. At the upper endpoint put $b = H - a$ and expand

$$L = \frac{L_2}{b^2} + \frac{L_1}{b} + O(1), \qquad T = \frac{T_2}{b^2} + \frac{T_1}{b} + O(1). \tag{3.597}$$

These coefficients are fixed algebraically, for example $L_2 = [b^2 L]_{b=0}$ and $L_1 = [\partial_b(b^2 L)]_{b=0}$. Because $\ln[1 + (H-b)t] = \ell - bt/(1+Ht) + O(b^2)$, the complete singular coefficients are

$$g_2 = L_2\ell - T_2, \qquad \ell = \ln(1+Ht), \tag{3.598}$$

$$g_1 = L_1\ell - \frac{L_2 t}{1+Ht} - T_1. \tag{3.599}$$

In particular, the derivative of the logarithm contributes to the simple pole and cannot be omitted. Thus $F = g_2/b^2 + g_1/b + O(1)$. Since $f(H,t) = 1$, direct expansion gives

$$a^\eta f(a,t)^\eta = H^\eta\left[1 + \eta b\left(-\frac{1}{H} + \frac{t^2}{1+Ht}\right) + O(b^2)\right]. \tag{3.600}$$

The ordinary finite part used for comparison is defined explicitly by

$$\mathrm{FP}\int_0^H F(a,t)\,\mathrm{d}a = \lim_{\delta\to 0^+}\left[\int_0^{H-\delta} F(a,t)\,\mathrm{d}a - \frac{g_2}{\delta} + g_1\ln\delta\right]. \tag{3.601}$$

After the two singular terms are subtracted, the remainder is bounded at $b=0$. Its integral is analytic in η near zero. The removed terms are integrated before expansion using

$$\int_0^H b^{\eta-2}\,\mathrm{d}b = \frac{H^{\eta-1}}{\eta-1}, \qquad \int_0^H b^{\eta-1}\,\mathrm{d}b = \frac{H^{\eta}}{\eta}, \tag{3.602}$$

first in their convergence half-planes and then by continuation. The double-pole kernel therefore does not by itself produce a pole at $\eta=0$. Its term proportional to ηb, however, multiplies the second integral and supplies a finite constant. The simple pole produces g_1/η. Relative to the ordinary finite part of the a integral, the finite matching contribution is

$$g_1\ln H + g_2\left(-\frac{1}{H} + \frac{t^2}{1+Ht}\right). \tag{3.603}$$

The second term arises because the factor ηb multiplying $b^{-2+\eta}$ leaves a finite limit after integration. It must not be lost by setting $\eta=0$ before integrating. The finite part of the unregulated integral already includes $-g_2/H$ and $g_1\ln H$. The matching term is additional to those terms. At $a=0$ the possible simple-pole coefficient cancels in the complete angular difference. Explicitly, $\ln(1+at) = at + O(a^2)$ implies

$$\lim_{a\to 0} aF(a,t) = t\lim_{a\to 0} a^2 L(a,t) - \lim_{a\to 0} aT(a,t) = 0. \tag{3.604}$$

The last equality is an exact identity of the rational coefficients. Therefore there is no matching term from that endpoint.

The cancellation at the other boundary can also be seen before numerical evaluation. Expanding $\ln(1+Ht)$ and the explicit rational endpoint coefficients gives, for every fixed $H>0$,

$$g_2(H,t) = \frac{64H^2(H+1)}{15}t + O(t^2), \tag{3.605}$$

$$g_1(H,t) = -\frac{64H(H+1)}{5}t + O(t^2). \tag{3.606}$$

All the coefficients of $t^{-4}, t^{-3}, t^{-2}, t^{-1}, t^0$ vanish in these complete combinations. Thus the functions entering the pole and finite endpoint matching are integrable at the lower angular boundary. This algebraic cancellation, rather than their numerical values at a few points, justifies treating that boundary as removable.

The remaining t integral of g_1 is rational times $\ln(1+Ht)$. The same finite-part polylogarithmic reduction gives seven terms.

Logarithmic lower endpoints. For the finite coefficient, the inner integral also contains $\ln t$ and weight-two functions. A word ending in zero can be rewritten by shuffle identities. In particular,

$$G(a,0;t) = \ln t\, G(a;t) - G(0,a;t). \tag{3.607}$$

For a longer word the same identity recursively removes its trailing zeros. The result is a polynomial in $\ln t$ whose coefficients are regular-origin polylogarithms. In the repeated-pole primitive, the finite-part constant is the coefficient of $t^{n-1}(\ln t)^0$. Terms containing $\ln t$

are therefore included in the reduction rather than incorrectly treated as ordinary Taylor coefficients. For example, the shuffle identity gives the exact check

$$\int_0^1 \frac{\ln t}{1+t}\,\mathrm{d}t = G(-1,0;1) = -G(0,-1;1) = \mathrm{Li}_2(-1) = -\frac{\pi^2}{12}. \tag{3.608}$$

Differentiation proves the primitive, and its lower endpoint vanishes.

For the finite inner result, use the following names for its eight logarithmic and dilogarithmic functions:

$$\ell = \ln(1+Ht), \qquad l_H = \ln H, \qquad l_2 = \ln(2+Ht), \tag{3.609}$$

$$l_P = \ln\Big(1+Ht-Ht^2\Big), \qquad l_t = \ln t, \tag{3.610}$$

$$d_1 = \mathrm{Li}_2(-Ht), \qquad d_2 = \mathrm{Li}_2(-1-Ht) + \frac{\pi^2}{12}, \tag{3.611}$$

$$d_3 = \mathrm{Li}_2\left(-\frac{(1-t)(1+Ht)}{t}\right) - \mathrm{Li}_2\left(-\frac{1-t}{t}\right). \tag{3.612}$$

All logarithm arguments are positive for $H > 0$ and $0 < t < 1$. The dilogarithm arguments are nonpositive, so their real branches follow from the original physical interval. Expanding the boundary formulas in (3.448) in these functions gives thirteen monomials. Write $R_f(H,t)$ for the rational coefficient of a monomial f, including R_1 for the constant monomial.

In the complete UW finite sum, the coefficient of $\ell \ln t$, with $\ell = \ln(1+Ht)$, is exactly the negative of the coefficient of the dilogarithm difference d_3. This is an identity of the rational coefficients for arbitrary H and t, not a numerical cancellation. It groups these terms into $d_3 - \ell \ln t$. Writing $a = -1/H$ and $t_\pm = (1 \pm \sqrt{1+4/H})/2$, the group is

$$d_3 - \ell \ln t = -\sum_{r=t_\pm} [G(1,r;t) + G(a,r;t) - G(0,r;t)] - G(0,a;t). \tag{3.613}$$

Every word on the right has a regular lower endpoint. This identity also explains why the apparent $\ln t$ terms disappear from the complete lower-endpoint expansion. The general finite-part rules remain useful when these terms are integrated separately.

The finite UW coordinate integral. Write the inner finite result as a finite sum $\sum_j R_j(t) M_j(t)$, where the R_j are its explicit rational coefficients and each M_j is a monomial in the eight logarithmic or dilogarithmic functions. Translating these functions to ordinary polylogarithms and applying shuffle products gives finite coefficients $c_{j\boldsymbol{a}}$ such that

$$M_j(t) = \sum_{\boldsymbol{a}} c_{j\boldsymbol{a}} G(\boldsymbol{a};t). \tag{3.614}$$

The final coordinate integral is therefore the finite endpoint expression

$$\mathcal{F}_{UW}(H) = \lim_{Z\to 1^-} \sum_{j,\boldsymbol{a}} c_{j\boldsymbol{a}} \mathfrak{T}[R_j;\boldsymbol{a}](Z). \tag{3.615}$$

All rational integrations in this formula are eliminated by the terminating partial-fraction rules, including the origin constants described above. The functions with a first letter equal to one are combined before the endpoint limit. Their separate logarithmic divergences need not cancel within a single term.

For evaluation, the complete inner finite function is analytic at $t = 1$. Indeed, its logarithm arguments are nonzero there and the two arguments in the dilogarithm difference

both tend to zero. The apparent rational poles cancel in the summed expression. With $d = 1 - t$, the remaining tail is consequently obtained algebraically,

$$\mathcal{F}_{UW}^{\text{inner}}(H, 1-d) = \sum_{n\geq 0} c_n d^n, \qquad \int_{1-\delta}^{1} \mathcal{F}_{UW}^{\text{inner}}(H,t)\,\mathrm{d}t = \sum_{n\geq 0} \frac{c_n \delta^{n+1}}{n+1}. \tag{3.616}$$

The coefficients are generated by rational power-series division and the Taylor series of the displayed logarithms and dilogarithms. No phase-space quadrature enters this endpoint evaluation. Individual terms near $t = 0$ require powers of t multiplied by powers of $\ln t$. The combined function is analytic there, as follows from the cancellation of $\ln t$ established above. The generated finite expression requires 52 nonempty polylogarithm words.

The order of the limiting operations. All endpoint statements here concern a fixed $H > 0$. They do not justify interchanging the photon-mass limit or the regulator expansion with the threshold limit $H \to 0$. The analytic regulator is first introduced in a common convergence half-plane. Only after the endpoint Taylor terms have been isolated are their Mellin integrals continued to $\eta = 0$. The bounded remainders can then be expanded under the integral.

There is also a uniform reason for the removable angular origins. The hyperbolic function entering each angular integral is analytic at $q = 0$, where

$$q = \cosh(\ln v) - 1 = \frac{(v-1)^2}{2v}. \tag{3.617}$$

Together with the Taylor expansion of its rational prefactors, its constant and linear terms give precisely the terms removed in $J-T$. For the UV map, $q = \zeta^2/[2(1+\zeta)]$ is independent of the inner energy coordinate. For the UW map, $q = a^2t^2/[2(1+at)]$ is uniformly small as $t \to 0$ for $0 \leq a \leq H$. Taylor's remainder is therefore uniform after the explicit soft endpoint powers have been factored out. Those powers are at most second order, and are controlled by $\operatorname{Re}\eta > 1$ before continuation. The local Mellin subtractions above preserve this uniform angular behavior because their coefficient functions and the regulator shape factors are analytic in the same angular neighborhood. This gives the reason for taking the Laurent coefficients before the last angular integration. A pointwise expansion of the un-subtracted endpoint integrand would not give that justification.

At the upper UW boundary, the logarithm arguments are positive and the two d_3 dilogarithm arguments tend to zero. At nonzero electron mass, the original hard angular denominators have no pole on the integration sphere for positive photon energies. The soft endpoint factors have already been isolated explicitly. Thus the complete soft-subtracted energy integral has an analytic neighborhood of $t = 1$, even when a particular angular partial-fraction representation becomes singular there. The largest apparent rational pole in the separated coefficient list is fifth order. To determine the tail through order d^n, the atom series are therefore expanded through at least order d^{n+5} before summing. All negative powers are canceled before the endpoint value is used. The resulting convergent power series may be integrated term by term inside its disk of convergence. These arguments establish the endpoint operations. The numerical verification chapter tests their implementation independently of these analytic arguments.

Exact removal of the finite UW endpoint poles

The cancellation of the apparent poles at $t = 1$ can be checked for arbitrary $H > 0$, without evaluating the remaining integral numerically. Write $d = 1 - t$, $E = \ln(1 + H)$,

and $C = (H+1)/H$ in this subsection. Here E is a logarithm, not an electron energy. Denote the rational coefficient multiplying an atom f in the thirteen-term inner result by R_f. Only six terms can have a pole at this endpoint:

$$R_1 + R_{d_3} d_3 + R_{l_P} l_P + R_\ell \ell + R_{\ell l_t} \ell l_t + R_{\ell l_P} \ell l_P. \tag{3.618}$$

Their rational pole orders are respectively $4, 3, 5, 2, 3, 3$. The remaining seven terms have rational denominators nonzero at $t = 1$ and analytic atoms there. Thus they cannot contribute a negative power of d.

The required Taylor coefficients follow from

$$\ell = E - \sum_{n=1}^{4} \frac{1}{n} \left(\frac{H}{1+H} \right)^n d^n + O(d^5), \tag{3.619}$$

$$l_t = -\sum_{n=1}^{4} \frac{d^n}{n} + O(d^5), \tag{3.620}$$

$$l_P = \sum_{n=1}^{4} \frac{(-1)^{n+1}}{n} (Hd - Hd^2)^n + O(d^5), \tag{3.621}$$

$$d_3 = \sum_{n=1}^{4} \frac{z_1^n - z_2^n}{n^2} + O(d^5), \qquad z_1 = -\frac{d(1+H-Hd)}{1-d}, \quad z_2 = -\frac{d}{1-d}. \tag{3.622}$$

Terms of degree above four in the last two finite sums are simply omitted when constructing the Taylor polynomial. The two dilogarithm arguments vanish at $d = 0$, so their ordinary convergent series applies. All these statements are at fixed $H > 0$ in a sufficiently small neighborhood of $d = 0$. There is no continuation through a dilogarithm cut.

For completeness, the rational expansion requires only polynomial arithmetic. If $R_f(1-d) = (-d)^{-m} N(d)/D(d)$ with $D(0) \neq 0$, write $N = \sum N_j d^j$, $D = \sum D_j d^j$. Its regular quotient coefficients obey

$$r_0 = \frac{N_0}{D_0}, \qquad r_n = \frac{N_n - \sum_{j=1}^{n} D_j r_{n-j}}{D_0}. \tag{3.623}$$

Multiplying this finite quotient series by the atom series determines every negative power. The largest pole is fifth order, so degree four is sufficient. No infinite-series truncation error enters a coefficient of a negative power.

Let $\mathrm{PP}_{d=0}$ denote the sum of strictly negative powers in a Laurent expansion. Substituting the actual rational coefficients of the inner result gives the particularly simple pair

$$\mathrm{PP}_{d=0}(R_1 + R_{l_P} l_P) = C \left[-\frac{16}{d^2} + \frac{4(H-10)}{d} \right], \tag{3.624}$$

$$\mathrm{PP}_{d=0}(R_{d_3} d_3) = C \left[\frac{16}{d^2} - \frac{4(H-10)}{d} \right]. \tag{3.625}$$

The apparent fourth- and third-order poles have already canceled within the first line. The fifth-order coefficient vanishes because $l_P(0) = 0$. For the other three terms, divide their principal parts by $8C$. The coefficients of d^{-2} and d^{-1} are

Term	$[d^{-2}]/(8C)$	$[d^{-1}]/(8C)$
$R_\ell \ell$	$2E(H+1)/H$	$-E(H-4-7/H)-2$
$R_{\ell l_t} \ell l_t$	$-2E/H$	$-7E/H + 2/(H+1)$
$R_{\ell l_P} \ell l_P$	$-2E$	$E(H-4) + 2H/(H+1)$

Each column sums to zero as a rational identity in H, E. Their third-order coefficients vanish because $l_t(0) = l_P(0) = 0$. Combining the two cancellations proves

$$[d^{-j}]F_{UW}(H, 1-d) = 0, \qquad j = 1, \dots, 5, \qquad H > 0. \tag{3.626}$$

Here $F_{UW}(H,t)$ denotes the finite inner integrand before the final t integration. Since each atom is analytic at the endpoint and the rational factors have finite pole order, the disappearance of their complete principal part proves that this endpoint is removable. It is an identity for the finite integrand, not only for its infrared pole.

The independent script `compton_soft_uv_uw_endpoint_certificate.py` reads the thirteen rational coefficients, computes the six contributing expansions, and checks every coefficient of E separately using exact arithmetic. `uw_endpoint_certificate.json` records the individual terms as well as the zero sums. This certificate establishes endpoint regularity. It does not provide a bound on the entire omitted numerical Taylor tail used when evaluating the final special functions.

Exact removal of the finite UW origin poles

The lower endpoint also admits an exact certificate for every fixed $H > 0$. The only explicit $\ln t$ in the thirteen-term expression comes from d_3 and the product $\ell \ln t$. Their rational coefficients satisfy

$$R_{d_3} + R_{\ell l_t} = 0. \tag{3.627}$$

It is therefore essential to combine these two terms first. Define

$$\mathcal{D}(t) = d_3(t) - \ell(t) \ln t, \qquad \ell(t) = \ln(1 + Ht). \tag{3.628}$$

The logarithm cancels exactly, rather than merely becoming integrable.

Its boundary constant follows from the real inversion identity

$$\mathrm{Li}_2(-z) = -\frac{\pi^2}{6} - \frac{1}{2}\ln^2 z - \mathrm{Li}_2(-1/z), \qquad z > 0. \tag{3.629}$$

Apply it to $z = (1-t)(1+Ht)/t$ and $z = (1-t)/t$. Both are positive on $0 < t < 1$, so the same negative-real-axis boundary value is used in both terms. The exact result is

$$\mathcal{D}(t) = -\ell \ln(1-t) - \frac{1}{2}\ell^2 - \mathrm{Li}_2\left(-\frac{t}{(1-t)(1+Ht)}\right) + \mathrm{Li}_2\left(-\frac{t}{1-t}\right). \tag{3.630}$$

Every term on this line is analytic at $t = 0$ and vanishes there. This proves $\mathcal{D}(0) = 0$ independently of any numerical evaluation. Differentiating either this expression or the original pair gives

$$\mathcal{D}'(t) = \frac{l_P}{1-t} - \frac{H l_P}{1+Ht} + \frac{l_P - \ell}{t}, \qquad l_P = \ln\left(1 + Ht - Ht^2\right). \tag{3.631}$$

The apparent $1/t$ is removable because $l_P(0) = \ell(0) = 0$. The resulting series starts as

$$\mathcal{D}(t) = -\frac{H^2}{2}t^2 + \frac{H^2(H+1)}{2}t^3 - \left(\frac{11H^4}{24} + \frac{2H^3}{3}\right)t^4 + O(t^5). \tag{3.632}$$

In particular, $d_3 = Ht \ln t + O(t^2[1 + |\ln t|])$ has a zero constant, fixed by the inversion identity, not chosen to enforce cancellation.

After this combination the entire kernel contains only atoms analytic at the origin. Its rational factors have poles of order at most five. Using the finite quotient recurrence of

the preceding subsection, expand the atom series through degree six. This determines all negative powers, the constant, and the coefficient of t exactly. The cancellation can be displayed without a long list of individual rational coefficients. Define the three groups

$$\mathcal{B}_0 = R_1 + R_{l_2} l_2 + R_{l_P} l_P + R_\ell \ell, \tag{3.633}$$

$$\mathcal{B}_1 = R_{d_3} \mathcal{D} + R_{\ell l_P} \ell l_P + R_{\ell^2} \ell^2, \tag{3.634}$$

$$\mathcal{B}_H = \ln H \, [R_{l_H} + R_{\ell l_H} \ell]. \tag{3.635}$$

Their expansions obtained from the explicit coefficient table are

$$\mathcal{B}_0 = (H+1) \left[-\frac{128}{H^2 t^2} + \frac{64}{Ht} + \frac{4(3H^2 - 67H - 72)}{3H} \right] + O(t), \tag{3.636}$$

$$\mathcal{B}_1 = (H+1) \left[\frac{128}{H^2 t^2} - \frac{64}{Ht} - \frac{4(3H^2 - 67H - 72)}{3H} \right] + O(t), \tag{3.637}$$

$$\mathcal{B}_H = O(t). \tag{3.638}$$

Their fifth-, fourth-, and third-order poles have already canceled inside these groups. The remaining three atom terms are individually $O(t)$ because their rational coefficients are regular and their atoms vanish. Thus all negative powers and the constant cancel in the full finite kernel. The first surviving coefficient is

$$F_{UW}(H,t) = -\frac{32H(H+1)}{15}(12 \ln H - 1)t + O(t^2), \qquad t \to 0^+, \quad H > 0 \text{ fixed}. \tag{3.639}$$

There is no residual $\ln t$ in this combined expression. The origin is an ordinary analytic endpoint, so the finite lower boundary of the GPL primitive equals the lower limit of the convergent integral.

The script `compton_soft_uv_uw_origin_certificate.py` first checks the exact rational coefficient identity removing $\ln t$. It then treats $\ln H$ and $\ln 2$ as independent symbols and checks the coefficients of $t^{-5}, \ldots, t^{-1}, t^0$ separately in exact arithmetic. Their values and the nonzero coefficient of t are stored in `uw_origin_certificate.json`. Together with the upper certificate, this establishes both finite endpoints for arbitrary $H > 0$. A rigorous error bound for the numerical special-function series is a separate issue.

3.6.3 An exact certificate for the integrated pole sum

This additional algebraic check tests the separately implemented sector primitives for every $H > 0$. The physical soft factorization was established earlier. No numerical comparison or published total enters this check. The UV pole density already uses $1 < v < H + 1$. In the UW pole density, set $v = 1 + Ht$, including $\mathrm{d}t = \mathrm{d}v/H$. The physical infrared angular integral has the same range after the substitution

$$T = -t_{\text{elastic}} = \frac{(v-1)^2}{v}, \qquad y_B = H - T, \tag{3.640}$$

$$b_e = \frac{v^2+1}{v^2-1} \ln v, \qquad \left| \frac{\mathrm{d}c}{\mathrm{d}v} \right| = \frac{2(H+1)}{H^2} \left(1 - \frac{1}{v^2} \right). \tag{3.641}$$

Here $y_B = 1 - u_{\text{elastic}}$ distinguishes the elastic invariant from the real-emission Dalitz variable. Let $\mathcal{B}(v) = \overline{|\mathcal{M}_0|^2}/e_{\mathrm{T}}^4$ be the Born square with its coupling factor removed, evaluated at $x = H$ and $y = y_B$. The target density for $8H\mathcal{C}_{\mathrm{IR}}$ is consequently

$$P(v) = \frac{16(H+1)}{H} \mathcal{B}(v) \left[\frac{v^2+1}{v^2} \ln v - 1 + \frac{1}{v^2} \right]. \tag{3.642}$$

Subtract this density from the sum of the UV density and twice the transformed UW density. The result is $L(v)\ln v + Q(v)$, with rational L and Q fixed directly by the sector coefficient tables. Integration by parts separates rational total derivatives from simple-pole remainders. This operation, called rational Hermite reduction, provides a short certificate for its integral. Define

$$D(v) = v^2 - (H+2)v + 1, \tag{3.643}$$

$$p = H^3 - 5H^2 - 28H - 24, \qquad d = H^2 + 6H + 4, \tag{3.644}$$

$$j_0 = H^5 + 4H^4 - 12H^3 + 48H^2 - 32H - 384, \tag{3.645}$$

$$j_1 = 2H^4 + 16H^3 - 40H^2 + 64H - 192, \tag{3.646}$$

$$j_2 = 3H^4 - 16H^3 - 184H^2 - 736H - 480. \tag{3.647}$$

Two rational functions suffice:

$$\mathcal{R}(v) = \frac{H+1}{H^3}\left\{\frac{128H^2}{(v-1)^2} + 16Hv^2 + \frac{128H(H+2)(v-1)}{D(v)} + \frac{32H(3H-4)}{v-1}\right. \tag{3.648}$$

$$\left. -\frac{16H}{v^2} - 32dv + \frac{192(H+2)^2}{(v+1)^2} - \frac{128(H+2)^2}{(v+1)^3} + \frac{32p}{v+1} + \frac{32d}{v}\right\}, \tag{3.649}$$

$$\mathcal{J}(v) = \frac{H+1}{3H^3}\left\{\frac{H^3(H+2)(H+6)(H^2-4H+8)[(H+2)v-1]}{D(v)^2} + \frac{H^2(j_0+j_1v)}{D(v)}\right. \tag{3.650}$$

$$-\frac{384H^2}{v-1} - 72Hv^2 - \frac{72H}{v^2} - (6H^4 - 368H^2 - 1472H - 960)v \tag{3.651}$$

$$\left. -\frac{1152(H+2)^2}{v+1} + \frac{1152(H+2)^2}{(v+1)^2} - \frac{2j_2}{v}\right\}. \tag{3.652}$$

The four constant coefficients needed below are

$$C = \frac{32(H+1)(H^2-4H-6)}{H^2}, \tag{3.653}$$

$$c_1 = \frac{2(H+1)}{3H^3}(3H^5 - 24H^4 - 112H^3 + 608H^2 + 2368H + 1920), \tag{3.654}$$

$$c_+ = \frac{256(H+1)p}{3H^3}, \qquad c_D = -\frac{2(H+1)(H^3-8H^2-8H+48)}{H}. \tag{3.655}$$

Direct differentiation and rational cancellation give the two identities

$$L = \mathcal{R}' + \frac{C}{v}, \tag{3.656}$$

$$Q - \frac{\mathcal{R}}{v} = \mathcal{J}' + \frac{c_1}{v} + \frac{c_+}{v+1} + c_D\frac{D'}{D}. \tag{3.657}$$

Thus an elementary primitive of the full difference is

$$\Phi(v) = \mathcal{R}(v)\ln v + \frac{C}{2}\ln^2 v + \mathcal{J}(v) + c_1\ln v + c_+\ln(v+1) + c_D\ln[-D(v)]. \tag{3.658}$$

There is no unresolved branch in this expression. On $1 \le v \le H+1$ the convex quadratic D is negative, since its endpoint values both equal $-H$. All displayed logarithm arguments are positive. In particular, the c_D logarithm has identical endpoint values.

The apparent pole at $v = 1$ must be combined before taking its limit. If the coefficients of $(v-1)^{-2}$ and $(v-1)^{-1}$ in $\mathcal{R}$ are r_{-2} and r_{-1}, the constant from $\mathcal{R}\ln v$ is $r_{-1} - r_{-2}/2$. Adding the constant from $\mathcal{J}$ gives

$$j_- = -\frac{H+1}{3H^3}(H^5 + 24H^4 - 28H^3 - 624H^2 - 1936H - 768), \tag{3.659}$$

$$j_+ = \frac{1}{3H^3(H+1)}\Big(H^9 + 6H^8 - 18H^7 - 36H^6 + 351H^5 \tag{3.660}$$

$$+2912H^4 + 6572H^3 + 6800H^2 + 3472H + 768\Big), \tag{3.661}$$

$$r_+ = -\frac{16(H^5 + 20H^4 + 78H^3 + 114H^2 + 72H + 16)}{H^3(H+1)}. \tag{3.662}$$

Here $j_+ = \mathcal{J}(H+1)$ and $r_+ = \mathcal{R}(H+1)$. With $L_s = \ln(H+1)$ and $L_2 = \ln[(H+2)/2]$, the integrated difference is

$$E(H) = j_+ - j_- + (r_+ + c_1)L_s + \frac{C}{2}L_s^2 + c_+ L_2. \tag{3.663}$$

The remaining rational, single-logarithm, and Bose-restoration pole expressions contain only the additional logarithms

$$\ln\frac{1}{H+1} = -L_s, \qquad \ln\left(1 + \frac{H^2}{4(H+1)}\right) = 2L_2 - L_s. \tag{3.664}$$

Substitution and polynomial collection in the two independent symbols L_s, L_2 gives zero for every rational coefficient in $E + A_{\rm rat} + A_{\log} + A_D$. Therefore

$$A_{\rm rat} + A_{\log} + A_{UV} + 2A_{UW} + A_D = 8H\mathcal{C}_{\rm IR}, \qquad H > 0. \tag{3.665}$$

The script `compton_soft_uv_pole_certificate.py` verifies both rational derivative identities, the endpoint constants, and the final zero polynomial using exact arithmetic. It also verifies that the combined UV and UW pole densities start at order $v-1$, so their lower endpoints are ordinary convergent integrals. The machine-readable certificate is `soft_pole_certificate.json` in this supplement.

3.7 Assembly in a common infrared convention

All quantities in this section use $m = 1$, so that s means $s_{\rm phys}/m^2$ and λ means $\lambda_{\rm phys}/m$. Put $H = s - 1 > 0$. Define

$$A(s) = \frac{1}{s}\int_{-1}^{1}\mathcal{B}(c)[b_e(c) - 1]\,\mathrm{d}c, \qquad \mathcal{C}_K(s) = \int_{-1}^{1}\mathcal{B}(c)K(s,c)\,\mathrm{d}c. \tag{3.666}$$

These are the integrated eikonal coefficient and the finite soft angular integral derived above. Let C_V be the complete virtual contribution, including both external-photon orderings and the on-shell subtractions, in units of $\alpha_{\rm T}^3/m^2$. Its photon-mass term is $A(s)\ln\lambda$.

The relation to the local subtraction follows by angular integration. For a future-directed timelike vector v, direct integration in polar coordinates gives $\int \mathrm{d}\Omega_1/(v\cdot n_1)^2 = 4\pi/v^2$. Combining the two denominators in the cross term with a Feynman parameter ξ therefore gives

$$\int \mathrm{d}\Omega_1\,\frac{p\cdot p_B'}{(p\cdot n_1)(p_B'\cdot n_1)} = 4\pi z_e\int_0^1\frac{\mathrm{d}\xi}{1 + 2(z_e-1)\xi(1-\xi)} = 4\pi b_e. \tag{3.667}$$

In the last step, the substitution $2\xi - 1$ gives the parameter integral

$$\frac{2\,\operatorname{artanh}\sqrt{(z_e-1)/(z_e+1)}}{\sqrt{z_e^2-1}} = \frac{\operatorname{arcosh} z_e}{\sqrt{z_e^2-1}}, \tag{3.668}$$

with continuous value one at $z_e = 1$. The two diagonal terms each give 4π, so $\int \mathrm{d}\Omega_1(-j_n^2) = 8\pi(b_e - 1)$. Using the factor $q/\sqrt{s} = (s-1)/(2s)$ in $\mathcal{A}_{\mathrm{loc}}$ then yields

$$\frac{1}{4\pi(s-1)}\int_{-1}^{1}\mathrm{d}c_2\int\mathrm{d}\Omega_1\,\mathcal{A}_{\mathrm{loc}} = \frac{1}{s}\int_{-1}^{1}\mathcal{B}(c_2)[b_e(c_2)-1]\,\mathrm{d}c_2 = A(s). \tag{3.669}$$

Thus the local soft subtraction integrates to the same infrared coefficient $A(1+X)$ used in Chapter 1.

The real integral regulated by $(xy)^\eta$ has the Laurent expansion

$$C_R(\eta) = \frac{A(s)}{\eta} + F_R(s) + O(\eta). \tag{3.670}$$

This auxiliary regulator is used to evaluate the energy integrals. Its finite part is converted to the same photon-mass convention as the virtual result before either contribution is combined. Near either soft corner the outgoing photon energy in the center-of-mass frame is $\omega = e/(2\sqrt{s})$, and $xy = He + O(e^2)$. Consequently a real-energy cut $\omega > \varepsilon$ gives

$$C_R^{\mathrm{hard}}(\varepsilon) = F_R - A\ln(2\sqrt{s}H\varepsilon) + o(1). \tag{3.671}$$

The soft integral with photon mass λ gives $A\ln(2\varepsilon/\lambda) - \mathcal{C}_K/s$. Adding it first to the hard integral removes ε. Adding the virtual term then gives

$$C_{\mathrm{total}} = C_V(s,\lambda) + F_R(s) + A(s)\ln\frac{1}{\sqrt{s}H\lambda} - \frac{\mathcal{C}_K(s)}{s}. \tag{3.672}$$

The $\ln\lambda$ terms cancel algebraically. This argument fixes both the coefficient of the regulator logarithm and the finite scale factor $\sqrt{s}H$.

For the energy-sector decomposition used here, let $F^z_{\mathrm{rat}}, F^z_{\log}, F^z_W, F^z_D$ denote finite parts of the integrated rational, single-logarithm, L_W, and restoring term D with $z > \zeta$. Power and logarithmic divergences are removed before taking $\zeta \to 0$. Write Δ_j for the finite conversion from this definition to $(xy)^\eta$. The UV and UW finite parts, F^η_{UV} and F^η_{UW}, have already been computed directly with the latter regulator. The required assembly is

$$\begin{aligned} F_R = \frac{1}{8sH}\big[&F^z_{\mathrm{rat}} + \Delta_{\mathrm{rat}} + F^z_{\log} + \Delta_{\log} + F^z_W \\ &+ F^z_D + \Delta_D + F^\eta_{UV} + 2F^\eta_{UW}\big]. \end{aligned} \tag{3.673}$$

Here $\Delta_W = 0$. The term D makes the replacement of the subtracted VW integral by a second UW integral exact. The factor $1/(8sH)$ includes the incident flux, the phase-space normalization, and the identical-photon factor already established in the real-emission derivation. It is applied once, after summing the unnormalized energy integrals.

Every energy integral in (3.673) is represented by a finite GPL primitive evaluated at its endpoints. The endpoint conversions are logarithms and dilogarithms of the explicit rational soft residues. Consequently the assembled answer has no unevaluated phase-space integral:

$$\boxed{\sigma^{\mathrm{NLO}}_{\mathrm{inc}} = \sigma_0 + \frac{\alpha_{\mathrm{T}}^3}{m^2}C_{\mathrm{total}}(s_{\mathrm{phys}}/m^2) + \frac{1}{m^2}O(\alpha_{\mathrm{T}}^4).} \tag{3.674}$$

The remainder is at fixed dimensionless kinematics. The preceding sector formulas and finite primitive recursions specify C_{total} constructively, without using a published total as an integration condition. Evaluating the GPLs by convergent series is an evaluation of these analytic special functions, not a numerical integration of the scattering phase space.

3.7.1 Scope of the analytic reduction

The complete virtual amplitude has been evaluated in closed scalar functions. The vertex and self-energy angular integrals are given explicitly in (3.155). The spanning and finite soft angular integrals have finite multiple-polylogarithm endpoint expressions, with explicit boundary constants. The coefficient of the infrared logarithm cancels by the rational identity (3.159). No published Compton total-cross-section coefficient is used in these reductions.

The real-energy integrations reduce to finite GPL primitives, with explicit endpoint constants and a common infrared definition. Their assembly with the virtual and soft terms is (3.674). The separate finite pieces depend on the subtraction convention, while their sum does not. Independent numerical integration checks the result without replacing the analytic primitives. The final verification chapter states the scope and estimated numerical accuracy of these checks.

Chapter 4

Simplified analytic cross section

4.1 The total cross section

Roman N. Lee, Matthew D. Schwartz and Xiaoyuan Zhang have already obtained an analytic expression for this total cross section [2]. The expression developed here is an independent recalculation for verification. The integrations of our virtual and real amplitudes are complete. This chapter simplifies their analytic sum in a common energy-variable basis. We begin with the sector assembly already derived, reduce its terms to that basis, and collect their coefficients. No published cross-section coefficient is used in this reduction. We use

$$X = \frac{s_{\rm phys}}{m^2} - 1, \qquad q = \frac{\sqrt{X+4} - \sqrt{X}}{\sqrt{X+4} + \sqrt{X}}, \qquad Y = \sqrt{\frac{X}{X+4}}, \qquad X > 0. \tag{4.1}$$

The square roots are positive. The Thomson coupling is denoted by $\alpha_{\rm T}$, and m is the electron mass. Define $s = 1 + X$ and $r_\omega = X/2$, so that the incident photon energy in the electron rest frame is mr_ω. The complete result is

$$\boxed{\sigma_{\rm inc}^{\rm NLO}(X) = \frac{\pi\alpha_{\rm T}^2}{m^2} B(X) + \frac{\alpha_{\rm T}^3}{m^2} C_1(X) + \frac{1}{m^2} O(\alpha_{\rm T}^4).} \tag{4.2}$$

Here the real contribution includes the complete two-photon final-state phase space, with its identical-photon factor. The remainder is at fixed $X > 0$. The Born coefficient is

$$\begin{aligned} B(X) = 2\Bigg[& \frac{1+r_\omega}{r_\omega^2} \left(\frac{2(1+r_\omega)}{1+2r_\omega} - \frac{\ln(1+2r_\omega)}{r_\omega} \right) \\ & + \frac{\ln(1+2r_\omega)}{2r_\omega} - \frac{1+3r_\omega}{(1+2r_\omega)^2} \Bigg]. \end{aligned} \tag{4.3}$$

For the NLO coefficient, put

$$\begin{aligned} \mathcal{R}_{\rm rem}(X) = {} & F_{\rm rat}^z + \Delta_{\rm rat} + F_{\rm log}^z + \Delta_{\rm log} + F_W^z \\ & + F_D^z + \Delta_D + 2F_{UW}^\eta. \end{aligned} \tag{4.4}$$

$$\begin{aligned} C_1(X) = {} & \widehat{C}_{V+{\rm UV}}(X) + \frac{\mathcal{R}_{\rm rem}(X)}{8X(1+X)} \\ & - A(1+X)\ln\!\big(X\sqrt{1+X}\big) - \frac{\mathcal{C}_K(1+X)}{1+X}. \end{aligned} \tag{4.5}$$

All finite parts in $\mathcal{R}_{\rm rem}$ use the unnormalized energy integrals of (3.673). Specifically, F_j^z is the constant term of the endpoint expansion with $z > \zeta$ after subtracting its power and logarithmic divergences. The finite GPL endpoint formula is (3.478). The term D restores the exact difference between VW and the second copy of UW, as defined in (3.498). The shifts $\Delta_{\rm rat}$ and $\Delta_{\log}$ are given by (3.493). The shift Δ_D includes the moving boundary and double-pole terms in (3.521). The W sector has no such shift. The finite part F_{UW}^{η} uses the analytic-regulator definition and closed primitives of (3.430)–(3.448). Thus $\mathcal{R}_{\rm rem}$ contains already integrated terms. It is not an omitted physical contribution or an unevaluated phase-space integral.

The infrared coefficient $A(s) = \mathcal{C}_{\rm IR}(s)/s$ has the integrated form (3.218). The finite soft term $\mathcal{C}_K(s)$ is given by (3.268). The three endpoint shifts are the finite regulator conversions derived in (3.521). To define the first term in (4.5), set

$$C_{V+\rm UV}(X;\ell_\lambda) = C_V(1+X, \lambda_{\rm phys}/m) + \frac{F_{UV}^{\eta}(X)}{8X(1+X)}. \tag{4.6}$$

Here UV labels the real angular denominator pair, not an ultraviolet counterterm. Separate the photon-mass logarithm as

$$C_{V+\rm UV}(X;\ell_\lambda) = A(1+X)\ell_\lambda + \widehat{C}_{V+\rm UV}(X), \qquad \ell_\lambda = \ln(\lambda_{\rm phys}/m). \tag{4.7}$$

Its 23 coefficients are displayed below. The hat denotes the constant coefficient in this polynomial in ℓ_λ, rather than a choice of a large photon mass. Equation (4.5) follows by canceling the photon-mass logarithm in (3.672). It is independent of $\lambda_{\rm phys}$. The reductions below put every finite contribution into the same alphabet before their coefficients are combined.

4.2 The virtual contribution in a common function basis

4.2.1 A constant alphabet

Use iterated integrals based at $X = 0$, with the following kernels:

$$G(a,w;X) = \int_0^X \omega_a(t) G(w;t), \qquad G(\varnothing;X) = 1, \tag{4.8}$$

$$\omega_a = \frac{\mathrm{d}X}{X-a} \quad (a = 0, 1, -1, -2, -4), \qquad \omega_r = \frac{\mathrm{d}X}{\sqrt{X(X+4)}}. \tag{4.9}$$

The root is positive for $X > 0$. Pure-zero words have the normalization $G(0,\ldots,0;X) = \ln^n X/n!$. The letter 1 occurs in combinations whose apparent pole is canceled by an inner function vanishing at $X = 1$, as in $G(1,0;X)$ with kernel $\ln X/(X-1)$.

Define a new variable q, unrelated to electric charge, by

$$q = \frac{\sqrt{X+4} - \sqrt{X}}{\sqrt{X+4} + \sqrt{X}}, \qquad X = \frac{(1-q)^2}{q}, \qquad 0 < q < 1. \tag{4.10}$$

The physical square roots are positive. Introduce

$$\rho_\pm = \frac{1 \pm \mathrm{i}\sqrt{3}}{2}, \qquad \gamma_\pm = \frac{3 \pm \sqrt{5}}{2}, \qquad \nu_a = \frac{\mathrm{d}q}{q-a}. \tag{4.11}$$

Direct differentiation gives the complete pullback of the kernels:

$$\frac{\mathrm{d}X}{X} = 2\nu_1 - \nu_0, \tag{4.12}$$

$$\frac{\mathrm{d}X}{X+1} = \nu_{\rho_+} + \nu_{\rho_-} - \nu_0, \tag{4.13}$$

$$\frac{\mathrm{d}X}{X+2} = \nu_{\mathrm{i}} + \nu_{-\mathrm{i}} - \nu_0, \tag{4.14}$$

$$\frac{\mathrm{d}X}{X+4} = 2\nu_{-1} - \nu_0, \tag{4.15}$$

$$\frac{\mathrm{d}X}{X-1} = \nu_{\gamma_+} + \nu_{\gamma_-} - \nu_0, \tag{4.16}$$

$$\frac{\mathrm{d}X}{\sqrt{X(X+4)}} = -\nu_0. \tag{4.17}$$

Replacing each letter by its linear combination in these equations converts every independently integrated word to constant-letter GPLs based at $q = 1$. The basepoint is approached from the left. Its logarithmic convention is fixed by the exact relation

$$\ln X = 2\ln(1-q) - \ln q. \tag{4.18}$$

Writing $q = 1 - \delta$ gives $X = \delta^2/(1-\delta)$ and $\ln X = 2\ln\delta + O(\delta)$. The leading coefficient of X in this relation is one, so it produces no finite logarithmic constant. This fixes words ending in the singular letter without an undetermined additive constant. All other boundary constants follow by induction from the vanishing lower-weight words. Complex letters occur in conjugate pairs, with the same real path and conjugate boundary values. The resulting alphabet is

$$\mathcal{A} = \{0, 1, -1, \mathrm{i}, -\mathrm{i}, \rho_+, \rho_-, \gamma_+, \gamma_-\}. \tag{4.19}$$

The transformed words have finite rational coefficient dictionaries. All six kernel identities in (4.17) are checked symbolically.

4.2.2 The vertex and self-energy contribution

The entire independently integrated vertex and self-energy contribution can already be reduced to constant letters. In the direct channel, the physical continuation gives

$$\mathrm{Re}\,\mathrm{Li}_2(1 + X + \mathrm{i}0) = \frac{\pi^2}{6} - G(-1, 0; X), \qquad \mathrm{Re}\ln(-X - \mathrm{i}0) = \ln X. \tag{4.20}$$

The imaginary part of the dilogarithm is fixed by its indicated boundary, not by a fit to the final answer. The interference coefficients in this channel are real, so these real parts suffice for this contribution. In the crossed channel the two angular endpoints are $y = X$ and $y = X/(1+X)$. At the latter endpoint, the substitution $y = t/(1+t)$ gives

$$\frac{\mathrm{d}y}{y} = \frac{\mathrm{d}t}{t} - \frac{\mathrm{d}t}{t+1}, \qquad \frac{\mathrm{d}y}{y-1} = -\frac{\mathrm{d}t}{t+1}. \tag{4.21}$$

Its derivative at $t = 0$ is one, so the logarithmic basepoint adds no constant: $\ln y = \ln t - \ln(1+t)$. Applying these replacements to the independent angular primitive, including its rational coefficients, gives exactly thirteen words:

$$C_{V+\Sigma}(s, \lambda) = \sum_{w \in \mathcal{W}_V} v_w(X, \widehat{c}_Z) G(w; X), \qquad \widehat{c}_Z = -4 - 4\ln\lambda. \tag{4.22}$$

Here λ is measured in units of m. The symbol $\widehat{c}_Z$ is the ultraviolet-finite field coefficient used in the angular primitive, not the full counterterm c_Z of Chapter 2. With $G(\varnothing; X) = 1$, the word set is

$$\mathcal{W}_V = \{\varnothing, (-1), (0), (-1,-1), (-1,0), (0,-1), (1,0), \tag{4.23}$$
$$(-1,-1,-1), (-1,-1,0), (-1,0,-1), (0,-1,-1), (0,-1,0), (0,1,0)\}. \tag{4.24}$$

Every v_w is rational in X and $\widehat{c}_Z$. All explicit π^2 constants cancel in this representation. For example,

$$v_{(0,1,0)} = v_{(0,-1,0)} = \frac{4(X+1)}{X^3}, \tag{4.25}$$
$$v_{(-1,-1,0)} = -\frac{4(X-1)}{X^3}, \qquad v_{(1,0)} = \frac{X^2 - 9X + 16}{X^3}. \tag{4.26}$$

The full coefficient and derivative dictionaries are obtained from the independent trace and angular-primitive files, without using the published total. This reduces the complete vertex and self-energy subset. It is not separately gauge invariant and must be combined with the spanning boxes.

4.2.3 The direct spanning box

The direct spanning-box channel can also be reduced completely. In its angular integral let

$$b = \sqrt{\frac{X}{X+4}}, \qquad y = \frac{X - (X+4)v^2}{1 - v^2}, \qquad 0 < v < V = \frac{X}{X+2}. \tag{4.27}$$

Its input GPL words use only $0, \pm 1$. Exact rational integration in v adds first letters $\pm b$, with no other roots. Setting $X = 4b^2/(1-b^2)$ makes every rational coefficient and partial-fraction pole rational in b. The resulting angular primitive has been checked by differentiation, with every GPL coefficient canceling identically for arbitrary b.

At the physical endpoint the variable q gives

$$b = \frac{1-q}{1+q}, \qquad V = \frac{(1-q)^2}{1+q^2}. \tag{4.28}$$

Both quantities are rational in q. Let $\mathcal{G}$ denote GPLs based at $q = 1$, with the same real left tangent used above. The weight-one transformations are

$$G(0; V) = 2\mathcal{G}(1; q) - \mathcal{G}(\mathrm{i}; q) - \mathcal{G}(-\mathrm{i}; q) - \ln 2, \tag{4.29}$$
$$G(1; V) = \mathcal{G}(0; q) - \mathcal{G}(\mathrm{i}; q) - \mathcal{G}(-\mathrm{i}; q), \tag{4.30}$$
$$G(-1; V) = \mathcal{G}(\rho_+; q) + \mathcal{G}(\rho_-; q) - \mathcal{G}(\mathrm{i}; q) - \mathcal{G}(-\mathrm{i}; q), \tag{4.31}$$
$$G(b; V) = 2\mathcal{G}(0; q) - \mathcal{G}(\mathrm{i}; q) - \mathcal{G}(-\mathrm{i}; q), \tag{4.32}$$
$$G(-b; V) = -\mathcal{G}(\mathrm{i}; q) - \mathcal{G}(-\mathrm{i}; q). \tag{4.33}$$

These follow by taking logarithms of the corresponding rational ratios, with their constants fixed at $q = 1$. In particular, for $q = 1 - \delta$ one has $V = \delta^2/2 + O(\delta^3)$, so the $-\ln 2$ in the first line is necessary.

Higher weights follow recursively from the total differential of a moving-letter GPL. All differences between $0, \pm 1, \pm b$ and V are rational functions of q. Their logarithmic derivatives have simple poles only in

$$\mathcal{A}_s = \{0, 1, -1, \mathrm{i}, -\mathrm{i}, \rho_+, \rho_-\}. \tag{4.34}$$

After a lower-weight word has been converted, integration of each term $\nu_a \mathcal{G}(w;q)$ prepends the letter a. Every original word containing a nonzero letter vanishes as $q \to 1$, since $V = O(\delta^2)$ while $b = O(\delta)$. Thus its integration constant is zero. The all-zero words are fixed by powers of the first weight-one identity. Coincident adjacent letters cancel in the total differential before individual logarithmic derivatives are taken. The transformation checks these cancellations exactly, including the cancellation of any spurious pure powers of $\ln(1-q)$ at the boundary.

Finally, the scalar factors multiplying the angular words satisfy

$$d_s = \frac{G(-1,0;X)}{X}, \qquad \ell_X = G(0;X). \tag{4.35}$$

Transforming these factors with the kernel map above and applying shuffle identities completes the direct-box result as 101 constant-letter words of weight at most three. Its rational coefficients depend only on q and $\ln\lambda$, with possible constant $\ln 2$ factors. The normalization is $C_{\mathrm{box},s} = J_s/(2X^2)$. This reduction uses only the independent box coefficients.

4.2.4 The crossed spanning box in the energy basis

Section 3.2.3 has already integrated the rational crossed subgroup in y, before introducing the angular square root. Its endpoint maps contain only the letters $-1, 0, 1$. Applying the kernel pullbacks above converts those words to the same q basis. The remaining angular integral has no auxiliary ρ_u letter.

4.2.5 The complete virtual contribution

After the crossed subgroup is integrated as just described, its remaining seventeen angular words involve only $0, \pm 1, \pm b$. The same total differential construction used for the direct box therefore applies. Combining it with the crossed subgroup gives 110 constant-letter words for J_u. The two independent constructions of the subgroup endpoint coefficients agree as exact rational functions. The remaining angular primitive has also been checked by an exact derivative, coefficient by coefficient, for arbitrary kinematics.

The entire independently computed virtual contribution is consequently in the comparison basis:

$$C_V(q,\lambda) = \sum_{w \in \mathcal{W}} V_w(q, \ln\lambda)\mathcal{G}(w;q). \tag{4.36}$$

There are 116 nonzero words. All explicit $\ln 2$ coefficients introduced by intermediate changes of endpoint cancel in this complete virtual sum. This is an algebraic cancellation of coefficients, not a numerical inference. The coefficients still depend on $\ln\lambda$, as expected for the virtual contribution.

4.2.6 Exact inversion of the kernel transformation

The common basis can be reduced without choosing functions from a published answer. Let M be the 9×6 matrix of coefficients in the six kernel pullbacks above. A row labels a letter in $\mathcal{A}$, and a column labels one of the six X kernels. Its six columns are linearly independent. Select six independent rows with a projection matrix P and define

$$L = (PM)^{-1}P, \qquad LM = I_6. \tag{4.37}$$

For weight n, let v_n denote the entire coefficient vector of the independently integrated q words, with zero entries for absent words. Applying the tensor product of n copies of L gives the candidate X coefficients c_n. The necessary and sufficient reconstruction test is

$$c_n = L^{\otimes n} v_n, \qquad M^{\otimes n} c_n - v_n = 0, \qquad 0 \le n \le 3. \tag{4.38}$$

Every component of the second equation is checked as an exact rational identity, including the rows not selected by P. Thus the choice of independent rows cannot hide an omitted word. The weight-zero equation includes constants. The common basepoint and logarithmic normalization established above make this an equality of integrals, including their boundary constants.

The remaining coefficient functions are rational in q. Polynomial reduction using $q^2 - (X+2)q + 1 = 0$ reduces each one to $A(X) + qB(X)$. The physical root is fixed by $0 < q < 1$:

$$q = \frac{X + 2 - \sqrt{X(X+4)}}{2}. \tag{4.39}$$

The other root exceeds one for $X > 0$. Substitution therefore gives coefficients rational in X and $Y = \sqrt{X/(X+4)}$, with no branch choice left to numerical evaluation.

4.3 The virtual and UV contribution

Common functions and exact alphabet reduction. The physical kernels required for the UV term are

$$\omega_0 = \frac{\mathrm{d}X}{X}, \qquad \omega_{-1} = \frac{\mathrm{d}X}{X+1}, \qquad \omega_{-4} = \frac{\mathrm{d}X}{X+4}, \qquad \omega_r = \frac{\mathrm{d}X}{\sqrt{X(X+4)}}. \tag{4.40}$$

Their exact pullbacks provide a finite algebraic test of whether the apparently larger endpoint alphabet is needed:

$$\omega_0 = 2\frac{\mathrm{d}q}{q-1} - \frac{\mathrm{d}q}{q}, \qquad \omega_{-4} = 2\frac{\mathrm{d}q}{q+1} - \frac{\mathrm{d}q}{q}, \tag{4.41}$$

$$\omega_{-1} = \frac{\mathrm{d}q}{q-\rho_+} + \frac{\mathrm{d}q}{q-\rho_-} - \frac{\mathrm{d}q}{q}, \qquad \omega_r = -\frac{\mathrm{d}q}{q}, \qquad \rho_\pm = \frac{1 \pm \mathrm{i}\sqrt{3}}{2}. \tag{4.42}$$

The endpoint $X = 0$ corresponds to the same left tangential base point $q = 1$ used in the preceding calculation. In particular, $\ln X = 2\ln(1-q) - \ln q$ has no additional constant there.

For each weight through three, form every tensor word in these four kernels and expand it with the displayed pullbacks. This gives a matrix with integer entries. Solving its coefficient equations against the independently integrated UV dictionary, and substituting the solution back into every equation, proves an identity of iterated integrals with the same base point. No comparison cross section enters this operation. The rational prefactors are simplified using

$$q^2 - (X+2)q + 1 = 0, \qquad q = \frac{X + 2 - (X+4)Y}{2}. \tag{4.43}$$

Thus they belong to $\mathbb{Q}(X) + Y\mathbb{Q}(X)$, without auxiliary roots. The two UV dictionaries each satisfy all 156 coefficient equations of this transformation. Their normalization remains that of the original real integral: the contribution to the coefficient of $\alpha_{\mathrm{T}}^3/m^2$ is obtained by multiplying the UV finite part by $1/[8X(1+X)]$.

Adding this transformed finite part to the full virtual contribution requires only the same 23 X words already obtained for the virtual amplitude. The files accompanying this calculation give the separate UV coefficients and the summed coefficients, so the normalization and every cancellation can be checked before other contributions are added. The 23-word sum by itself is not infrared finite. The other terms appear with their complete endpoint definitions in (4.5). The following sections reduce them to the same small alphabet, so that their coefficients can be added algebraically.

After exact removal of vanishing coefficients, the UV finite dictionary contains twelve nonzero X words (rather than its original 41 q words), and its pole dictionary contains three. All twelve finite words already occur in the full virtual dictionary. Thus their sum still contains 23 words. Its coefficients are displayed explicitly below.

Explicit partial sum. Write $G_{a_1,\ldots,a_n} = G(a_1,\ldots,a_n;X)$ and $G_\varnothing = 1$, with the kernels above, $\omega_{-2} = \mathrm{d}X/(X+2)$ and $\omega_1 = \mathrm{d}X/(X-1)$. These iterated integrals have base point zero:

$$G_{a,w}(X) = \int_0^X \omega_a(t) G_w(t), \qquad G_{\underbrace{0,\ldots,0}_{n}}(X) = \frac{\ln^n X}{n!}. \tag{4.44}$$

The words ending in zero that contain the letter 1 are understood by their continuous real extension at $X = 1$. Let $\ell_\lambda = \ln(\lambda_{\mathrm{phys}}/m)$. The following expression includes the complete virtual contribution and the finite UV real contribution:

$$C_{V+\mathrm{UV}}(X;\ell_\lambda) = \sum_{w\in\mathcal{W}} c_w(X;\ell_\lambda) G_w(X). \tag{4.45}$$

The set $\mathcal{W}$ and every coefficient are specified by the following equations.

$$\begin{aligned} 144X^4(X+1)^2\, c_\varnothing = {} & (-5X^9 - 2X^8 + 141X^7) \\ & + (486X^6 - 720X^5\ell_\lambda + 1380X^5) \\ & + (-8640X^4\ell_\lambda + 2128X^4 - 14400X^3\ell_\lambda) \\ & + (9392X^3 - 6912X^2\ell_\lambda + 15616X^2) \\ & + (10752X + 3072). \end{aligned} \tag{4.46}$$

$$\begin{aligned} 24X^5(X+1)^2(X+4)\, c_{-1} = {} & (-5X^9 - 32X^8 - 48X^7\ell_\lambda) \\ & + (29X^7 + 1056X^6\ell_\lambda + 80X^6) \\ & + (8832X^5\ell_\lambda - 1656X^5 - 7504X^4) \\ & + (21312X^4\ell_\lambda + 20352X^3\ell_\lambda - 17088X^3) \\ & + (6912X^2\ell_\lambda - 18944X^2 - 9728X) \\ & + (-2048). \end{aligned} \tag{4.47}$$

$$\begin{aligned} 36X^4(X-1)(X+1)^3\, c_0 = {} & (3X^{11} + 18X^{10} - 42X^9) \\ & + (-12X^8 + 699X^7 + 1336X^6) \\ & + (-3694X^5 - 10404X^4 - 3680X^3) \\ & + (7616X^2 + 6912X + 1536). \end{aligned} \tag{4.48}$$

$$\begin{aligned} 12X^5\,(X+1)\,(X+4)\;c_{-1,-1} = {} & \left(X^8 - 51X^7 - 48X^6\ell_\lambda\right) \\ & + \left(-596X^6 - 48X^5\ell_\lambda - 900X^5\right) \\ & + \left(1056X^4\ell_\lambda + 3772X^4 + 2208X^3\ell_\lambda\right) \\ & + \left(8448X^3 + 1152X^2\ell_\lambda + 5824X^2\right) \\ & + \left(2816X + 1024\right). \end{aligned} \tag{4.49}$$

$$\begin{aligned} 6X^5\,(X+1)^2\,(X+4)\;c_{-1,0} = {} & \left(X^9 - 2X^8 - 77X^7\right) \\ & + \left(-410X^6 - 650X^5 + 1558X^4\right) \\ & + \left(7372X^3 + 9744X^2 + 5376X\right) \\ & + \left(1024\right). \end{aligned} \tag{4.50}$$

$$\begin{aligned} X^3\,(X+1)^2\,(X+4)\;c_{-2,-1} = {} & \left(-2X^5 - 52X^4 - 292X^3\right) \\ & + \left(-632X^2 - 576X - 192\right). \end{aligned} \tag{4.51}$$

$$\begin{aligned} 6X^5\,(X+1)^2\,(X+4)\;c_{0,-1} = {} & \left(X^9 + 10X^8 + 7X^7\right) \\ & + \left(4X^6 + 1162X^5 + 5536X^4\right) \\ & + \left(11008X^3 + 10944X^2 + 5376X\right) \\ & + \left(1024\right). \end{aligned} \tag{4.52}$$

$$X^3\,c_{1,0} = \left(-X^2 + X - 2\right). \tag{4.53}$$

$$\begin{aligned} X^4\,(X+4)\;c_{r,-1} = {} & \left(3X^5Y + 12X^4Y\ell_\lambda + 36X^4Y\right) \\ & + \left(24X^3Y\ell_\lambda + 104X^3Y - 120X^2Y\right) \\ & + \left(-288X^2Y\ell_\lambda - 960XY\ell_\lambda - 768XY\right) \\ & + \left(-576Y\ell_\lambda - 256Y\right). \end{aligned} \tag{4.54}$$

$$X^3\,c_{-1,-1,-1} = \left(2X^2 + 16X + 22\right). \tag{4.55}$$

$$X^3\,c_{-1,-1,0} = \left(12X^2 - 20X - 48\right). \tag{4.56}$$

$$X^3\,c_{-1,-2,-1} = \left(4X^2 - 16X - 24\right). \tag{4.57}$$

$$X^3\,c_{-1,0,-1} = \left(4X^2 - 4X - 12\right). \tag{4.58}$$

$$\begin{aligned} X^4\,(X+4)\;c_{-4,r,-1} = {} & \left(-18X^4Y - 48X^3Y + 360X^2Y\right) \\ & + \left(1296XY + 672Y\right). \end{aligned} \tag{4.59}$$

$$X^3\,c_{0,-1,-1} = \left(-X^2 - 8X + 8\right). \tag{4.60}$$

$$X^3\,c_{0,-1,0} = \left(2X^2 + 4X - 12\right). \tag{4.61}$$

$$X^3\,c_{0,1,0} = \left(2X^2 + 4X - 12\right). \tag{4.62}$$

$$\begin{aligned} X^4\,(X+4)\;c_{0,r,-1} = {} & \left(3X^4Y - 6X^3Y - 144X^2Y\right) \\ & + \left(-384XY - 480Y\right). \end{aligned} \tag{4.63}$$

$$\begin{aligned} X^4\,(X+4)\;c_{r,-1,-1} = {} & \left(21X^4Y + 54X^3Y - 432X^2Y\right) \\ & + \left(-1536XY - 768Y\right). \end{aligned} \tag{4.64}$$

$$\begin{aligned} X^4\,(X+4)\;c_{r,-1,0} = {} & \left(-6X^4Y - 36X^3Y + 192XY\right) \\ & + \left(-192Y\right). \end{aligned} \tag{4.65}$$

$$X^4\,(X+4)\;c_{r,-2,-1} = (-8X^4Y - 16X^3Y + 192X^2Y) + (640XY + 384Y). \tag{4.66}$$

$$X^4\,(X+4)\;c_{r,0,-1} = (6X^4Y + 4X^3Y - 192X^2Y) + (-576XY - 576Y). \tag{4.67}$$

$$X^3\,c_{r,r,-1} = (15X^2 - 12X - 96). \tag{4.68}$$

4.4 A shorter infrared coefficient

The two roots in (3.218) are reciprocal. This relation reduces its endpoint sum to one dilogarithmic function. Set

$$X = s - 1 > 0, \qquad Y = \sqrt{\frac{X}{X+4}}, \qquad q = \frac{X + 2 - \sqrt{X(X+4)}}{2}, \qquad L = \ln(1+X). \tag{4.69}$$

Then $0 < q < 1$, $X = (1-q)^2/q$, and $z_- = q$, $z_+ = q^{-1}$. Define

$$\mathcal{G}(X) = \frac{1}{2}\ln^2 q + \frac{1}{3}\,\mathrm{Li}_2(-q^3) - \mathrm{Li}_2(-q) - \frac{\pi^2}{18}. \tag{4.70}$$

Both dilogarithm arguments lie between -1 and zero. All functions in this expression are consequently real on the physical interval, with their branches fixed without a continuation across a cut. The integrated infrared coefficient is

$$A(1+X) = -\frac{2(X^2-4X-6)}{X^3}L^2 - \frac{2P(X)}{X^3(1+X)^2(X+4)}L + \frac{12YQ(X)}{X^4(X+4)}\mathcal{G}(X) - \frac{R(X)}{X^2(1+X)^2}, \tag{4.71}$$

where

$$P(X) = X^5 - 22X^4 - 184X^3 - 444X^2 - 424X - 144, \tag{4.72}$$

$$Q(X) = X^4 + 2X^3 - 24X^2 - 80X - 48, \tag{4.73}$$

$$R(X) = 5X^3 + 60X^2 + 100X + 48. \tag{4.74}$$

The normalization is $A(s) = \mathcal{C}_{\mathrm{IR}}(s)/s$, so $A(s)\ln(\lambda_{\mathrm{phys}}/m)$ is the logarithmic term of the complete virtual coefficient in units of $\alpha_{\mathrm{T}}^3/m^2$. Equation (4.71) is exact for every $X > 0$. Its apparent threshold poles cancel in the combined expression.

To derive it directly from the angular integral, denote the endpoint difference of a primitive by $\Delta J = J(s) - J(1)$. The two logarithmic pole primitives in (3.218) satisfy

$$\Delta H_1(q) = -\ln q, \qquad \Delta H_1(q^{-1}) = \ln q, \tag{4.75}$$

$$\Delta H_2(q) = 1, \qquad \Delta H_2(q^{-1}) = 1. \tag{4.76}$$

Their residues obey $c_{-,q} = c_{-,q^{-1}}$ and $d_{-,q} + d_{-,q^{-1}} = 8$. Thus the logarithms of the negative-sign sector cancel, while its double poles give the constant eight. For the positive-sign sector, $c_{+,q} = -c_{+,q^{-1}}$. Use the real dilogarithm inversion identity

$$\mathrm{Re}\,\mathrm{Li}_2(u) + \mathrm{Re}\,\mathrm{Li}_2(u^{-1}) = \frac{\pi^2}{3} - \frac{1}{2}\ln^2 u, \qquad u > 0, \tag{4.77}$$

to combine its simple-pole endpoint functions into

$$\begin{aligned}\Delta J_1(q) - \Delta J_1(q^{-1}) = 2\,\mathrm{Li}_2(q) - \mathrm{Li}_2(q/s) - \mathrm{Li}_2(qs) \\ - 2L\ln q - \frac{1}{2}L^2 \equiv \mathcal{F}(q).\end{aligned} \tag{4.78}$$

Here $qs = 1 - q + q^2 < 1$ and $q/s = q^2/(1 - q + q^2) < 1$. Differentiation and the identity $(1 + q)(1 - q + q^2) = 1 + q^3$ give

$$\frac{\mathrm{d}\mathcal{F}}{\mathrm{d}q} = -\frac{3L}{q}, \qquad \frac{\mathrm{d}\mathcal{G}}{\mathrm{d}q} = -\frac{L}{q}, \qquad \mathcal{F}(1) = \mathcal{G}(0) = 0. \tag{4.79}$$

In the last boundary value the argument of $\mathcal{G}$ is X, which tends to zero as $q \to 1$. Therefore $\mathcal{F}(q) = 3\mathcal{G}(X(q))$, with no undetermined constant. The double-pole primitives supply only elementary terms,

$$\Delta J_2(q) = -\frac{L}{s - q} - \frac{L + \ln q}{q}, \tag{4.80}$$

$$\Delta J_2(q^{-1}) = -\frac{L}{s - q^{-1}} + q(\ln q - L). \tag{4.81}$$

Substituting these differences and the elementary power primitives into (3.218) yields (4.71). This derivation uses only the original Born-weighted soft angular integral.

An independent internal check differentiates the complete virtual 23-word result with respect to $\ln(\lambda_{\mathrm{phys}}/m)$. Its four nonzero coefficients agree algebraically with (4.71), using $G(-1; X) = L$, $G(-1, -1; X) = L^2/2$ and $G(r, -1; X) = \mathcal{G}(X)$. This check does not set any integration constant in the soft calculation.

4.5 Reduction of the virtual root functions

4.5.1 Collecting the coefficients

A compact expression need not eliminate the G functions. It is useful first to combine their coefficients, before expanding any function into classical polylogarithms. For the six root words in the preceding table, the shuffle product gives

$$G_{0,r,-1} = \ln X\, G_{r,-1} - G_{r,0,-1} - G_{r,-1,0}. \tag{4.82}$$

Thus $G_{0,r,-1}$ is redundant. After this substitution, the coefficients of $G_{r,0,-1}$ and $G_{r,-1,0}$ are proportional with ratio $1 : -3$. Define

$$p_F(X) = 7X^4 + 18X^3 - 144X^2 - 512X - 256, \tag{4.83}$$

$$p_K(X) = 3X^4 + 10X^3 - 48X^2 - 192X - 96, \tag{4.84}$$

$$p_W(X) = 3X^4 + 8X^3 - 60X^2 - 216X - 112, \tag{4.85}$$

$$p_Z(X) = X^4 + 2X^3 - 24X^2 - 80X - 48, \tag{4.86}$$

$$p_0(X) = X^4 - 2X^3 - 48X^2 - 128X - 160. \tag{4.87}$$

The contribution C_{six} of these six original words to $\widehat{C}_{V+\mathrm{UV}}$ becomes

$$\begin{aligned}C_{\mathrm{six}}(X) = \frac{Y}{X^4(X+4)}\{3p_F G_{r,-1,-1} + p_K(G_{r,0,-1} - 3G_{r,-1,0}) \\ - 6p_W G_{-4,r,-1} - 8p_Z G_{r,-2,-1} + 3p_0 \ln X\, G_{r,-1}\}.\end{aligned} \tag{4.88}$$

All coefficients are obtained by exact rational collection of the original table. This removes one weight-three function and gives the other five a common denominator. The G notation keeps this relation shorter than its fully expanded classical form. The identities below provide alternative representations for evaluation and further algebraic reductions.

4.5.2 Auxiliary classical identities

The rational-letter terms of the preceding coefficient list do not require separate GPL evaluations. Put $L = \ln(1+X)$, $M = \ln X$ and $v = 1/(1+X)$, with $X > 0$. The elementary reductions are

$$G_{-1} = L, \qquad G_0 = M, \qquad G_{-1,-1} = \tfrac{1}{2}L^2, \qquad G_{-1,-1,-1} = \tfrac{1}{6}L^3, \tag{4.89}$$

$$G_{0,-1} = -\operatorname{Li}_2(-X), \qquad G_{-1,0} = LM + \operatorname{Li}_2(-X), \tag{4.90}$$

$$G_{1,0} = \frac{\pi^2}{6} - \operatorname{Li}_2(1-X), \tag{4.91}$$

$$G_{0,-1,0} = -M\operatorname{Li}_2(-X) + 2\operatorname{Li}_3(-X), \tag{4.92}$$

$$G_{0,1,0} = -M\mathcal{L}_2(X) + 2\mathcal{L}_3(X). \tag{4.93}$$

Here $\mathcal{L}_n(X) = \operatorname{Re}\operatorname{Li}_n(X + \mathrm{i}0)$. The real boundary value is fixed by integrating $\mathrm{d}\mathcal{L}_2/\mathrm{d}X = -\ln|1-X|/X$ and $\mathrm{d}\mathcal{L}_3/\mathrm{d}X = \mathcal{L}_2/X$ from the interval $0 < X < 1$. It is not selected by fitting numerical values. The combination for $G_{1,0}$ is regular at $X = 1$.

One additional classical combination suffices for the three words with two letters -1 and one zero:

$$T(X) = \tfrac{1}{2}L^2M - \tfrac{1}{3}L^3 - L\operatorname{Li}_2(v) - \operatorname{Li}_3(v) + \zeta(3), \tag{4.94}$$

$$G_{0,-1,-1} = T(X), \tag{4.95}$$

$$G_{-1,0,-1} = -L\operatorname{Li}_2(-X) - 2T(X), \tag{4.96}$$

$$G_{-1,-1,0} = \tfrac{1}{2}L^2M + L\operatorname{Li}_2(-X) + T(X). \tag{4.97}$$

For example, substituting $v = 1/(1+X)$ into $\frac{1}{2}\int_0^X \ln^2(1+t)\,\mathrm{d}t/t$ gives the displayed T. Its derivative is $L^2/(2X)$ and its lower limit is zero. The last two lines then follow from the shuffle product. All dilogarithms and trilogarithms with argument v are on $0 < v < 1$. These substitutions reduce the number of nonclassical words in the partial sum without changing any rational coefficient or the photon-mass convention.

The repeated square-root kernel can also be integrated classically. Use the same q as in (4.70). Since $\omega_r = -\mathrm{d}q/q$, integrating that independently derived expression once more gives

$$G_{r,r,-1} = \operatorname{Li}_3(-q) - \frac{1}{9}\operatorname{Li}_3(-q^3) - \frac{1}{6}\ln^3 q + \frac{\pi^2}{18}\ln q + \frac{2}{3}\zeta(3). \tag{4.98}$$

Differentiating with respect to q gives $-\mathcal{G}(X(q))/q$. At $q = 1$ the expression vanishes, using $\operatorname{Li}_3(-1) = -3\zeta(3)/4$, which fixes the constant. Its two trilogarithms have negative real arguments. The remaining rational-letter word also has a short classical form. With $u = 1 + X$, its inner weight-two function is $\operatorname{Li}_2(-u) + \ln u \ln(1+u) + \pi^2/12$. Integrating it with $\mathrm{d}u/u$ gives

$$G_{-1,-2,-1} = 2\operatorname{Li}_3(-1-X) - L\operatorname{Li}_2(-1-X) + \frac{\pi^2}{12}L + \frac{3}{2}\zeta(3). \tag{4.99}$$

The last constant follows from the zero value at $X = 0$. Together with (4.70), these reductions leave only six nonclassical words in this partial sum:

$$\{G_{-4,r,-1}, G_{0,r,-1}, G_{r,-1,-1}, G_{r,-1,0}, G_{r,-2,-1}, G_{r,0,-1}\}\,. \tag{4.100}$$

A classical primitive for the squared logarithm. The word $G_{r,-1,-1}$ can be removed from this list without introducing another iterated integral. Define

$$a = e^{\mathrm{i}\pi/3}, \quad b = \overline{a}, \quad X = (1-q)^2/q, \quad 0 < q < 1, \tag{4.101}$$

$$A = 1 - aq, \quad B = 1 - bq, \quad R = A/B. \tag{4.102}$$

The quadratic factorization gives $AB = 1-q+q^2$ and hence $\ln(1+X) = \log A + \log B - \ln q$. Here and below log is the principal complex logarithm. Introduce the classical function

$$\mathcal{J}(z,\rho) = \log^2 z \log(1 - z/\rho) + 2\log z \,\mathrm{Li}_2(z/\rho) - 2\,\mathrm{Li}_3(z/\rho). \tag{4.103}$$

Differentiation of the polylogarithms cancels the terms proportional to Li_2 and $\log(1-z/\rho)$, leaving

$$\frac{\partial \mathcal{J}(z,\rho)}{\partial z} = \frac{\log^2 z}{z-\rho}. \tag{4.104}$$

For the ratio R, the inverse transformation is $q = (1-R)/(a-bR)$. Consequently

$$\frac{\mathrm{d}q}{q} = \frac{\mathrm{d}R}{R-1} - \frac{\mathrm{d}R}{R-a/b}. \tag{4.105}$$

This second pole is essential. Applying this identity to $2\log A \log B = \log^2 A + \log^2 B - \log^2 R$ reduces the product of logarithms to the same primitive. The remaining mixed term follows from

$$\frac{\mathrm{d}}{\mathrm{d}q}\left[-\ln q\,\mathrm{Li}_2(aq) + \mathrm{Li}_3(aq)\right] = \frac{\ln q \log(1-aq)}{q}. \tag{4.106}$$

Combining these operations gives

$$\mathcal{F}(q) = 2\mathcal{J}(A,1) + 2\mathcal{J}(B,1) - \mathcal{J}(R,1) + \mathcal{J}(R,a/b) + \mathcal{D}(q), \tag{4.107}$$

$$\mathcal{D}(q) = 2\ln q\left[\mathrm{Li}_2(aq) + \mathrm{Li}_2(bq)\right] - 2\left[\mathrm{Li}_3(aq) + \mathrm{Li}_3(bq)\right] + \frac{1}{3}\ln^3 q, \tag{4.108}$$

$$G_{r,-1,-1}(X) = -\frac{1}{2}\left[\mathcal{F}(q) - \mathcal{F}(1)\right]. \tag{4.109}$$

Indeed, $\mathcal{F}'(q) = \ln^2[(1-q+q^2)/q]/q$. The minus sign follows from $\omega_r = -\mathrm{d}q/q$, and the subtraction at $q=1$ enforces the original lower limit $X = 0$.

The branches in this formula follow directly from the physical interval. The arguments of A and B lie respectively in $[-\pi/3, 0]$ and $[0, \pi/3]$, so $\log R = \log A - \log B$ throughout the interval. The ratio R follows the lower unit-circle arc from 1 to $e^{-2\pi\mathrm{i}/3}$. Neither this arc nor the rotated arc $R/(a/b)$ crosses the polylogarithm cut $[1,\infty)$ in the open interval. The other polylogarithm arguments are A, B, aq and bq, which also avoid that cut. The logarithms in $\mathcal{J}$ are continuous on these paths. Thus principal values suffice, with no numerically chosen branch corrections. Although individual terms are complex, the derivative of $\mathcal{F}$ is real, so the endpoint difference in (4.109) is real. This reduction leaves five nonclassical words in the partial sum.

Classical forms for the five remaining words. The same primitive also integrates the mixed logarithms generated by integration by parts. This gives a common classical representation for all six words, rather than six unrelated integration formulas. For $d \in \{a, b\}$ and $h \in \{1, -1\}$, define

$$B_d(q) = 1 - dq, \qquad R_d(q) = q/B_d(q), \qquad T_{hd}(q) = (1-hq)/B_d(q), \tag{4.110}$$

$$N_d(q) = -\ln q\,\mathrm{Li}_2(dq) + \mathrm{Li}_3(dq), \tag{4.111}$$

$$P_{hd}(q) = \tfrac{1}{2}\left[\mathcal{J}(1-hq,1) + \mathcal{J}(B_d,1) - \mathcal{J}(T_{hd},1) + \mathcal{J}(T_{hd},h/d)\right], \tag{4.112}$$

$$H_{dc}(q) = \tfrac{1}{2}\left[\mathcal{J}(q,c) + \mathcal{J}(B_d,1-dc) - \mathcal{J}(R_d,c/(1-dc)) + \mathcal{J}(R_d,-1/d)\right]. \tag{4.113}$$

The definition of N_d is also used for $d = 1, -1$. In H_{dc} only $c = 1, \mathrm{i}, -\mathrm{i}$ is needed. To derive these expressions, apply the square-of-a-ratio identity used above to each pair of logarithms. The transformed measures include both the finite pole and the image of infinity. Direct differentiation then gives

$$N'_d(q) = \frac{\ln q \log(1 - dq)}{q}, \tag{4.114}$$

$$P'_{hd}(q) = \frac{\log(1 - hq) \log(1 - dq)}{q}, \tag{4.115}$$

$$H'_{dc}(q) = \frac{\ln q \log(1 - dq)}{q - c}. \tag{4.116}$$

These identities specify the primitives independently of any cross-section formula from the literature.

Some individual terms have a singular expression at $q = 1$, although the required combinations are finite. Their boundary values must therefore be evaluated before numerical subtraction. The necessary limits are

$$\mathcal{J}(0, \rho) = 0 \quad (\rho \neq 0), \qquad \mathcal{J}(1, 1) = -2\zeta(3), \tag{4.117}$$

$$P_{1d}(1) = \tfrac{1}{2}\mathcal{J}(1 - d, 1), \tag{4.118}$$

$$H_{d1}(1) = \tfrac{1}{2}\big[-2\zeta(3) + \log^2(1 - d) \log(-d) + 4\zeta(2) \log(1 - d) + \mathcal{J}(1/(1 - d), -1/d)\big]. \tag{4.119}$$

For example, the two divergent logarithms in H_{d1} occur in $\mathcal{J}(B_d, 1 - d) - \mathcal{J}(R_d, 1/(1 - d))$. As $q \to 1^-$, their difference tends to $\log(-d)$. Their coefficients both tend to $\log^2(1 - d)$ because $\log(1/(1 - d)) = -\log(1 - d)$. The dilogarithms tend to $\zeta(2)$, giving the displayed $4\zeta(2) \log(1 - d)$ term. This determines the boundary constant analytically.

Put $S(q) = N_a(q) + N_b(q)$ and define

$$U(q) = -2 \sum_{d=a,b} P_{1d}(q) + S(q) + 2N_1(q) - \tfrac{1}{3} \ln^3 q, \tag{4.120}$$

$$V(q) = \ln q \, \mathrm{Li}_2(-X) + 2 \sum_{d=a,b} H_{d1}(q) - S(q) - 2\mathcal{J}(q, 1) + \tfrac{1}{3} \ln^3 q, \tag{4.121}$$

$$W(q) = 2 \sum_{d=a,b} P_{-1,d}(q) - S(q) - 2N_{-1}(q) + \tfrac{1}{3} \ln^3 q + \ln 4 \, E(q), \tag{4.122}$$

$$E(q) = \mathrm{Li}_2(aq) + \mathrm{Li}_2(bq) + \tfrac{1}{2} \ln^2 q, \tag{4.123}$$

$$Z(q) = -\ln q \, D(X) + \sum_{c=\mathrm{i},-\mathrm{i}} \left[\sum_{d=a,b} H_{dc}(q) - \mathcal{J}(q, c) \right] - S(q) + \tfrac{1}{3} \ln^3 q, \tag{4.124}$$

$$D(X) = \mathrm{Li}_2(-1 - X) + \ln(1 + X) \ln(2 + X) + \frac{\pi^2}{12}. \tag{4.125}$$

All occurrences of X here mean $(1 - q)^2/q$. Write $\Delta f = f(q) - f(1)$, with the finite boundary values just derived. The five reductions are

$$G_{r,0,-1}(X) = \Delta V, \tag{4.126}$$

$$G_{r,-1,0}(X) = \Delta U - \Delta V, \tag{4.127}$$

$$G_{0,r,-1}(X) = \ln X \, \mathcal{G}(X) - \Delta U, \tag{4.128}$$

$$G_{-4,r,-1}(X) = \ln\!\left(\frac{X + 4}{4}\right) \mathcal{G}(X) + \Delta W, \tag{4.129}$$

$$G_{r,-2,-1}(X) = \Delta Z. \tag{4.130}$$

Here $\mathcal{G}(X) = G_{r,-1}(X)$ is the classical dilogarithm combination in (4.70).

For completeness, the integration steps fixing these formulas can be checked without manipulating the full rational coefficients of the cross section. Since $G_{0,-1} = -\operatorname{Li}_2(-X)$, integration by parts gives

$$G_{r,0,-1}(X) = \ln q \operatorname{Li}_2(-X) + \int_1^q \ln t \ln(1+X(t)) \left(\frac{2}{t-1} - \frac{1}{t}\right) \mathrm{d}t. \tag{4.131}$$

Inserting $\ln(1+X(t)) = \log(1-at) + \log(1-bt) - \ln t$ and using (4.116) gives ΔV. The shuffle identity gives $G_{-1,0} + G_{0,-1} = \ln X \ln(1+X)$, whose integral with ω_r is ΔU. Multiplication of $G_{r,-1}$ by $\ln X$ then gives the third line of (4.130). For the fourth line, integrate the outer -4 kernel by parts and use

$$\ln\left(\frac{X(t)+4}{4}\right) = 2\ln(1+t) - \ln t - \ln 4. \tag{4.132}$$

Finally, $D(X) = G_{-2,-1}(X)$ satisfies $\mathrm{d}D/\mathrm{d}X = \ln(1+X)/(X+2)$. The factorization $X(t)+2 = (t-\mathrm{i})(t+\mathrm{i})/t$ gives

$$\frac{\mathrm{d}D(X(t))}{\mathrm{d}t} = \ln(1+X(t)) \left(\frac{1}{t-\mathrm{i}} + \frac{1}{t+\mathrm{i}} - \frac{1}{t}\right). \tag{4.133}$$

Integrating $-D(X(t))\,\mathrm{d}t/t$ by parts now gives ΔZ. Thus all constants follow from the original zero lower limit, and no GPL remains among these six functions.

The complex arguments in the mixed primitives also have fixed physical branches. For $d = a$, B_d lies below the real axis, while q/B_d and $(1 \pm q)/B_d$ lie above it. Their arguments stay between 0 and $\pi/3$, so each logarithm of a ratio is the difference of the corresponding logarithms. The $d = b$ expressions are their conjugate continuations. For $c = \pm\mathrm{i}$, the arguments $B_a/(1-ac)$ and $R_a(1-ac)/c$ can cross the real axis, but only below 1. This follows by multiplying numerator and denominator by the conjugate denominator. For instance the possible crossing is at $q = 2-\sqrt{3}$, where $B_a/(1-a\mathrm{i}) = 2\sqrt{3}-3 < 1$ and $R_a(1+a\mathrm{i})/(-\mathrm{i}) = 1 - 2\sqrt{3}/3 < 0$. Thus these paths do not cross the polylogarithm cut. The sole real argument on that cut is $1+q$ in $\mathcal{J}(1+q,1)$. There the real boundary value is used:

$$\mathcal{J}(z,1) = \ln^2 z \ln(z-1) + 2\ln z \operatorname{Re}\operatorname{Li}_2(z) - 2\operatorname{Re}\operatorname{Li}_3(z), \qquad z > 1. \tag{4.134}$$

Its derivative is $\ln^2 z/(z-1)$ and its limit at $z = 1$ is $-2\zeta(3)$. This fixes the continuation without a fitted imaginary part. Equations (4.109) and (4.130) complete the classical reduction of this virtual and UV partial sum. Further simplification of their combined coefficients and arguments remains a separate step.

Combining the root terms in the cross section. The preceding conversion to classical functions does not by itself shorten the cross section. Their coefficients must also be combined. Let $C_{\mathrm{six}}(X)$ denote just the contribution of the six words listed above to $\widehat{C}_{V+\mathrm{UV}}$. Their original rational coefficients have the common factor $Y/[X^4(X+4)]$. After inserting their classical forms, the coefficients of ΔU and ΔV are in the exact ratio $-3:4$. Define $K(q) = 4V(q) - 3U(q)$ and use the polynomials in (4.87). The six contributions then combine into

$$\begin{aligned} C_{\mathrm{six}}(X) = \frac{Y}{X^4(X+4)} \Big\{ &-\frac{3}{2} p_F \Delta\mathcal{F} + p_K \Delta K - 6 p_W \Delta W - 8 p_Z \Delta Z \\ &+ \left[3p_0 \ln X - 6 p_W \ln\left(\frac{X+4}{4}\right)\right] \mathcal{G}(X) \Big\}. \end{aligned} \tag{4.135}$$

Thus four boundary differences suffice for this contribution. In particular, U and V need not be evaluated separately. There is also a cancellation inside K: the $8\mathcal{J}(q,1)$ generated by $8\sum_d H_{d1}$ cancels its explicit $-8\mathcal{J}(q,1)$ term. This follows by expanding (4.113), before evaluating any special function. Equation (4.135) is an exact coefficient collection of the six original terms in this diagram subset. The other virtual, soft, and real terms must still be added using the common-regulator assembly (3.672).

4.6 Combining the finite regulator conversions

The three finite shifts in (4.4) can be combined before evaluating the real contribution. Define

$$M(X) = \Delta_{\rm rat} + \Delta_{\log} + \Delta_D, \qquad L = \ln(1+X), \quad J = \ln(1+X/2), \quad X > 0. \tag{4.136}$$

The shifts include the finite displacement of the double-pole boundary in the Bose-restoring term. Combining their independently integrated endpoint moments gives

$$\begin{aligned} M(X) = &-\frac{P_0}{3X} + \frac{2P_1}{X(1+X)}\,\mathrm{Li}_2(-X) - \frac{8(1+X)(X^2-4X-6)}{X^2}L^3 \\ &+ \frac{16(1+X)(X^2-4X-6)}{X^2}L\,\mathrm{Li}_2(-X) - \frac{256(1+X)(X+2)^2}{X^3}J \\ &+ \frac{128(1+X)P_2}{3X^3}\left[\mathrm{Li}_2\!\left(-\frac{X^2}{4(1+X)}\right) - 2JL\right] - \frac{P_3L^2+P_4L}{3X^3(1+X)}. \end{aligned} \tag{4.137}$$

Here the polynomials are

$$P_0 = X^5 - X^4 - 8X^3 + 196X^2 + 1104X + 864, \tag{4.138}$$

$$P_1 = X^5 - 6X^4 - 43X^3 - 32X^2 + 32X + 32, \tag{4.139}$$

$$P_2 = X^3 - 5X^2 - 28X - 24, \tag{4.140}$$

$$P_3 = 3X^7 - 18X^6 - 233X^5 - 96X^4 + 3008X^3 + 8240X^2 + 8000X + 2688, \tag{4.141}$$

$$P_4 = X^9 + 6X^8 - 18X^7 - 24X^6 + 363X^5 + 1652X^4 + 412X^3 - 3984X^2 - 4704X - 1536. \tag{4.142}$$

In particular, the explicit $\ln X$ cancels in this sum. The reduction uses positive-real logarithm identities and

$$\mathrm{Li}_2\!\left(\frac{X}{1+X}\right) = -\,\mathrm{Li}_2(-X) - \frac{1}{2}\ln^2(1+X). \tag{4.143}$$

The last identity follows by differentiating both sides and fixing the zero value at $X = 0$. Both sides are real for $X > 0$. The remaining dilogarithm arguments in (4.137) are negative, so their real branches are fixed throughout the physical interval. Coefficient collection then requires only rational identities in X. Thus the real remainder can be written more compactly as

$$\mathcal{R}_{\rm rem}(X) = F^z_{\rm rat} + F^z_W + F^z_{\log} + F^z_D + 2F^\eta_{UW} + M(X). \tag{4.144}$$

The script `compton_matching_simplify.py` constructs this sum from the separate endpoint moments and verifies the rational coefficients exactly. The overall factor $1/[8X(1+X)]$ is still applied only in (4.5).

4.7 The finite soft contribution

The kernel was simplified before integration in (3.247). Its real-letter endpoint expression is (3.270), with the integration constants fixed by vanishing boundary terms. We now reduce that completed expression to classical functions and then to the common energy basis.

4.7.1 The finite soft coefficient in classical polylogarithms

All weight-three terms in the preceding soft coefficient arise from products of two ordinary logarithms. This permits a classical polylogarithmic endpoint expression for the complete coefficient, without a GPL primitive operator.

For $z > 0$ and real $\rho \neq 0$, define

$$J(z,\rho) = \ln^2 z \ln|1 - z/\rho| + 2\ln z \,\mathrm{Re}\,\mathrm{Li}_2(z/\rho) - 2\,\mathrm{Re}\,\mathrm{Li}_3(z/\rho), \tag{4.145}$$

$$K(z,\rho) = \ln z \ln|1 - z/\rho| + \mathrm{Re}\,\mathrm{Li}_2(z/\rho). \tag{4.146}$$

The real parts denote the real boundary values on the standard polylogarithm cut. They enter because the original logarithms are real. Their derivatives are listed below. At the degenerate values $\rho = 0, \infty$, the primitive is defined up to an additive constant, which cancels in every normalized boundary difference. We use

$$\partial_z J(z,\rho) = \frac{\ln^2 z}{z-\rho}, \qquad J(z,0) = \frac{1}{3}\ln^3 z, \qquad J(z,\infty) = 0, \tag{4.147}$$

$$\partial_z K(z,\rho) = \frac{\ln z}{z-\rho}, \qquad K(z,0) = \frac{1}{2}\ln^2 z. \tag{4.148}$$

The convention at infinity specifies a constant primitive with zero derivative. It will occur when a pole maps to infinity.

For a real letter a outside the open physical interval, put $A_a(z) = 1 - z/a$ and $\ell_a(z) = \ln A_a(z)$. All factors used below are positive for $0 < z < Z$. The normalized single-log primitive is

$$\mathcal{K}_{a;c}(z) = K(A_a(z), A_a(c)) - K(1, A_a(c)). \tag{4.149}$$

For $a \neq b$ define

$$\begin{aligned}\mathcal{H}_{a,b;c}(z) = \frac{1}{2}\Big[&J(A_a(z), A_a(c)) + J(A_b(z), A_b(c)) \\ &- J\left(\frac{A_a(z)}{A_b(z)}, \frac{A_a(c)}{A_b(c)}\right) + J\left(\frac{A_a(z)}{A_b(z)}, \frac{b}{a}\right)\Big] - (z \to 0).\end{aligned} \tag{4.150}$$

If $c = b$, the third term has its second argument at infinity and vanishes. If $a = b$, use $\mathcal{H}_{a,a;c}(z) = J(A_a(z), A_a(c)) - J(1, A_a(c))$. These definitions obey

$$\mathcal{K}'_{a;c}(z) = \frac{\ell_a(z)}{z-c}, \qquad \mathcal{H}'_{a,b;c}(z) = \frac{\ell_a(z)\ell_b(z)}{z-c}, \qquad \mathcal{K}_{a;c}(0) = \mathcal{H}_{a,b;c}(0) = 0. \tag{4.151}$$

To verify the product primitive, first use $2\ln A \ln B = \ln^2 A + \ln^2 B - \ln^2(A/B)$. Under $v = A_a(z)/A_b(z)$, the rational differential transforms as

$$\frac{\mathrm{d}z}{z-c} = \frac{\mathrm{d}v}{v - A_a(c)/A_b(c)} - \frac{\mathrm{d}v}{v - b/a}. \tag{4.152}$$

The second kernel is the image of infinity and is essential. This identity derives both ratio terms in (4.150). Positivity of A_a, A_b justifies the logarithmic identity without additional multiples of $i\pi$. Subtraction at zero fixes every integration constant.

Here is a finite formula for including the rational coefficients. For any rational $R(z)$ in the four soft kernels, write

$$R = P_R' + \sum_c \frac{r_c(R)}{z-c}, \qquad P_R = \sum_{n\geq 0} \frac{p_n z^{n+1}}{n+1} - \sum_c \sum_{j\geq 2} \frac{r_{c,j}}{(j-1)(z-c)^{j-1}}. \tag{4.153}$$

The sums use its polynomial and partial-fraction coefficients. They are finite algebraic derivatives of rational functions. Let $\mathcal{E}[R](z)$ be the elementary primitive of R normalized to zero at the origin,

$$\mathcal{E}[R](z) = P_R(z) - P_R(0) + \sum_c r_c(R) \ln\left|\frac{z-c}{-c}\right|. \tag{4.154}$$

No pole here lies at the integration origin. Define the following explicit combinations of classical functions:

$$\mathcal{I}_a[R] = P_R \ell_a + \sum_c r_c(R) \mathcal{K}_{a;c} - \mathcal{E}\left[\frac{P_R}{z-a}\right]. \tag{4.155}$$

For the product, put $R_a = P_R/(z-a)$ and $R_b = P_R/(z-b)$. Then

$$\begin{aligned} \mathcal{I}_{a,b}[R] = {} & P_R \ell_a \ell_b + \sum_c r_c(R) \mathcal{H}_{a,b;c} - P_{R_a} \ell_b - P_{R_b} \ell_a \\ & - \sum_c r_c(R_a) \mathcal{K}_{b;c} - \sum_c r_c(R_b) \mathcal{K}_{a;c} \\ & + \mathcal{E}\left[\frac{P_{R_a}}{z-b} + \frac{P_{R_b}}{z-a}\right]. \end{aligned} \tag{4.156}$$

These are nonrecursive formulas. Direct differentiation gives $\mathcal{I}_a'[R] = R\ell_a$ and $\mathcal{I}_{a,b}'[R] = R\ell_a\ell_b$. Their zero values at the origin follow from the normalized $\mathcal{E}, \mathcal{K}, \mathcal{H}$ and $\ell_a(0) = 0$.

Denote the weights of the three specified logarithm lists by b_a for L_β, u_a for L_1 and k_a for L_k. The complete classical primitive is

$$\begin{aligned} \Psi_{\mathrm{cl}}(z) = {} & \sum_a b_a \mathcal{I}_a[\widetilde{R}_0] + \sum_{a,b} b_a u_b \mathcal{I}_{a,b}[\widetilde{R}_1] \\ & + \sum_{a,b} b_a k_b \mathcal{I}_{a,b}[\widetilde{R}_k] + \sum_a k_a \mathcal{I}_a[\widetilde{R}_f]. \end{aligned} \tag{4.157}$$

Each index ranges over its four explicitly given real letters. Finally,

$$\boxed{\mathcal{C}_K(s) = \lim_{z \to Z^-} \Psi_{\mathrm{cl}}(z) - \frac{\operatorname{artanh}\beta}{\beta} \mathcal{C}_0(s).} \tag{4.158}$$

This expression uses only rational functions, real logarithms, Li_2 and Li_3. The endpoint limit is taken after combining the terms. It is the same finite limit already established for the soft coefficient.

There is a further exact simplification before evaluation. Each logarithm list has weights summing to zero. Hence the first two J terms in (4.150) cancel in both weighted double sums, including their base-point constants. Only the two ratio terms are needed there. This cancellation reduces the number of classical function evaluations without changing the endpoint or its branch choice.

4.7.2 Collection before the soft endpoint evaluation

The soft coefficient can be shortened further by collecting the rational integration-by-parts terms before expanding the logarithms into GPLs. This also removes separate divergent boundary terms. Keep $X = s - 1 > 0$, $\beta = X/(X+2)$, and use the original angular variable $0 < U < 1$. Define

$$k = \sqrt{\frac{1+XU^2}{s}}, \qquad d = 1 - \beta^2 U^2, \tag{4.159}$$

$$a = \operatorname{artanh}(\beta U), \quad L = \operatorname{artanh} U, \quad f = \operatorname{artanh}(U/k). \tag{4.160}$$

The square root is positive. Since $k^2 - U^2 = (1-U^2)/s > 0$, both inverse hyperbolic tangents involving U are real in the open interval. Their derivatives are

$$a' = \frac{\beta}{d}, \qquad L' = \frac{1}{1-U^2}, \qquad f' = \frac{1}{k(1-U^2)}, \qquad k' = \frac{XU}{sk}. \tag{4.161}$$

Primes here mean differentiation with respect to U.

Rational integration by parts. Let $\Phi(U)$ be the soft integrand after the pole removal in (3.247), including its common factor $4/(\beta E^2)$. Rational division and the derivatives above give

$$\Phi(U) = \frac{\mathrm{d}\mathcal{B}}{\mathrm{d}U} + \sum_i T_i(U) f_i(U), \qquad (f_i) = (aL, af, a, L, f, 1), \tag{4.162}$$

where the rational boundary can be chosen as

$$\mathcal{B} = Q_{aL} aL + Q_{af} af + Q_a a + Q_L L + Q_1. \tag{4.163}$$

Its coefficients are

$$Q_{aL} = \frac{2U(X+2)}{X^3 s}(U^2X^3 + X^3 + 24X^2 + 36X + 16), \tag{4.164}$$

$$Q_{af} = \frac{128 s^2 U k}{X^3(X+4)d}, \tag{4.165}$$

$$Q_a = -\frac{2U^2(X+2)}{s}, \qquad Q_L = \frac{U^2(X+2)^2}{Xs}, \tag{4.166}$$

$$Q_1 = -\frac{U(X+2)^2}{Xs}. \tag{4.167}$$

For example, the root-dependent part follows directly from

$$\frac{\mathrm{d}}{\mathrm{d}U}\frac{Uk}{d} = \frac{1 + (2X + \beta^2)U^2}{skd^2}. \tag{4.168}$$

Matching its constant and U^2 coefficients removes the double pole from the coefficient of af. Substituting f' then gives a rational coefficient of a, because the factor k in Q_{af} cancels. The other descendants follow by differentiating aL, a, and L. Each step is a rational identity after replacing k^2 by $(1+XU^2)/s$.

Vanishing logarithmic boundaries. Replace each of Q_{aL}, Q_{af}, Q_a, Q_L by $Q_i(U) - Q_i(1)$ and add $Q_i(1)f_i'$ to the remainder. This leaves the full integrand unchanged. All these differences are $O(1-U)$. At the upper endpoint a is finite, while L and f grow logarithmically. Every logarithmic boundary therefore vanishes. At $U = 0$, all three logarithms vanish. The only integrated boundary is

$$Q_1(1) - Q_1(0) = -\frac{(X+2)^2}{Xs}. \tag{4.169}$$

The cancellation of the apparent pole in the coefficient of a uses

$$\frac{1/k - 1}{1 - U^2} = \frac{X}{sk(1+k)}. \tag{4.170}$$

This identity follows by multiplying by $1 + k$ and using the displayed expression for k^2. It also fixes the finite endpoint value without subtracting two separately divergent numerical quantities.

Define the seven energy-dependent coefficients

$$A_L = \frac{4(X+2)(X^3+12X^2+18X+8)}{X^3 s}, \tag{4.171}$$

$$A_f = \frac{32s(X+2)^2}{X^3(X+4)}, \tag{4.172}$$

$$P = -\frac{16s(X^2-4X-6)}{X^2(X+2)}, \tag{4.173}$$

$$B = \frac{16s(X^4+2X^3-24X^2-80X-48)}{X^3(X+2)(X+4)}, \tag{4.174}$$

$$C_L = -\frac{4(X^3+14X^2+20X+8)}{X^2 s}, \tag{4.175}$$

$$M = -\frac{32s(X+2)}{X^2(X+4)}, \qquad C_0 = \frac{8(X+2)}{X^2}. \tag{4.176}$$

Here $A_L = Q_{aL}(1)$ and $A_f = Q_{af}(1)$. The resulting remainder is the short expression

$$\begin{aligned}\widehat{\Phi}(U) = \frac{1}{d}\Bigg[PaL + \frac{Baf}{k} + (A_L + A_f)\frac{1+\beta^2 U}{1+U}a \\ + (C_L U + A_L\beta)L + \left(\frac{MU}{k} + A_f\beta\right)f + C_0\beta\Bigg] + \frac{A_f X a}{sk(1+k)}.\end{aligned} \tag{4.177}$$

More explicitly, define the endpoint-normalized boundary

$$\begin{aligned}\widetilde{\mathcal{B}}(U) = [Q_{aL}(U) - A_L]aL + [Q_{af}(U) - A_f]af \\ + [Q_a(U) - Q_a(1)]a + [Q_L(U) - Q_L(1)]L + Q_1(U).\end{aligned} \tag{4.178}$$

The identity that implements both integrations by parts is

$$\boxed{\Phi(U) = \frac{\mathrm{d}\widetilde{\mathcal{B}}}{\mathrm{d}U} + \widehat{\Phi}(U).} \tag{4.179}$$

It can be checked without integrating: differentiate the displayed boundary with the four derivatives above and collect the coefficients of aL, af, a, L, f, and 1. Subtracting them and the corresponding coefficients of (4.177) from the pole-removed integrand gives six

rational zero identities using $k^2 = (1 + XU^2)/s$. This states the complete relation without requiring the intermediate T_i separately. None of its rational coefficients has a pole on the closed physical interval. The logarithmic growth at $U = 1$ is integrable. In particular, the constant-function term integrates immediately to

$$\int_0^1 \frac{C_0\beta}{d}\,\mathrm{d}U = C_0\,\mathrm{artanh}\,\beta = \frac{4(X+2)}{X^2}\ln s. \tag{4.180}$$

Thus the already integrated soft coefficient admits the exact reorganization

$$\mathcal{C}_K(s) = -\frac{(X+2)^2}{Xs} + \int_0^1 \widehat{\Phi}(U)\,\mathrm{d}U - \frac{\mathrm{artanh}\,\beta}{\beta}\mathcal{C}_0(s). \tag{4.181}$$

Equation (4.181) is an identity for collecting the existing analytic endpoint expression. Its remaining terms are evaluated with the logarithmic primitives of the preceding subsection. Under $U = (w - w^{-1})/(2\sqrt{X})$, all their coefficients are proper rational functions with simple poles, so only the single-pole primitives are needed. The square root has not been expanded into additional energy-dependent letters at the collection step.

Finite analytic endpoint sum. The reorganized coefficient can also be stated without a remaining integral. Put

$$W = \sqrt{s} + \sqrt{X}, \qquad Z = W - 1, \qquad r = \sqrt{1 + X/\beta^2} - \frac{\sqrt{X}}{\beta}. \tag{4.182}$$

Define three lists of pairs consisting of a letter and its weight,

$$\mathcal{A} = \{(r, \tfrac{1}{2}), (-r^{-1}, \tfrac{1}{2}), (r^{-1}, -\tfrac{1}{2}), (-r, -\tfrac{1}{2})\}, \tag{4.183}$$

$$\mathcal{L} = \{(W^{-1}, \tfrac{1}{2}), (-W, \tfrac{1}{2}), (W, -\tfrac{1}{2}), (-W^{-1}, -\tfrac{1}{2})\}, \tag{4.184}$$

$$\mathcal{F} = \{(W^{-1}, \tfrac{1}{2}), (-W^{-1}, \tfrac{1}{2}), (W, -\tfrac{1}{2}), (-W, -\tfrac{1}{2})\}. \tag{4.185}$$

For example, $a(w) = \sum_{(b,h)\in\mathcal{A}} h \ln[(w-b)/(1-b)]$. The other two lists give $L(w)$ and $f(w)$. Use $U = (w - w^{-1})/(2\sqrt{X})$ and $k = (w + w^{-1})/(2\sqrt{s})$ in the five rational functions

$$R_{aL} = U'\frac{P}{d}, \qquad R_{af} = U'\frac{B}{kd}, \tag{4.186}$$

$$R_a = U'\left[\frac{(A_L + A_f)(1+\beta^2 U)}{(1+U)d} + \frac{A_f X}{sk(1+k)}\right], \tag{4.187}$$

$$R_L = U'\frac{C_L U + A_L\beta}{d}, \qquad R_f = U'\frac{MU/k + A_f\beta}{d}. \tag{4.188}$$

Here $U' = \mathrm{d}U/\mathrm{d}w$. Each function is proper rational and has only simple poles. If $\mathcal{P}_R$ is its pole set, its coefficients are given by the finite algebraic limits

$$R(w) = \sum_{c\in\mathcal{P}_R} \frac{\rho_c}{w-c}, \qquad \rho_c = \lim_{w\to c}(w-c)R(w). \tag{4.189}$$

For a list $\mathcal{B}$ and two lists $\mathcal{B}, \mathcal{D}$, define the following finite sums of ordinary GPLs:

$$\mathcal{S}[R;\mathcal{B}] = \sum_{c\in\mathcal{P}_R}\sum_{(b,h)\in\mathcal{B}} \rho_c h\, G(c-1, b-1; Z), \tag{4.190}$$

$$\mathcal{S}[R;\mathcal{B},\mathcal{D}] = \sum_{c\in\mathcal{P}_R}\sum_{(b,h)\in\mathcal{B}}\sum_{(e,j)\in\mathcal{D}} \rho_c hj\,[G(c-1, b-1, e-1; Z) + G(c-1, e-1, b-1; Z)]. \tag{4.191}$$

These follow by substituting the partial fractions and using the shuffle identity for the product of the two logarithms. The soft coefficient is therefore

$$\begin{aligned}\mathcal{C}_K(s) = &-\frac{(X+2)^2}{Xs}+\frac{4(X+2)}{X^2}\ln s-\frac{\ln s}{2\beta}\mathcal{C}_0(s)\\ &+\mathcal{S}[R_{aL};\mathcal{A},\mathcal{L}]+\mathcal{S}[R_{af};\mathcal{A},\mathcal{F}]\\ &+\mathcal{S}[R_a;\mathcal{A}]+\mathcal{S}[R_L;\mathcal{L}]+\mathcal{S}[R_f;\mathcal{F}].\end{aligned} \tag{4.192}$$

All the residues in this expression are algebraic functions specified by the five short rational functions above. No endpoint extrapolation or numerical integration is part of its definition. The possible poles belong to $\{0,\pm r,\pm r^{-1},W^{-1},-W,-W^{-1}\}$. Since $0<r<W^{-1}<1<W<r^{-1}$, no outer letter lies on the integration interval. The inner letter W produces only the integrable endpoint logarithms. Thus every GPL in this endpoint sum is finite separately, and all logarithms follow the real physical branch established above.

4.7.3 The soft coefficient in the energy variable

The auxiliary endpoint letters of the preceding soft expression can be removed entirely. The result uses the same energy variable X and the same root kernel as the virtual contribution. Throughout this subsection $G(a_1,\dots,a_n)$ means $G(a_1,\dots,a_n;X)$, with

$$\omega_a=\frac{\mathrm{d}X}{X-a}\quad(a=0,-1,-2,-4),\qquad \omega_r=\frac{\mathrm{d}X}{\sqrt{X(X+4)}}. \tag{4.193}$$

The root is positive for $X>0$, and every function is continued from $X=0$ along the positive real axis. Set $s=1+X$ and $Y=\sqrt{X/(X+4)}$ as above. Define the combinations

$$\mathcal{D}=G(0,-1)+G(-2,-1), \tag{4.194}$$

$$\mathcal{V}=G(-1,-1,-1)-G(-1,-2,-1)-G(-1,0,-1), \tag{4.195}$$

$$\mathcal{R}=3G(0,r,-1)-3G(-1,r,-1)-3G(-4,r,-1)+4G(r,-2,-1)+4G(r,0,-1). \tag{4.196}$$

Then the complete finite soft coefficient is

$$\boxed{\mathcal{C}_K(1+X)=c_0+c_1G(-1)+c_{11}G(-1,-1)+c_D\mathcal{D}+c_V\mathcal{V}+c_rG(r,-1)+c_R\mathcal{R}.} \tag{4.197}$$

The seven coefficients are

$$c_0=-\frac{(X+2)^2}{Xs}, \tag{4.198}$$

$$c_1=-\frac{(X+2)(X^3+14X^2+28X+16)}{X^3s}, \tag{4.199}$$

$$c_{11}=-\frac{(X+2)(7X^4+32X^3+24X^2-16X-16)}{X^4s}, \tag{4.200}$$

$$c_D=\frac{2(X+2)(X^4+24X^3+98X^2+120X+48)}{X^3s(X+4)}, \tag{4.201}$$

$$c_V=\frac{4s(X^2-4X-6)}{X^3}, \tag{4.202}$$

$$c_r=-\frac{24Ys(X+2)^3}{X^4(X+4)}, \tag{4.203}$$

$$c_R=\frac{2Ys(X^4+2X^3-24X^2-80X-48)}{X^4(X+4)}. \tag{4.204}$$

This formula contains fourteen GPL words when the three combinations are expanded, including the constant term. All coefficients are rational in X and Y. In particular, the auxiliary root $\sqrt{s}$ and all separate $\ln 2$ terms have disappeared. In the total coefficient (4.5), this expression enters as $-\mathcal{C}_K(s)/s$.

Reduction of the lower-weight integrals. Use the functions a, L, f, k, d in the preceding subsection and define

$$I_1 = \int_0^1 \frac{aL}{d}\,\mathrm{d}U, \qquad I_2 = \int_0^1 \frac{af}{kd}\,\mathrm{d}U, \tag{4.205}$$

$$I_3 = \int_0^1 \frac{a(1+\beta^2 U)}{(1+U)d}\,\mathrm{d}U, \qquad I_4 = \int_0^1 \frac{a}{k(1+k)}\,\mathrm{d}U, \tag{4.206}$$

$$I_5 = \int_0^1 \frac{UL}{d}\,\mathrm{d}U, \qquad I_6 = \int_0^1 \frac{L}{d}\,\mathrm{d}U, \tag{4.207}$$

$$I_7 = \int_0^1 \frac{Uf}{kd}\,\mathrm{d}U, \qquad I_8 = \int_0^1 \frac{f}{d}\,\mathrm{d}U. \tag{4.208}$$

These integrals organize the derivation of the closed expression, rather than define functions that remain to be integrated. Put $a_* = a(1) = \frac{1}{2}\ln s$. Since $(f-L)' = X/[sk(1+k)]$, integration by parts gives

$$\frac{X}{s} I_4 + \beta I_8 = a_*^2 + \beta I_6. \tag{4.209}$$

At $U = 0$ the boundary vanishes. At $U = 1$ the finite difference $f - L$ tends to $\frac{1}{2}\ln s = a_*$, because $1 - U/k = (1-U)/s + O((1-U)^2)$. This determines the finite boundary constant in (4.209).

The rational-letter integrals are

$$\beta I_1 = -\tfrac{1}{4}\mathcal{V}, \tag{4.210}$$

$$I_3 = \tfrac{1}{2}G(0,-1) - \tfrac{1}{2}\ln 2\, G(-1) - \tfrac{1}{2}G(-1,-2) - \tfrac{1}{4}G(-1,-1) + \tfrac{1}{2}G(-2,-1), \tag{4.211}$$

$$\beta^2 I_5 = \tfrac{1}{4}G(-1,-1), \tag{4.212}$$

$$\beta I_6 = \tfrac{1}{2}\ln 2\, G(-1) - \tfrac{1}{4}G(-1,-1) + \tfrac{1}{2}G(-1,-2). \tag{4.213}$$

To obtain these expressions, factor d into its two linear factors and write a and L as differences of logarithms. The resulting GPLs have endpoint one and letters $\pm 1/\beta, \pm 1$. Their total differentials with respect to X contain only the logarithmic differentials of X, $X+1$, and $X+2$. Integration of those differentials gives the displayed energy-variable words. Every word containing a moving letter vanishes as $X \to 0$, which fixes its integration constant. An arbitrary tangential constant for the inner letter one cancels before the result is collected. Thus no value is assigned to a divergent logarithm in obtaining these identities.

Energy equations for the root integrals. The two remaining integrals I_2 and I_7 can be integrated by differentiating with respect to X. At fixed U,

$$\partial_X a = \frac{2U}{(X+2)^2 d}, \qquad \partial_X f = \frac{U}{2sk}, \qquad \partial_X k^2 = \frac{U^2-1}{s^2}. \tag{4.214}$$

For any compact positive interval of X, these derivatives have at most integrable logarithmic endpoint behavior. Differentiation under the integral is therefore allowed. Introduce

$$Q_7 = p_7\frac{k}{d} + h_7\frac{1}{k}, \qquad p_7 = \frac{2(X+2)}{X^2(X+4)}, \qquad h_7 = -\frac{(X+2)^2}{2X^2 s(X+4)}, \tag{4.215}$$

$$Q_2 = U\left(p_2\frac{k}{d} + h_2\frac{1}{k}\right), \qquad p_2 = \frac{2}{(X+2)(X+4)}, \qquad h_2 = \frac{(X+2)^2}{2Xs(X+4)}. \tag{4.216}$$

Direct differentiation gives the two rational identities

$$\partial_X\frac{U}{kd} = \partial_U Q_7 - \frac{2(X+6)}{X(X+2)(X+4)}\frac{U}{kd}, \tag{4.217}$$

$$\partial_X\frac{1}{kd} = \partial_U Q_2 - \frac{4}{X(X+2)(X+4)}\frac{1}{kd}. \tag{4.218}$$

They are verified by substituting $k^2 = (1+XU^2)/s$ and matching rational coefficients. For the first equation, integrate the total derivative against f using $Q_7 - Q_7(1)$, so its boundary vanishes. The remaining rational terms and the integral of $f' - L'$ give

$$I_7' = -\frac{2(X+6)}{X(X+2)(X+4)}I_7 + \frac{3(X+2)^2}{2X^2(X+4)}a_*. \tag{4.219}$$

The integrating factor is $X\sqrt{X(X+4)}/(X+2)^2$, hence

$$\frac{\mathrm{d}}{\mathrm{d}X}\left[\frac{X\sqrt{X(X+4)}}{(X+2)^2}I_7\right] = \frac{3}{4}\frac{\ln(1+X)}{\sqrt{X(X+4)}}. \tag{4.220}$$

At $X = 0$, $I_7 \to \int_0^1 U\,\mathrm{artanh}\,U\,\mathrm{d}U = 1/2$, so the product on the left vanishes. Therefore

$$I_7 = \frac{3(X+2)^2}{4X\sqrt{X(X+4)}}G(r,-1). \tag{4.221}$$

For the second equation the endpoint cancellation is implemented by subtracting $Q_2(1)a_*$ from $Q_2 a$ before multiplying by f'. Define the two lower-weight integrals

$$J_{Ua} = \int_0^1 \frac{Ua}{d}\,\mathrm{d}U, \qquad J = \int_0^1 \frac{a_* - Ua}{1-U^2}\,\mathrm{d}U. \tag{4.222}$$

The subtracted numerator in J vanishes at the upper endpoint. The differential equation becomes

$$\begin{aligned} I_2' = &-\frac{4}{X(X+2)(X+4)}I_2 - \frac{X-2}{2s(X+4)}I_7 \\ &+ \frac{X}{(X+2)(X+4)}J_{Ua} + \frac{X+2}{X(X+4)}(J + a_*^2). \end{aligned} \tag{4.223}$$

The rational-letter reductions give

$$\beta^2 J_{Ua} = -\tfrac{1}{4}G(-1,-1) + \tfrac{1}{2}G(-2,-1), \tag{4.224}$$

$$J = \tfrac{1}{2}G(0,-1) - \tfrac{1}{4}G(-1,-1), \tag{4.225}$$

$$\beta^2 J_{Ua} + J + a_*^2 = \tfrac{1}{2}\mathcal{D}. \tag{4.226}$$

Using (4.221) and the integrating factor $\gamma = \sqrt{X(X+4)}/(X+2)$ now yields

$$\frac{\mathrm{d}(\gamma I_2)}{\mathrm{d}X} = \frac{3}{8}\left(\frac{1}{X} - \frac{1}{X+1} - \frac{1}{X+4}\right)G(r,-1) + \frac{\mathcal{D}}{2\sqrt{X(X+4)}}. \tag{4.227}$$

Each term is a defining GPL differential. Since $I_2 = O(X)$ at small X, the constant at $X = 0$ is zero, and

$$\gamma I_2 = \frac{1}{8}\mathcal{R}. \tag{4.228}$$

Substituting these evaluated integrals into (4.181) proves (4.197). The two terms proportional to A_f in (4.209) cancel the remaining $G(-1,-2)$ and $\ln 2$ terms exactly.

4.8 A shorter finite UW expression

The finite UW contribution already has the integrated endpoint form (3.615). Its thirteen inner monomials contain repeated weight-two functions. Combining them before the rational primitive reduces the endpoint expression. Throughout this section $H = s - 1 > 0$, the electron mass is one, and $0 < t < 1$. Define

$$P = 1 + Ht - Ht^2, \qquad \ell = \ln(1 + Ht), \qquad p = \ln P, \qquad b = \ln(1 + Ht/2). \tag{4.229}$$

All three logarithms are real. The two weight-two combinations needed below are

$$J_2(H,t) = \mathrm{Li}_2(-1 - Ht) + \frac{\pi^2}{12} + \ell \ln(2 + Ht), \tag{4.230}$$

$$J(H,t) = \mathrm{Li}_2(t/P) - \mathrm{Li}_2(t) - p \ln(1 - t) + \frac{p^2}{2}. \tag{4.231}$$

The dilogarithm arguments in J obey $0 < t/P \le t < 1$. The argument in J_2 is negative. These expressions therefore fix the physical real branches directly.

Exact coefficient combinations. Let $r_f(H,t)$ denote the coefficient of the atom monomial f in the thirteen-term inner result. Its rational coefficients obey

$$r_{d_3} = -r_{\ell \ln t} = r_{\ell p}, \qquad r_{d_2} = r_{\ell \ln(2+Ht)}, \qquad [\ln 2] r_1 = -r_{\ln(2+Ht)}. \tag{4.232}$$

The last notation means the coefficient of $\ln 2$ in r_1. These are polynomial identities after multiplication by their common rational denominators, for arbitrary H and t. Here $d_2 = \mathrm{Li}_2(-1 - Ht) + \pi^2/12$ and d_3 is the two-dilogarithm difference defined in the inner integration. Thus its finite kernel is

$$\Phi_{UW} = A_0 + A_1 \ell + A_2 \ell^2 + A_p p + A_b b + A_d \,\mathrm{Li}_2(-Ht) + A_{J_2} J_2 + A_J J, \tag{4.233}$$

with the following factored sums of the original rational coefficients:

$$A_0 = r_1|_{\ln 2 = 0} + r_{\ln H} \ln H, \qquad A_1 = r_\ell + r_{\ell \ln H} \ln H, \tag{4.234}$$

$$A_2 = r_{\ell^2}, \qquad A_p = r_p, \tag{4.235}$$

$$A_b = r_{\ln(2+Ht)}, \qquad A_d = r_{d_1}, \tag{4.236}$$

$$A_{J_2} = r_{d_2}, \qquad A_J = r_{d_3}. \tag{4.237}$$

Here $d_1 = \mathrm{Li}_2(-Ht)$. Keeping these sums factored avoids expanding large common numerators. The explicit eight coefficients are stored with the calculation in `uw_simplified_kernel.json`.

Dilogarithm reduction. To obtain J without guessing a branch, first apply the negative-real-axis identity $\mathrm{Li}_2(-z)+\mathrm{Li}_2(-1/z)=-\pi^2/6-\ln^2 z/2$, with $z>0$, to d_3. It gives the analytic combination $D=d_3-\ell\ln t$ derived at the lower endpoint. Next use

$$\mathrm{Li}_2(-z)=-\mathrm{Li}_2\!\left(\frac{z}{1+z}\right)-\frac{1}{2}\ln^2(1+z),\qquad z>0. \tag{4.238}$$

The two positive arguments are $t/[(1-t)(1+Ht)]$ and $t/(1-t)$. Their transformed arguments are t/P and t. The logarithms combine exactly to give $J=D+\ell p$, hence (4.231). In particular no explicit $\ln t$ survives. Both $J(0)=0$ and $J_2(0)=0$. At the upper endpoint their analyticity follows from the original $d_3-\ell\ln t+\ell p$ and $d_2+\ell\ln(2+Ht)$ representations. This observation also fixes the value at $t=1$, where separate terms of (4.231) have logarithmic derivatives.

The integrated result. Introduce the letters

$$a=-\frac{1}{H},\qquad c=-\frac{2}{H},\qquad r_\pm=\frac{1\pm\sqrt{1+4/H}}{2}. \tag{4.239}$$

In the same GPL convention as the rational primitive,

$$J_2(H,t)=G(c,a;t), \tag{4.240}$$

$$J(H,t)=\sum_{r=r_\pm}\big[G(0,r;t)-G(1,r;t)+G(r,a;t)\big]-G(0,a;t). \tag{4.241}$$

For example $J_2'=H\ell/(2+Ht)$ and $J_2(0)=0$ prove its single-word form. The formula for J follows from the shuffle product $G(a;t)G(r;t)=G(a,r;t)+G(r,a;t)$ and the expression for D. The complete finite integral is therefore

$$\begin{aligned}\mathcal{F}_{UW}(H)=\lim_{Z\to1^-}\Big\{&\mathfrak{T}[A_0;\varnothing](Z)+\mathfrak{T}[A_1;a](Z)+2\mathfrak{T}[A_2;a,a](Z)\\&+\mathfrak{T}[A_b;c](Z)+\mathfrak{T}[A_{J_2};c,a](Z)-\mathfrak{T}[A_d+A_J;0,a](Z)\\&+\sum_{r=r_\pm}\Big(\mathfrak{T}[A_p;r](Z)+\mathfrak{T}[A_J;0,r](Z)\\&\qquad-\mathfrak{T}[A_J;1,r](Z)+\mathfrak{T}[A_J;r,a](Z)\Big)\Big\}.\end{aligned} \tag{4.242}$$

The operator $\mathfrak{T}$ is the finite rational-to-GPL primitive already evaluated by the terminating rules above, including its zero finite part at the lower endpoint. Thus this is a shorter representation of the completed integral, with fourteen primitive contributions. It is not a replacement by an unevaluated quadrature. The terms with first letter one must be summed before taking the upper limit. The total cross section uses this sector with the same factor $1/(8sH)$ and the second Bose-related contribution as before.

Removal of apparent endpoint poles. The certificates in Sections 3.6.2 and 3.6.2 also give a form in which each summand is finite at both endpoints. For any one of the eight functions f_i in (4.233), decompose its coefficient as

$$A_i(t)=A_i^{\mathrm{reg}}(t)+\sum_{e=0,1}\sum_{k=1}^{m_{ie}}\frac{c_{iek}}{(t-e)^k}. \tag{4.243}$$

The regular coefficient has no pole at zero or one. Since every f_i is analytic there, define its finite Taylor subtraction

$$f_i^{[e,k]}(t) = \frac{f_i(t) - \sum_{n=0}^{k-1} f_i^{(n)}(e)(t-e)^n/n!}{(t-e)^k}. \tag{4.244}$$

Then the same finite kernel satisfies the exact identity

$$\Phi_{UW}(t) = \sum_i A_i^{\mathrm{reg}}(t) f_i(t) + \sum_{i,e,k} c_{iek} f_i^{[e,k]}(t). \tag{4.245}$$

Indeed, the omitted Taylor pieces are exactly the principal parts of the full kernel at zero and one, which the two symbolic certificates show to vanish. These divided differences are explicit logarithms and dilogarithms minus finite polynomials. They introduce no new integral. The factored form (4.242) is shorter for presenting the integrated result.

4.9 Reciprocal symmetry of the real-emission integral

After the first energy integration, write $s = 1 + H$, with $H > 0$, and $z = s + 1 - u - s/u$. The two algebraic boundaries of the first integral are $x_- = s - u$ and $x_+ = s - s/u$. Under $u \mapsto s/u$, z is unchanged and these boundaries exchange places. The oriented boundary difference therefore changes sign. Its Jacobian $j = 1 - s/u^2$ also changes sign, so the complete last integrand obeys

$$\frac{s}{u^2} f(s/u) = f(u). \tag{4.246}$$

This identity holds separately for the regular rational, energy-logarithm, W, and Bose-restoring sectors. It can also be checked coefficient by coefficient: L_1, L_2, L_3 change sign, whereas $L_1 L_2$ is invariant under the reciprocal transformation.

The physical upper branch has $\sqrt{s} < u < s$. Its reflected partner is $1 < u < \sqrt{s}$. A common cut $z > \epsilon$ selects reciprocal endpoints $u_-(\epsilon)u_+(\epsilon) = s$. Consequently

$$\int_{\sqrt{s}}^{u_+(\epsilon)} f(u)\,\mathrm{d}u = \frac{1}{2}\int_{u_-(\epsilon)}^{u_+(\epsilon)} f(u)\,\mathrm{d}u. \tag{4.247}$$

Thus the square-root lower boundary can be eliminated exactly, without changing the infrared subtraction or making an asymptotic approximation.

Let A be the coefficient of $1/(s-u)$ at the upper endpoint. Reciprocal symmetry makes it also the coefficient of $1/(u-1)$ at the lower endpoint. Define

$$\mathrm{FP}_{[1,s]} \int f = \lim_{\delta \to 0^+} \left[\int_{1+\delta}^{s-\delta} f(u)\,\mathrm{d}u + 2A \ln \delta \right]. \tag{4.248}$$

Since $z = H(u-1) + O((u-1)^2)$ at the lower boundary and $z = H(s-u)/s + O((s-u)^2)$ at the upper boundary, the finite part in the original z convention is

$$F_z = \frac{1}{2}\,\mathrm{FP}_{[1,s]} \int f + A \ln \frac{H}{\sqrt{s}}. \tag{4.249}$$

The explicit logarithm is essential. The replacement of the lower boundary does not permit changing the finite subtraction constant.

The Bose-restoring term also has a double endpoint pole. If

$$f(s-b) = D/b^2 + A/b + O(1), \tag{4.250}$$

reciprocal symmetry gives $f(1+a) = D/(sa^2) + A/a + O(1)$. The finite-part definition then also subtracts $D(1+1/s)/\delta$. Expanding both cut boundaries to second order gives the corrected general formula

$$F_z = \frac{1}{2}\,\mathrm{FP}_{[1,s]} \int f + A \ln \frac{H}{\sqrt{s}} - \frac{D(s+1)}{2sH}. \tag{4.251}$$

The previous expression is the case $D = 0$. The extra constant comes from the double pole multiplied by the second-order change of the endpoint, and must be kept for the Bose term.

Eliminating the apparent threshold at $s = 9$. The regular rational primitive contains ten words whose first letters are

$$q_\pm = \frac{s - 3 \pm \sqrt{(s-1)(s-9)}}{2}. \tag{4.252}$$

They can be replaced by a single real dilogarithm and GPLs whose letters already occur in the other terms. This replacement follows from the coefficients of our primitive. It does not use a total-cross-section formula. Throughout this paragraph the GPL basepoint is $r = \sqrt{s}$, as in the original primitive. Define

$$A(u) = \frac{(u-1)(u+s)^2}{u(u+1)^2(s-u)}, \qquad Q(u) = u^2 - (s-3)u + s, \tag{4.253}$$

$$(k_0, k_1, k_s, k_{-s}, k_{-1}) = (-1, 1, -1, 2, -2), \qquad C(s) = \frac{128s(s+5)}{3(s-1)}. \tag{4.254}$$

The ten coefficients are precisely $C(s)k_b$ for each of the two first letters $q_\pm$ and each $b \in \{0, 1, s, -s, -1\}$. Since $A(r) = 1$,

$$L_3(u) = \ln A(u) = \sum_b k_b G(b; u). \tag{4.255}$$

Factorization of $1 - A$ gives the rational differential identity

$$\frac{Q'(u)}{Q(u)} = \frac{\mathrm{d}}{\mathrm{d}u} \ln(1 - A(u)) - \frac{1}{u-r} - \frac{1}{u+r} + \frac{1}{u} + \frac{1}{u-s} + \frac{2}{u+1}. \tag{4.256}$$

Here the logarithmic derivative denotes the rational quotient $-A'(u)/(1 - A(u))$ wherever its separate terms have removable poles. Multiplying by L_3 and using $\mathrm{d}\,\mathrm{Li}_2(1-A) = -\ln A\,\mathrm{d}\ln(1-A)$ yields

$$\sum_{\epsilon=\pm} \sum_b k_b G(q_\epsilon, b; u) = -\mathrm{Li}_2(1 - A(u)) \tag{4.257}$$

$$+ \sum_b k_b \big[-G(r, b; u) - G(-r, b; u) + G(0, b; u)\big] \tag{4.258}$$

$$+ \sum_b k_b \big[G(s, b; u) + 2G(-1, b; u)\big]. \tag{4.259}$$

Both sides vanish at $u = r$, which fixes the integration constant. Although r is a first letter on the right, the summed second-letter function L_3 vanishes there, so the corresponding

lower endpoint is regular. For $r < u < s$, $A(u) > 0$. Thus $1 - A(u) < 1$, and the dilogarithm in this expression lies on its real branch. At a zero of Q, $A = 1$ and this dilogarithm is regular. The formula also holds by continuity at $s = 9$, without choosing a side of an artificial threshold.

Replacing the ten words by (4.259) removes $q_\pm$ from the primitive. It reintroduces the letters $\pm r$, so it is a trade between two representations rather than a simultaneous elimination of every auxiliary root. The accompanying coefficient record contains the full replaced primitive. Its ten input coefficients, rational differential identity, and lower-endpoint constant are checked exactly by `compton_real_equivalence_qpair.py`.

Combining the rational and W primitives. The two primitives should be added before selecting a function basis. Their original lists contain 48 and 19 nonempty GPL words. Collecting identical words and canceling their exact rational coefficients leaves 43 words. In particular, the six Gram words discussed above disappear. The coefficients in this combined list can be reduced further by shuffle identities. For weight two,

$$G(a,b) + G(b,a) = G(a)G(b), \qquad G(a,a) = \frac{1}{2}G(a)^2. \tag{4.260}$$

Applying these identities only to pairs present in the combined list leaves 40 GPL words and three elementary logarithm products.

There is a further reduction in the weight-three part. Define

$$K = \frac{32s(2s+1)}{(s-1)^2}, \qquad g = G(0;u) = \ln(u/\sqrt{s}). \tag{4.261}$$

The entire weight-three contribution of the combined primitive is

$$K\big[G(0,0,s) + G(0,s,0) - G(0,0,1) - G(0,1,0) - 2G(0,0,0)\big] \tag{4.262}$$

$$= K\big[g\{G(0,s) - G(0,1)\} - G(0,0,s) + G(0,0,1) - g^3/3\big]. \tag{4.263}$$

This follows from $G(0)G(0,b) = 2G(0,0,b) + G(0,b,0)$ for $b = 1, s$. Consequently the combined representation requires 37 nonempty GPL words, three logarithm products, two logarithm–weight-two products, and one logarithm cube. These are representation counts, not a claim of functional independence of all remaining GPLs. The rational term is included throughout.

By contrast, substituting (4.259) before the same combination gives 50 GPL words. Weight-two shuffle reduction leaves 42 words, eight logarithm products, and the additional dilogarithm. The root-pair identity is therefore useful for displaying regularity at $s = 9$, but the original root-pair representation gives the shorter combined formula. The full coefficient dictionary for the 37-word form and exact reconstruction checks are recorded by `compton_real_equivalence_combine.py`.

Classical trilogarithms for the weight-three term. For positive real z, define $\mathcal{L}_n(z)$ here as $\mathrm{Re}\,\mathrm{Li}_n(z + \mathrm{i}0)$, for $n = 2, 3$. The two boundary values are complex conjugates, so their real parts agree. The differential relations are

$$\frac{\mathrm{d}\mathcal{L}_2(z)}{\mathrm{d}z} = -\frac{\ln|1-z|}{z}, \qquad \frac{\mathrm{d}\mathcal{L}_3(z)}{\mathrm{d}z} = \frac{\mathcal{L}_2(z)}{z}. \tag{4.264}$$

They determine the continuation needed on each of the intervals used below. For $b = 1, s$ and $r < u < s$, direct differentiation and the zero value at $u = r$ give

$$G(0,b;u) = -\mathcal{L}_2(u/b) + \mathcal{L}_2(r/b) - \ln|1 - r/b|\, g, \tag{4.265}$$

$$G(0,0,b;u) = -\mathcal{L}_3(u/b) + \mathcal{L}_3(r/b) + \mathcal{L}_2(r/b)g - \tfrac{1}{2}\ln|1 - r/b|\, g^2. \tag{4.266}$$

No singularity at $u = b$ is crossed along either open integration interval. The complete weight-three contribution therefore reduces to

$$\mathcal{F}^{(3)}_{R_0+W}(u) = K\big[\mathcal{L}_3(u/s) - \mathcal{L}_3(u) - \mathcal{L}_3(1/r) + \mathcal{L}_3(r) \tag{4.267}$$

$$+ g\{\mathcal{L}_2(u) - \mathcal{L}_2(u/s)\} + \tfrac{1}{2}g^2 \ln r - \tfrac{1}{3}g^3\big]. \tag{4.268}$$

The logarithmic coefficient uses $|1 - r|/|1 - r/s| = r$. Thus the five original weight-three words require only classical trilogarithms, dilogarithms, and logarithms. The lower endpoint still vanishes exactly.

The remaining weight-two GPLs admit a uniform classical representation. For distinct letters a, b write

$$z_a(u; b) = \frac{a - u}{a - b}, \qquad z_a(r; b) = \frac{a - r}{a - b}. \tag{4.269}$$

Then

$$G(a, b; u) = -\operatorname{Li}_2(z_a(u; b)) + \operatorname{Li}_2(z_a(r; b)) \tag{4.270}$$

$$- \operatorname{Log}(1 - z_a(r; b))G(a; u). \tag{4.271}$$

The logarithm and dilogarithm on the right must be continued together from $u = r$ along the same path as the original GPL. More explicitly, their logarithms satisfy

$$\operatorname{Log}(1 - z_a(u; b)) - \operatorname{Log}(1 - z_a(r; b)) = G(b; u). \tag{4.272}$$

This condition prevents independent principal-branch substitutions at the two endpoints. Differentiation of (4.271) then gives $G(b; u)/(u-a)$, and its initial value is zero. It therefore proves the identity, including the continuation around removable letters. Conjugate letters are combined before taking the real physical value. Together with (4.268), this eliminates all GPL notation from the combined primitive if a classical-function representation is preferred. Expanding every such identity separately need not shorten the displayed expression, so conjugate pairs and common coefficients should remain grouped.

At the upper endpoint the weight-three part simplifies further. Put $\ell = \frac{1}{2}\ln s$. The real inversion identities for $z > 1$ are

$$\mathcal{L}_2(z) + \operatorname{Li}_2(z^{-1}) = \frac{\pi^2}{3} - \frac{1}{2}\ln^2 z, \tag{4.273}$$

$$\mathcal{L}_3(z) - \operatorname{Li}_3(z^{-1}) = \frac{\pi^2}{3}\ln z - \frac{1}{6}\ln^3 z. \tag{4.274}$$

Their derivatives agree on $z > 1$, and their constants follow from $\operatorname{Li}_2(1) = \pi^2/6$ and $\operatorname{Li}_3(1) = \zeta(3)$. Applying them at $z = s, r$ gives

$$\mathcal{F}^{(3)}_{R_0+W}(s) = K\left[\zeta(3) - \operatorname{Li}_3(s^{-1}) - \ell\operatorname{Li}_2(s^{-1}) - \frac{\pi^2\ell}{6} - \frac{2\ell^3}{3}\right]. \tag{4.275}$$

The lower endpoint is zero. The resulting expression depends only on s and contains no auxiliary square root or continued polylogarithm.

Finite energy endpoints before further combination. For the combined rational and W primitive, cancellation of common factors makes every rational coefficient regular at $u = s$. Thus its finite upper endpoint needs only the coefficients evaluated at s. No derivative of a coefficient multiplying a power-divergent GPL is required. This statement

follows by exact cancellation of their rational denominators, whose endpoint factors are powers of s, $s-1$, and $s+3$.

For a one-letter word define its constant endpoint term by

$$h_a = \begin{cases} \mathrm{Log}[(s-a)/(r-a)], & a \neq s, \\ -\ln(s-r), & a = s. \end{cases} \tag{4.276}$$

The second line removes the term $\ln(s-u)$ from $G(s;u)$. For a weight-two word write d_{ab} for the corresponding constant term. For $a \neq s$, it is the boundary value of (4.271). For $a = s$ it is explicitly

$$d_{sb} = \mathrm{Li}_2\left(\frac{s-r}{s-b}\right) + \mathrm{Log}\left(\frac{r-b}{s-b}\right)\ln(s-r). \tag{4.277}$$

Indeed, the omitted divergent term is $-\mathrm{Log}[(r-b)/(s-b)]\ln(s-u)$. All logarithms use the continuation specified above. The only logarithm products in the reduced primitive are $G(0)^2$, $G(0)G(1)$, and $G(0)G(s)$, so their constant terms are simply the corresponding products of the h_a. There is no product of two divergent one-letter functions in this representation.

Replacing each word in the coefficient table by these constant endpoint terms, and replacing its entire weight-three block by (4.275), gives a function $E_{R_0+W}(s)$ of energy alone. Let A_{R_0} be the pole coefficient of this pair, with the normalization preceding the common factor $1/(8sH)$. Its z-cut finite part is

$$F_{R_0+W,z} = E_{R_0+W}(s) - \frac{2s(-s^5-5s^4+18s^3+122s^2+151s-45)}{3(s^2+2s-3)} + A_{R_0}\ln\frac{s-1}{s}. \tag{4.278}$$

The rational subtraction is the complete lower-basepoint constant. The W part has zero lower constant and zero soft pole. Finally, the previously derived rational-sector matching shift converts this expression to the common $(xy)^\eta$ convention. This order of operations preserves all finite endpoint terms before combining the other real sectors.

4.9.1 Explicit combined rational and W primitive

Put $r = \sqrt{s} > 1$. All GPLs below have basepoint r and endpoint u. The following finite sum specifies the complete combined primitive, including its rational and elementary terms.

$$\mathcal{F}_{R_0+W}(u) = \sum_{j=0}^{37} c_j(u)\,\mathcal{G}_j(u) + \sum_{j=1}^{3} d_j(u)\mathcal{L}_j(u) + K\left[G(0)\{G(0,s)-G(0,1)\} - G(0)^3/3\right]. \tag{4.279}$$

Here $K = 32s(2s+1)/(s-1)^2$. The two weight-three GPL terms already occur in the coefficient table, so the last term contains only their shuffle products. Write $q_\pm = (r^2-3\pm\sqrt{r^4-10r^2+9})/2$ and $a_\pm = 1\pm\sqrt{1-r^2}$, $b_\pm = -1\pm\sqrt{1-r^2}$. Define the polynomial abbreviations before listing the coefficients.

$$P_1 = 540 - 1380s - 537s^5 - 48s^6 + 12s^7 + 610s^4 \tag{4.280}$$
$$+ 2334s^3 + 12701s^2. \tag{4.281}$$

$$P_2 = 252 - 1128s - 609s^6 - 24s^7 + 12s^8 + 784s^5 \tag{4.282}$$
$$+ 1872s^4 + 2705s^2 + 14424s^3. \tag{4.283}$$

$$P_3 = 5 - s^4 - s^5 - 49s + 20s^2 + 26s^3 + 107u - 109su \tag{4.284}$$
$$- 47s^2u + 25s^3u. \tag{4.285}$$

$$P_4 = 90 - 42s - 42s^2 - 6s^3 + 117u + s^4u + 24s^3u \tag{4.286}$$
$$+ 130s^2u + 208su. \tag{4.287}$$

$$P_5 = 2268 - 227127s^3 - 184189s^4 - 36221s^5 - 13500s \tag{4.288}$$
$$- 2693s^8 - 2367s^2 - 597s^9 - 319s^7 + 12s^{11} \tag{4.289}$$
$$+ 36s^{10} + 6201s^6. \tag{4.290}$$

$$P_6 = - 648 - 5122s^4 - 3759s^5 - 1589s^3 - 1246s^6 \tag{4.291}$$
$$- 187s^7 + 2s^8 + 3s^9 + 972s + 2358s^2. \tag{4.292}$$

$$P_7 = 1755 - 10593s^2 - 6172s - 4076s^3 - 55s^4 + 5s^6 \tag{4.293}$$
$$+ 128s^5. \tag{4.294}$$

$$P_8 = - 162 + s^9 - 2476s^3 - 1980s^2 - 1409s^4 - 689s^5 \tag{4.295}$$
$$- 286s^6 - 70s^7 - 3s^8 + 162s. \tag{4.296}$$

$$P_9 = - 162 + s^9 - 2692s^3 - 2153s^4 - 1008s^2 - 989s^5 \tag{4.297}$$
$$- 322s^6 - 70s^7 - 3s^8 + 486s. \tag{4.298}$$

$$\mathcal{G}_0 = 1, \tag{4.299}$$
$$\mathcal{G}_1 = G(0; u), \tag{4.300}$$
$$\mathcal{G}_2 = G(1; u), \tag{4.301}$$
$$\mathcal{G}_3 = G(r^2; u). \tag{4.302}$$

$$c_0 = \frac{P_1 u}{18s\left(s-1\right)^2} \tag{4.303}$$

$$- \frac{u^2\left(12s^5+261s^4-700s^3+1027s^2+2040s-396\right)}{36s\left(s-1\right)^2} \tag{4.304}$$

$$+ \frac{u^3\left(12s^4+24s^3-79s^2+787s+48\right)}{54\left(s-1\right)^2} \tag{4.305}$$

$$- \frac{u^4\left(24s^2+185s-4\right)}{24\left(s-1\right)^2} \tag{4.306}$$

$$+ \frac{2su^5\left(s+7\right)}{15\left(s-1\right)^2} \tag{4.307}$$

$$- \frac{P_1}{18u\left(s-1\right)^2} \tag{4.308}$$

$$- \frac{2s^6\left(s+7\right)}{15u^5\left(s-1\right)^2} \tag{4.309}$$

$$- \frac{s^3\left(12s^4+24s^3-79s^2+787s+48\right)}{54u^3\left(s-1\right)^2} \tag{4.310}$$

$$+ \frac{s^4\left(24s^2+185s-4\right)}{24u^4\left(s-1\right)^2} \tag{4.311}$$

$$+ \frac{s\left(12s^5+261s^4-700s^3+1027s^2+2040s-396\right)}{36u^2\left(s-1\right)^2} \tag{4.312}$$

$$+ \frac{8s\left(s+5\right)\left(s+u\right)}{\left(s+3\right)\left(s+u^2+2u\right)} \tag{4.313}$$

$$+ \frac{4s\left(-s+u\right)\left(s+5\right)\left(s^3-3s^2-9s+1\right)}{3\left(s-1\right)\left(s+u^2-2u\right)}. \tag{4.314}$$

$$c_1 = \frac{P_2}{18s\left(s-1\right)^2} \tag{4.315}$$
$$+ \frac{4u\left(s^5-13s^4+12s^3-123s^2+45s-9\right)}{3s\left(s-1\right)^2} \tag{4.316}$$
$$+ \frac{u^2\left(3s^4+4s^3+169s^2+12s-6\right)}{3s\left(s-1\right)^2} \tag{4.317}$$
$$- \frac{4su^3\left(s+26\right)}{9\left(s-1\right)^2} \tag{4.318}$$
$$+ \frac{5su^4}{6\left(s-1\right)^2} \tag{4.319}$$
$$+ \frac{4\left(s^5-13s^4+12s^3-123s^2+45s-9\right)}{3u\left(s-1\right)^2} \tag{4.320}$$
$$+ \frac{5s^5}{6u^4\left(s-1\right)^2} \tag{4.321}$$
$$- \frac{4P_3 s}{3\left(s-1\right)\left(s+u^2-2u\right)} \tag{4.322}$$
$$- \frac{4s^4\left(s+26\right)}{9u^3\left(s-1\right)^2} \tag{4.323}$$
$$+ \frac{s\left(3s^4+4s^3+169s^2+12s-6\right)}{3u^2\left(s-1\right)^2} \tag{4.324}$$
$$+ \frac{4P_4 s}{3\left(s+3\right)^2\left(s+u^2+2u\right)} \tag{4.325}$$
$$+ \frac{4s\left(-s+2u\right)\left(s+5\right)\left(s^3-3s^2-9s+1\right)}{3\left(s+u^2-2u\right)^2} \tag{4.326}$$
$$+ \frac{8s\left(s-1\right)\left(s+5\right)\left(s+2u\right)}{\left(s+3\right)\left(s+u^2+2u\right)^2}. \tag{4.327}$$

$$
\begin{aligned}
c_2 = & -\frac{P_5}{18s\left(s-1\right)^3\left(s+3\right)^2} && (4.328)\\
& +\frac{su^4}{2\left(s-1\right)^2} && (4.329)\\
& +\frac{4u\left(7s^6-31s^5-82s^4-138s^3-415s^2-30s-3\right)}{3s\left(s-1\right)^3} && (4.330)\\
& -\frac{u^2\left(17s^5-39s^4-37s^3-237s^2-102s+18\right)}{3s\left(s-1\right)^3} && (4.331)\\
& +\frac{4su^3\left(3s-10\right)}{9\left(s-1\right)^2} && (4.332)\\
& +\frac{s^5}{2u^4\left(s-1\right)^2} && (4.333)\\
& +\frac{4\left(7s^6-31s^5-82s^4-138s^3-415s^2-30s-3\right)}{3u\left(s-1\right)^3} && (4.334)\\
& -\frac{4P_3s}{3\left(s-1\right)\left(s+u^2-2u\right)} && (4.335)\\
& -\frac{s\left(17s^5-39s^4-37s^3-237s^2-102s+18\right)}{3u^2\left(s-1\right)^3} && (4.336)\\
& +\frac{4s^4\left(3s-10\right)}{9u^3\left(s-1\right)^2} && (4.337)\\
& +\frac{4s\left(-s+2u\right)\left(s+5\right)\left(s^3-3s^2-9s+1\right)}{3\left(s+u^2-2u\right)^2}. && (4.338)
\end{aligned}
$$

$$c_3 = \frac{P_5}{18s\left(s-1\right)^3\left(s+3\right)^2} \tag{4.339}$$
$$- \frac{su^4}{2\left(s-1\right)^2} \tag{4.340}$$
$$- \frac{4u\left(7s^6-31s^5-82s^4-138s^3-415s^2-30s-3\right)}{3s\left(s-1\right)^3} \tag{4.341}$$
$$+ \frac{u^2\left(17s^5-39s^4-37s^3-237s^2-102s+18\right)}{3s\left(s-1\right)^3} \tag{4.342}$$
$$- \frac{4su^3\left(3s-10\right)}{9\left(s-1\right)^2} \tag{4.343}$$
$$- \frac{4\left(7s^6-31s^5-82s^4-138s^3-415s^2-30s-3\right)}{3u\left(s-1\right)^3} \tag{4.344}$$
$$- \frac{s^5}{2u^4\left(s-1\right)^2} \tag{4.345}$$
$$- \frac{4s^4\left(3s-10\right)}{9u^3\left(s-1\right)^2} \tag{4.346}$$
$$+ \frac{s\left(17s^5-39s^4-37s^3-237s^2-102s+18\right)}{3u^2\left(s-1\right)^3} \tag{4.347}$$
$$+ \frac{4P_3 s}{3\left(s-1\right)\left(s+u^2-2u\right)} \tag{4.348}$$
$$- \frac{4s\left(-s+2u\right)\left(s+5\right)\left(s^3-3s^2-9s+1\right)}{3\left(s+u^2-2u\right)^2}. \tag{4.349}$$

$$\mathcal{G}_4 = G(b_-, 0; u), \tag{4.350}$$
$$\mathcal{G}_5 = G(b_+, 0; u), \tag{4.351}$$
$$\mathcal{G}_6 = G(a_-, 0; u), \tag{4.352}$$
$$\mathcal{G}_7 = G(a_+, 0; u). \tag{4.353}$$

$$c_4 = \frac{4s\left(s^5+9s^4+116s^3+516s^2+307s-693\right)}{3\left(s-1\right)\left(s+3\right)^3}. \tag{4.354}$$

$$c_5 = \frac{4s\left(s^5+9s^4+116s^3+516s^2+307s-693\right)}{3\left(s-1\right)\left(s+3\right)^3}. \tag{4.355}$$

$$c_6 = -\frac{4\left(s^6-s^5-33s^4+75s^3+18s^2+24s-12\right)}{3s\left(s-1\right)^2}. \tag{4.356}$$

$$c_7 = -\frac{4\left(s^6-s^5-33s^4+75s^3+18s^2+24s-12\right)}{3s\left(s-1\right)^2}. \tag{4.357}$$

$$\mathcal{G}_8 = G(q_-, 0; u), \tag{4.358}$$
$$\mathcal{G}_9 = G(q_+, 0; u), \tag{4.359}$$
$$\mathcal{G}_{10} = G(a_-, 1; u), \tag{4.360}$$
$$\mathcal{G}_{11} = G(a_+, 1; u). \tag{4.361}$$

$$c_8 = -\frac{128s\,(s+5)}{3\,(s-1)}. \tag{4.362}$$

$$c_9 = -\frac{128s\,(s+5)}{3\,(s-1)}. \tag{4.363}$$

$$c_{10} = -\frac{4\left(s^6 - s^5 - 33s^4 + 75s^3 + 18s^2 + 24s - 12\right)}{3s\,(s-1)^2}. \tag{4.364}$$

$$c_{11} = -\frac{4\left(s^6 - s^5 - 33s^4 + 75s^3 + 18s^2 + 24s - 12\right)}{3s\,(s-1)^2}. \tag{4.365}$$

$$\mathcal{G}_{12} = G(0, 1; u), \tag{4.366}$$
$$\mathcal{G}_{13} = G(q_-, 1; u), \tag{4.367}$$
$$\mathcal{G}_{14} = G(q_+, 1; u), \tag{4.368}$$
$$\mathcal{G}_{15} = G(a_-, r^2; u). \tag{4.369}$$

$$c_{12} = -\frac{4\left(s^8 - 14s^7 - 93s^6 + 30s^5 + 403s^4 - 2002s^3 - 8747s^2 - 8334s + 324\right)}{3\,(s-1)^2\,(s+3)^3}. \tag{4.370}$$

$$c_{13} = \frac{128s\,(s+5)}{3\,(s-1)}. \tag{4.371}$$

$$c_{14} = \frac{128s\,(s+5)}{3\,(s-1)}. \tag{4.372}$$

$$c_{15} = \frac{4\left(s^6 - s^5 - 33s^4 + 75s^3 + 18s^2 + 24s - 12\right)}{3s\,(s-1)^2}. \tag{4.373}$$

$$\mathcal{G}_{16} = G(a_+, r^2; u), \tag{4.374}$$
$$\mathcal{G}_{17} = G(0, r^2; u), \tag{4.375}$$
$$\mathcal{G}_{18} = G(q_-, r^2; u), \tag{4.376}$$
$$\mathcal{G}_{19} = G(q_+, r^2; u). \tag{4.377}$$

$$c_{16} = \frac{4\left(s^6 - s^5 - 33s^4 + 75s^3 + 18s^2 + 24s - 12\right)}{3s\,(s-1)^2}. \tag{4.378}$$

$$c_{17} = -\frac{4P_6}{3s\,(s-1)^2\,(s+3)^3}. \tag{4.379}$$

$$c_{18} = -\frac{128s\,(s+5)}{3\,(s-1)}. \tag{4.380}$$

$$c_{19} = -\frac{128s\,(s+5)}{3\,(s-1)}. \tag{4.381}$$

$$\mathcal{G}_{20} = G(-r^2; u), \tag{4.382}$$
$$\mathcal{G}_{21} = G(b_-, -r^2; u), \tag{4.383}$$
$$\mathcal{G}_{22} = G(b_+, -r^2; u), \tag{4.384}$$
$$\mathcal{G}_{23} = G(0, -r^2; u). \tag{4.385}$$

$$c_{20} = -\frac{2P_7 s}{3(s-1)^3(s+3)^2} \tag{4.386}$$
$$-\frac{4su\left(s^3+45s^2-405s+43\right)}{3(s-1)^3} \tag{4.387}$$
$$+\frac{2su^2\left(s^2-26s+63\right)}{3(s-1)^3} \tag{4.388}$$
$$-\frac{4P_4 s}{3(s+3)^2(s+u^2+2u)} \tag{4.389}$$
$$-\frac{4s^2\left(s^3+45s^2-405s+43\right)}{3u(s-1)^3} \tag{4.390}$$
$$+\frac{2s^3\left(s^2-26s+63\right)}{3u^2(s-1)^3} \tag{4.391}$$
$$-\frac{8s(s-1)(s+5)(s+2u)}{(s+3)(s+u^2+2u)^2}. \tag{4.392}$$

$$c_{21} = -\frac{4s\left(s^5+9s^4+116s^3+516s^2+307s-693\right)}{3(s-1)(s+3)^3}. \tag{4.393}$$

$$c_{22} = -\frac{4s\left(s^5+9s^4+116s^3+516s^2+307s-693\right)}{3(s-1)(s+3)^3}. \tag{4.394}$$

$$c_{23} = \frac{4s\left(s^4-2s^3-678s^2+150s-431\right)}{3(s-1)^3}. \tag{4.395}$$

$$\mathcal{G}_{24} = G(1,-r^2;u), \tag{4.396}$$
$$\mathcal{G}_{25} = G(r^2,-r^2;u), \tag{4.397}$$
$$\mathcal{G}_{26} = G(q_-,-r^2;u), \tag{4.398}$$
$$\mathcal{G}_{27} = G(q_+,-r^2;u). \tag{4.399}$$

$$c_{24} = -\frac{256s\left(s^6-53s^4-231s^3-388s^2-253s-36\right)}{3(s-1)^3(s+3)^3}. \tag{4.400}$$

$$c_{25} = -\frac{256s\left(s^6-53s^4-231s^3-388s^2-253s-36\right)}{3(s-1)^3(s+3)^3}. \tag{4.401}$$

$$c_{26} = \frac{256s(s+5)}{3(s-1)}. \tag{4.402}$$

$$c_{27} = \frac{256s(s+5)}{3(s-1)}. \tag{4.403}$$

$$\mathcal{G}_{28} = G(-1;u), \tag{4.404}$$
$$\mathcal{G}_{29} = G(b_-,-1;u), \tag{4.405}$$
$$\mathcal{G}_{30} = G(b_+,-1;u), \tag{4.406}$$
$$\mathcal{G}_{31} = G(0,-1;u). \tag{4.407}$$

$$c_{28} = \frac{2P_7 s}{3\left(s-1\right)^3\left(s+3\right)^2} \tag{4.408}$$
$$+ \frac{4su\left(s^3+45s^2-405s+43\right)}{3\left(s-1\right)^3} \tag{4.409}$$
$$- \frac{2su^2\left(s^2-26s+63\right)}{3\left(s-1\right)^3} \tag{4.410}$$
$$- \frac{2s^3\left(s^2-26s+63\right)}{3u^2\left(s-1\right)^3} \tag{4.411}$$
$$+ \frac{4P_4 s}{3\left(s+3\right)^2\left(s+u^2+2u\right)} \tag{4.412}$$
$$+ \frac{4s^2\left(s^3+45s^2-405s+43\right)}{3u\left(s-1\right)^3} \tag{4.413}$$
$$+ \frac{8s\left(s-1\right)\left(s+5\right)\left(s+2u\right)}{\left(s+3\right)\left(s+u^2+2u\right)^2}. \tag{4.414}$$

$$c_{29} = \frac{4s\left(s^5+9s^4+116s^3+516s^2+307s-693\right)}{3\left(s-1\right)\left(s+3\right)^3}. \tag{4.415}$$

$$c_{30} = \frac{4s\left(s^5+9s^4+116s^3+516s^2+307s-693\right)}{3\left(s-1\right)\left(s+3\right)^3}. \tag{4.416}$$

$$c_{31} = -\frac{4s\left(s^4-2s^3-678s^2+150s-431\right)}{3\left(s-1\right)^3}. \tag{4.417}$$

$$\mathcal{G}_{32} = G(q_-, -1; u), \tag{4.418}$$
$$\mathcal{G}_{33} = G(q_+, -1; u), \tag{4.419}$$
$$\mathcal{G}_{34} = G(1, -1; u), \tag{4.420}$$
$$\mathcal{G}_{35} = G(r^2, -1; u). \tag{4.421}$$

$$c_{32} = -\frac{256s\left(s+5\right)}{3\left(s-1\right)}. \tag{4.422}$$

$$c_{33} = -\frac{256s\left(s+5\right)}{3\left(s-1\right)}. \tag{4.423}$$

$$c_{34} = \frac{256s\left(s^6-53s^4-231s^3-388s^2-253s-36\right)}{3\left(s-1\right)^3\left(s+3\right)^3}. \tag{4.424}$$

$$c_{35} = \frac{256s\left(s^6-53s^4-231s^3-388s^2-253s-36\right)}{3\left(s-1\right)^3\left(s+3\right)^3}. \tag{4.425}$$

$$\mathcal{G}_{36} = G(0, 0, 1; u), \tag{4.426}$$
$$\mathcal{G}_{37} = G(0, 0, r^2; u). \tag{4.427}$$

$$c_{36} = \frac{32s\left(2s+1\right)}{\left(s-1\right)^2}. \tag{4.428}$$

$$c_{37} = -\frac{32s\left(2s+1\right)}{\left(s-1\right)^2}. \tag{4.429}$$

$$\mathcal{L}_1 = G(0;u)G(0;u). \tag{4.430}$$

$$\begin{aligned}
d_1 = & -\frac{2\left(s^5-22s^4+22s^3-398s^2+73s+36\right)}{3\left(s-1\right)^2} && (4.431)\\
& -\frac{16su\left(s+1\right)}{\left(s-1\right)^2} && (4.432)\\
& -\frac{8s^3}{u^2\left(s-1\right)^2} && (4.433)\\
& +\frac{8su^2}{\left(s-1\right)^2} && (4.434)\\
& -\frac{32s\left(su-3s+3u-1\right)}{\left(-s+u^2\right)\left(s-1\right)} && (4.435)\\
& +\frac{16s^2\left(s+1\right)}{u\left(s-1\right)^2}. && (4.436)
\end{aligned}$$

$$\mathcal{L}_2 = G(0;u)G(1;u). \tag{4.437}$$

$$\begin{aligned}
d_2 = & \frac{8P_8}{3s\left(s-1\right)^2\left(s+3\right)^3} && (4.438)\\
& -\frac{16su\left(s+1\right)}{\left(s-1\right)^2} && (4.439)\\
& -\frac{8s^3}{u^2\left(s-1\right)^2} && (4.440)\\
& +\frac{8su^2}{\left(s-1\right)^2} && (4.441)\\
& -\frac{32s\left(su-3s+3u-1\right)}{\left(-s+u^2\right)\left(s-1\right)} && (4.442)\\
& +\frac{16s^2\left(s+1\right)}{u\left(s-1\right)^2}. && (4.443)
\end{aligned}$$

$$\mathcal{L}_3 = G(0;u)G(r^2;u). \tag{4.444}$$

$$\begin{aligned}
d_3 = & \frac{8P_9}{3s\left(s-1\right)^2\left(s+3\right)^3} && (4.445)\\
& +\frac{16su\left(s+1\right)}{\left(s-1\right)^2} && (4.446)\\
& -\frac{8su^2}{\left(s-1\right)^2} && (4.447)\\
& +\frac{8s^3}{u^2\left(s-1\right)^2} && (4.448)\\
& -\frac{16s^2\left(s+1\right)}{u\left(s-1\right)^2} && (4.449)\\
& +\frac{32s\left(su-3s+3u-1\right)}{\left(-s+u^2\right)\left(s-1\right)}. && (4.450)
\end{aligned}$$

All coefficients follow by collecting the original two primitives and applying the displayed shuffle identities. The generating script checks their exact coefficient reconstruction before this table is used. Endpoint constants and the common regulator conversion are applied to the combined primitive as before.

The grouped upper endpoint. The energy-only weight-two result can be grouped into seven coefficient blocks. Put $\bar{c}_j = c_j(s)$ and $\bar{d}_j = d_j(s)$, using the explicit coefficient table, and define

$$D_b = \sum_{\epsilon=\pm}(d_{b_\epsilon,0} - d_{b_\epsilon,-s} + d_{b_\epsilon,-1}), \tag{4.451}$$

$$D_a = \sum_{\epsilon=\pm}(d_{a_\epsilon,0} + d_{a_\epsilon,1} - d_{a_\epsilon,s}), \tag{4.452}$$

$$D_q = \sum_{\epsilon=\pm}(d_{q_\epsilon,0} - d_{q_\epsilon,1} + d_{q_\epsilon,s} - 2d_{q_\epsilon,-s} + 2d_{q_\epsilon,-1}). \tag{4.453}$$

Then all 30 weight-two terms have the compact form

$$E^{(2)}_{R_0+W} = \bar{c}_4 D_b + \bar{c}_6 D_a + \bar{c}_8 D_q + \bar{c}_{12} d_{0,1} + \bar{c}_{17} d_{0,s} \tag{4.454}$$

$$+ \bar{c}_{23}(d_{0,-s} - d_{0,-1}) + \bar{c}_{24}(d_{1,-s} + d_{s,-s} - d_{1,-1} - d_{s,-1}). \tag{4.455}$$

The coefficient identities are rational identities for arbitrary $s > 1$. The elementary part is

$$E^{(\mathrm{el})}_{R_0+W} = \bar{c}_0 + \bar{c}_1\ell + \bar{c}_2 \ln[r(s-1)] + \bar{c}_{20} \ln \frac{2r}{s+1} \tag{4.456}$$

$$+ \bar{d}_1\ell^2 + \bar{d}_2\ell \ln(r+1) - \bar{d}_3\ell \ln[r(r-1)]. \tag{4.457}$$

Here the cancellations $\bar{c}_3 = -\bar{c}_2$ and $\bar{c}_{28} = -\bar{c}_{20}$ have been performed before displaying the answer. The complete constant upper endpoint is $E^{(\mathrm{el})}_{R_0+W} + E^{(2)}_{R_0+W} + \mathcal{F}^{(3)}_{R_0+W}(s)$.

4.9.2 Cancellation between the four real sectors

Section 3.5.3 has already removed the $b_\pm$ pole family directly from the common integrand. The following identities also recover that cancellation from the separately integrated sectors and reduce the remaining functions. The common primitive becomes shorter when the rational, W, energy-logarithm and Bose-restoring contributions are combined before reducing individual functions. Denote these primitives by F_{R_0}, F_W, $F_{\log}$ and F_D, respectively. The factor $\ln s$ multiplying the energy-logarithm primitive is included in $F_{\log}$. All G functions in this subsection have the same basepoint $r = \sqrt{s}$ and the same final argument u. The subscript R_0 denotes the rational kernel. Taking its integrated finite part with the same cutoff $z > \zeta$ as in the sector assembly gives

$$F^z_{R_0} = F^z_{\mathrm{rat}}. \tag{4.458}$$

This is a change of notation for the same finite part, with no additional matching shift or normalization factor.

Consider the pair of letters

$$b_\pm = -1 \pm \sqrt{1-s}, \qquad \sqrt{1-s} = \mathrm{i}\sqrt{s-1}, \qquad s > 1. \tag{4.459}$$

Their complete contribution to the rational primitive is

$$F_{R_0}|_{b_\pm} = c_b(s) \sum_{b=b_+,b_-} [G(b,0;u) - G(b,-s;u) + G(b,-1;u)], \tag{4.460}$$

$$c_b(s) = \frac{4s\,(s^5+9s^4+116s^3+516s^2+307s-693)}{3(s-1)(s+3)^3}. \tag{4.461}$$

The independently integrated Bose-restoring primitive has exactly the opposite six coefficients:

$$F_D|_{b_\pm} = -\,F_{R_0}|_{b_\pm}\,, \qquad F_W|_{b_\pm} = F_{\log}|_{b_\pm} = 0. \tag{4.462}$$

Thus neither letter is needed in the combined primitive. This is a cancellation of identical functions with identical paths, so it introduces no change of branches or boundary constants.

The letters r and $-r$ cancel between the rational and W sectors. Define

$$Q_c(u) = G(c,0;u) + G(c,1;u) - G(c,s;u). \tag{4.463}$$

Their coefficients are

$$F_{R_0}|_{\pm r} = -\frac{16(r-1)^2}{r+1}Q_r(u) + \frac{16(r+1)^2}{r-1}Q_{-r}(u), \tag{4.464}$$

$$F_W|_{\pm r} = -\,F_{R_0}|_{\pm r}\,, \qquad F_{\log}|_{\pm r} = F_D|_{\pm r} = 0. \tag{4.465}$$

Combining all four sectors therefore removes twelve weight-two words with these four first letters. The remaining alphabet is

$$\left\{0,1,-1,s,-s,1+\sqrt{1-s},1-\sqrt{1-s},q_+,q_-\right\}, \qquad q_\pm = \frac{s-3\pm\sqrt{(s-1)(s-9)}}{2}. \tag{4.466}$$

The $q_\pm$ pair has the alternative representation derived above. It is not necessary to expand every remaining G function to make these cancellations explicit.

There are 44, 19, 14 and 28 nonzero GPL terms in the four separate primitive dictionaries. After collecting their coefficients, 37 distinct words remain. Three shuffle reductions leave 34 words and three products of weight-one functions, together with the rational term. The coefficient collection uses $H = s-1$ and $r^2 = s$ before identifying equal letters. Both the change of variables in each coefficient and the shuffle reconstruction are checked as exact algebraic identities. This collection precedes the finite endpoint operation. The lower-limit rational constants, the moving-boundary finite shifts, and the soft contribution must still be combined to obtain a compact total cross section.

The finite endpoint after collection. Let $F(u) = F_{R_0}(u) + F_W(u) + F_{\log}(u) + F_D(u)$ and $b = s-u$. The collected primitive has only two rational coefficients with a pole at $b=0$: its constant coefficient $c_\varnothing$ and its coefficient c_0 multiplying $G(0;u) = \ln(u/\sqrt{s})$. Their residues are

$$c_\varnothing(s-b) = \frac{d_0(s)}{b} + O(1), \qquad d_0(s) = \frac{8s(s^3+6s^2-s-2)}{3}, \tag{4.467}$$

$$c_0(s-b) = \frac{d_1(s)}{b} + O(1), \qquad d_1(s) = -\frac{32s^2(s-5)}{3(s-1)}. \tag{4.468}$$

All other function coefficients are regular at $u=s$. The product with the first-order expansion of $G(0;u)$ is essential:

$$G(0;s-b) = \tfrac{1}{2}\ln s - \frac{b}{s} + O(b^2). \tag{4.469}$$

It contributes the finite term

$$T_{\rm end}(s) = -\frac{d_1(s)}{s} = \frac{32s(s-5)}{3(s-1)}. \tag{4.470}$$

Thus substituting $u = s$ in the regular coefficients alone would miss a finite contribution. The coefficient of the double pole in the integrand is the coefficient of $1/b$ in its primitive:

$$D_4(s) = d_0(s) + \tfrac{1}{2} d_1(s) \ln s. \tag{4.471}$$

At the lower limit, every GPL vanishes. The coefficients of the products of two weight-one functions have at most a simple pole at $u = \sqrt{s}$, whereas the products vanish quadratically, so these terms also give zero. The rational term has the finite limit

$$F(\sqrt{s}) = -\frac{2(s+1)(s^4 - 3s^3 - 41s^2 + 27s - 8)}{3(s-1)}. \tag{4.472}$$

This combines the four lower limits before evaluating any polylogarithm.

To state the remaining change of cutoff, define $A_4(s)$ by the integrand expansion

$$F'(s-b) = \frac{D_4(s)}{b^2} + \frac{A_4(s)}{b} + O(\ln b). \tag{4.473}$$

In particular the primitive contains $D_4/b - A_4 \ln b$. Writing $\mathcal{E}_4(s)$ for its finite coefficient, including (4.470), the four-sector finite part in the original z convention is

$$F_4^z = \mathcal{E}_4(s) - F(\sqrt{s}) + A_4(s) \ln \frac{s-1}{s} - \frac{D_4(s)}{s(s-1)}. \tag{4.474}$$

Indeed, $z = (s-1)b/s - b^2/s^2 + O(b^3)$ gives

$$\frac{1}{b} = \frac{s-1}{sz} - \frac{1}{s(s-1)} + O(z), \qquad \ln b = \ln z + \ln \frac{s}{s-1} + O(z). \tag{4.475}$$

The second-order boundary relation supplies the last finite term in (4.474). The finite shifts from the z convention to the common real-emission convention are subsequently added as in (4.137).

Removing a conjugate pair at the endpoint. The remaining letters $a_\pm = 1 \pm \mathrm{i}\sqrt{s-1}$ occur only in the combination

$$\mathcal{A}_{\rm pair}(u) = \sum_{a=a_+,a_-} \big[G(a,0;u) + G(a,1;u) - G(a,s;u)\big]. \tag{4.476}$$

Their common coefficient in the four-sector primitive is

$$\gamma_a(s) = -\frac{32s(s^2-3s-1)}{(s-1)^2} \ln s - \frac{4(s^5 + s^4 - 25s^3 + 47s^2 - 12s + 12)}{3s(s-1)}. \tag{4.477}$$

It is shorter to integrate the summed logarithmic derivative than to evaluate the six complex-letter GPLs separately. Put

$$B(u) = \frac{u(u-1)}{s-u}, \qquad P(u) = u^2 - 2u + s = (u-a_+)(u-a_-). \tag{4.478}$$

For $\sqrt{s} < u < s$, $B(u) > 0$ and $B(\sqrt{s}) = 1$. The second-letter combination is exactly $\ln B(u)$, so

$$\mathcal{A}'_{\text{pair}}(u) = \frac{P'(u)}{P(u)} \ln B(u). \tag{4.479}$$

The factorization $P(u) = (s-u)(1+B(u))$ gives

$$\frac{P'(u)}{P(u)} = \frac{B'(u)}{1+B(u)} + \frac{1}{u-s}. \tag{4.480}$$

Since the derivative of $\ln B \ln(1+B) + \text{Li}_2(-B)$ is $\ln B\, B'/(1+B)$, the primitive is

$$\begin{aligned} \mathcal{A}_{\text{pair}}(u) = {} & \ln B(u) \ln(1+B(u)) + \text{Li}_2(-B(u)) + \frac{\pi^2}{12} \\ & + G(s,0;u) + G(s,1;u) - G(s,s;u). \end{aligned} \tag{4.481}$$

The constant follows from $\mathcal{A}_{\text{pair}}(\sqrt{s}) = 0$ and $\text{Li}_2(-1) = -\pi^2/12$. All arguments in the explicit dilogarithm term are negative real, fixing its branch throughout the integration interval.

The separate terms contain logarithms at the upper endpoint, but their sum has a finite limit. With $r = \sqrt{s}$ and $b = s - u$, the required expansions are

$$\ln B \ln(1+B) + \text{Li}_2(-B) = \tfrac{1}{2} \ln^2\!\left(\frac{s(s-1)}{b}\right) - \frac{\pi^2}{6} + O(b|\ln b|), \tag{4.482}$$

$$G(s,0;s-b) = \ln r \ln \frac{b}{s-r} + \text{Li}_2(1-1/r) + O(b), \tag{4.483}$$

$$G(s,1;s-b) = \ln(r+1) \ln \frac{b}{s-r} + \text{Li}_2\!\left(\frac{r}{r+1}\right) + O(b), \tag{4.484}$$

$$G(s,s;s-b) = \tfrac{1}{2} \ln^2 \frac{b}{s-r}. \tag{4.485}$$

Both powers of $\ln b$ cancel exactly. The limit first takes the form

$$\mathcal{A}_{\text{pair}}(s) = \text{Li}_2(1-1/r) + \text{Li}_2\!\left(\frac{r}{r+1}\right) + \tfrac{1}{2} \ln^2[r(r+1)] - \frac{\pi^2}{12}. \tag{4.486}$$

The reflection and inversion identities of the dilogarithm then give

$$\boxed{\mathcal{A}_{\text{pair}}(s) = \frac{\pi^2}{4} + \text{Li}_2(-1/\sqrt{s}) - \text{Li}_2(1/\sqrt{s}) + \frac{1}{2} \ln s \ln(s-1).} \tag{4.487}$$

For example, use $\text{Li}_2(r/(r+1)) = -\text{Li}_2(-r) - \frac{1}{2}\ln^2(r+1)$ and $\text{Li}_2(-r) + \text{Li}_2(-1/r) = -\pi^2/6 - \frac{1}{2}\ln^2 r$, together with the reflection identity for $\text{Li}_2(1-1/r)$. The final two dilogarithm arguments lie in $(-1,1)$ for every $s > 1$. Thus the complete contribution is $\gamma_a(s)\mathcal{A}_{\text{pair}}(s)$, and the six complex-letter endpoint values are unnecessary.

4.9.3 The matched real contribution

The four real sectors and their finite matching shifts can now be combined into a formula with two dilogarithms and one trilogarithm. Define

$$\mathcal{R}_4(s) = F^z_{\text{rat}} + F^z_W + F^z_{\log} + F^z_D + \Delta_{\text{rat}} + \Delta_{\log} + \Delta_D. \tag{4.488}$$

Here $s > 1$, $H = s - 1$ and $r = \sqrt{s}$. Use the real logarithms

$$L = \ln s, \qquad M = \ln(s-1), \qquad J = \ln \frac{s+1}{2}, \tag{4.489}$$

and the following three special-function combinations:

$$D(s) = \mathrm{Li}_2(1/s), \tag{4.490}$$

$$V(s) = \mathrm{Li}_2\left(\frac{2}{s+1}\right) + \ln 2\, J, \tag{4.491}$$

$$T(s) = \zeta(3) - \mathrm{Li}_3(1/s). \tag{4.492}$$

Both dilogarithm arguments and the trilogarithm argument lie strictly between zero and one. The complete matched expression is

$$\begin{aligned} \mathcal{R}_4(s) = {} & A_0 + A_1 L + A_2 L^2 + A_3 L^3 + \pi^2(B_0 + B_1 L) \\ & + (C_0 + C_1 L)D + KT + NV + EJ^2 + FJ \\ & + M(H_0 + H_1 L + H_2 L^2 + H_3 J). \end{aligned} \tag{4.493}$$

Every coefficient is a rational function of s. Explicitly,

$$A_0 = \frac{\begin{array}{c} 30s^7 - 198s^6 - 785s^5 + 4079s^4 \\ - 22927s^3 + 22309s^2 - 10692s + 4428 \end{array}}{108sH}. \tag{4.494}$$

$$A_1 = \frac{\begin{array}{c} 39s^7 - 333s^6 + 174s^5 + 10076s^4 \\ + 6798s^3 - 116s^2 + 222s + 36 \end{array}}{9sH^3}. \tag{4.495}$$

$$A_2 = \frac{\begin{array}{c} -5s^7 + 77s^6 + 102s^5 - 630s^4 \\ - 641s^3 - 151s^2 + 132s - 36 \end{array}}{3sH^3}. \tag{4.496}$$

$$A_3 = -\frac{8s\left(3s^2 - 25s - 2\right)}{3H^2}. \tag{4.497}$$

$$B_0 = \frac{2\left(s^4 - 17s^3 + 25s^2 + 219s - 12\right)}{9H}. \tag{4.498}$$

$$B_1 = -\frac{8s\left(5s+1\right)}{3H^2}. \tag{4.499}$$

$$C_0 = -\frac{4\left(s^4 - 17s^3 - 39s^2 - 101s - 12\right)}{3H}. \tag{4.500}$$

$$C_1 = \frac{16s}{s-1}. \tag{4.501}$$

$$K = \frac{32s\left(2s+1\right)}{H^2}. \tag{4.502}$$

$$N = -\frac{256s\left(s+5\right)}{3H}. \tag{4.503}$$

$$E = -\frac{128s\left(2s^3 - 5s^2 - 24s + 3\right)}{3H^3}. \tag{4.504}$$

$$F = \frac{256s\,(s+1)\,(s^2-19s-4)}{3H^3}. \tag{4.505}$$

$$H_0 = \frac{\begin{array}{r}-2s^7+2s^6+64s^5-260s^4\\ -314s^3-2638s^2+220s-144\end{array}}{3sH}. \tag{4.506}$$

$$H_1 = \frac{\begin{array}{r}-8s^7+80s^6+24s^5-1800s^4\\ -2336s^3-2248s^2+240s-96\end{array}}{3sH^3}. \tag{4.507}$$

$$H_2 = -\frac{16s\,(2s^2-9s-2)}{H^2}. \tag{4.508}$$

$$H_3 = -\frac{256s\,(s^3-19s^2-21s-9)}{3H^3}. \tag{4.509}$$

This is the unnormalized real finite contribution. Its relation to the remainder in (4.5) is

$$\mathcal{R}_{\rm rem}(X) = \mathcal{R}_4(1+X) + 2F^{\eta}_{UW}(X). \tag{4.510}$$

Thus (4.493) replaces the four separate endpoint evaluations and their three matching shifts. The UV term is already included with the virtual contribution in that master formula. The remaining UW and finite soft terms are separate.

Reduction of the function basis. After the endpoint identities above, real reflection and inversion relations put the dilogarithm arguments in $(0,1)$. Two five-term identities reduce their number further:

$$\mathrm{Li}_2\!\left(\frac{2}{r+1}\right) + \mathrm{Li}_2\!\left(\frac{r+1}{2r}\right) = \mathrm{Li}_2(1/r) + \mathrm{Li}_2\!\left(\frac{1}{r+1}\right) + \frac{\pi^2}{12} - \frac{1}{2}\ln^2 2 + \ln 2 \ln\frac{r+1}{r}. \tag{4.511}$$

The same identity with r replaced by s reduces the half-argument function in the matching shift. For the second identity, define $\mathcal{L}(z) = \mathrm{Li}_2(z) + \frac{1}{2}\ln z \ln(1-z)$ for $0 < z < 1$. Then

$$\mathcal{L}\!\left(\frac{r+1}{r^2+1}\right) - \mathcal{L}\!\left(\frac{r-1}{r(r+1)}\right) = \mathcal{L}(1/r) + \mathcal{L}\!\left(\frac{1}{r^2+1}\right) - \mathcal{L}\!\left(\frac{1}{r+1}\right). \tag{4.512}$$

Both identities follow by differentiation using $\mathrm{d}\,\mathrm{Li}_2(z)/\mathrm{d}z = -\ln(1-z)/z$, with the integration constant fixed as $r \to \infty$. The remaining square-root arguments occur in a common-coefficient difference. Landen transformation and duplication give

$$\mathrm{Li}_2\!\left(\frac{1}{r+1}\right) - \mathrm{Li}_2(1/r) = -\frac{1}{2}\,\mathrm{Li}_2(1/r^2) - \frac{1}{2}\ln^2\!\left(\frac{r+1}{r}\right). \tag{4.513}$$

After this substitution, all coefficients of $\ln(r+1)$ cancel exactly when $\ln(r-1) = \ln(s-1) - \ln(r+1)$ is used. The result therefore has no square-root argument. Adding the matching shift also cancels the entire coefficient of $\mathrm{Li}_2(1/(s+1))$. The matching shifts are reduced with the explicit identities

$$\mathrm{Li}_2(1-s) = D - LM + \tfrac{1}{2}L^2 - \frac{\pi^2}{6}, \tag{4.514}$$

$$\mathrm{Li}_2\!\left(-\frac{(s-1)^2}{4s}\right) = 4\,\mathrm{Li}_2\!\left(\frac{2}{s+1}\right) - 2D - 2\,\mathrm{Li}_2\!\left(\frac{1}{s+1}\right) \tag{4.515}$$

$$+ J^2 + 2J(L+\ln 2) - \tfrac{3}{2}L^2 + 2M(L-2J) - \frac{\pi^2}{6} - \ln^2 2. \tag{4.516}$$

For the second line, put $v = (s-1)/(s+1)$, apply Landen transformation to the negative argument, and then use $\mathrm{Li}_2(v^2) = 2\,\mathrm{Li}_2(v) + 2\,\mathrm{Li}_2(-v)$. Reflection and the preceding half-argument identity give the displayed expression. All intermediate arguments used in these identities are real and their logarithms have the signs fixed by $s > 1$. Equivalently, denote the two right-hand sides by $Q_1(s)$ and $Q_2(s)$. Differentiation and the positive-factor logarithm identities for $s > 1$ give

$$Q_1'(s) = -\frac{\ln s}{s-1}, \qquad Q_2'(s) = -\frac{2(s+1)}{s(s-1)} \ln \frac{s+1}{2\sqrt{s}}. \tag{4.517}$$

These are precisely the derivatives of the two left-hand sides. At $s \to 1^+$ both sides vanish. Here the coefficient of $M = \ln(s-1)$ vanishes at least linearly, and $\mathrm{Li}_2(1/2) = \pi^2/12 - \ln^2 2/2$ fixes the remaining constant. Equality of derivatives and this boundary value establish the identities on the connected domain $s > 1$.

4.9.4 The integrated UW contribution

Put $H = X > 0$ and use Y and q defined at the beginning of this chapter. The completed UW integral can be expressed in the same functions as the virtual contribution. Define the real logarithms

$$L_0 = \ln X, \qquad L_1 = \ln(1+X), \qquad L_2 = \ln(1+X/2). \tag{4.518}$$

In the following formula every G has endpoint X. Its rational letters have kernels $\mathrm{d}z/(z-a)$, and the letter r has kernel $\mathrm{d}z/\sqrt{z(z+4)}$, with positive square root for $z > 0$. Define

$$\mathcal{U}(X) = G(-4, r, -1; X) + G(r, -1, -1; X) - G(r, 0, -1; X). \tag{4.519}$$

The unnormalized finite contribution is

$$\begin{aligned} F^{\eta}_{UW}(X) = {} & a_0 + a_1 L_1 + a_L L_0 + a_{11} L_1^2 + a_{10} L_0 L_1 \\ & + a_D\,\mathrm{Li}_2(-X) + a_2 L_2 + a_{D2}\,\mathrm{Li}_2(-X/2) \\ & + a_{20}(L_0 L_2 + L_2^2/4) + (a_R + 2K L_0) G(r, -1; X) \\ & + K\mathcal{U}(X) + a_{011} G(0, -1, -1; X) + a_{RR} G(r, r, -1; X). \end{aligned} \tag{4.520}$$

Its contribution to C_1 is $F^{\eta}_{UW}/[4X(1+X)]$, including the second Bose-related sector. The coefficients are as follows.

$$a_0 = \frac{\begin{array}{c} 497X^6 - 1647X^5 + 46X^4 + 51270X^3 \\ + 18924X^2 - 46080X - 18432 \end{array}}{216X^3}. \tag{4.521}$$

$$a_1 = \frac{\begin{array}{l} -24X^8 + 24X^7 + 1548X^6 - 7712X^5 \\ -91001X^4 - 208525X^3 - 177316X^2 - 44544X \\ +6144 \end{array}}{18X^4\,(X+4)}. \tag{4.522}$$

$$a_L = \frac{\begin{array}{c} -12X^6 - 24X^5 + 48X^4 + 2912X^3 \\ + 6656X^2 + 5376X + 1536 \end{array}}{9X^3}. \tag{4.523}$$

$$a_{11} = \frac{\begin{array}{c} X^8 - 3X^7 - 106X^6 + 210X^5 \\ + 3762X^4 + 8930X^3 + 7656X^2 + 1664X \\ - 512 \end{array}}{3X^4\,(X+4)}. \tag{4.524}$$

$$a_{10} = \frac{\begin{array}{c} 4X^8 - 12X^7 - 352X^6 + 144X^5 \\ + 8160X^4 + 21248X^3 + 19200X^2 + 3584X \\ - 2048 \end{array}}{3X^4\,(X+4)}. \tag{4.525}$$

$$a_D = \frac{\begin{array}{c} 2X^6 - 6X^5 - 224X^4 - 632X^3 \\ + 352X^2 + 1920X + 1152 \end{array}}{3X^2\,(X+4)}. \tag{4.526}$$

$$a_2 = -\frac{128\,(X+1)\,(X+2)\,\left(X^2 - 17X - 22\right)}{3X^3}. \tag{4.527}$$

$$a_{D2} = \frac{128\,(X+1)\,(X+6)}{3X}. \tag{4.528}$$

$$a_{20} = \frac{256\,(X+1)\,\left(X^3 - 5X^2 - 28X - 24\right)}{3X^3}. \tag{4.529}$$

$$a_R = \frac{\begin{array}{c} - 24X^5Y - 72X^4Y + 528X^3Y + 2112X^2Y \\ + 768XY - 768Y \end{array}}{X^3\,(X+4)}. \tag{4.530}$$

$$K = \frac{\begin{array}{c} 24X^5Y + 72X^4Y - 528X^3Y - 2496X^2Y \\ - 2688XY - 768Y \end{array}}{X^3\,(X+4)}. \tag{4.531}$$

$$a_{011} = -\frac{16\,(X+1)\,(7X+3)}{X^2}. \tag{4.532}$$

$$a_{RR} = -\frac{24\,(X+1)\,\left(X^2 - 2X - 12\right)}{X^2}. \tag{4.533}$$

Integration and endpoint constants. Starting from the eight-function kernel above, rational integration by parts gives

$$\Phi_{UW}(t) = \frac{\mathrm{d}}{\mathrm{d}t}\sum_i Q_i(t) f_i(t) + \sum_i T_i(t) f_i(t). \tag{4.534}$$

Here Q_i are rational functions, and T_i have only simple poles. The identity follows by substituting the derivatives of the eight functions and matching their rational coefficients. At each endpoint the negative powers in the Laurent expansion of the sum $\sum_i Q_i f_i$ cancel. Its constant term must include products of the poles of Q_i with the Taylor coefficients of f_i. Thus the integrated boundary is the difference of the two constant terms, not the difference of the constant coefficients of the rational functions alone. The simple-pole terms integrate to GPLs with endpoint $t = 1$. Set $X = (1-q)^2/q$ and $\delta = 1 - q$, with $0 < q < 1$. The moving letters are

$$a = -\frac{q}{(1-q)^2}, \qquad c = -\frac{2q}{(1-q)^2}, \qquad r_- = -\frac{q}{1-q}, \qquad r_+ = \frac{1}{1-q}. \tag{4.535}$$

They lie outside the interval $[0,1]$. Together with the fixed letters $0,1$, they specify the endpoint functions whose constants are needed.

For a word w, define $g_w(q)$ as the constant coefficient in the $t \to 1^-$ expansion of $G(w;t)$ in powers of $\ln(1-t)$. The endpoint is regularized at fixed q before taking $q \to 1$. The remaining terms vanish as $(1-t)$ times powers of its logarithm, so this constant-term operation respects shuffle products. In particular, $g_\varnothing = 1$, $g_0 = g_1 = 0$, and all nonempty pure-zero or pure-one words have value zero. The nonzero fixed-letter values through weight three are

$$g_{0,1} = -\zeta(2), \qquad g_{1,0} = \zeta(2), \tag{4.536}$$
$$g_{0,0,1} = -\zeta(3), \qquad g_{0,1,0} = 2\zeta(3), \tag{4.537}$$
$$g_{1,0,0} = -\zeta(3), \qquad g_{0,1,1} = \zeta(3), \tag{4.538}$$
$$g_{1,0,1} = -2\zeta(3), \qquad g_{1,1,0} = \zeta(3). \tag{4.539}$$

Three convergent integrals fix this table:

$$G(0,1;1) = \int_0^1 \frac{\ln(1-t)}{t}\,\mathrm{d}t = -\zeta(2), \tag{4.540}$$
$$G(0,0,1;1) = -\operatorname{Li}_3(1) = -\zeta(3), \tag{4.541}$$
$$G(0,1,1;1) = \frac{1}{2}\int_0^1 \frac{\ln^2(1-t)}{t}\,\mathrm{d}t = \frac{1}{2}\int_0^1 \frac{\ln^2 u}{1-u}\,\mathrm{d}u = \zeta(3). \tag{4.542}$$

The first two follow from the convergent polylogarithm series. For the third, expanding $1/(1-u) = \sum_{n=0}^{\infty} u^n$ and using $\int_0^1 u^n \ln^2 u\,\mathrm{d}u = 2/(n+1)^3$ gives the stated value. The remaining entries follow from the regularized shuffle relations

$$0 = g_{0,1} + g_{1,0}, \tag{4.543}$$
$$0 = 2g_{0,0,1} + g_{0,1,0}, \tag{4.544}$$
$$0 = g_{1,0,0} + g_{0,1,0} + g_{0,0,1}, \tag{4.545}$$
$$0 = g_{1,0,1} + 2g_{0,1,1}, \tag{4.546}$$
$$0 = g_{0,1,1} + g_{1,0,1} + g_{1,1,0}. \tag{4.547}$$

These are the products $g_0 g_1$, $g_0 g_{0,1}$, $g_1 g_{0,0}$, $g_1 g_{0,1}$ and $g_0 g_{1,1}$, respectively.

The words containing a moving letter have zero boundary value as $q \to 1$. To establish this before interchanging limits, first use shuffle products to remove initial letters 1 and terminal letters 0. The factors g_1 and g_0 vanish. Every surviving word therefore starts with a letter other than 1 and ends with a letter other than 0, and still contains each moving letter of the original word. Its ordered integral is absolutely convergent at both endpoints. For $0 < \delta < 1/4$, each moving letter α satisfies

$$\left|\frac{1}{t-\alpha(q)}\right| \le \frac{2}{|\alpha(q)|}, \qquad 0 \le t \le 1. \tag{4.548}$$

Replacing the moving kernels by these bounds leaves only the fixed kernels $1/t$, $1/(1-t)$ and the constant kernel in the ordered integral. The endpoint conditions just established make this integral finite. Consequently, for each surviving word w,

$$|G(w;1)| \le C_w \prod_{j\ \text{moving}} \frac{2}{|\alpha_j(q)|}, \qquad C_w < \infty, \tag{4.549}$$

where C_w is independent of q. The factors $1/a$ and $1/c$ are $O(\delta^2)$, while $1/r_\pm = O(\delta)$. The bound therefore tends to zero, and so does the finite shuffle combination defining g_w. This

proves the moving-letter boundary value through every weight required here. It applies to each special function before multiplication by possibly singular rational energy coefficients.

Differentiating the endpoint GPLs by the total-differential recursion of Section 4.2 now gives logarithmic differentials in a fixed alphabet with the explicitly fixed constants above. Add the complete Hermite boundary difference, including its Laurent–Taylor cross terms, before returning to the X kernels. The result contains eighteen words with rational coefficients in X and Y. Exact substitution of the kernel maps reproduces every coefficient of the fixed-alphabet expression.

Collection of the functions. The shuffle identity gives

$$G(0,r,-1;X) = L_0 G(r,-1;X) - G(r,0,-1;X) - G(r,-1,0;X). \tag{4.550}$$

The five root words occur with coefficients proportional to $1,2,1,2,1$, in the order $(-4,r,-1)$, $(0,r,-1)$, $(r,-1,-1)$, $(r,-1,0)$, $(r,0,-1)$. Applying this identity reduces their sum to $K[\mathcal{U} + 2L_0 G(r,-1;X)]$. For the rational letters, use

$$G(-1,0;X) = L_0 L_1 + \mathrm{Li}_2(-X), \tag{4.551}$$

$$G(-2,0;X) = L_0 L_2 + \mathrm{Li}_2(-X/2), \tag{4.552}$$

$$G(-j,-j;X) = \tfrac{1}{2}\ln^2(1+X/j), \qquad j = 1,2. \tag{4.553}$$

These are shuffle identities and the defining dilogarithm integral, so no continuation across a cut is needed. All kernels and logarithms in the final expression are real for $X > 0$.

4.10 The collected total in one function basis

All contributions can now be combined in the same energy variable. This section gives the complete coefficient of $\alpha_{\mathrm{T}}^3/m^2$, including the virtual correction and the integrated two-photon final state. It is obtained from the independently evaluated sectors above. Define $X = s_{\mathrm{phys}}/m^2 - 1 > 0$ and $Y = \sqrt{X/(X+4)}$. Use the GPL kernels and real branches in Section 4.2, including the removable combinations containing the letter 1. The complete result is

$$\sigma_{\mathrm{inc}}^{\mathrm{NLO}} = \frac{\pi\alpha_{\mathrm{T}}^2}{m^2} B(X) + \frac{\alpha_{\mathrm{T}}^3}{m^2} C_1(X) + \frac{1}{m^2} O(\alpha_{\mathrm{T}}^4), \tag{4.554}$$

where B is given in (4.3) and

$$\boxed{C_1(X) = \sum_{j=0}^{22} c_j(X,Y) F_j(X).} \tag{4.555}$$

The functions and their full coefficients follow. No unevaluated phase-space or Feynman-parameter integral occurs in this expression.

$$F_0 = 1, \qquad c_0 = \frac{-273X^3 + 982X^2 + 2960X + 1744}{24X^2\,(X+1)^2}. \tag{4.556}$$

$$F_1 = G(-1;X), \qquad c_1 = \frac{37X^4 - 54X^3 - 339X^2 - 428X - 184}{4X^3\,(X+1)^2}. \tag{4.557}$$

$$F_2 = G(0;X), \qquad c_2 = \frac{-14X^4 - 17X^3 + 17X^2 + 22X + 8}{2X\,(X-1)\,(X+1)^3}. \tag{4.558}$$

$$F_3 = G(-1,-1;X), \qquad c_3 = \frac{\begin{array}{c}-4X^6-35X^5+31X^4+755X^3\\+1765X^2+1506X+440\end{array}}{X^3\left(X+1\right)^2\left(X+4\right)}. \tag{4.559}$$

$$F_4 = G(-1,0;X), \qquad c_4 = \frac{X^4+7X^3+X^2-3X-2}{X^3\left(X+1\right)^2}. \tag{4.560}$$

$$F_5 = G(-2,-1;X), \qquad c_5 = \frac{\begin{array}{c}-4X^5-104X^4-584X^3-1264X^2\\-1152X-384\end{array}}{X^3\left(X+1\right)^2\left(X+4\right)}. \tag{4.561}$$

$$F_6 = G(0,-1;X), \qquad c_6 = \frac{\begin{array}{c}X^6+8X^5-17X^4-210X^3\\-448X^2-352X-96\end{array}}{X^3\left(X+1\right)^2\left(X+4\right)}. \tag{4.562}$$

$$F_7 = G(1,0;X), \qquad c_7 = \frac{-X^2+X-2}{X^3}. \tag{4.563}$$

$$F_8 = G(r,-1;X), \qquad c_8 = \frac{Y\left(3X^4+18X^3+44X^2-8X-64\right)}{X^4}. \tag{4.564}$$

$$F_9 = G(-1,-1,-1;X), \qquad c_9 = \frac{-2X^2+32X+46}{X^3}. \tag{4.565}$$

$$F_{10} = G(-1,-1,0;X), \qquad c_{10} = \frac{\left(2X+3\right)\left(4X-8\right)}{X^3}. \tag{4.566}$$

$$F_{11} = G(-1,-2,-1;X), \qquad c_{11} = \frac{8X^2-32X-48}{X^3}. \tag{4.567}$$

$$F_{12} = G(-1,0,-1;X), \qquad c_{12} = \frac{4X^2-14X-24}{X^3}. \tag{4.568}$$

$$F_{13} = G(-4,r,-1;X), \qquad c_{13} = -\frac{6Y\left(X^3-12X-8\right)}{X^4}. \tag{4.569}$$

$$F_{14} = G(0,-1,-1;X), \qquad c_{14} = \frac{-5X^2-32X+8}{X^3}. \tag{4.570}$$

$$F_{15} = G(0,-1,0;X), \qquad c_{15} = \frac{2X^2+4X-12}{X^3}. \tag{4.571}$$

$$F_{16} = G(0,1,0;X), \qquad c_{16} = \frac{2X^2+4X-12}{X^3}. \tag{4.572}$$

$$F_{17} = G(0,r,-1;X), \qquad c_{17} = -\frac{3Y\left(X-2\right)\left(X+4\right)}{X^3}. \tag{4.573}$$

$$F_{18} = G(r,-1,-1;X), \qquad c_{18} = \frac{3Y\left(5X^4+14X^3-96X^2-352X-128\right)}{X^4\left(X+4\right)}. \tag{4.574}$$

$$F_{19} = G(r,-1,0;X), \qquad c_{19} = -\frac{6Y\left(X-2\right)\left(X+4\right)}{X^3}. \tag{4.575}$$

$$F_{20} = G(r,-2,-1;X), \qquad c_{20} = -\frac{16Y\left(X^4+2X^3-24X^2-80X-48\right)}{X^4\left(X+4\right)}. \tag{4.576}$$

$$F_{21} = G(r,0,-1;X), \qquad c_{21} = -\frac{8Y\left(X^4+3X^3-18X^2-68X-24\right)}{X^4\left(X+4\right)}. \tag{4.577}$$

$$F_{22} = G(r,r,-1;X), \qquad c_{22} = \frac{9X^2-24}{X^3}. \tag{4.578}$$

Collection of the real terms. Put $L = \ln(1+X)$, $M = \ln X$, and $J = \ln(1+X/2)$. The classical functions in the four-sector real expression satisfy

$$\mathrm{Li}_2\!\left(\frac{1}{1+X}\right) = \zeta(2) + G(-1,0;X) - G(-1,-1;X), \tag{4.579}$$

$$\mathrm{Li}_2\!\left(\frac{2}{2+X}\right) + \ln 2\, J = \zeta(2) + G(-2,0;X) - G(-2,-2;X), \tag{4.580}$$

$$\zeta(3) - \mathrm{Li}_3\!\left(\frac{1}{1+X}\right) = \zeta(2)G(-1;X) + G(-1,-1,0;X) - G(-1,-1,-1;X). \tag{4.581}$$

Each identity follows by differentiating with respect to X. The constants are fixed as $X \to 0^+$, where the dilogarithms and trilogarithm take their values at one. Every nonconstant GPL on these right-hand sides vanishes in this limit. In particular, the logarithmic lower endpoint in $G(-1,0;X)$ and $G(-2,0;X)$ is integrable because $\int_0^X |\ln t|\,\mathrm{d}t = O(X[1 + |\ln X|])$. The remaining products of L, M, and J are expanded with the shuffle identity, counting each interleaving even when repeated letters give the same word. All classical arguments above lie in $(0,1)$ for $X > 0$, so this conversion introduces no continuation across a cut.

Assembly. Substitute these identities, the completed UW expression, and (4.197) into (4.5). The finite virtual coefficient is its constant term in $\ell_\lambda = \ln(\lambda_{\mathrm{phys}}/m)$ after the previously proved infrared cancellation. Taking this coefficient does not set the photon mass to m. The finite conversion term remains $-A(1+X)[\ln X + \frac{1}{2}\ln(1+X)]$. Expanding this product gives the last coefficients needed in the sum above. There are five contributions to this collection: $\widehat{C}_{V+\mathrm{UV}}$, $F^{\eta}_{UW}/[4X(1+X)]$, $\mathcal{R}_4(1+X)/[8X(1+X)]$, $-\mathcal{C}_K(1+X)/(1+X)$, and the finite conversion term. The factor $1/4$ for UW follows from $\mathcal{R}_{\mathrm{rem}} = \mathcal{R}_4 + 2F^{\eta}_{UW}$, so its Bose-related second copy is already included. Expand each contribution in the same words and add their rational coefficients. This is an algebraic reconstruction and does not require a claim that the functions are linearly independent. The infrared coefficient used in the last contribution is (4.71), with $L^2 = 2G(-1,-1;X)$ and $\mathcal{G}(X) = G(r,-1;X)$. The coefficients of five words vanish identically in the complete sum. In particular the coefficient of $G(-1,r,-1;X)$ from the finite soft term cancels its coefficient from the infrared conversion term. All coefficients proportional to π^2 outside the GPLs also cancel. The remaining coefficients are the rational functions listed above.

The apparent point $X = 1$. The factor $1/(X-1)$ multiplies $G(0;X) = \ln X$, so it has a removable limit. The product involving this coefficient tends to $+1$. The two GPLs containing the rational letter $+1$ are also real and finite:

$$G(1,0;X) = \ln X \ln|1-X| + \mathrm{Re}\,\mathrm{Li}_2(X), \tag{4.582}$$

$$G(0,1,0;X) = 2\,\mathrm{Re}\,\mathrm{Li}_3(X) - \ln X\,\mathrm{Re}\,\mathrm{Li}_2(X). \tag{4.583}$$

Their derivatives are $\ln X/(X-1)$ and $G(1,0;X)/X$. At $X = 1$ their values are $\zeta(2)$ and $2\zeta(3)$. Thus the apparent inner pole does not require a contour choice for the physical answer.

4.10.1 A product form of the total

Products of elementary logarithms can make the collected result shorter without introducing additional polylogarithms. Define $L = \ln(1+X)$ and $M = \ln X$. The two GPLs with the

same coefficient combine into

$$\mathcal{P}(X) = G(0,-1,0;X) + G(0,1,0;X) = \tfrac{1}{2}\left[\operatorname{Re}\operatorname{Li}_3(X^2) - M\operatorname{Re}\operatorname{Li}_2(X^2)\right]. \tag{4.584}$$

For the derivation, first take $0 < X < 1$. The shuffle relation $G(0)G(0,a) = 2G(0,0,a) + G(0,a,0)$, with $G(0,a;X) = -\operatorname{Li}_2(X/a)$ and $G(0,0,a;X) = -\operatorname{Li}_3(X/a)$, gives

$$G(0,a,0;X) = 2\operatorname{Li}_3(X/a) - M\operatorname{Li}_2(X/a), \qquad a = \pm 1. \tag{4.585}$$

Summing the two signs and using $\operatorname{Li}_n(z) + \operatorname{Li}_n(-z) = 2^{1-n}\operatorname{Li}_n(z^2)$ gives the stated result. Both shuffle multiplicities and this duplication factor are needed for its factor $1/2$. For $X > 1$, continue all functions on the same edge of the principal cut. On either edge, the imaginary parts of $\operatorname{Li}_2(X)$ and $\operatorname{Li}_3(X)$ are respectively $\pm\pi\ln X$ and $\pm\pi\ln^2 X/2$, with the same sign. These follow by integrating their derivatives from $X = 1$, and they cancel in $2\operatorname{Li}_3(X) - M\operatorname{Li}_2(X)$. The analogous cancellation holds for the paired expression at argument X^2. Thus both edges give the same real function and the continuation adds no branch constant. The following equivalent form uses only real polylogarithms with arguments in $(0,1]$:

$$\mathcal{P}(X) = \begin{cases} \tfrac{1}{2}[\operatorname{Li}_3(X^2) - M\operatorname{Li}_2(X^2)], & 0 < X \le 1, \\ \tfrac{1}{2}[\operatorname{Li}_3(X^{-2}) + M\operatorname{Li}_2(X^{-2})] + \tfrac{1}{3}M^3 + \tfrac{1}{6}\pi^2 M, & X \ge 1. \end{cases} \tag{4.586}$$

To derive the second line, put $t = \ln X > 0$, $D(t) = \operatorname{Li}_2(e^{-2t})$ and $T(t) = \operatorname{Li}_3(e^{-2t})$. Their derivatives are $D'(t) = 2\ln(1-e^{-2t})$ and $T'(t) = -2D(t)$. The identity $\ln(e^{2t}-1) = 2t + \ln(1-e^{-2t})$ then gives

$$\operatorname{Re}\operatorname{Li}_2(e^{2t}) = \frac{\pi^2}{3} - 2t^2 - D(t), \tag{4.587}$$

$$\operatorname{Re}\operatorname{Li}_3(e^{2t}) = T(t) - \frac{4}{3}t^3 + \frac{2\pi^2}{3}t. \tag{4.588}$$

Indeed, the derivatives of the two sides are respectively $-2\ln(e^{2t}-1)$ and $2\operatorname{Re}\operatorname{Li}_2(e^{2t})$. Write $\zeta(n) = \sum_{k=1}^{\infty} k^{-n}$ for the Riemann zeta function at integers $n > 1$. Their values at $t = 0$ agree because $\operatorname{Li}_2(1) = \pi^2/6$ and $\operatorname{Li}_3(1) = \zeta(3)$. These derivative and boundary conditions fix both identities and give $\mathcal{P}(1) = \zeta(3)/2$. The sole apparent rational pole at $X = 1$ can also be removed before evaluation. Define the continuous elementary function

$$R_{\log}(X) = \begin{cases} \ln X/(X-1), & X \ne 1, \\ 1, & X = 1, \end{cases} \qquad R_{\log}(1+h) = 1 - \frac{h}{2} + \frac{h^2}{3} + O(h^3). \tag{4.589}$$

The series follows by dividing the Taylor series of $\ln(1+h)$ by h, which converges for $|h| < 1$. Extracting the common power X^{-3} gives the exact total coefficient as

$$C_1(X) = \frac{1}{X^3}\left[R_{\log}(X) + \mathcal{E}(X,L,M) + \sum_{j=1}^{13} b_j(X,Y,L,M)\mathcal{F}_j(X)\right]. \tag{4.590}$$

The elementary part is

$$\mathcal{E} = e_0 L^3 + e_1 L^2 M + e_2 L^2 + e_3 LM + e_4 L + e_5 M + e_6. \tag{4.591}$$

$$e_0 = -\frac{X^2 - 16X - 23}{3}. \tag{4.592}$$

$$e_1 = 4X^2 - 2X - 12. \tag{4.593}$$

$$e_2 = -2X^3 - \frac{11X^2}{2} + \frac{133X}{2} + 36 + \frac{96}{X+4} - \frac{6}{X+1} + \frac{1}{(X+1)^2}. \tag{4.594}$$

$$e_3 = X^2 + 5X - 10 + \frac{12}{X+1} - \frac{4}{(X+1)^2}. \tag{4.595}$$

$$e_4 = \frac{37X^2}{4} - 32X - 30 - \frac{15}{X+1} - \frac{1}{(X+1)^2}. \tag{4.596}$$

$$e_5 = -\frac{14X^5 + 31X^4 + 14X^3 - 6X^2 - 8X - 2}{2(X+1)^3}. \tag{4.597}$$

$$e_6 = -\frac{X\left(273X^3 - 982X^2 - 2960X - 1744\right)}{24(X+1)^2}. \tag{4.598}$$

The thirteen remaining functions and their coefficients are

$$\mathcal{F}_1 = G(0, -1; X). \tag{4.599}$$

$$b_1 = X^3 + X^2 - 43X + 6 - \frac{96}{X+4} - \frac{6}{X+1} + \frac{2}{(X+1)^2} - 2LX(2X+5). \tag{4.600}$$

$$\mathcal{F}_2 = G(-2, -1; X). \tag{4.601}$$

$$b_2 = -4X^2 - 80X - 68 - \frac{128}{X+4} + \frac{8}{X+1} - \frac{4}{(X+1)^2}. \tag{4.602}$$

$$\mathcal{F}_3 = G(1, 0; X), \qquad b_3 = -X^2 + X - 2. \tag{4.603}$$

$$\mathcal{F}_4 = G(r, -1; X). \tag{4.604}$$

$$b_4 = Y\left(3X^3 + 18X^2 + 44X - 8 - \frac{64}{X}\right) - 3MY(X-2)(X+4). \tag{4.605}$$

$$\mathcal{F}_5 = G(0, -1, -1; X), \qquad b_5 = -5X^2 - 8X + 32. \tag{4.606}$$

$$\mathcal{F}_6 = G(-1, -2, -1; X), \qquad b_6 = 8\left(X^2 - 4X - 6\right). \tag{4.607}$$

$$\mathcal{F}_7 = G(-4, r, -1; X), \qquad b_7 = Y\left(-6X^2 + 72 + \frac{48}{X}\right). \tag{4.608}$$

$$\mathcal{F}_8 = \mathcal{P}(X), \qquad b_8 = 2X^2 + 4X - 12. \tag{4.609}$$

$$\mathcal{F}_9 = G(r, 0, -1; X), \qquad b_9 = Y\left(-5X^2 + 14X + 88 + \frac{48}{X+4} + \frac{48}{X}\right). \tag{4.610}$$

$$\mathcal{F}_{10} = G(r, -1, 0; X), \qquad b_{10} = -3Y(X-2)(X+4). \tag{4.611}$$

$$\mathcal{F}_{11} = G(r, -1, -1; X), \qquad b_{11} = Y\left(15X^2 - 18X - 216 - \frac{96}{X+4} - \frac{96}{X}\right). \tag{4.612}$$

$$\mathcal{F}_{12} = G(r, -2, -1; X), \qquad b_{12} = Y\left(-16X^2 + 32X + 256 + \frac{64}{X+4} + \frac{192}{X}\right). \tag{4.613}$$

$$\mathcal{F}_{13} = G(r, r, -1; X), \qquad b_{13} = 9X^2 - 24. \tag{4.614}$$

This form follows from the complete coefficient table using only the identities

$$G(-1,0) = LM - G(0,-1), \tag{4.615}$$
$$G(-1,0,-1) = L\,G(0,-1) - 2G(0,-1,-1), \tag{4.616}$$
$$G(-1,-1,0) = \tfrac{1}{2}ML^2 - L\,G(0,-1) + G(0,-1,-1), \tag{4.617}$$
$$G(0,r,-1) = M\,G(r,-1) - G(r,0,-1) - G(r,-1,0). \tag{4.618}$$

The common endpoint is X. In addition, repeated letters -1 give powers of L divided by factorials. Expanding every product back by the shuffle identity reproduces all twenty-three original coefficients exactly. The two words combined into $\mathcal{P}$ have the same coefficient $c_{15} = c_{16}$, so this last grouping replaces their sum without another factor of two. The extraction of $R_{\log}$ is fixed by the coefficient c_2 of M in the collected table. Direct polynomial division gives

$$X^3 c_2(X,Y) = \frac{1}{X-1} + e_5(X), \qquad X^3 c_2(X,Y)M = R_{\log}(X) + e_5(X)M. \tag{4.619}$$

The second equality at $X = 1$ means its continuous limit. All other coefficients are multiplied by X^3 before the common factor X^{-3} is placed outside the bracket. Thus this extraction changes neither the normalization of C_1 nor the coefficient of its removable term. Every displayed rational coefficient is regular at $X = 1$, as follows by exact polynomial cancellation. The function $G(1,0;X) = \zeta(2) - \mathrm{Li}_2(1-X)$ is also explicitly real and regular for every $X > 0$. Its derivative is $\ln X/(X-1)$ and its limit as $X \to 0^+$ is zero, which fixes the identity without numerical continuation.

Chapter 5

Asymptotic expansions

5.1 Limits of the independent virtual contribution

This section and the following soft and real-radiation sections expand the separate contributions to the same total cross section. None of the separate contributions is the physical total. They are combined with the finite regulator conversion in (3.672), so that the photon-mass logarithm cancels before a limit of the total is interpreted. Define

$$X = \frac{s_{\mathrm{phys}}}{m^2} - 1 = \frac{2\omega}{m}, \qquad L = \ln X, \qquad \ell_\lambda = \ln \frac{\lambda_{\mathrm{phys}}}{m}. \tag{5.1}$$

The contribution considered here includes the complete virtual amplitude, with the on-shell subtractions, but excludes real emission. Its normalization is

$$\sigma_{\mathrm{virt},1} = \frac{\alpha_{\mathrm{T}}^3}{m^2} C_V(1 + X, \lambda_{\mathrm{phys}}/m). \tag{5.2}$$

Thus its photon-mass logarithm has not yet canceled. All coefficients below are obtained from the independently integrated amplitude.

A smaller exact basis. The 116 constant-letter coefficients of the independent virtual answer can be transformed back to 23 words in X, with letters $0, 1, -1, -2, -4, r$. Inverting the constant kernel-map matrix and substituting the result verifies all 116 original coefficient equations exactly. The resulting prefactors have the form

$$R_w(X) + \sqrt{\frac{X}{X+4}}\, S_w(X), \tag{5.3}$$

where R_w and S_w are rational functions, also linear in ℓ_λ. Only the change-of-basis matrix is used in this inversion. The published cross-section coefficients are not inputs.

5.1.1 Low energy

The defining kernels have convergent expansions near zero:

$$\frac{1}{X - a} = -\sum_{n=0}^{\infty} \frac{X^n}{a^{n+1}} \quad (a \neq 0), \tag{5.4}$$

$$\frac{1}{\sqrt{X(X+4)}} = \frac{1}{2\sqrt{X}} \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} \frac{X^n}{4^n}. \tag{5.5}$$

These expansions, together with $G(0;X)=\ln X$, recursively determine all required words as powers and logarithms. At each integration step, for $p>-1$, one uses

$$\int_0^X t^p \ln^k t\,\mathrm{d}t = X^{p+1}\sum_{j=0}^{k}\frac{(-1)^j k!}{(k-j)!(p+1)^{j+1}}\ln^{k-j} X. \tag{5.6}$$

The all-zero words are fixed separately by $G(0,\ldots,0;X)=\ln^n X/n!$. Multiplication by the exact rational prefactors and expansion of the remaining square root produce only rational coefficients and ℓ_λ. All negative powers and all half-integer powers cancel in the sum. The result is

$$C_V = X^2\left[-\frac{8}{3}\ln X+\frac{8}{9}\ell_\lambda+\frac{22}{9}\right] \tag{5.7}$$

$$+X^3\left[\frac{16}{5}\ln X-\frac{28}{15}\ell_\lambda-\frac{16}{3}\right]+O\Big(X^4|\ln X|\Big). \tag{5.8}$$

This expands the leading small-photon-mass expression obtained at fixed $X>0$, with ℓ_λ kept symbolic. The omitted power corrections in the photon mass are not uniform as $X\to 0$. It is therefore not a threshold expansion of the exact amplitude at fixed nonzero photon mass. The regulator logarithm cancels when this expression is combined with the real contribution. It supplies a direct check on the infrared coefficient:

$$A(1+X)=\frac{8}{9}X^2-\frac{28}{15}X^3+O(X^4). \tag{5.9}$$

5.1.2 High energy

The leading rational prefactor of every word is $O(1/X)$, so only the logarithmic polynomial and constant of each word at infinity are needed. The rational-letter words reduce to classical polylogarithms and shuffle products. Their inversion formulas fix, for example,

$$G(-1,0;X)=\tfrac{1}{2}L^2-\tfrac{\pi^2}{6}+O(L/X), \tag{5.10}$$

$$G(0,-1;X)=\tfrac{1}{2}L^2+\tfrac{\pi^2}{6}+O(L/X), \tag{5.11}$$

$$G(1,0;X)=\tfrac{1}{2}L^2+\tfrac{\pi^2}{3}+O(L/X). \tag{5.12}$$

For the root word, its independently established dilogarithmic form gives

$$G(r,-1;X)=\tfrac{1}{2}L^2-\tfrac{\pi^2}{18}+O(L/X). \tag{5.13}$$

Consequently the coefficient of the photon-mass logarithm is fixed completely at this order:

$$A(1+X)=\frac{4L^2-2L-2\pi^2/3-5}{X}+O\left(\frac{L^2}{X^2}\right). \tag{5.14}$$

The logarithmic terms of the finite virtual contribution have also been obtained. Its leading expansion can be written as

$$C_V=\frac{1}{X}\left[-\frac{2}{3}L^3+(4\ell_\lambda+1)L^2+\left(-2\ell_\lambda+\frac{5\pi^2}{3}\right)L-\left(\frac{2\pi^2}{3}+5\right)\ell_\lambda+B_V\right]+O\left(\frac{L^3}{X^2}\right). \tag{5.15}$$

To evaluate B_V, define the finite endpoint constants

$$K_a = \lim_{X\to\infty}\Big[G(r,-1,-1;X) - L^3/6\Big], \tag{5.16}$$

$$K_b = \lim_{X\to\infty}\Big[G(r,-1,0;X) - L^3/6 + \pi^2 L/6\Big], \tag{5.17}$$

$$K_c = \lim_{X\to\infty}\Big[G(r,0,-1;X) - L^3/6 - \pi^2 L/6\Big], \tag{5.18}$$

$$K_d = \lim_{X\to\infty}\Big[G(r,-2,-1;X) - L^3/6 + \pi^2 L/12\Big], \tag{5.19}$$

$$K_e = \lim_{X\to\infty}\Big[G(-4,r,-1;X) - L^3/6 + \pi^2 L/18\Big], \tag{5.20}$$

$$K_f = \lim_{X\to\infty}\Big[G(0,r,-1;X) - L^3/6 + \pi^2 L/18\Big]. \tag{5.21}$$

The remaining finite constant is

$$B_V = 9K_a - 18K_b - 6K_c - 8K_d - 12K_e - 9K_f - \frac{19}{2} - \frac{2\pi^2}{3} + 19\zeta(3). \tag{5.22}$$

We evaluate these six limits directly from the root primitives before substituting them into B_V.

In this calculation put $\ell = \ln X$, $s = 1 + X$, and $\beta = X/(X+2)$. The letter r denotes the positive square-root kernel. The functions used in the soft master are

$$k = \sqrt{\frac{1+XU^2}{s}}, \qquad d = 1 - \beta^2 U^2, \tag{5.23}$$

$$a(U) = \operatorname{artanh}(\beta U), \qquad f(U) = \operatorname{artanh}(U/k), \qquad 0 < U < 1. \tag{5.24}$$

Here d is a scalar function, unrelated to the spacetime dimension. The symbol a without an argument in the unit-circle identities below denotes a complex number, not the function $a(U)$.

Root-of-unity endpoints. Let $\theta = \pi/3$, $a = \exp(\mathrm{i}\theta)$, and define the real Clausen constant

$$\mathcal{C} = \operatorname{Im}\operatorname{Li}_2(a) = \sum_{n=1}^{\infty}\frac{\sin(n\pi/3)}{n^2}. \tag{5.25}$$

The rationalized root primitives have endpoints $q = 1$ and $q = 0$. Their unit-circle values are obtained from

$$\operatorname{Li}_2(e^{\mathrm{i}t}) = \frac{\pi^2}{6} - \frac{\pi|t|}{2} + \frac{t^2}{4} + \mathrm{i}\,\operatorname{Cl}_2(t), \tag{5.26}$$

$$\operatorname{Im}\operatorname{Li}_3(e^{\mathrm{i}t}) = \frac{\pi^2 t}{6} - \frac{\pi t|t|}{4} + \frac{t^3}{12}, \qquad -2\pi \le t \le 2\pi. \tag{5.27}$$

Here $\operatorname{Cl}_2(t) = \sum_{n\ge1}\sin(nt)/n^2$ is the Clausen function. These Fourier identities follow by integrating the geometric-series logarithm, with constants fixed at $t = 0$. Splitting the defining series into residue classes also gives

$$\operatorname{Re}\operatorname{Li}_3(a) = \frac{\zeta(3)}{3}, \quad \operatorname{Re}\operatorname{Li}_3(a^2) = -\frac{4\zeta(3)}{9}, \quad \operatorname{Cl}_2(2\theta) = \frac{2}{3}\mathcal{C}. \tag{5.28}$$

For explicit endpoint bookkeeping, write

$$J_{mn} = (\mathrm{i}m\theta)^2\ln(1-a^{m-n}) + 2\mathrm{i}m\theta\operatorname{Li}_2(a^{m-n}) - 2\operatorname{Li}_3(a^{m-n}), \qquad J_{00} = -2\zeta(3). \tag{5.29}$$

This is the earlier logarithm-squared primitive at a unit-circle endpoint. All phases are the continuous ones inherited from $0 < q < 1$. For $m = \pm 1, \pm 2$, the principal phases give those same endpoint values. With $S_3 = \mathrm{Li}_3(a) + \mathrm{Li}_3(a^{-1})$, the constants of the corresponding primitives are

$$P_A = 2J_{-1,0} + 2J_{1,0} - J_{-2,0} + J_{-2,2} - 2S_3, \tag{5.30}$$

$$M_n = \frac{1}{2}\left[-2\zeta(3) + (-\mathrm{i}n\theta)^2(-2\mathrm{i}n\theta) - 4\mathrm{i}n\theta\zeta(2) + J_{n,2n}\right], \quad n = \pm 1, \tag{5.31}$$

$$P_c = 2(M_1 + M_{-1}) - S_3 + 4\zeta(3), \tag{5.32}$$

$$P_f = -J_{1,0} - J_{-1,0} + S_3 + 2\zeta(3). \tag{5.33}$$

Evaluating the $q \to 0$ end of the same primitives gives

$$K_a = 3\zeta(3) + \mathrm{Li}_3(a^{-2}) + \frac{P_A}{2}, \quad K_c = -2S_3 - P_c, \quad K_f = -4\zeta(3) - 2S_3 + P_f. \tag{5.34}$$

The imaginary terms cancel in each final constant. The shuffle identity gives $K_b = -K_c - K_f$.

For the outer letter -4, the only additional dilogarithmic value is

$$\mathrm{Im}\,\mathrm{Li}_2(2a) = \frac{5}{6}\mathcal{C} + \frac{\pi}{2}\ln 2. \tag{5.35}$$

One way to establish it is to use $D(z) = \mathrm{Im}\,\mathrm{Li}_2(z) + \arg(1-z)\ln|z|$. Its differential is $\mathrm{d}D = \ln|z|\,\mathrm{d}\arg(1-z) - \ln|1-z|\,\mathrm{d}\arg z$. Integration gives one half of the sum of Cl_2 at twice the three angles of the triangle with vertices $0, 1, z$. For $z = 2a$ those angles are $\pi/3, \pi/2, \pi/6$, and $\arg(1-2a) = -\pi/2$. This yields the displayed value with its branch fixed. In the paired endpoint primitives all $\mathrm{Li}_3(2a)$ and $\mathrm{Li}_3(2a^{-1})$ terms cancel. The remaining pair contributes

$$-4\theta\pi\ln 2 + 8\theta\,\mathrm{Im}\,\mathrm{Li}_2(2a) = \frac{20}{3}\theta\mathcal{C}. \tag{5.36}$$

Together with $J(2,1) = -7\zeta(3)/4$, this fixes K_e. The value of $J(2,1)$ follows from trilogarithm inversion and $\mathrm{Li}_3(1/2) = 7\zeta(3)/8 - \pi^2\ln 2/12 + \ln^3 2/6$.

The remaining constant from the soft master. The exact energy equation (4.228) determines K_d once the finite high-energy constant of I_2 is known. It can be extracted without root-valued endpoint functions. At large X, use

$$I_2^{\mathrm{app}}(X) = \int_0^1 \frac{a}{Ud}\left[\frac{1}{2}\ln X + \ln 2 + \ln U - \frac{1}{2}\ln\left(1-U^2\right)\right]\mathrm{d}U, \qquad I_2 - I_2^{\mathrm{app}} = o(1). \tag{5.37}$$

In this expression $a = \mathrm{artanh}(\beta U)$ and $d = 1 - \beta^2 U^2$ keep their exact X dependence. To justify the error estimate, split the interval at $U = X^{-1/4}$. The lower pieces of both integrals are $O(X^{-1/4}\ln X)$. In the upper piece, use the exact identity

$$f = \ln\frac{(k+U)\sqrt{s}}{\sqrt{1-U^2}}, \qquad \frac{k}{U} = \sqrt{1 + \frac{1-U^2}{sU^2}}. \tag{5.38}$$

Expanding these positive quantities bounds the integrated difference by $O(X^{-3/4}\ln^2 X) + O(X^{-1}\ln^2 X)$. Thus the approximation has the same finite constant. No approximation has been made to the endpoint-regulating factor d.

Partial fractions and the fixed-letter differential reduction give

$$\begin{aligned} I_2^{\mathrm{app}} = {} & \frac{\ell}{4}G(0,-1) - \frac{\ell}{8}G(-1,-1) + \frac{1}{2}G(-2,0,-1) + \frac{1}{2}G(0,-2,-1) \\ & - \frac{1}{4}G(-2,-1,-1) - \frac{1}{4}G(0,-1,-1) - \frac{1}{4}G(-1,-2,-1) \\ & - \frac{1}{4}G(-1,0,-1) + \frac{1}{4}G(-1,-1,-1). \end{aligned} \tag{5.39}$$

Every endpoint is X. The few additional rational-letter constants can be evaluated by elementary parameter differentiation. Let $k_{012}(b)$ be the finite large-X constant of $G(0,-1,-b;X)$. Differentiating its defining integrals with respect to b gives

$$\partial_b G(0,-1,-b;X) = -\frac{\mathrm{Li}_2(-X)}{b(b-1)} + \frac{\mathrm{Li}_2(-X/b)}{b-1}, \tag{5.40}$$

so dilogarithm inversion yields

$$k'_{012}(b) = -\frac{\ln^2 b}{2(b-1)} - \frac{\zeta(2)}{b}. \tag{5.41}$$

Since $k_{012}(1) = \zeta(3)$ and $\int_1^2 \ln^2 b\,\mathrm{d}b/(b-1) = \zeta(3)/4$, one obtains $k_{012}(2) = 7\zeta(3)/8 - \zeta(2)\ln 2$. The latter definite integral follows from the trilogarithm value at one half stated above. Explicitly it is $2\zeta(3) - 2\,\mathrm{Li}_3(1/2) - \pi^2\ln 2/6 + \ln^3 2/3$. For $G(-2,-1,-1;X)$, shifting the integration variable by one and then inverting it gives the constant

$$-\frac{1}{2}\int_1^\infty \frac{\ln^2 v}{v(1+v)}\,\mathrm{d}v = -\frac{1}{2}\int_0^1 \frac{\ln^2 t}{1+t}\,\mathrm{d}t = -\frac{3}{4}\zeta(3). \tag{5.42}$$

The last equality follows by integrating the alternating geometric series. The shuffle product $G(-1)G(-2,-1)$ then fixes the constant of $G(-1,-2,-1)$ to $3\zeta(3)/2$. Finally the shuffle identity fixes the sum of the constants of $G(-2,0,-1)$ and $G(0,-2,-1)$ to $-7\zeta(3)/8$. Therefore

$$I_2 = \frac{5\ell^3}{48} + \frac{\pi^2\ell}{16} - \frac{3}{8}\zeta(3) + o(1). \tag{5.43}$$

Only this constant is extracted with the approximation. The sharper inverse-power remainder of the cross section follows from its exact GPL kernels, rather than from the approximation error bound.

Collecting the endpoint evaluations gives

$$K_a = \frac{5}{9}\zeta(3) - \frac{4\pi}{9}\mathcal{C}, \qquad K_b = \frac{2}{3}\zeta(3), \tag{5.44}$$

$$K_c = \frac{2}{3}\zeta(3) - \frac{4\pi}{3}\mathcal{C}, \qquad K_d = \frac{1}{12}\zeta(3) + \frac{\pi}{3}\mathcal{C}, \tag{5.45}$$

$$K_e = \frac{16}{9}\zeta(3) - \frac{8\pi}{9}\mathcal{C}, \qquad K_f = -\frac{4}{3}\zeta(3) + \frac{4\pi}{3}\mathcal{C}. \tag{5.46}$$

For K_d, the constant term of the exact I_2 identity is $3(K_f + 2K_a - K_e)/8 + (K_d + K_c)/2 = -3\zeta(3)/8$.

Substitution of these constants into B_V cancels the Clausen terms and gives

$$B_V = -\frac{19}{2} - \frac{2\pi^2}{3} - 2\zeta(3). \tag{5.47}$$

This determines the finite virtual constant without fitting numerical values or using the published high-energy total.

5.1.3 The Born term

For comparison, direct expansion of the Klein–Nishina total gives

$$\sigma_0 = \frac{\pi\alpha_{\mathrm{T}}^2}{m^2}\left[\frac{8}{3} - \frac{8}{3}X + \frac{52}{15}X^2 - \frac{133}{30}X^3 + O(X^4)\right], \tag{5.48}$$

$$\sigma_0 = \frac{\pi\alpha_{\mathrm{T}}^2}{m^2}\left[\frac{2\ln X + 1}{X} + \frac{18 - 8\ln X}{X^2} + O\!\left(\frac{\ln X}{X^3}\right)\right]. \tag{5.49}$$

These formulas use the same X in both limits and keep the cross-section dimension explicit.

5.2 Asymptotic expansions of the soft contribution

Use the dimensionless variable $X = s_{\mathrm{phys}}/m^2 - 1 = 2\omega/m$. In the intermediate formulas below $m = 1$, so $s = 1 + X$. The infrared residue is $A = \mathcal{C}_{\mathrm{IR}}/s$, and the finite soft angular term in the matched total is $-\mathcal{C}_K/s$. These quantities are expanded from the eikonal parameter integral and the Born trace, independently of any published total cross section.

5.2.1 Uniform threshold expansion

At fixed physical angle $-1 \le c \le 1$ and Feynman parameter $0 \le u \le 1$, write

$$t = -\frac{X^2(1-c)}{2(1+X)}, \qquad E^2 = \frac{(X+2)^2}{4(1+X)}, \qquad M_u = 1 - tu(1-u), \tag{5.50}$$

$$v_u^2 = 1 - \frac{M_u}{E^2}, \qquad \beta^2 = \frac{X^2}{(X+2)^2}. \tag{5.51}$$

Both v_u^2 and β^2 are uniformly $O(X^2)$, and $M_u \ge 1$. Consequently the convergent real series

$$\frac{\operatorname{artanh} v}{v} = 1 + \frac{v^2}{3} + \frac{v^4}{5} + O(v^6) \tag{5.52}$$

can be integrated term by term on the compact parameter and angular intervals. Its remainder is uniform for sufficiently small positive X, since $0 \le v_u \le \beta < 1$. Substitution in the defining integral for K gives

$$K(s,c) = \frac{5}{36}(1-c)(X^2 - X^3) \tag{5.53}$$

$$+ \left(-\frac{3c^2}{200} - \frac{437c}{3600} + \frac{491}{3600}\right)X^4 \tag{5.54}$$

$$+ \left(\frac{3c^2}{100} + \frac{187c}{1800} - \frac{241}{1800}\right)X^5 + O(X^6). \tag{5.55}$$

For the infrared factor, put $d = -t/2$. Its corresponding analytic series is

$$b_e - 1 = \frac{2d}{3} - \frac{d^2}{5} + O(d^3). \tag{5.56}$$

The apparent inverse powers of X in the invariant Born expression cancel when $y = X + t$ is substituted before expansion. Its leading term is $2(1 + c^2)$. Multiplication followed by exact polynomial integration over c yields

$$A(1+X) = \frac{8}{9}X^2 - \frac{28}{15}X^3 + \frac{238}{75}X^4 - \frac{7618}{1575}X^5 + O(X^6), \tag{5.57}$$

$$\frac{\mathcal{C}_K(1+X)}{s} = \frac{20}{27}X^2 - \frac{14}{9}X^3 + \frac{6023}{2250}X^4 - \frac{195133}{47250}X^5 + O(X^6). \tag{5.58}$$

All coefficients are rational because the uniform expansion leaves only polynomials on the compact integration domain.

The term matching the symmetric analytic real regulator to the photon mass is

$$\mathcal{M}(X;\lambda) = -A(1+X)\left[\ln X + \frac{1}{2}\ln(1+X) + \ln\lambda\right] - \frac{\mathcal{C}_K(1+X)}{s}. \tag{5.59}$$

Its first nonzero order is therefore

$$\mathcal{M}(X;\lambda) = -X^2\left[\frac{8}{9}\ln(X\lambda) + \frac{20}{27}\right] + O\Big(X^3[1 + |\ln X| + |\ln\lambda|]\Big). \tag{5.60}$$

This is a component of the matched answer, not the total correction. The photon-mass logarithm cancels against the virtual term before the physical limit $\lambda \to 0$. These component expansions refer to the leading small-λ expression obtained at fixed $X > 0$, with $\ln\lambda$ kept symbolic. They do not describe a threshold limit at fixed nonzero photon mass. The exact algebraic infrared identity allows the cancellation to be made before expanding if a regulator-independent remainder is required. The script `compton_soft_asymptotic_low.py` performs the polynomial steps and saves their exact coefficients in `soft_asymptotic_low.json`.

5.2.2 The high-energy infrared residue

The leading large-X residue can also be obtained directly from its elastic angular integral. Use $T = -t = X - y$, where $X/(X+1) \le y \le X$. The Born factor is $\mathcal{B}(X,y) = \overline{|\mathcal{M}_0|^2}/e_{\mathrm{T}}^4$, as in the angular integral above. Since $\mathrm{d}c = 2s\,\mathrm{d}y/X^2$,

$$A(1+X) = \frac{2}{X^2}\int_{X/(X+1)}^{X} \mathcal{B}(X,y)[b_e(X-y) - 1]\,\mathrm{d}y. \tag{5.61}$$

For large T, the exact rapidity expression gives $b_e(T) = \ln T + O((1+\ln T)/T)$. Its replacement by $\ln T$ is not pointwise valid at $T = 0$, but the endpoint error is integrable. Indeed $\mathcal{B}$ is bounded near $y = X$, while $\ln T$ is integrable at zero. For $T \ge 1$, splitting the remaining interval at $y = X/2$ bounds the integrated replacement error by $O(1 + \ln^2 X)$ before the prefactor X^{-2}. The subleading Born terms likewise contribute at most $O(\ln^2 X/X^2)$. The lower endpoint can be replaced by one at this accuracy. Therefore the leading term is

$$A(1+X) = \frac{4}{X^2}\left\{X\int_1^X \frac{\ln(X-y)-1}{y}\,\mathrm{d}y + \frac{1}{X}\int_0^X y[\ln(X-y)-1]\,\mathrm{d}y\right\} + O\left(\frac{\ln^2 X}{X^2}\right). \tag{5.62}$$

The first integral follows from $y = Xz$ and $\int_0^1 \ln(1-z)\,\mathrm{d}z/z = -\pi^2/6$. The second uses $\int_0^1 z\ln(1-z)\mathrm{d}z = -3/4$. Putting $L = \ln X$ gives

$$A(1+X) = \frac{1}{X}\left(4L^2 - 2L - \frac{2\pi^2}{3} - 5\right) + O\left(\frac{L^2}{X^2}\right), \qquad X \to \infty. \tag{5.63}$$

The logarithmic part of the real-regulator matching is consequently

$$-A\ln(\sqrt{s}X\lambda) = -\frac{4L^2 - 2L - 2\pi^2/3 - 5}{X}\left(\frac{3}{2}L + \ln\lambda\right) + O\left(\frac{L^3 + L^2|\ln\lambda|}{X^2}\right). \tag{5.64}$$

The finite term $-\mathcal{C}_K/s$ must still be included. As at threshold, the photon-mass logarithm is canceled algebraically before removing the regulator in the total cross section.

5.2.3 The high-energy finite soft term

A separate derivation is needed for the finite function K, because its parameter endpoints generate additional logarithms. Put $T = -t$ and $r = T/s$. Away from the forward region $T = O(1)$, subtract the endpoint approximation $\frac{1}{2}\ln(s/M_u)$ from $\operatorname{artanh} v_u/v_u$ before taking the massless limit of the remaining parameter integral. The subtracted part uses the already integrated functions b_e and $\mathcal{H}(T)$,

$$K = \frac{1}{2}\ln s\,(b_e - 1) - \frac{1}{2}z_e\mathcal{H}(T) + F(r) + \text{power-suppressed terms}. \tag{5.65}$$

Here $z_e\mathcal{H}(T) = \frac{1}{2}\ln^2 T - \pi^2/6 + O((1+\ln^2 T)/T)$. The finite shape function is

$$F(r) = \frac{1}{2}\int_0^1 \frac{\mathrm{d}u}{u(1-u)}\left[\frac{\operatorname{artanh}\sqrt{1-4ru(1-u)}}{\sqrt{1-4ru(1-u)}} - \frac{1}{2}\ln\frac{1}{ru(1-u)}\right], \qquad 0 < r \le 1. \tag{5.66}$$

The subtraction removes both parameter endpoint singularities. It also makes $F(r) \to 0$ as $r \to 0$. Differentiating the convergent difference and setting $u = (1+w)/2$ gives

$$F'(r) = \int_0^1 \mathrm{d}w\,\frac{\operatorname{artanh} v/v - 1}{v^2}, \qquad v^2 = 1 - r + rw^2. \tag{5.67}$$

To perform this integral, use the exact identity

$$(1-r)\frac{\operatorname{artanh} v/v - 1}{v^2} = \frac{\mathrm{d}}{\mathrm{d}w}\left(w\frac{\operatorname{artanh} v}{v}\right) - \frac{1}{1-w^2}. \tag{5.68}$$

The two endpoint logarithms must be combined. Their difference as $w \to 1^-$ is $-\frac{1}{2}\ln r$, giving

$$F'(r) = -\frac{\ln r}{2(1-r)}, \qquad F(r) = \frac{1}{2}\left[\mathrm{Li}_2(r) + \ln r\ln(1-r)\right]. \tag{5.69}$$

All arguments lie on the real interval, and the value at $r = 1$ is the continuous limit $F(1) = \pi^2/12$.

With $L = \ln X$, the finite angular dependence at leading power is

$$K = \frac{1}{4}L^2 - \frac{1}{2}L + f(r) + \text{power-suppressed terms}, \tag{5.70}$$

$$f(r) = \frac{\pi^2}{12} - \frac{1}{4}\ln^2 r + F(r), \qquad f(1) = \frac{\pi^2}{6}. \tag{5.71}$$

Use $z = y/X$, so $r = 1 - z$ at leading power. The Born weight becomes $4(X^{-1})(z^{-1} + z)\mathrm{d}z$ after the cross-section prefactor. The logarithmic lower endpoint is $z = 1/X$ to the required accuracy. The finite constant needed in the angular integration is

$$C_f = \int_0^1 \left[\frac{f(r) - f(1)}{1-r} + (1-r)f(r)\right]\mathrm{d}r \tag{5.72}$$

$$= -\zeta(3) + \frac{\pi^2}{24} - \frac{1}{4}. \tag{5.73}$$

For the first integral, integration by parts and $f'(r) = -\ln r/[2r(1-r)]$ reduce it to $-\zeta(3)$. Expanding $-\ln(1-r) = \sum_{n\ge1} r^n/n$ proves the logarithmic moments by absolutely convergent integration. The second integral uses only elementary logarithmic moments and one integration by parts of $\mathrm{Li}_2(r)$. The result is

$$\frac{\mathcal{C}_K}{s} = \frac{1}{X}\left[L^3 - \frac{3}{2}L^2 + \left(\frac{2\pi^2}{3} - 1\right)L + \frac{\pi^2}{6} - 1 - 4\zeta(3)\right] + O\left(\frac{L^3}{X^2}\right). \tag{5.74}$$

The forward interval $T \lesssim 1$ cannot use the fixed-r expansion. Its width in y is bounded, and its integrated difference contributes at most $O(L^2/X^2)$. On $T \geq 1$, the endpoint-subtracted parameter integral bounds the omitted terms by a constant times $(1+L^2)/T + (1+L^2)/s$. Splitting the angular interval at $y = X/2$ then gives the stated $O(L^3/X^2)$ remainder. Thus the angular integral has been controlled in the endpoint region as well as at fixed angle. The exact differential identity and moment algebra are checked by `compton_soft_asymptotic_high.py`.

5.3 Asymptotic expansions of real radiation

Throughout this section $X = s_{\mathrm{phys}}/m^2 - 1 > 0$ and $s = 1 + X$. The labels UV and UW denote the angular denominator pairs defined in the real-emission calculation.

5.3.1 Threshold terms of the UV sector

The analytic inner integral permits a uniform threshold expansion on a fixed interval. Set $v = 1 + Xw$, with $0 \leq w \leq 1$ and $\mathrm{d}v = X\,\mathrm{d}w$. Then

$$\ln v = \ln(1 + Xw), \tag{5.75}$$

$$\ln(1-k) = \ln[1 + Xw(1-w)] - \ln(1+Xw), \qquad k = \frac{Xw^2}{1+Xw}. \tag{5.76}$$

Both logarithms have uniformly convergent Taylor series for $0 < X < 1$. The rational denominators left after the soft subtraction contain $1 + Xw$ and $1 + Xw(1-w)$, which are bounded away from zero. The apparent powers of w^{-1} cancel when the Laurent coefficients are combined. This cancellation is performed before integrating the coefficient polynomials. Keeping the Jacobian is essential to obtain the correct leading power.

Let F_{UV} denote the finite Laurent coefficient before multiplication by the real cross-section prefactor $1/[8sX]$. Direct expansion and polynomial integration give

$$F_{UV}(X) = \frac{128}{X} + \frac{608}{3} - \frac{64}{3}\ln X \tag{5.77}$$

$$+ X\left(\frac{1568}{15} + \frac{64}{15}\ln X\right) + X^2\left(\frac{380}{9} + \frac{56}{15}\ln X\right) \tag{5.78}$$

$$+ X^3\left(\frac{1364}{315} - \frac{400}{63}\ln X\right) + O\Big(X^4[1 + |\ln X|]\Big). \tag{5.79}$$

The inverse power belongs to the sector decomposition, not to the physical threshold cross section. It must be added to the other real sectors before their common prefactor is expanded. The stated remainder becomes $O(X^3[1 + |\ln X|])$ after that prefactor. The exact coefficient calculation is recorded in `soft_asymptotic_UV_low.json`.

5.3.2 Threshold terms of the UW sector

For the UW sector the existing rectangle $0 < a < X$, $0 < t < 1$ has already been integrated in a. Its logarithms are expanded uniformly in X. The difference of dilogarithms in its remaining kernel can be expanded without separately expanding either divergent argument at $t = 0$:

$$d_3(X,t) = \mathrm{Li}_2\!\left(-\frac{(1-t)(1+Xt)}{t}\right) - \mathrm{Li}_2\!\left(-\frac{1-t}{t}\right), \tag{5.80}$$

$$\frac{\partial d_3}{\partial X} = -\frac{t}{1+Xt}\left[-\ln t + \ln(1 + Xt(1-t))\right], \qquad d_3(0,t) = 0. \tag{5.81}$$

Both dilogarithm arguments are negative, which fixes their real branches. The Taylor series of this derivative is uniform apart from the integrable factor $t \ln t$. The analogous derivative of $\mathrm{Li}_2(-1 - Xt) + \pi^2/12$ is $-t \ln(2 + Xt)/(1 + Xt)$, also with a zero boundary value at $X = 0$. The apparent endpoint poles cancel when the complete coefficient at each power of X is collected, before its final t integration.

To control the remainder, first apply the endpoint Taylor subtractions of Section 4.8 to the eight-function kernel $\Phi_{UW}(X, t)$. Separate its explicit dependence on $\ln X$, which is at most linear. The accompanying rational coefficients have poles of order at most four at $X = 0$. Define

$$X^4 \Phi_{UW}(X, t) = \Psi_0(X, t) + \ln X \, \Psi_1(X, t). \tag{5.82}$$

For any fixed $0 < X_0 < 1$, both functions Ψ_j extend analytically to a neighborhood of $[0, X_0] \times [0, 1]$. Indeed, the nonendpoint denominators $1 + Xt$, $2 + Xt$ and $1 + Xt(1 - t)$ are bounded away from zero there. Each endpoint Taylor subtraction is an analytic divided difference. For the function J, its short dilogarithm expression establishes joint analyticity near $t = 0$, and its original $d_3 - \ell \ln t + \ell p$ expression does so near $t = 1$. The eighth X derivatives of the Ψ_j are consequently bounded on this compact rectangle. Taylor expansion through X^7 has a remainder bounded uniformly in t by $CX^8[1 + |\ln X|]$, with C independent of X and t. Dividing by X^4 and integrating over the fixed unit interval gives the remainder below. The coefficient integrals give, for one UW sector,

$$F_{UW}(X) = X \left(\frac{16}{15} - \frac{64}{5} \ln X \right) + X^2 \left(-\frac{4}{3} - \frac{16}{5} \ln X \right) \tag{5.83}$$

$$+ X^3 \left(\frac{788}{315} - \frac{236}{105} \ln X \right) + O\Big(X^4 [1 + |\ln X|] \Big) . \tag{5.84}$$

The normalization is again before $1/[8sX]$. The complete real answer contains $2F_{UW}$ and the separate rational Bose-restoration term. The exact coefficients are saved in `soft_asymptotic_UW_low.json`.

5.3.3 The combined real contribution at low energy

Put $X = s - 1 > 0$. The finite real coefficient F_{R}^{η} in this paragraph is the constant term obtained with the symmetric regulator $(X_1 X_2)^{\eta}$, before matching to the photon-mass calculation. With $P = p + k$, these are exactly the dimensionless Dalitz variables

$$X_1 = x = \frac{2P \cdot k_1}{m^2}, \qquad X_2 = y = \frac{2P \cdot k_2}{m^2}. \tag{5.85}$$

In the center-of-mass frame, $X_i = 2\sqrt{s}\, \omega_i / m$. Thus $(X_1 X_2)^{\eta} = (xy)^{\eta}$ uses the same finite-part normalization as the earlier integration. The subscripts distinguish the two outgoing energies from the incident-energy parameter X.

For the regular rational, energy-logarithm, W, and Bose terms, use reciprocal symmetry to replace the last energy interval by $1 < u < s$, with an overall factor $1/2$. Set

$$u = 1 + Xw, \qquad 0 < w < 1, \qquad \ell = \ln \frac{w}{1 - w}. \tag{5.86}$$

The logarithms in the exact kernels become

$$L_1 = 2 \ln(1 + Xw) - \ln(1 + X), \tag{5.87}$$

$$L_2 = \ell + \ln(1 + Xw), \tag{5.88}$$

$$L_3 = \ell + 2 \ln[2 + X(1 + w)] - \ln(1 + Xw) - 2 \ln(2 + Xw). \tag{5.89}$$

Their expansions now have a fixed integration interval. The removable Gram factors are combined before expanding. For example, $\ln(a/b)/(a-b)$ is analytic at $a=b>0$. Such divided differences remove the apparent interior pole near $w=1/2$. After the four sectors are added, every coefficient through the required order is a polynomial in w multiplying ℓ, plus a rational function whose only poles are at the fixed endpoints. This cancellation is an exact rational identity.

Subtract the simple and double endpoint terms before interchanging the expansion and the integral. The remaining rational denominators are bounded away from zero for sufficiently small positive X on the compact w interval. Their subtracted remainders are bounded by a constant times $1+|\ln w|+|\ln(1-w)|$. This integrable bound justifies the coefficientwise expansion. The endpoint counterterms themselves are integrated exactly in the same finite-part convention. The needed polynomial moment is

$$\int_0^1 w^n \ell \,\mathrm{d}w = \frac{H_n}{n+1}, \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}, \qquad H_0 = 0. \tag{5.90}$$

The finite parts of w^{-1} and $(w-1)^{-1}$ vanish. Those of w^{-2} and $(w-1)^{-2}$ both equal -1.

Let $\Phi = \mathrm{FP}\int_0^1 Xf(1+Xw)\,\mathrm{d}w$ for the sum of these four sectors. Let A be its upper simple-pole coefficient and D_w its upper double-pole coefficient. Their exact expansions are

$$\Phi = -\frac{192}{X} - \frac{1024}{3} - \frac{1792}{9}X - \frac{532}{9}X^2 - \frac{7052}{135}X^3 + O(X^4), \tag{5.91}$$

$$A = \frac{32}{3} + \frac{32}{3}X + \frac{4}{3}X^2 + \frac{188}{15}X^3 + O(X^4), \tag{5.92}$$

$$D_w = \frac{32}{X} + \frac{224}{3} + \frac{544}{9}X + \frac{208}{9}X^2 + \frac{172}{45}X^3 + O(X^4). \tag{5.93}$$

The double pole belongs to the Bose-restoring contribution. Its finite boundary correction must be included even though the complete real integral eventually has only a logarithmic soft divergence.

Denote the sum of the already derived z-cut to η-scheme matching constants by M_η. Expanding their exact logarithms and dilogarithms gives

$$M_\eta = \frac{80}{3} + \frac{112}{3}X + \frac{62}{9}X^2 + \frac{352}{45}X^3 + O(X^4\ln X). \tag{5.94}$$

The unnormalized four-sector finite part is therefore

$$F_4 = \frac{\Phi}{2} + 2A\ln X - \frac{A}{2}\ln(1+X) - \frac{D_w(X+2)}{2(X+1)} + M_\eta. \tag{5.95}$$

The logarithmic terms follow from the common z boundary. The term containing D_w follows from the second-order displacement of that boundary. Substitution yields

$$F_4 = -\frac{128}{X} - \frac{608}{3} + \frac{64}{3}\ln X \tag{5.96}$$

$$+ X\left(\frac{64}{3}\ln X - \frac{320}{3}\right) + X^2\left(\frac{8}{3}\ln X - \frac{356}{9}\right) \tag{5.97}$$

$$+ X^3\left(\frac{376}{15}\ln X - \frac{2596}{135}\right) + O(X^4\ln X). \tag{5.98}$$

Combining this with the independently expanded UV and twice the UW contributions cancels every X^{-1}, constant, X, and X^2 term, including their logarithms. The exact remaining unnormalized expression is

$$F_4 + F_{UV} + 2F_{UW} = X^3\left(\frac{128}{9}\ln X - \frac{1336}{135}\right) + O(X^4\ln X). \tag{5.99}$$

Restoring the common phase-space factor $1/[8X(1+X)]$ gives

$$\boxed{F_{\mathrm{R}}^{\eta} = X^2\left(\frac{16}{9}\ln X - \frac{167}{135}\right) + O(X^3\ln X).} \tag{5.100}$$

The remainder follows from the uniform subtracted expansion and the single explicit logarithm in the regulator matching. It is not inferred by fitting numerical values. In the center-of-mass local-subtraction convention, using $A_{\mathrm{phys}} = 8X^2/9 + O(X^3)$ and $\mathcal{C}_{\mathrm{R}} = F_{\mathrm{R}}^{\eta} - A_{\mathrm{phys}}\ln[2\sqrt{1+X}\,X]$, the same result is

$$\mathcal{C}_{\mathrm{R}} = X^2\left[\frac{8}{9}\ln\frac{X}{2} - \frac{167}{135}\right] + O(X^3\ln X). \tag{5.101}$$

5.3.4 The high-energy UV contribution

Put $L = \ln X$. The constant-letter representation of the UV finite coefficient gives an independent route to its large-X expansion. In that representation, $q = (\sqrt{X+4} - \sqrt{X})/(\sqrt{X+4} + \sqrt{X})$ tends to zero and

$$q = \frac{1}{X} - \frac{2}{X^2} + \frac{5}{X^3} - \frac{14}{X^4} + \frac{42}{X^5} + O(X^{-6}). \tag{5.102}$$

The GPL differential equation expands each word in powers of q and $\ln q$. The only singular kernel at this endpoint is $\mathrm{d}q/q$. All other kernels have convergent Taylor series about zero. Words of weight at most three therefore have remainders bounded by powers of q times at most $|\ln q|^3$. The rational coefficients have poles of order at most three, so the word expansion is carried through q^4 before extracting the physical $1/X$ term.

The integration constants in this expansion are the finite parts $\mathrm{FP}\,G_1(w;0)$, where the subscript denotes basepoint one and the same real left tangent as above. They are evaluated analytically, not determined by fitting. Let $a_\pm = (1 \pm \mathrm{i}\sqrt{3})/2$ and $D(t) = 1 - t + t^2$. The weight-one sum is

$$G_1(a_+;t) + G_1(a_-;t) = \ln D(t). \tag{5.103}$$

Shuffle identities remove most higher-weight endpoint constants. The remaining weight-two sum is

$$\sum_{a=a_\pm} G_1(1,a;0) = \int_0^1 \frac{\ln D(t)}{1-t}\mathrm{d}t = -\frac{\pi^2}{18}. \tag{5.104}$$

The substitution $t \mapsto 1-t$ reduces the integral to $-\mathrm{Li}_2(a_+) - \mathrm{Li}_2(a_-)$. The standard real boundary values follow directly by integrating the Fourier series for $\ln\left(1 - e^{\mathrm{i}\theta}\right)$, or by the dilogarithm reflection identity on the conjugate pair.

Define the only remaining weight-three combination by

$$U = \sum_{a,b=a_\pm} G_1(0,a,b;0) = -\frac{1}{2}\int_0^1 \frac{\ln^2 D(t)}{t}\mathrm{d}t, \tag{5.105}$$

$$V = \sum_{a=a_\pm} G_1(-1,0,a;0) = \int_0^1 \frac{\ln D(t)\ln(1+t)}{t}\mathrm{d}t. \tag{5.106}$$

Both are convergent real integrals. The algebraic identity $(1+t)D(t) = 1 + t^3$ gives

$$V - U = \frac{1}{2}\int_0^1 \frac{\ln^2(1+t^3) - \ln^2(1+t)}{t}\mathrm{d}t = -\frac{1}{3}\int_0^1 \frac{\ln^2(1+t)}{t}\mathrm{d}t = -\frac{\zeta(3)}{12}. \tag{5.107}$$

For the last elementary polylogarithmic moment, set $w = 1/(1+t)$. A primitive of $\ln^2 w/(1-w)$ is $-\ln^2 w \ln(1-w) - 2\ln w\,\mathrm{Li}_2(w) + 2\,\mathrm{Li}_3(w)$. Its values at $w = 1/2, 1$, together with

$$\mathrm{Li}_2(1/2) = \frac{\pi^2}{12} - \frac{1}{2}\ln^2 2, \tag{5.108}$$

$$\mathrm{Li}_3(1/2) = \frac{7}{8}\zeta(3) - \frac{\pi^2}{12}\ln 2 + \frac{1}{6}\ln^3 2, \tag{5.109}$$

prove that the moment is $\zeta(3)/4$. Thus every boundary constant needed here is fixed analytically.

Writing $\widehat{F}_{UV} = F_{UV}/[8sX]$, the result is

$$\widehat{F}_{UV} = X^3\left(\frac{L}{12} - \frac{5}{144}\right) + X^2\left(\frac{L}{3} + \frac{1}{18}\right) \tag{5.110}$$

$$+ X\left(\frac{5}{24}L^2 - \frac{49}{24}L + \frac{65}{72}\right) - \frac{7}{6}L^2 + \frac{11}{3}L + \frac{47}{36} + \frac{\pi^2}{6} \tag{5.111}$$

$$+ \frac{1}{X}\left[7L^3 - \frac{73}{6}L^2 + \left(\frac{133}{8} - \frac{\pi^2}{3}\right)L + \frac{2335}{144} - \frac{\pi^2}{6} - 4\zeta(3)\right] + O\left(\frac{L^3}{X^2}\right). \tag{5.112}$$

The growing terms are artifacts of the sector decomposition and must cancel in the complete real contribution. They are included explicitly so that this cancellation can be checked rather than assumed. The recurrence and the exact endpoint-constant reductions are implemented in `compton_soft_asymptotic_uv_high.py`, with the reduced result saved in `soft_asymptotic_UV_high.json`.

5.4 The total cross section at low energy

The preceding expansions determine the first nonzero radiative correction to the total cross section. Here $X = s_{\mathrm{phys}}/m^2 - 1 = 2\omega/m$, with ω the incident photon energy in the electron rest frame, and $\ell_\lambda = \ln(\lambda_{\mathrm{phys}}/m)$. Before taking the low-energy limit, cancel the exact photon-mass logarithm in (3.672) by writing

$$C_1 = [C_V - A\ell_\lambda] + F_{\mathrm{R}}^{\eta} - A\ln[X\sqrt{1+X}] - \frac{\mathcal{C}_K}{1+X}. \tag{5.113}$$

Each term in this expression is independent of the photon mass. Consequently the following expansion does not interchange a fixed-regulator threshold expansion with the infrared limit:

$$C_V - A\ell_\lambda = X^2\left[-\frac{8}{3}\ln X + \frac{22}{9}\right] + O\Big(X^3|\ln X|\Big), \tag{5.114}$$

$$F_{\mathrm{R}}^{\eta} = X^2\left[\frac{16}{9}\ln X - \frac{167}{135}\right] + O\Big(X^3|\ln X|\Big), \tag{5.115}$$

$$-A\ln[X\sqrt{1+X}] - \frac{\mathcal{C}_K}{1+X} = X^2\left[-\frac{8}{9}\ln X - \frac{20}{27}\right] + O\Big(X^3|\ln X|\Big). \tag{5.116}$$

The logarithmic coefficients sum to $-16/9$, and the constants sum to $22/9 - 167/135 - 20/27 = 7/15$. Thus

$$\boxed{\sigma_1 = \frac{\alpha_{\mathrm{T}}^3}{m^2}\left\{X^2\left[-\frac{16}{9}\ln X + \frac{7}{15}\right] + O\Big(X^3|\ln X|\Big)\right\}, \qquad X \to 0^+.} \tag{5.117}$$

This is the order-α_{T}^3 contribution. The NLO total is $\sigma_0 + \sigma_1$, with the Born expansion

$$\sigma_0 = \frac{\pi\alpha_{\mathrm{T}}^2}{m^2}\left[\frac{8}{3} - \frac{8}{3}X + \frac{52}{15}X^2 - \frac{133}{30}X^3 + O(X^4)\right]. \tag{5.118}$$

The remainder follows by adding the remainders of the independently expanded, regulator-independent terms. No numerical fit or published total is used to fix either coefficient. Since $X^2 \ln X \to 0$, this order of the radiative correction vanishes in the Thomson limit.

5.4.1 Expansion of the collected analytic formula

The complete product form (4.590) gives two further orders beyond the leading result above. The external energy variable remains X. Direct expansion gives

$$\begin{aligned}\sigma_1 = \frac{\alpha_{\mathrm{T}}^3}{m^2}\Bigg\{ & X^2\left[-\frac{16}{9}\ln X + \frac{7}{15}\right] + X^3\left[\frac{4}{3}\ln X - \frac{113}{90}\right] \\ & + X^4\left[-\frac{25}{9}\ln X + \frac{203}{125}\right] + O\Big(X^5|\ln X|\Big)\Bigg\}, \qquad X \to 0^+.\end{aligned} \tag{5.119}$$

No individual virtual or real integral is expanded again here. The starting point is the complete infrared-finite analytic formula.

The expansion can be performed recursively from the defining GPL kernels. For a nonzero rational letter a, and for the root letter,

$$\frac{1}{X-a} = -\sum_{j=0}^{\infty}\frac{X^j}{a^{j+1}}, \qquad |X| < |a|, \tag{5.120}$$

$$\frac{1}{\sqrt{X(X+4)}} = \frac{1}{2\sqrt{X}}\sum_{j=0}^{\infty}\binom{-1/2}{j}\frac{X^j}{4^j}, \qquad |X| < 4. \tag{5.121}$$

For the zero letter the kernel is $1/X$. Starting with the empty word $G(\varnothing; X) = 1$, multiply the kernel series by the series for the shorter word and integrate each term. For $p \neq -1$,

$$\int X^p \ln^k X \,\mathrm{d}X = X^{p+1}\sum_{j=0}^{k}\frac{(-1)^j k!}{(k-j)!(p+1)^{j+1}}\ln^{k-j} X. \tag{5.122}$$

For $p = -1$ the primitive is $\ln^{k+1} X/(k+1)$. The zero-letter words are normalized as $G(\underbrace{0,\ldots,0}_{k}; X) = \ln^k X/k!$. All other integration constants follow from their vanishing limit at the lower endpoint. This gives the required half-integer power series without a numerical fit.

The rational coefficients of the total have poles of order at most four at $X = 0$. Accordingly the GPL series are computed through X^8 before multiplication, which suffices for the total through X^4. On summing all terms, the negative powers, the constant and linear terms, every half-integer power, and powers of $\ln X$ above one cancel exactly. The first surviving term is the X^2 term displayed in (5.119). The local kernel series converge in a sufficiently small interval $0 < X < 1$. After the displayed terms are removed, the next possible contributions have the form X^5 times an analytic function or $X^5 \ln X$ times an analytic function, giving the stated remainder. Indeed, every non-elementary word in the product form has at most one trailing zero, and the only explicit factor $\ln X$ multiplying a GPL multiplies a word with no trailing zero. The elementary part is also at most linear in $\ln X$. Thus no higher logarithmic power can first appear in the omitted orders.

Only after deriving these coefficients do we compare the leading term with Lee, Schwartz and Zhang [2], Equation (10). Their variable is the same $X = (s_{\mathrm{phys}} - m^2)/m^2$, and their order-$\alpha^3$ normalization is α^3/m^2. The logarithmic coefficient $-16/9$ and the finite coefficient $7/15$ agree separately. That displayed literature equation does not give the X^3 and X^4 coefficients, so no comparison of those two orders is inferred from it.

5.5 The collected total at high energy

The complete analytic coefficient gives the high-energy expansion

$$\sigma_1 = \frac{\alpha_{\mathrm{T}}^3}{m^2}\left\{\frac{1}{X}\left[\frac{\ln^3 X}{3} - \frac{\ln^2 X}{2} + \frac{17}{4}\ln X - \frac{75}{8} - \frac{\pi^2}{2} + 4\zeta(3)\right] + O\left(\frac{\ln^3 X}{X^2}\right)\right\}, \quad X \to \infty. \tag{5.123}$$

Here $X = s_{\mathrm{phys}}/m^2 - 1$. This expansion is taken after all virtual and real contributions have been integrated and combined. The finite constant in the square brackets is determined below from our GPL primitives, independently of a comparison cross section.

Terms needed from the kernels. Set $\ell = \ln X$. Most coefficients in the collected expression are $O(1/X)$, so their GPLs are needed through the finite constant. Three weight-two functions multiply coefficients of order one and require an additional inverse-power term:

$$G(-1,-1;X) = \frac{\ell^2}{2} + \frac{\ell}{X} + O\left(\frac{\ell}{X^2}\right), \tag{5.124}$$

$$G(0,-1;X) = \frac{\ell^2}{2} + \frac{\pi^2}{6} - \frac{1}{X} + O(X^{-2}), \tag{5.125}$$

$$G(r,-1;X) = \frac{\ell^2}{2} - \frac{\pi^2}{18} + \frac{2\ell+1}{X} + O\left(\frac{\ell}{X^2}\right). \tag{5.126}$$

The first two follow from logarithmic expansion and dilogarithm inversion. The third follows from (4.70) with $q = X^{-1} - 2X^{-2} + O(X^{-3})$. Keeping these subleading terms is essential because the order-one contributions cancel in the sum.

For the six remaining root words, define the constants by

$$G(r,-1,-1;X) = \ell^3/6 + K_a + O(\ell^2/X), \tag{5.127}$$

$$G(r,-1,0;X) = \ell^3/6 - \pi^2\ell/6 + K_b + O(\ell^2/X), \tag{5.128}$$

$$G(r,0,-1;X) = \ell^3/6 + \pi^2\ell/6 + K_c + O(\ell^2/X), \tag{5.129}$$

$$G(r,-2,-1;X) = \ell^3/6 - \pi^2\ell/12 + K_d + O(\ell^2/X), \tag{5.130}$$

$$G(-4,r,-1;X) = \ell^3/6 - \pi^2\ell/18 + K_e + O(\ell^2/X), \tag{5.131}$$

$$G(0,r,-1;X) = \ell^3/6 - \pi^2\ell/18 + K_f + O(\ell^2/X). \tag{5.132}$$

The logarithmic terms follow by integrating their defining kernels at large X. Rational-letter functions follow by the same recursion, with constants fixed by their classical polylogarithmic expressions. Before evaluating the six constants, collecting every term gives

$$\begin{aligned} C_1(X) = \frac{1}{X}\Bigg[&\frac{\ell^3}{3} - \frac{\ell^2}{2} + \frac{17\ell}{4} - \frac{75}{8} - \frac{\pi^2}{2} + 13\zeta(3) \\ &+ 15K_a - 6K_b - 8K_c - 16K_d - 6K_e - 3K_f\Bigg] + O(\ell^3/X^2). \end{aligned} \tag{5.133}$$

Using the endpoint constants already evaluated in (5.46), we obtain

$$15K_a - 6K_b - 8K_c - 16K_d - 6K_e - 3K_f = -9\zeta(3). \tag{5.134}$$

The Clausen constant cancels exactly. Substitution proves (5.123), including its finite constant. The rational kernels have inverse-power expansions at infinity, so the next order is bounded by $O(\ln^3 X/X^2)$.

Only now do we compare with Lee, Schwartz and Zhang, Equation (11) [2]. Replacing X by s_{phys}/m^2 inside the leading term changes only the displayed remainder. All three logarithmic coefficients and the finite coefficient agree with that published expression.

5.6 Summary

Use $X = (s_{\text{phys}} - m^2)/m^2 = 2\omega/m$, where ω is the incident photon energy in the electron rest frame. Write the NLO total as $\sigma_0 + \sigma_1$, with σ_0 of order α_{T}^2 and σ_1 of order α_{T}^3. The low-energy results are

$$\sigma_0 = \frac{\pi\alpha_{\text{T}}^2}{m^2}\left[\frac{8}{3} - \frac{8}{3}X + \frac{52}{15}X^2 - \frac{133}{30}X^3 + O(X^4)\right]. \tag{5.135}$$

$$\begin{aligned}\sigma_1 = \frac{\alpha_{\text{T}}^3}{m^2}\Bigg\{&X^2\left[-\frac{16}{9}\ln X + \frac{7}{15}\right] + X^3\left[\frac{4}{3}\ln X - \frac{113}{90}\right] \\ &+ X^4\left[-\frac{25}{9}\ln X + \frac{203}{125}\right] + O\Big(X^5|\ln X|\Big)\Bigg\}, \qquad X \to 0^+.\end{aligned} \tag{5.136}$$

The combination of virtual, real, and soft contributions is derived in Section 5.4, and the further orders follow directly from the collected formula in Section 5.4.1. The photon-mass logarithms cancel before taking this limit. The radiative correction vanishes, while the Born cross section tends to $8\pi\alpha_{\text{T}}^2/(3m^2)$.

At high energy, the Born result is

$$\sigma_0 = \frac{\pi\alpha_{\text{T}}^2}{m^2}\left[\frac{2\ln X + 1}{X} + \frac{18 - 8\ln X}{X^2} + O\left(\frac{\ln X}{X^3}\right)\right], \qquad X \to \infty. \tag{5.137}$$

The NLO correction is

$$\sigma_1 = \frac{\alpha_{\text{T}}^3}{m^2}\left\{\frac{1}{X}\left[\frac{\ln^3 X}{3} - \frac{\ln^2 X}{2} + \frac{17}{4}\ln X - \frac{75}{8} - \frac{\pi^2}{2} + 4\zeta(3)\right] + O\left(\frac{\ln^3 X}{X^2}\right)\right\}. \tag{5.138}$$

Section 5.5 derives this expansion directly from the collected total, including the finite constants of its GPLs. The comparison with the published expansion is made after those constants are determined.

Chapter 6

Comparison with the published analytic total

6.1 The analytic total of Lee, Schwartz and Zhang

The formula reproduced in this section is Equation (6) of Lee, Schwartz and Zhang's *The Compton Scattering Total Cross Section at Next-to-Leading Order* [2]. It provides a reference for checking the independent integration of the amplitudes developed above.

Their observable includes $\gamma e^- \to \gamma e^-$ and $\gamma e^- \to \gamma\gamma e^-$ with all photon energies accepted. It excludes the distinct pair-production channel $\gamma e^- \to e^- e^+ e^-$. Define

$$X = \frac{s_{\text{phys}} - m^2}{m^2} = \frac{2\omega}{m}, \qquad Y = \sqrt{\frac{X}{X+4}}, \qquad M = \ln X, \qquad L = \ln(1+X). \tag{6.1}$$

Here $s_{\text{phys}} = (p+k)^2$ is dimensionful. These are the same X, Y as in the independent derivation. The reference denotes them by lowercase x, y. The correction, as distinct from Born plus correction, is

$$\sigma_1 = \frac{\alpha_{\text{T}}^3}{m^2 X^3}\mathcal{H}(X), \qquad \sigma_{\text{inc}}^{\text{NLO}} = \sigma_0 + \sigma_1, \qquad \mathcal{H} = \mathcal{H}_{\text{cl}} + \mathcal{H}_{\text{rat}} + \mathcal{H}_{\text{root}}. \tag{6.2}$$

This is Equation (6) of Lee, Schwartz and Zhang [2], regrouped into classical polylogarithms, iterated rational kernels and a square-root kernel. Their notation σ_{NLO} denotes the correction σ_1, not the sum in (6.2). The function $\mathcal{H}(X)$ is the published numerator. We reserve $H = s - 1$ for the energy variable used in the independent integration. The following numbered terms specify the entire function.

Define the classical polylogarithm by $\text{Li}_n(z) = \sum_{j=1}^{\infty} z^j/j^n$ for $|z| < 1$ and analytic

continuation elsewhere. Then

$$\mathcal{H}_{\rm cl} = \sum_{j=1}^{9} C_j^{\rm ref}, \tag{6.3}$$

$$C_1^{\rm ref} = -\frac{X(273X^3 - 982X^2 - 2960X - 1744)}{24(X+1)^2}, \tag{6.4}$$

$$C_2^{\rm ref} = \frac{37X^4 - 54X^3 - 339X^2 - 428X - 184}{4(X+1)^2}\, L, \tag{6.5}$$

$$C_3^{\rm ref} = \frac{X^2(14X^4 + 17X^3 - 17X^2 - 22X - 8)}{2(1+X)^3}\frac{M}{1-X}, \tag{6.6}$$

$$C_4^{\rm ref} = -\frac{4X^6 + 35X^5 - 31X^4 - 755X^3 - 1765X^2 - 1506X - 440}{2(X+1)^2(X+4)}\, L^2, \tag{6.7}$$

$$C_5^{\rm ref} = -\frac{X^6 + 7X^5 - 28X^4 - 239X^3 - 449X^2 - 338X - 88}{(X+1)^2(X+4)}\, \mathrm{Li}_2(-X), \tag{6.8}$$

$$C_6^{\rm ref} = -\frac{X^2 - 16X - 23}{3}\, L^3, \tag{6.9}$$

$$C_7^{\rm ref} = (X^2 - X + 2)\left[\mathrm{Li}_2(1-X) - \frac{\pi^2}{6}\right], \tag{6.10}$$

$$C_8^{\rm ref} = \frac{X^4 + 7X^3 + X^2 - 3X - 2}{(X+1)^2}\, LM, \tag{6.11}$$

$$C_9^{\rm ref} = (X^2 + 2X - 6)[\mathrm{Li}_3(X^2) - M\,\mathrm{Li}_2(X^2)]. \tag{6.12}$$

The quotient $M/(1-X)$ has the continuous value -1 at $X = 1$. For $X > 1$ the two polylogarithms in $C_9^{\rm ref}$ individually have imaginary parts. On a common branch they cancel in this combination, which is real for positive X.

To define the remaining functions, let $\boldsymbol{b}$ denote an ordered list of letters and put

$$G(a, \boldsymbol{b}; X) = \int_0^X \mathrm{d}z\, w_a(z) G(\boldsymbol{b}; z), \qquad G(\varnothing; X) = 1, \tag{6.13}$$

$$w_a(z) = \frac{1}{z-a} \quad (a = 0, -1, -2, -4), \qquad w_r(z) = \frac{1}{\sqrt{z(z+4)}}, \tag{6.14}$$

$$G(\underbrace{0, \ldots, 0}_{n}; X) = \frac{\ln^n X}{n!}. \tag{6.15}$$

The letter r is a kernel label. It corresponds to the letter denoted y in Reference [2], and is not the value Y held constant inside an integration. The purely rational-kernel terms are

$$\mathcal{H}_{\rm rat} = \sum_{j=1}^{5} R_j, \tag{6.16}$$

$$R_1 = -\frac{4(X^5 + 26X^4 + 146X^3 + 316X^2 + 288X + 96)}{(X+1)^2(X+4)} G(-2, -1; X), \tag{6.17}$$

$$R_2 = 8(X^2 - 4X - 6) G(-1, -2, -1; X), \tag{6.18}$$

$$R_3 = 4(2X^2 - X - 6) G(-1, -1, 0; X), \tag{6.19}$$

$$R_4 = 2(2X^2 - 7X - 12) G(-1, 0, -1; X), \tag{6.20}$$

$$R_5 = -(5X^2 + 32X - 8) G(0, -1, -1; X). \tag{6.21}$$

Finally the square-root-kernel terms are

$$\mathcal{H}_{\rm root} = \sum_{j=1}^{7} T_j, \tag{6.22}$$

$$T_1 = -3(X-2)(X+4)Y[G(0,r,-1;X) + 2G(r,-1,0;X)], \tag{6.23}$$

$$T_2 = -\frac{8Y(X^4+3X^3-18X^2-68X-24)}{(X+4)X} G(r,0,-1;X), \tag{6.24}$$

$$T_3 = \frac{3Y(5X^4+14X^3-96X^2-352X-128)}{(X+4)X} G(r,-1,-1;X), \tag{6.25}$$

$$T_4 = -\frac{16Y(X^4+2X^3-24X^2-80X-48)}{(X+4)X} G(r,-2,-1;X), \tag{6.26}$$

$$T_5 = \frac{3X^4+18X^3+44X^2-8X-64}{X} YG(r,-1;X), \tag{6.27}$$

$$T_6 = -\frac{6Y(X^3-12X-8)}{X} G(-4,r,-1;X), \tag{6.28}$$

$$T_7 = 3(3X^2-8)G(r,r,-1;X). \tag{6.29}$$

All G functions displayed here are real for positive X. The exact formula includes the mass dependence of both phase spaces. The logarithmic soft factor alone does not determine its finite real-emission contribution.

6.2 Elementary reductions

Several nested integrals can be checked without solving a numerical system. Direct integration gives

$$G(-1;X) = L, \qquad G(0,-1;X) = -\operatorname{Li}_2(-X), \tag{6.30}$$

$$G(-1,0;X) = LM + \operatorname{Li}_2(-X), \qquad G(-1,-1;X) = \tfrac{1}{2}L^2, \tag{6.31}$$

$$G(-2,-1;X) = \operatorname{Li}_2(-1-X) + L\ln(X+2) + \frac{\pi^2}{12}, \tag{6.32}$$

$$G(-1,-2,-1;X) = 2\operatorname{Li}_3(-1-X) - L\operatorname{Li}_2(-1-X) + \frac{\pi^2 L}{12} + \frac{3}{2}\zeta(3). \tag{6.33}$$

For example, differentiating the third line gives $L/(X+2)$, and its value at $X=0$ is zero. This verifies both its kernel and integration constant.

For the square root choose

$$z = \frac{1-Y}{1+Y} = \frac{4}{(\sqrt{X+4}+\sqrt{X})^2}, \qquad X = \frac{(1-z)^2}{z}, \qquad \frac{\mathrm{d}X}{\sqrt{X(X+4)}} = -\frac{\mathrm{d}z}{z}. \tag{6.34}$$

The rationalized expression for z avoids subtraction of nearly equal numbers at large X. Since $1+X = (1+z^3)/[z(1+z)]$, the integral of the logarithm reduces to dilogarithms. With the integration constant fixed at $z=1$,

$$G(r,-1;X) = \frac{1}{3}\operatorname{Li}_2(-z^3) - \operatorname{Li}_2(-z) + \frac{1}{2}\ln^2 z - \frac{\pi^2}{18}, \tag{6.35}$$

$$G(r,r,-1;X) = \operatorname{Li}_3(-z) - \frac{1}{9}\operatorname{Li}_3(-z^3) - \frac{1}{6}\ln^3 z + \frac{\pi^2}{18}\ln z + \frac{2}{3}\zeta(3). \tag{6.36}$$

The second line follows by a further integration with $-\mathrm{d}z/z$. These are checks on the iterated-integral evaluation, independent of its differential-equation solver.

6.3 Limits of the integrated result

The low-energy expansion of Lee, Schwartz and Zhang's total cross section is given in Equation (10) of Reference [2]. Its order-α_{T}^3 term is

$$\sigma_1 = \frac{\alpha_{\mathrm{T}}^3}{m^2} X^2 \left[-\frac{16}{9} \ln X + \frac{7}{15} + \mathcal{O}(X \ln X) \right]. \tag{6.37}$$

The Born Thomson limit therefore remains fixed by the physical low-energy charge. With the same variable $X = (s_{\mathrm{phys}} - m^2)/m^2$ and normalization $\alpha_{\mathrm{T}}^3/m^2$, our independently assembled result (5.117) agrees with both displayed NLO coefficients in their Equation (10): $-16/9$ for $X^2 \ln X$ and $7/15$ for X^2. The cited equation writes the higher terms as an ellipsis. The explicit remainder order above follows from our separate virtual, real, and soft expansions. Agreement of these two coefficients is a low-energy check, not a proof of equality of the complete cross sections at arbitrary energy.

At large s_{phys}/m^2, with $L = \ln(s_{\mathrm{phys}}/m^2)$, Equation (11) of the same reference gives

$$\sigma_1 = \frac{\alpha_{\mathrm{T}}^3}{s_{\mathrm{phys}}} \left[\frac{L^3}{3} - \frac{L^2}{2} + \frac{17L}{4} - \frac{75}{8} - \frac{\pi^2}{2} + 4\zeta(3) + \mathcal{O}\left(\frac{m^2}{s_{\mathrm{phys}}} L^3 \right) \right]. \tag{6.38}$$

This expansion tests the high-energy magnitude and logarithmic structure. The exact expression is used for the data comparison.

6.4 Analytic comparison with the published total

Agreement at finitely many energies is a useful numerical check, but does not prove that two analytic functions coincide. This section compares the complete independent result in Equation (4.555) with Equation (6) of Lee, Schwartz and Zhang [2]. The published coefficients enter only this comparison and are not used to construct the independent amplitude integral.

Observable and normalization. Write $s = s_{\mathrm{phys}}/m^2$ and $X = s - 1$. The comparison concerns unpolarized $e^-\gamma \to e^-\gamma$ together with $e^-\gamma \to e^-\gamma\gamma$, integrated over the complete photon phase space. It excludes the separate charged-pair final state. Both calculations use the on-shell mass and physical low-energy charge. The published calculation uses dimensional regularization, whereas the independent construction combines photon-mass and auxiliary Mellin regulators before removing them. The comparison is between their finite sums, not their individual real or virtual finite parts. The precise normalization is

$$\sigma_1 = \frac{\alpha_{\mathrm{T}}^3}{m^2} C_{\mathrm{total}}(s), \qquad C_{\mathrm{LSZ}}(s) = \frac{\mathcal{H}_{\mathrm{LSZ}}(X)}{X^3}, \qquad X = s - 1. \tag{6.39}$$

Here $\mathcal{H}_{\mathrm{LSZ}}(X) = \mathcal{H}(X)$ is the numerator defined in the preceding section. It is not the energy variable denoted by H in the independent integration.

Classical functions in one iterated-integral basis. Let G have the kernels defined in the preceding reference section. Several changes of basis remove apparent differences

immediately:

$$\ln X = G(0;X), \qquad \ln(1+X) = G(-1;X), \tag{6.40}$$

$$\mathrm{Li}_2(-X) = -G(0,-1;X), \tag{6.41}$$

$$\mathrm{Li}_2(1-X) - \frac{\pi^2}{6} = -G(1,0;X), \tag{6.42}$$

$$\mathrm{Li}_3(X^2) - \ln X\,\mathrm{Li}_2(X^2) = 2G(0,1,0;X) + 2G(0,-1,0;X). \tag{6.43}$$

For the last identity, apply $\mathrm{Li}_n(X^2) = 2^{n-1}[\mathrm{Li}_n(X) + \mathrm{Li}_n(-X)]$ and the shuffle product

$$G(0;X)G(0,a;X) = 2G(0,0,a;X) + G(0,a,0;X). \tag{6.44}$$

Alternatively, differentiating both sides of (6.43) gives the same expression, and both sides vanish at $X = 0$. These arguments initially apply for $0 < X < 1$ and extend continuously through $X = 1$. Indeed, the inner kernel of $G(1,0;X)$ is $\ln X/(X-1)$, which is finite at one. Thus the right-hand side is real on the whole positive axis. On the left-hand side the imaginary parts cancel when both polylogarithms are continued on the same side of their cut. No choice of an imaginary part based on numerical agreement is required.

After reducing products by shuffle identities, the complete reference is a linear combination of words of weight at most three with letters

$$\{0,1,-1,-2,-4,r\}, \qquad \omega_r = \frac{\mathrm{d}X}{\sqrt{X(X+4)}}. \tag{6.45}$$

The letter 1 is needed for the classical combinations above, although it is absent from the thirteen nonclassical words displayed separately in the published formula.

Comparison in the constant alphabet. Section 4.2 derives the constant-letter representation of the complete virtual contribution from its own angular primitives. Applying the kernel map (4.17) to the published total gives exactly the same set of 116 words. Both expressions use basepoint $q = 1$ and the logarithmic convention fixed there.

The comparison is performed on the complete infrared-finite sums. No difference between the published total and a partial contribution is used to reconstruct the remaining sectors.

Equality of the complete finite sum. The finite soft reduction in Section 4.7.3, the integrated UW term in Equation (4.520), and the other real-emission sectors complete the independent result. Their sum is Equation (4.555). In particular, the intermediate radical $\sqrt{s}$ no longer appears in its kernels or coefficients. The only square root in the energy representation is $Y = \sqrt{X/(X+4)}$, with its positive value for $X > 0$. This cancellation follows from the sector reductions before the published expression is compared.

Let $\mathcal{W}_{23}$ denote the twenty-three words listed in Equation (4.555), including the empty word. Expanding the published expression with the classical identities above and the shuffle rule gives exactly the same word set. Thus

$$C_1(X) - C_{\mathrm{LSZ}}(1+X) = \sum_{w\in\mathcal{W}_{23}} \Delta_w(X,Y)G(w;X), \qquad \Delta_w = c_w^{\mathrm{ind}} - c_w^{\mathrm{LSZ}}. \tag{6.46}$$

All twenty-three differences vanish as rational functions of X and Y. No numerical value of X, expansion around an endpoint, or relation between distinct GPL words is used for

this final subtraction. For example, the common coefficients of three words are

$$c_{(1,0)} = -\frac{X^2 - X + 2}{X^3}, \tag{6.47}$$

$$c_{(0,-1,0)} = c_{(0,1,0)} = \frac{2(X^2 + 2X - 6)}{X^3}, \tag{6.48}$$

$$c_{(r,r,-1)} = \frac{3(3X^2 - 8)}{X^3}. \tag{6.49}$$

The coefficient comparison includes the empty word, so a residual additive constant is also excluded. Linear independence of the words is unnecessary: equality of each coefficient is sufficient.

It remains to identify their definitions on the physical axis. The root weight in Equations (7)–(8) of the reference is $Y\,\mathrm{d}X/X = \mathrm{d}X/\sqrt{X(X+4)}$, exactly the kernel ω_r above. Both definitions start at zero, use $G(0;X) = \ln X$, and integrate along positive real X. All other nonzero letters are negative, except for the two classical combinations containing the letter 1. For those combinations $\ln X/(X-1)$ has a removable singularity. Their continuous real values were established above by differentiation and the zero-endpoint boundary values. The apparently singular elementary term at $X = 1$ is likewise fixed by $\lim_{X\to 1} \ln X/(X-1) = 1$. Consequently the coefficient identity gives

$$\boxed{C_1(X) = C_{\mathrm{LSZ}}(1+X), \qquad X > 0,} \tag{6.50}$$

including $X = 1$ by continuity. No integration constant or physical branch has been adjusted to enforce this equality.

The comparison can be reproduced by `compton_total_reference_identity.py`. It reads the independent coefficient dictionary and separately expands the transcription of Equation (6) of the reference. The report records every coefficient difference and hashes of both inputs. The transcription was checked term by term against the published equation, including its overall X^{-3} normalization and root weight. The independent calculation does not import that transcription.

A shorter product representation. Equality with the published formula does not require keeping its organization. Section 4.10.1 applies further shuffle identities to the independently obtained coefficients. Three weight-three words are replaced by products involving lower-weight functions, and two equal-coefficient words are combined into $\mathcal{P}(X)$. The result has an elementary logarithmic polynomial and thirteen remaining grouped functions. Expanding its products recovers all twenty-three coefficients exactly. This reduction decreases the number of separate special-function terms, although some coefficients then contain elementary logarithms. It makes no claim that thirteen is a minimal function count.

Chapter 7

Numerical verification

7.1 Scope and reproducibility of the checks

The algebraic reductions and physical branches in the preceding chapters are fixed before any numerical comparison. The command `make check` regenerates its required analytic intermediates and runs the standard verification suite. It checks source consistency, exact coefficient identities, amplitudes, integrated sectors, endpoint matching, asymptotic expansions and numerical implementations. Each check records its exit status and its output in `build/checks/`. The command `make check-list` lists the preparation steps and tests. The more expensive full phase-space audits have separate commands in the build instructions. Their saved results below are distinguished from the standard suite.

For example, `compton_dilog_identities.py` establishes the two real dilogarithm reductions by differentiating both sides and fixing their common boundary values at $s = 1$. Only after that proof does `check_compton_dilog_identities.py` compare their implementations at $s = 1.0001, 1.02, 1.6, 10, 1000$, using 70-digit arithmetic and an absolute tolerance of 10^{-60}. These samples determine neither the identity nor its integration constant.

The regular product form has a separate implementation in `compton_total_regular.py`. Its exact checks reconstruct every coefficient and verify the real-axis inversion relations by derivatives and boundary values. The numerical program `check_compton_total_regular.py` then compares the full result with the original collected expression at ten energies from $X = 0.0002$ to $X = 200000$, including $X = 1$ and offsets of 10^{-20} on either side. Its tolerance is $10^{-30}\max(1, |C_1|)$. The paired trilogarithm is also checked against its original boundary-value representation at those points. These tests check the implementation of already established identities.

Near threshold, subtracting primitive endpoints and dividing by X^3 amplifies rounding errors. The regular evaluator adds six working digits per decade of $X < 1$ to its base guard precision, then repeats the evaluation with thirty additional digits. It checks both the change and the imaginary residual relative to the result. A further increase is allowed, and failure to converge raises an error. At $X = 10^{-18}$ and 10^{-24}, the same test compares the result with the independently derived low-energy expansion through X^4. The relative differences are below 7×10^{-55} and 7×10^{-73}, respectively. These tests validate the evaluation of the exact expression, without determining its coefficients.

The same program checks the shared root-function evaluator in `compton_root_classical.py`. It evaluates the common endpoint differences once and uses the shuffle identity for the outer-zero word. The original primitives, endpoint constants and complex branches are unchanged. At the same ten energies, seven words are compared with their sep-

arate implementations using 80-digit arithmetic and tolerance $10^{-70}\max(1,|G|)$. The word $G(r,r,-1;X)$ is checked through its defining derivative, with tolerance $10^{-65}\max(1,|\partial_X G|)$.

The direction averages have a separate check in `compton_real_analytic_angular_check.py`. It compares the closed masters with direct sphere quadrature at four regular kinematic points. Near threshold, small Gram determinants and cancellation between terms require additional working precision. At $H = 0.02$, the check repeats the master sum at increasing precision and the sphere integral at two angular resolutions. Both quadratures agree with the master sum to a relative tolerance of 10^{-30}. These evaluations share the same 25 coefficients, so the comparison tests the direction averages and their implementation. Precision and angular-resolution changes provide stability checks, not rigorous error bounds or independent checks of the original trace.

7.2 Independent amplitude checks

The program `check_compton_amplitudes.py` constructs explicit Dirac matrices with $m = e_{\mathrm{T}} = 1$. It does not insert a known squared Compton amplitude. It sums physical transverse photon polarizations and performs the spin traces. Three Born points with $(\omega/m, \cos\theta) = (0.1, -0.4), (1, 0.3), (20, -0.7)$ agree with Klein–Nishina to a relative error below 5×10^{-16}.

For the real amplitude it constructs on-shell three-body events at $\omega/m = 0.3, 1, 7$. At every event it replaces each of the three photon polarizations in turn by its momentum. The spinor-sandwiched matrix vanishes with a relative residual below 6×10^{-14}. Interchanging the two outgoing photons changes the kernel by less than 3×10^{-16} relative to its norm. A separate test of the contracted off-shell vertex integrand verifies (2.40) before counterterms, with a relative residual below 2×10^{-16}.

The soft check is more restrictive than a Ward identity. Keep a hard-photon direction fixed and compare the exact three-body square with $e_{\mathrm{T}}^2(-J^2)B_{\mathrm{amp}}$, projecting the hard system onto on-shell two-body kinematics in the denominator. Table 7.1 shows the approach to the derived limit. Corrections are linear in the small photon energy at this fixed event. The program `check_compton_virtual.py` separately integrates the full subtracted loop

Soft photon energy divided by m	Exact square / soft approximation
10^{-3}	0.9980259445
10^{-4}	0.9998019242
10^{-5}	0.9999801857
10^{-6}	0.9999980193

Table 7.1: Soft-limit check of the six real-emission diagrams at incident energy $\omega = m$. The event directions are fixed in the verification script.

tensor by Gauss quadrature. It uses $\omega/m = 0.3$, $\cos\theta = 0.2$ and $\lambda/m = 0.7$ so that the parameter denominators stay away from physical cuts. This finite regulator is a test of the matrix identities, not a physical small-λ prediction. Including all eight diagrams, the two external Ward residuals decrease from 4.1×10^{-9} at quadrature order 12 to 7.6×10^{-15} at order 28. The loop matrix changes by 5.3×10^{-15} relative to its norm between orders 20 and 28. This independently tests the parameter reduction, finite counterterms and relative signs. It does not evaluate the inclusive total or the physical small-λ loop across its cuts.

7.3 Cross-check of the analytic scalar box

The script `check_compton_scalar_analytic.py` evaluates (3.30) at 50 decimal digits and compares it with quadrature of (3.25). At $-t = 0.001, 0.09, 5, 7800$, the absolute differences are below 3×10^{-50}. Quadrature serves only to check the derived dilogarithm identity. This test checks the scalar box identity, rather than the total phase-space integration.

7.4 Cross-check of the closed vertex

The script `check_compton_vertex_analytic.py` compares the six closed moments in (3.39) with their parameter integrals at $S = -4, 0, 0.49, 1.6, 10$. Both real and imaginary parts are included. At 60 decimal digits, the largest absolute difference is below 10^{-58}. The continuation in the integrand and in the closed expression is fixed by the Feynman denominator as derived above.

The same program compares all four vertex insertions in the Compton amplitude with parameter quadrature at $(s, c) = (1.6, 0.2), (10, -0.4), (100, 0.7)$. Increasing the quadrature order from 32 to 64 and 128 decreases the maximum matrix discrepancy at the last point from 8.33×10^{-9} to 1.37×10^{-10} and 2.19×10^{-12}. Here the discrepancy is the Frobenius norm of the matrix difference divided by the larger of one and the norm of the closed matrix. The analytic expression itself has no quadrature order. These tests concern the vertex contribution, rather than the complete total-cross-section integration.

7.5 Analytic integration of a virtual subset

The trace coefficients for the vertex and self-energy terms are generated by `compton_virtual_trace_analytic.py` using Clifford anticommutation. Explicit covariant polarization sums of the Dirac matrices reproduce those coefficients to relative errors below 3×10^{-15} at the three points used above. The covariant sum is used on both sides of this check because this subset alone is not gauge invariant.

The primitive in (3.154) has an exact symbolic derivative equal to the crossed-ordering trace. Independent 50-digit quadrature of that trace at $s = 1.6, 10, 100$ differs from the primitive's endpoint difference by less than 10^{-45}. The script `check_compton_virtual_angular_analytic.py` also verifies the complete vertex-plus-self-energy angular integral, including its phase-space normalization, against a separate double-precision trace quadrature. At $\lambda = 10^{-6}$ the analytic coefficients $\mathcal{C}_{\text{vertex}+\Sigma}$ are

$$s = 1.6: \quad 52.913883157178253462497154 4028, \tag{7.1}$$

$$s = 10: \quad 15.111450203374370597352074 5024, \tag{7.2}$$

$$s = 100: \quad 2.819710687579985786981396 25584. \tag{7.3}$$

These regulator-dependent numbers are diagram-subset checks. They are not physical NLO correction factors and must be combined with the spanning and real terms before comparison with a total cross section.

7.6 Ward identities of the complete closed amplitude

The script `compton_amplitude_analytic.py` combines the closed self-energy, vertex and spanning matrices. No loop-parameter quadrature is used. Replacing either external polarization by its photon momentum gives normalized Ward residuals below 1.1×10^{-13}

at $(s,c) = (1.6, 0.2), (10, -0.4), (100, 0.7)$, for both $\lambda = 10^{-5}$ and 10^{-8}. The normalization is the larger of one and the norm of a transverse spinor-sandwiched amplitude, as in the earlier Ward check. This tests the relative signs and finite terms of the complete analytic virtual amplitude in the physical region. It is distinct from the real-radiation energy integrals in the total cross section.

The script `check_compton_infrared_analytic.py` separately changes λ from 10^{-5} to 10^{-8} in this same closed amplitude. At the three points above it verifies

$$\frac{\operatorname{Re}\mathcal{T}(10^{-5}) - \operatorname{Re}\mathcal{T}(10^{-8})}{4\ln(10^3)} = \mathcal{B}(b_e - 1) \tag{7.4}$$

with absolute residuals below 2×10^{-14}. This is the coefficient required for pointwise cancellation against the massive-photon soft term. It checks the independently derived infrared factor, rather than determining that factor by fitting.

7.7 Checks of the integrated amplitudes

The following checks test the analytic integrations derived in Chapter 3. Numerical values do not fix their constants or physical branches.

7.7.1 Checks of the spanning matrix

Numerical integration of the earlier two-parameter formula provides a check of the analytic box, not its definition. At $(s,c) = (1.6, 0.2), (10, -0.4), (100, 0.7)$, comparison of all four physical polarization matrices in both photon orderings with 128-point integration in each parameter gives maximal relative matrix differences 1.9×10^{-13}, 7.1×10^{-13} and 5.0×10^{-12}, respectively. Separately, evaluating the rational interference coefficients agrees with explicit covariant polarization contractions of the closed 4×4 matrices to relative precision better than 7.1×10^{-15} at these points. Neither comparison uses a published total-cross-section formula.

7.7.2 Evaluation of the spanning angular primitives

The numerical evaluation of the GPL expression can be performed by convergent Taylor expansions without integrating the original angle-dependent correction. Around a point v_0 away from a letter a, write the Taylor coefficients of the shorter word as t_n and those of the derivative of $G(a,w)$ as d_n. Its differential equation gives the recursion

$$d_{-1} = 0, \qquad d_n = \frac{t_n - d_{n-1}}{v_0 - a}, \qquad [(v - v_0)^{n+1}]G(a,w;v) = \frac{d_n}{n+1}. \tag{7.5}$$

Overlapping disks along the upper path then continue the same analytic function to the endpoint. This operation evaluates the closed special functions on the branch already fixed by the Feynman pole displacement and the continuation path derived above. No cross-section comparison enters that choice.

As a check, take $\lambda = 10^{-5}$ and define $J_j = \mathcal{F}_j(v_{\max}; x) - \mathcal{L}_j(x)$ for $j = s, u$. This is the quantity before multiplication by $1/(2x^2)$. Evaluation of the closed expressions gives

$$J_s(0.6) = -18.610539663050517256626078724120481781643390076 9, \tag{7.6}$$

$$J_u(0.6) = -14.139127852215781722776948997533748438351661621 7, \tag{7.7}$$

$$J_u(9) = -3080.097759497736575300830540132374613663922186 50. \tag{7.8}$$

Independent quadrature of the already derived angular interference agrees with these values to absolute differences 9×10^{-47}, 10^{-62} and 2×10^{-58}, respectively. The last GPL evaluation crosses the apparent η pole by the specified upper path. Its residual imaginary part is below 3×10^{-64} before taking any real part. These checks test the analytic integrations and the causal branch choice. They do not use a published total cross section.

7.7.3 Checks of the finite soft function

The script `compton_soft_analytic.py` evaluates (3.193) directly and separately integrates (3.176) for a check. At 70 decimal digits, checks at $s = 1.6$, 10, 100, and 19570.511835591835, with $a^2/q^2 = 0.0001$, 0.2, 0.8, and 0.9999, give absolute differences below 1.2×10^{-66}. These quadratures check the derived closed expression. They do not define it. The result completes the finite soft parameter integral at fixed angle. Its Born-weighted elastic angular integral is a further part of the total-cross-section calculation.

7.7.4 Check of the integrated infrared coefficient

Direct checks of the original angular integral at 65-digit working precision give $\mathcal{C}_{\rm IR} = 0.2034909958483591$, 10.14476601141435, 64.23179863499513, and 359.2271977981074 at $s = 1.6$, 10, 100, and 19570.511835591835. The largest absolute difference between the dilogarithmic expression and quadrature is 1.1×10^{-61}.

7.7.5 Independent numerical angular integration

The numerical angular check evaluates the covariant Dirac trace before integrating over the scattering angle. Its scalar one-loop functions are the logarithms and dilogarithms obtained from the parameter integrals. It does not evaluate an integrated GPL primitive inside the numerical integrand. In units $m = 1$, put $x = s - 1$, $y = x + t$, and use the spin-averaged interference trace $\mathcal{T}$ and the Born factor $\mathcal{B} = \overline{|\mathcal{M}_0|^2}/e_{\rm T}^4$ from (3.172). The finite virtual-plus-soft coefficient in the local real-emission subtraction scheme is

$$\mathcal{C}_{\rm VS}(s) = \frac{1}{s}\int_{-1}^{1} \mathrm{d}c \left\{ \frac{\mathrm{Re}\,\mathcal{T}(s,c;\lambda)}{4} + \mathcal{B}(s,c)\left[(b_e-1)\ln\frac{2}{\lambda} - K(s,c)\right]\right\}. \tag{7.9}$$

The logarithm of the photon mass cancels inside this combined integrand. This coefficient must be combined with the real contribution defined by the same local subtraction. The finite coefficient obtained with the analytic factor $(xy)^\eta$ has a different soft matching term. Here x, y in the regulator are the real-emission Dalitz variables, not the elastic invariants used in this angular check.

To resolve both angular endpoints at large s, write $L = \ln s$ and introduce $0 < u < 1$ by

$$c = \left(1+\frac{2}{x}\right)\tanh\!\left[L(u-\tfrac{1}{2})\right], \qquad y = x - \frac{x^2}{2s}(1-c), \tag{7.10}$$

$$\frac{dc}{du} = \left(1+\frac{2}{x}\right)\frac{L}{\cosh^2\!\left[L(u-\tfrac{1}{2})\right]}. \tag{7.11}$$

The identity $\tanh(L/2) = x/(x+2)$ proves that the endpoints map exactly to $c = -1$ and $c = 1$. Positive Gauss–Legendre nodes lie strictly inside this interval, so no removable endpoint expression is evaluated directly. The check uses arbitrary-precision arithmetic and three successively doubled node counts. Ten times the larger of the final two differences is an empirical discretization error estimate, not a rigorous error bound. The reported uncertainty also has the conservative arithmetic floor 10^{20-d} when using d decimal digits. Increasing

the arithmetic precision provides a separate check of cancellation error, particularly near threshold. A second variable map, $y = (x/s)\exp(u \ln s)$, checks the angular Jacobian independently. The scope of this test is the angular integration and its implementation. The earlier comparison with parameter quadrature tests the scalar one-loop functions themselves.

7.7.6 Checks of the soft angular primitive

The algebraic reduction has also been implemented independently of the quadrature check. At $s = 10$ it generates 306 terms involving 305 distinct nonempty polylogarithm words, including words needed to evaluate their lower-weight derivatives. Partial-fraction reconstruction of the rational kernels agrees to 1.3×10^{-60} at 60-digit precision. At $z = 0.9Z$, evaluating the generated polylogarithms through their defining differential equations gives 17.00560385716334, compared with 17.00560385716422 from direct quadrature of the rationalized kernel. These two numerical evaluations check the algebraically integrated expression.

7.7.7 Series evaluation of the soft angular endpoint

Arbitrary-precision polylogarithm evaluation. The generated expression can be checked beyond machine precision by convergent power series, without integrating the Compton kernel. For a word $(a, \boldsymbol{b})$ expand $G(a, \boldsymbol{b}; z_0 + h)$ in powers of h with coefficients c_n, and write the tail word as $G(\boldsymbol{b}; z_0 + h) = \sum_n d_n h^n$. Its differential equation gives

$$c_{n+1} = \frac{d_n - n c_n}{(z_0 - a)(n+1)}, \qquad z_0 \neq a. \tag{7.12}$$

Words are processed in increasing length, so all d_n are already known. At the regular origin of a word beginning with zero, $d_0 = 0$ and $c_n = d_n/n$ for $n \geq 1$. Steps smaller than the distance to the nearest letter analytically continue these series along the real physical path. At 60-digit working precision, the 306-term expression at $s = 10$ and $z = 0.9Z$ agrees with direct quadrature to 1.0×10^{-59}. Thus the finite polylogarithmic reduction itself, rather than only the preceding integral transformations, has been checked independently. For the total angular coefficient, taking $z/Z = 1 - 10^{-25}$ gives $d = 8.8303688 \times 10^{-26}$. Adding the analytically derived endpoint tail then yields $\mathcal{C}_K(10) = 10.058462232631976837745\ldots$. Its difference from the original integral is 7.8×10^{-48}, consistent with the $O(d^2 \ln d)$ tail remainder. At $s = 1.6$ the interior endpoint check agrees to 2.8×10^{-58}. These checks use only the amplitudes, soft integral, and algebraic reductions derived here, with no published total cross section as input.

7.7.8 Check of the inner-energy Laurent coefficients

The script `compton_real_uv_endpoint.py` verifies the Laurent coefficients against the exact beta–hypergeometric expressions at 80 decimal digits. This check evaluates analytic functions and does not use numerical quadrature.

7.7.9 Checks of the UV difference

At $H = 9$, the generated endpoint expression for this pole coefficient contains six terms, with words drawn from $G(r_\pm - 1, -1; 9)$, $G(r_\pm - 1; 9)$, and $G(-1; 9)$, together with a

rational endpoint term. Evaluating this analytic expression gives

$$\int_1^{10} \mathcal{P}_{\mathrm{UV}}(9,v)\,\mathrm{d}v = 27551.455004902211436469664024448 3\ldots. \tag{7.13}$$

The difference from independent quadrature is 4.4×10^{-61} at 65-digit working precision. The quadrature uses the analytic local series near $v=1$ to avoid subtracting individually divergent numerical terms. The integrated result includes the complete difference with its Taylor subtraction, rather than either divergent component in isolation.

The finite coefficient is reduced by the same rules to 26 terms. Its origin expansions at two independent values of H are

$$\mathcal{F}_{\mathrm{UV}}(9,1+\zeta) = \left(\frac{24064}{729} - \frac{28672\ln 3}{243}\right)\zeta + O(\zeta^2), \tag{7.14}$$

$$\mathcal{F}_{\mathrm{UV}}(3/5,1+\zeta) = \left(\frac{34816}{15} + \frac{114688}{135}\ln\frac{5}{3}\right)\zeta + O(\zeta^2). \tag{7.15}$$

The integrated endpoint values are

$$\int_1^{10} \mathcal{F}_{\mathrm{UV}}(9,v)\,\mathrm{d}v = 114586.931264374645722061177843892 4\ldots, \tag{7.16}$$

$$\int_1^{8/5} \mathcal{F}_{\mathrm{UV}}(3/5,v)\,\mathrm{d}v = 504.446288248308891415453736676 4\ldots. \tag{7.17}$$

At 65-digit precision, independent quadrature differs by 6.3×10^{-60} and 4.2×10^{-62}, respectively. These values concern this angular difference sector before the overall normalization and the other sectors are assembled. The callable routine `uv_difference_closed` returns its pole or finite coefficient from the generated polylogarithmic expression. It performs no quadrature.

7.7.10 Check of the UW pole integral

At $H=9$ its integrand begins as $-1152t+O(t^2)$, and

$$\int_0^1 g_1(9,t)\,\mathrm{d}t = -15.922740461335295244261781552 3\ldots. \tag{7.18}$$

At 60-digit working precision this expression agrees with independent quadrature to 2.6×10^{-56}.

7.7.11 Checks of the UW function basis

The eight functions in the inner finite result have also been checked against their real logarithm and dilogarithm definitions. At $H=9$ and $t=0.1,0.5,0.9$, their polylogarithmic translations agree to better than 2.4×10^{-45} at 45-digit working precision. This includes the difference of two dilogarithms with large negative arguments near $t=0$. The identity is fixed by differentiation and its value at the origin, so the numerical check does not choose a physical branch.

7.7.12 Checks of the finite UW integral

Its endpoint values before the common overall normalization are

$$\mathcal{F}_{UW}(9) = -599.278509012081875840715039026856 3\ldots, \tag{7.19}$$

$$\mathcal{F}_{UW}(3/5) = 5.333721426195491037519313275871 36\ldots. \tag{7.20}$$

Independent coordinate quadrature differs by 2.4×10^{-74} and 1.3×10^{-65}, respectively. These differences are consistent with the estimated analytic tail remainder. The production routine `real_uw_closed` returns the pole and finite coefficients using only the polylogarithmic expression and convergent endpoint series. It does not include the crossed angular sector.

7.7.13 Check of the UW inner finite coefficient

The independent numerical check integrates the original a expression only after subtracting its two endpoint poles. It agrees with (3.448) at an interior point $H = 0.6, t = 0.3$ to better than 10^{-43} relatively. The numerical integral is a check, not a definition of the answer.

7.7.14 Checks of the soft-boundary conversion

For the kernels $K = x^{-2}$ and $K = x^{-3}$ of Section 3.5.2, the comparison is as follows. At $H = 9$, evaluation of the exact Mellin formulas at $\eta = \pm10^{-22}$ with 90-digit arithmetic reproduces these finite terms to more than 42 digits. The exact cutoff formulas at $\zeta = 10^{-30}$ agree with their displayed asymptotic finite parts to more than 30 digits. These evaluations test analytic formulas and do not replace the integration by quadrature.

7.7.15 Checks of the Bose subtraction correction

The moments P, A, B, C and finite parts F_z, F_η use the definitions of Section 3.5.2. Here $H = s - 1$, and D is the rational correction in (3.498). An independent evaluation at $H = 9$ gives

$$P = 37497.977308534921150928535612 1\ldots, \tag{7.21}$$

$$A = 3813.015415377579449859084099 43\ldots, \tag{7.22}$$

$$B = -1194.395178431005510190399282 60\ldots, \tag{7.23}$$

$$C = 875.197982980881880082536498852\ldots, \tag{7.24}$$

$$F_\eta - F_z = 8697.248379879614621637256975 50\ldots. \tag{7.25}$$

These values are evaluations of the closed moments, with no numerical integration. They provide the finite matching contribution of this individual subtraction correction.

At $H = 9$, the resulting closed endpoint value is

$$F_z[D] = -34968.099264817270724925437024207650294723 1\ldots, \tag{7.26}$$

$$F_\eta[D] = -26270.850884937656103288180048711825381197 9\ldots. \tag{7.27}$$

Evaluations requested at 45 and 70 decimal digits agree in all the displayed digits. As an independent check only, quadrature of the one-dimensional kernel up to $u = s - 0.02$ agrees with its generalized-polylogarithm primitive to an absolute difference below 1.1×10^{-79} at 85-digit precision. The finite endpoint value itself uses the analytic asymptotic expansion, not quadrature or a fit to a cutoff sequence.

7.7.16 Finite parts of the real-energy primitives

Exact symbolic differentiation verifies the displayed GPL primitives for all three sectors. As independent checks, at $H = 9$ their z-cut finite parts are

$$F_{z,W} = -199.124507902257128044039631623365 9, \tag{7.28}$$

$$F_{z,\ln s} = 371.201221607027861054082102311937 4, \tag{7.29}$$

$$F_{z,\mathrm{reg}} = -13063.031140945009575350282664403 29. \tag{7.30}$$

These numbers do not include the common factor $1/(8sH)$ or the conversion from the z cut to the symmetric analytic regulator. Separate real integrations check the interior primitive boundary differences to better than fifty decimal places at this point. Continuing the full regular-sector GPL combination above and below its artificial interior letter agrees to eighty decimal places. These checks concern evaluation of the analytic primitives and do not supply coefficients to them.

7.7.17 Checks of the real soft residues

The following checks evaluate the analytic residue moments and their integrated sum. At $H = 0.6, 9, 99$, separate 70-digit quadratures of these residue moments agree with the analytic expressions to better than 10^{-60} relatively.

At $H = 9$ this correction is

$$A_D = 3813.01541537757944985908409942886783767351749\ldots. \tag{7.31}$$

Evaluating the analytic expressions on both sides of (3.509) gives

H	Common value	Relative difference
0.6	0.976756780072123721541847283885	7.0×10^{-68}
9	730.42315282183326937494670874 3	5.0×10^{-70}
99	50871.58451891614048348018690 55	6.0×10^{-66}

This is a numerical check of the separately derived analytic coefficients at three energies. It does not replace an identity proof for arbitrary H, nor does it verify the finite total.

7.7.18 Fixed-energy checks of endpoint subtraction

At $H = 9$ the pole integrand has the local expansion

$$\mathcal{P}_{\mathrm{UV}}(9, 1+\zeta) = -\frac{7168}{243}\zeta + O(\zeta^2). \tag{7.32}$$

Thus its apparent origin poles cancel and the lower endpoint is regular.

As an independent check, the coefficient of its analytic-regulator pole at $H = 9$ integrates to

$$P_{\mathrm{shift}}(9) = 20\ln 10 + \frac{2913280}{59319}\ln\frac{11}{2} + \frac{5602208}{1521}. \tag{7.33}$$

This term checks the subtraction bookkeeping independently of the finite-angle polylogarithm evaluation.

7.8 Checks of the analytic simplification

These checks compare the simplified expressions with the original independent sector results. The algebraic derivations above fix all coefficients and boundary values before any numerical comparison.

7.8.1 Checks of the classical root functions

The reduction was also checked against the defining integrals at $X = 10^{-6}$, 10^{-2}, 0.6, 1, 10 and 10^4, using 70-digit arithmetic. The substitution $X = 4\sinh^2(y/2)$ removes the square-root kernel from those numerical checks. All 36 endpoint comparisons have absolute differences below 1.2×10^{-68}. Independent differentiation checks of all six classical expressions at five of these points also reproduce their defining kernels. These numerical checks supplement the primitive identities and boundary derivations above. They do not supply any part of the analytic expressions.

7.8.2 Checks of the simplified soft primitive

At an interior endpoint, the old primitive equals $\widetilde{\Psi} + 4apf/(\beta E^2 dk)$, which permits a direct check without a numerical integral. At $z = 0.7Z$, evaluating both GPL expressions with 65 decimal digits gives differences below 2.6×10^{-63} for $s = 1.6$ and $s = 10$. These checks support the implementation of the exact relation (3.247).

7.8.3 Checks of the collected soft kernel

As a cross-check of this reorganization, numerical integration of the regular kernel gives

$$\mathcal{C}_K(1.6) = 0.17122655099905486957125621095221829796 61\ldots, \tag{7.34}$$

$$\mathcal{C}_K(10) = 10.05846223263197683774454833268801090816 43\ldots. \tag{7.35}$$

These values agree with the previous analytic soft expression. The numerical integrations test the algebraic reorganization and do not replace the analytic expression.

7.8.4 Checks of the soft energy-variable integrals

Direct quadrature of the defining I_1, I_2, I_7 integrals at $X = 10^{-3}, 0.6, 9, 100$ agrees with their energy-variable analytic expressions to better than 3×10^{-72} in absolute value for I_2 and I_7. The integral evaluations in this check are not used to determine any coefficient or boundary constant.

7.8.5 Checks of the integrated UW expression

Evaluation of the fixed-alphabet result and the original finite GPL primitive gives

$$F^{\eta}_{UW}(0.6) = 5.33372142619549103751931327587135840926047\ldots, \tag{7.36}$$

$$F^{\eta}_{UW}(9) = -599.27850901208187584071503902685630877154 3\ldots. \tag{7.37}$$

The absolute differences are 1.4×10^{-65} and 3.1×10^{-63}, respectively. These checks concern the UW contribution. The conversion to (4.520) uses the exact coefficient and shuffle identities above.

7.8.6 Checks of the matched real contribution

Fresh evaluations of the original four GPL primitives at $s = 1.6$ and 10 agree with the endpoint formula before matching. Adding the original finite shifts gives agreement with (4.493) through the 45 significant digits recorded for those original values. These checks use our own integrated expressions. No published total is substituted.

7.8.7 Checks of the four-sector endpoints

The coefficients A_4, D_4 extracted from the collected primitive were also checked against the Laurent expansions of the four original integrands at $s = 1.01$, 1.6 and 10, using 70-digit arithmetic. This checks the endpoint signs and normalization independently of the primitive coefficient collection.

The conjugate-pair endpoint

The function $\mathcal{A}_{\mathrm{pair}}(s)$ is defined in (4.487). Direct quadrature of $P' \ln B/P$ at six points from $s = 1.000001$ to 10^4 agrees with this expression to an absolute error below 6×10^{-70} in 70-digit arithmetic.

7.8.8 Checks of the direct-box basis conversion

The independent direct-box conversion is derived in Section 4.2. As a check of its implementation, evaluation at $s = 1.6$ and $s = 10$ differs from the original closed endpoint representation by less than 3.4×10^{-78}. The exact derivative and boundary arguments, rather than these two numbers, establish the basis transformation.

7.9 Numerical evaluation of the published reference

Only after completing the amplitude integration do we evaluate the published total in Equation (6) of Lee, Schwartz and Zhang [2]. Its thirteen iterated-integral contributions are combined into one integral in $t = \ln z$, so the lower-weight functions at a common argument are computed once. Near $z = 0$, convergent power series evaluate the lower-weight functions without cancellation between constant terms. This numerical integration evaluates the special functions in the published reference. It is not an input to the independent derivation or to its closed endpoint formulas.

We compare evaluations with 75 and 90 decimal digits and lower endpoints $t = -140$ and $t = -165$. The omitted tails have analytic absolute bounds. For example, for $0 < z < 1$ the lower-weight words obey

$$|G(-1, 0; z)| \le z(1 + |\ln z|), \tag{7.38}$$

$$0 \le G(-2, -1; z) \le z^2/4, \tag{7.39}$$

$$0 \le G(r, -1; z) \le z^{3/2}/3. \tag{7.40}$$

Here the letter r denotes the kernel $1/\sqrt{z(z+4)}$, as in the published formula, rather than the angular variable used earlier. Multiplication by the outer kernels and integration of these elementary bounds controls the lower-endpoint truncation. The reported error estimate adds these tail bounds, ten times the difference between the two evaluations, and a rounding estimate based on working precision and the absolute sizes of the individual coefficients. It is an independent numerical error estimate, not an interval-arithmetic proof and not an estimate inferred from agreement with our result. For the seven values $s = 1.0002$, 1.02, 1.6, 10, 100, 2001, and 200001, all reference error estimates are below 5.6×10^{-58}. At $s = 1.6$, evaluating each published iterated integral separately with 75-digit arithmetic and the same lower endpoint $t = -140$ differs from the combined evaluation by 7.74×10^{-75}. This checks the combination independently of the comparison with our cross section.

7.9.1 Evaluation of the reference words

The implementation evolves the thirteen words in $\mathcal{H}$ and uses the elementary expressions for $G(-1;X)$, $G(0,-1;X)$, $G(-1,0;X)$ and $G(-1,-1;X)$ from Section 6.2 for their lower-weight derivatives. At $\tau = \ln X$ the coefficient for a rational letter is $X/(X-a)$ and that for r is $\sqrt{X/(X+4)}$. Each derivative involves a lower-weight word. All thirteen unknowns vanish at $X = 0$. Starting at $\tau = -36$ with zero initial values neglects only endpoint powers, and repeating at $\tau = -40$ checks this approximation. Large cancellations among the terms of $\mathcal{H}$ make the accuracy of the final correction weaker than the accuracy of each individual G. The verification reports the error of the combined cross section as well as that of the functions.

Numerically the double-precision implementation is intended for $X \geq 0.01$ and the tested energies, rather than arbitrarily close to threshold or at arbitrarily large energy.

7.10 Checks of the collected total

At $X = 0.6, 9, 99$, the collected formula gives

$$C_1(0.6) = 0.39497522084295373429747165489072725825168951498178\ldots, \tag{7.41}$$

$$C_1(9) = 0.77515165401174580343102580969237768056470578000273\ldots, \tag{7.42}$$

$$C_1(99) = 0.35308838117224168652124506257313479270540259204215\ldots. \tag{7.43}$$

The absolute differences from the original independent analytic evaluations are 2.2×10^{-79}, 8.8×10^{-81}, and 1.1×10^{-85}, respectively. These are checks between two representations of our own calculation. No published total is used in the collection or in these comparisons.

7.11 Integrated benchmarks

The program `check_compton_nlo.py` tests four elementary reductions of the G functions at $X = 0.07, 0.5, 1, 10, 20000$. The maximum absolute error divided by $\max(1, |G|)$ is 7.5×10^{-14}. Tightening both differential-equation tolerances and lowering the starting point changes the correction at the three tested PrimEx energies by less than 2.5×10^{-8} of that correction. This is far below the experimental uncertainty.

For a head-on laser of energy 1.165 eV, transform an electron beam energy E to the rest-frame photon energy by

$$\omega = \omega_{\text{laser}} \frac{E + \sqrt{E^2 - m^2}}{m}. \tag{7.44}$$

Using $m = 0.51099906\,\text{MeV}$ and $\alpha_{\text{T}}^{-1} = 137.0359895$ gives Table 7.2. The reference values are theoretical predictions from Table 3 of Denner and Dittmaier [1], used here to check the numerical implementation. They are not experimental input to the PrimEx figure.

At center-of-mass energy $\sqrt{s_{\text{phys}}} = 1\,\text{GeV}$ the correction is 8.17%, which reproduces the 8.2% quoted in Reference [2]. These comparisons verify the implementation of Equation (6) of Lee, Schwartz and Zhang [2].

7.12 Experimental input and reproducibility

The data source is Table 2 of the PrimEx publication [3], the same thin-carbon measurements used in the book's tree-level comparison. An unchanged copy of the raw table is included

E [GeV]	$100\sigma_1/\sigma_0$ [%], evaluation	Reference [1], Table 3 [%]
4	0.0023092	0.0023
6	0.0044867	0.0045
8	0.0070869	0.0071

Table 7.2: Inclusive photonic corrections with no acceptance cuts. The laser energy is 1.165 eV in all three rows.

as `data/primex_carbon_raw.dat`. Its columns give the run period, photon energy in GeV, cross section in mb per electron, and separate statistical and systematic uncertainties in the same cross-section units. The 29 values are copied without fitting or rescaling.

The comparison uses $m = 0.51099895\,\mathrm{MeV}$, $\alpha_{\mathrm{T}}^{-1} = 137.035999084$, and $1\,\mathrm{GeV}^{-2} = 0.3893793721\,\mathrm{mb}$. The plotted error is the quadrature sum of the two tabulated errors. The published cross sections include background, luminosity and acceptance corrections and are not Born-unfolded quantities. A detector simulation is therefore not reapplied to the reported total. The experiment's extraction already uses a radiative-event model. The comparison does not redo that extraction.

At 5 GeV the tree and inclusive NLO predictions are 0.264648 mb and 0.273692 mb respectively.

All calculation programs are in `scripts/`. `compton_nlo.py` evaluates the exact total in Equation (6) of Lee, Schwartz and Zhang [2].

`generate_compton_nlo_comparison.py` evaluates the independently integrated total through `compton_total_independent_closed.py`, without calling the published formula. It writes sampled theory, experimental points and their errors as `.dat` files in the local `tex/figures/` directory. The vector figure is drawn by PGFPlots during the chapter build. Verification programs write their numeric reports in the `reports/` directory. The build instructions give the commands and required Python libraries.

Figure 7.1 compares the measured totals with both predictions. The inclusive radiative correction raises the curve over this energy range. The lower panel displays this change relative to the Born prediction using the same normalization for data and theory.

7.13 Independent numerical check of the real contribution

The numerical check integrates the original six-diagram spin sum, using explicit 4×4 Dirac matrices and transverse photon polarizations. It does not evaluate the GPL primitives or insert the published total cross section. We use the ordered-photon, locally subtracted integral in (3.167). Thus its output is precisely $\mathcal{C}_{\mathrm{R}}$ in (3.172), including the $\mathcal{A}_{\mathrm{loc}} \ln \omega_*$ term. No finite photon-energy cut remains.

To resolve the two electron directions, the angular sampling uses a normalized mixture of two positive densities. With $\beta = q/E$ and $\widehat{\boldsymbol{v}}_{i,f}$ the incoming and elastic outgoing electron directions, define

$$g_{i,f}(\widehat{\boldsymbol{n}}) = \frac{\beta}{2\pi \ln s} \frac{1}{1 - \beta \widehat{\boldsymbol{v}}_{i,f} \cdot \widehat{\boldsymbol{n}}}, \qquad g = \tfrac{1}{2}(g_i + g_f). \tag{7.45}$$

Each density integrates to one because $\ln[(1+\beta)/(1-\beta)] = \ln s$. Each channel is mapped to a unit square by using the logarithm of $1 - \beta \widehat{\boldsymbol{v}}_{i,f} \cdot \widehat{\boldsymbol{n}}$ and a uniform azimuth. The two channel contributions are averaged with weight $1/g$. The elastic polar angle uses the analogous logarithmic map for $E + qc_2$, and the last coordinate is ω/ω_*. The resulting finite four-dimensional integral is evaluated by adaptive cubature.

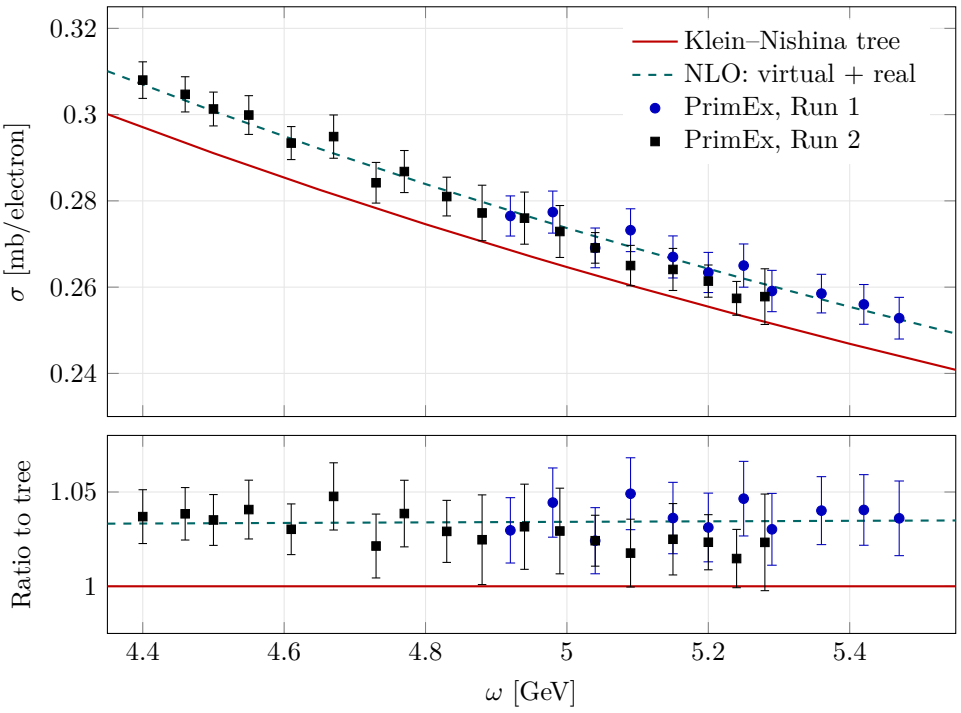


Figure 7.1: The PrimEx thin-carbon cross sections per electron, compared with the tree prediction and our analytically integrated NLO result, including virtual corrections and double Compton emission. Points are from Table 2 of Reference [3]. Run 1 and Run 2 denote the two data-taking periods. Error bars combine the reported statistical and systematic uncertainties in quadrature. The lower panel divides both data and theory by the tree prediction.

At each energy we repeat the calculation with increasingly small requested tolerances. The numerical uncertainty assigned to the last result is twice the largest of the last two cubature error estimates and the change between those two results. It is an empirical convergence estimate, not a rigorous bound. A run that exhausts its evaluation budget is explicitly distinguished from a run satisfying the requested tolerance. For the low-energy endpoints and $s = 2001$, the matrix and kinematic arithmetic use `long double`, while the cubature weights and accumulated results use double precision. At $s = 200001$ the double-precision calculation is also repeated independently with `long double` at the same requested tolerance. The saved audit records identify the source, compiler command, binary hash, requested tolerances, evaluation counts, and every refinement result.

For comparison with the symmetric analytic regulator, if its real coefficient is $A/\eta + F$, the precise conversion is

$$\mathcal{C}_{\mathrm{R}} = F - A\ln[2\sqrt{s}(s-1)]. \tag{7.46}$$

This follows from $xy \to 2\sqrt{s}(s-1)\omega$ at a soft endpoint. The factor for the identical photons is already accounted for by the ordering. Consequently this numerical result can be added directly to the independently integrated $\mathcal{C}_{\mathrm{VS}}$, without importing an analytic finite constant from the total cross section.

At the highest energy the successive real coefficients are

$$0.01360181858, \quad 0.01361650195, \quad 0.01362436493, \quad 0.01362842773. \tag{7.47}$$

The refinement changes decrease from 1.47×10^{-5} to 7.86×10^{-6} and then 4.06×10^{-6}. Accordingly the assigned uncertainty is 8.13×10^{-6}, larger than the final cubature error estimate. At the penultimate tolerance, the independent double-precision and `long double` runs each use 4 157 469 evaluations and differ by 2.31×10^{-15}. The nodes were not recorded, so equal evaluation counts do not establish an identical adaptive grid. This precision comparison is consistent with integration convergence dominating the reported uncertainty. These numerical tolerances do not limit the precision of the independently evaluated analytic formula.

7.14 Three-way comparison over the energy range

The coefficient compared here is

$$C(s) = C_1(s-1) = \frac{m^2\sigma_1}{\alpha_{\mathrm{T}}^3}, \qquad s = \frac{s_{\mathrm{phys}}}{m^2} = 1 + \frac{2\omega}{m}. \tag{7.48}$$

The photon energy ω is measured in the initial electron rest frame. The seven energies span $10^{-4} \le \omega/m \le 10^5$. The first value at each energy in Table 7.3 is our independent analytic coefficient, freshly evaluated from the collected expression in Equation (4.555). Its 50 displayed digits also agree with the original assembled sector primitives. All 50 displayed significant digits agree between two runs at 65 and 85 requested decimal digits, each with 40 guard digits. This agreement checks numerical stability. The endpoint series used to evaluate our primitives have truncation-error estimates, but a rigorous bound on every omitted tail has not been established. The displayed digits therefore do not constitute a certified 50-digit error bound.

The two subsequent entries are absolute coefficient differences,

$$\Delta C_{\mathrm{num}} = C_{\mathrm{num}} - C(s), \qquad \Delta C_{\mathrm{ref}} = C_{\mathrm{ref}} - C(s). \tag{7.49}$$

The differences use the full saved analytic value before rounding it to the 50 digits displayed in the table. For C_{num}, the virtual and soft terms are integrated numerically over the elastic angle from the pointwise amplitude. The real term is a separate four-dimensional phase-space integral of the six-diagram double-Compton matrix element, with its local soft subtraction. Neither calculation calls the analytic total or its integrated GPL primitives.

The real quadrature error estimate includes both the integrator's reported error and the change upon refinement. The virtual estimate includes angular refinement and an arithmetic accuracy floor. We add these estimates rather than treating them as independent statistical errors. Thus the ± entries are numerical error estimates, not confidence intervals or experimental uncertainties. The numerical integration is not required to reproduce 50 digits.

The reference coefficient is Equation (6) of Lee, Schwartz and Zhang [2]. It is evaluated separately, after our analytic result is fixed. Its error estimate uses two working precisions, changes in the special-function integration endpoint, an analytic bound on the omitted tail, and a rounding scale based on the sizes of its terms. It does not use the difference from our answer to estimate its own error. This comparison tests numerical evaluation of the two analytic expressions. Section 6.4 separately establishes their exact equality by comparing every coefficient and fixing their common physical continuation.

ω/m	Calculation	Coefficient or difference
0.000100	Our analytic	0.00000062423288400014031134169341516503703552376453828153
	Numerical difference	$5.30 \times 10^{-15} \pm 2.0 \times 10^{-12}$
	Reference difference	$-4.03 \times 10^{-80} \pm 5.6 \times 10^{-58}$
0.0100	Our analytic	0.00291875646655687689956024812442444316283020677482 37
	Numerical difference	$1.11 \times 10^{-10} \pm 2.0 \times 10^{-8}$
	Reference difference	$2.37 \times 10^{-86} \pm 6.4 \times 10^{-65}$
0.300	Our analytic	0.39497522084295373429747165489072725825168951498178
	Numerical difference	$2.72 \times 10^{-9} \pm 6.8 \times 10^{-7}$
	Reference difference	$8.37 \times 10^{-90} \pm 1.1 \times 10^{-69}$
4.50	Our analytic	0.77515165401174580343102580969237768056470578000273
	Numerical difference	$5.71 \times 10^{-8} \pm 1.2 \times 10^{-6}$
	Reference difference	$2.40 \times 10^{-91} \pm 3.7 \times 10^{-72}$
49.5	Our analytic	0.35308838117224168652124506257313479270540259204215
	Numerical difference	$-3.86 \times 10^{-8} \pm 1.2 \times 10^{-6}$
	Reference difference	$-6.91 \times 10^{-89} \pm 1.0 \times 10^{-71}$
1.00×10^{3}	Our analytic	0.07046389661013719899011704900339950390684838336 0028
	Numerical difference	$3.33 \times 10^{-6} \pm 1.5 \times 10^{-5}$
	Reference difference	$-4.10 \times 10^{-87} \pm 6.7 \times 10^{-71}$
1.00×10^{5}	Our analytic	0.00287045288320063582221889949830774466811293889457 86
	Numerical difference	$-1.34 \times 10^{-6} \pm 8.1 \times 10^{-6}$
	Reference difference	$-9.38 \times 10^{-87} \pm 5.5 \times 10^{-70}$

Table 7.3: Three independent evaluations of the NLO Compton correction $\sigma_1 = (\alpha_{\mathrm{T}}^3/m^2)C_1$, from low to high incident energy in the electron rest frame. Our analytic coefficient C_1 is evaluated from the collected formula and printed to 50 significant digits. Each difference subtracts this coefficient from the indicated calculation. The ± entries estimate numerical error, not experimental uncertainty. The reference is Equation (6) of Lee, Schwartz and Zhang [2].

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